# Practical Manual on

# Crop Production



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# Practical Crop Production

ICAR e-Course

For

B.Sc (Agriculture) and B.Tech (Agriculture)



**AGRO202, 203 Practical Crop Production- I & II**

**(0+2)**

Crop Choice is according to season

The crops are:

Irrigated puddled lowland rice / any irrigated dry crop (maize / sorghum / pearl millet / finger millet / cotton / groundnut / sunflower / sesame

The student him/herself is doing the cultivation practices.

*If transplanted crop:*

Selection of nursery area - Preparation of nursery - Application of manures to nursery - seed treatment with fungicide and bio-fertilizers - seed soaking and incubation - forming nursery beds and sowing seeds - weed control and plant protection to nursery - preparation of main field - application of organic manures - basal application of fertilizers and bio-fertilizers - pulling out seedlings and transplanting - application of herbicides - after cultivation practices - top dressing of fertilizers, plant protection measures - harvesting, threshing, drying and cleaning the produce - working out cost of cultivation and economics.

*If direct sown crop:*

Selection of main field - Preparation of main field - land configuration for sowing - application of manures and fertilizers - seed treatment with fungicide and bio-fertilizers - sowing and irrigation - application of herbicides - after cultivation practices - top dressing of fertilizers, plant protection measures - harvesting, threshing, drying and cleaning the produce - working out cost of cultivation and economics

# Crop Production Guide

## RICE (*Oryza sativa* L.)

### I. SEASON AND VARIETIES

| District/Season | Month | Varieties |
|---|---|---|
| **1. Kanchipuram/Tiruvallur** | | |
| Sornavari | (April -May) | ADT 36, IR 36, IR 50, ADT 37, ASD 16, ASD 17, IR 64, ASD 18, ADT 42, MDU 5, ASD 20, ADT43, CO 47, TRY (R)2*, ADT (R) 45, ADTRH 1, ADT (R) 47 |
| Samba | (Aug) | IR 20, White Ponni, CO 43, ADT 40, PY 4, ADT 39, TRY 1, ASD 19, ADT(R) 44, CORH 2 |
| Late Samba | (Sep - Oct) | IR 20, White Ponni, ADT 39, CO 43, TRY 1, ADT (R)46, CORH 2 |
| Navarai | (Dec -Jan) | ADT 36, ADT 37, ASD 16, IR 64, ASD 18, ADT 42, ADT 43 MDU 5, ASD 20 |
| Dry | (July - Aug) | PMK 2, MDU 5, TKM 11, PMK (R) 3, TKM (R) 12 |
| Semi-dry | (July - Aug) | IR 20, TKM 10, PMK 2, MDU 5, TKM 11, TKM (R) 12, PMK (R) 3 |
| **2. Vellore/Tiruvannamalai** | | |
| Sornavari | (April-May) | IR 64, ADT 36, IR 50, ADT 37, ASD 16, ASD 17, ASD 18, ADT 42, MDU 5, ASD 20, ADT 43, CO 47, ADT (R) 45, ADT RH1, ADT (R) 47 |
| Samba | (Aug) | Ponmani, ADT 40, Bhavani, IR 20, White Ponni, CO 43, Paiyur 1, PY 4, CO 45, TRY 1, ASD 19, CORH 2 |
| Navarai | (Dec -Jan) | ADT 36, IR 20, ADT 39, CO 43, IR 64, ASD 16, ASD 18, ADT 42, MDU 5, CO 47, ASD 20, TRY (R)2* |
| **3. Cuddalore/ Villupuram** | | |
| Sornavari | (April -May) | ADT 36, IR 50, ASD 16, IR 64, ASD 18, ADT 42, MDU 5, ASD 20, ADT 43, CO 47, ADT (R) 45, TRY (R)2*, ADTRH 1, ADT (R) 47 |
| Samba | (Aug) | IR 20, White Ponni, CO 43, Ponmani, PY 4, ADT 38, TRY 1, ASD 19, ADT (R) 44, CORH 2 |
| Navarai | (Dec-Jan) | ADT 36, IR 20, IR 36, IR 64, ADT 39, ASD 16, ASD 18, ADT 42, MDU 5, ASD 20, TRY (R)2* |
| **4. Tiruchirappalli/Karur/Perambalur** | | |
| Kuruvai | (Jun -Jul) | ADT 36, IR 50, IR 64, ASD 16, ADT 37, ASD 18, ADT 42, MDU 5, ADT 43, CO 47, ADT (R) 45 (except Karur), TRY (R)2*, ADTRH 1, ADT (R) 47 |
| Samba | (Aug) | IR 20, White Ponni, CO 43, ADT 40, Ponmani, TRY 1, ASD 19, ADT (R) 44 |
| Late Samba / Thaladi | (Sep -Oct) | IR20, WhitePonni, ADT39, CO43, TRY1, ASD19, ADT(R)46, |
| Navarai | (Dec -Jan) | ADT 36, IR 64, ASD 16, ASD 18, ADT 42, MDU 5, ASD 20, TRY (R)2* |
| **5. Thanjavur/Nagapattinam/Tiruvarur** | | |
| Kuruvai | (Jun -Jul) | ADT 36, IR 50, IR 64, ADT 37, ASD 16, ASD 18, ADT 42, MDU 5, ADT 43, ADT (R) 45, TRY (R) 2*, ADTRH 1, ADT (R) 47, ,ADT (R) 48 |
| Samba | (Aug) | IR 20, White Ponni, CO 43, Ponmani, ADT 38, TRY 1, ASD 19, ADT (R) 44, CORH 2 |
| Late Samba / Thaladi | (Sep -Oct) | ADT 38, IR 20, CO 43, Ponmani, ADT 39, TRY 1, ASD 19, ADT (R)46, |
| Navarai (Kullankar) | (Dec -Jan) | ADT 36, ADT 37, IR 64, ASD 16, ASD 18, ADT 42, MDU 5, ASD 20, TRY (R)2* |

| | | |
|---|---|---|
| **6. Pudukottai** | | |
| Kuruvai | (Jun -Jul) | ADT 36, IR 50, IR 64, ASD 16, ADT 42, MDU 5, ASD 20, ADT 43, ADT (R) 45, TRY (R) 2*, ADTRH 1, ADT (R) 47 |
| Samba | (Aug) | IR 20, White Ponni, CO 43, Ponmani, TRY 1, ASD 19, ADT (R) 44, CORH 2 |
| Late Samba/Thaladi | (Sep - Oct) | IR 20, ADT 38, ADT 39, TRY 1, ASD 19, CO 43, ADT (R)46 |
| Dry | (Jul -Aug) | ADT 36, PMK 2, TKM 10, TKM (R) 12, PMK (R) 3 |
| Semi-dry | (Jul -Aug) | ADT 36, PMK 2, TKM 10, TKM (R) 12, PMK (R) 3 |
| **7. Madurai/Dindigul/Theni** | | |
| Kar | (May - Jun) | ADT 36, IR 50, IR 36, IR 64, ADT 37, ASD 16, ASD 18, ADT 42, MDU 5, ASD 20, ADT 43, CO 47, ADT (R) 45 (Dindigul only), TRY (R) 2*, ADTRH 1, ADT (R) 47 |
| Samba | (Aug) | IR 20, White Ponni, CO 42, CO 43, ADT 38, ADT 40, MDU 4, TRY 1, ASD 19, ADT (R) 44, CORH 2 |
| Late Samba/Thaladi | (Sep - Oct) | IR 20, White Ponni, MDU 3, ADT 39, MDU 4, CO 43, ASD 19, TRY 1, ADT (R)46 |
| Navarai | (Dec -Jan) | IR 64, ADT 36, ADT 37, ASD 16, ASD 18, ADT 42, MDU 5, ASD 20, TRY (R) 2* |
| Semi-dry | (Jul -Aug) | PMK 2, TKM 10, MDU 5, TKM (R) 12, PMK (R) 3 |
| **8. Ramanathapuram** | | |
| Samba | (Aug) | IR 20, White Ponni, CO 43, MDU 3, ASD 19, TRY 1, ADT (R) 44, CORH 2 |
| Rainfed & Semidry | (Jul -Aug) | ASD 17, ADT 36, PMK 2, MDU 5, TKM (R) 12, PMK (R) 3 |
| **9. Virudhunagar** | | |
| Samba | (Sep-Oct) | CO 43, TRY 1, IR 20, ADT (R)46, ADT 39, CORH 2 |
| Dry | (Jul -Aug) | ADT 36, PMK 2, MDU 5, TKM (R) 12, PMK (R) 3 |
| **10. Sivaganga** | | |
| Semi-dry | (Jul –Aug) | ADT 36, IR 36, ADT 39, PMK 2, MDU 5, TKM (R) 12, PMK (R) 3 |
| **11. Tirunelveli, Thoothukudi** | | |
| Early Kar | (Apr - May) | IR 50, ADT 36, IR 64, ADT 42, ADT 43, ADT 45, CO 47, ADT (R) 47 |
| Kar | (May -Jun) | ASD 16, ASD 17, ASD 18, ADT 42, ADT 43, CO 47, ADT (R) 45, TRY (R) 2*, ADTRH 1, ADT (R) 47 |
| Late Samba/Thaladi | (Sep - Oct) | White Ponni, IR 20, ADT 39, ASD 19, TRY 1, ADT (R)46, CORH 2 |
| Pishanam/Late Pishanam | (Sep-Oct.) | ASD 18, ASD 16, ASD 19, CO 43, TRY 1, ADT (R)46 |
| Semi Dry | (July- Aug) | MDU 5, ADT 36, TKM (R) 12, PMK (R) 3 |
| **12. Kanyakumari** | | |
| Kar | (May –Jun) | ADT 36, IR 50, IR 64, ASD 16, ASD 17, ASD 18, ADT 42, MDU 5, ASD 20, ADT 43, ADT 45, CO 47, ADTRH 1, ADT (R) 47 |
| Pishanam / Late Samba / Thaladi | (Sep – Oct) | White Ponni, IR 20, Ponmani, CO 43, TRY 1, TPS 2, TPS 3, ADT (R) 44, ADT 39, ASD 18, ASD 19, MDU 5, ADT (R) 46 |
| Semi-dry | (Jul – Aug) | ADT 36, ASD 17, PMK 2, TKM (R) 12, PMK (R) 3 |
| **13. Salem, Namakkal** | | |
| Kar | (May - Jun) | IR 50, ADT 36,IR 64, ADT 37, ASD 16, ASD18, ADT 42, MDU 5, ASD 20, ADT 43, CO 47, ADT (R) 45, TRY (R)2*, ADTRH 1, ADT (R) 47 |
| Samba | (Aug) | IR 20, White Ponni, Bhavani, CO 43, MDU 4, TRY 1, ASD 19, ADT (R) 44 |
| Navarai | (Dec - Jan) | IR 20, ADT 36, IR 64, ASD 18, ASD 19, ADT 42, MDU 5, ASD 20, TRY (R)2* |
| **14. Dharmapuri/ Krishnagiri** | | |
| Kar | (May -Jun) | IR 50, IR 64, ASD 16, Bhavani, IR 20, White Ponni, CO 43, ASD 18, MDU 4, ASD 19, PAIYUR 1, ADT 42, TRY 1, |

| | | |
|---|---|---|
| | | MDU 5, ASD 20, ADT 43, CO 47, ADTRH 1, TRY (R)2*, ADT (R) 47 |
| Navarai | (Dec- Jan) | IR 64, ADT 37, ASD 16, ADT 36, ASD 18, ADT 42, MDU 5, ASD 20, TRY (R)2* |
| Samba/Late Samba | (Aug - Oct) | TRY 1, Bhavani, IR 20, White Ponni, CO 43, MDU 4, ASD 19, ADT (R) 44, ADT (R) 46 |
| **15. Coimbatore** | | |
| Kar | (May - Jun) | IR 50, ADT 36, ASD 16, IR 64, ASD 18, ADT 42, MDU 5, ASD 20, ADT 43, CO 47, ADT (R) 45, TRY (R)2*, ADTRH 1, ADT (R) 47 |
| Samba | (Aug) | IR 20, CO 43, White Ponni, ADT 39, MDU 4, TRY 1, ASD 19, Bhavani, ADT(R) 44, CORH 2 |
| Late Samba/Thaladi | (Sep - Oct) | IR 20, ADT 39, ADT(R) 46, CORH 2 |
| Navarai | (Dec -Jan) | IR 20, ADT 36, IR 64, ASD 16, ASD 18, TRY1, MDU 5, ASD 20, TRY (R) 2* |
| **16. Erode** | | |
| Kar | (May - Jun) | IR 50, ASD 16, IR 64, ADT 36, ASD 18, ADT 42, MDU 5, ASD 20, ADT 43, CO, 47, ADT (R) 45, TRY (R)2*, ADTRH 1, ADT (R) 47 |
| Samba | (Aug) | IR 20, Bhavani, CO 43, White Ponni, ADT 39, TRY 1, CO 46, ADT (R) 44 |
| Late Samba | (Sep - Oct) | IR 20, White Ponni, ADT 39, CO 43, TRY 1, CO 46, ADT (R) 46, CORH 2 |
| Navarai | (Dec – Jan) | IR 20, ADT 36, IR 64, ASD 16, ASD 18, ADT 42, MDU 5, ASD 20 |
| **17. The Nilgiris** | | |
| Samba | (Jul -Aug) | IR 20, CO 43, TRY 1, ADT (R) 44 |

* suitable for salt affected soils

*Note of Caution of the varieties: IR 50 and ADT43 are recommended for Kar, Sornavari and Kuruvai seasons and should not be grown during cold weather period. ASD19 and White Ponni are also susceptible to blast and care should be taken on plant protection measures.*

## II. PARTICULARS OF RICE VARIETIES

| PARTICULARS | PY 4 (JAWAHAR) | IR 20 | BHAVANI |
|---|---|---|---|
| Parentage | IR 8 X H 4 | IR 262 X TKM 6 | Peta x BPI 76 |
| Duration (Days) | 145 - 150 | 130 -135 | 130 - 135 |
| Average Yield (kg/ha) | 5330 | 5000 | 5000 |
| 1000 grain wt (g) | 24.8 | 19 | 21.5 |
| Grain L/B ratio | 3.1 | 3 | 4.72 |
| Grain type | Long bold | Medium slender | Long Slender |
| **Morphological Characters** | | | |
| Habit | Semi dwarf, erect | Semi-dwarf | Medium tall |
| Leaf sheath | Green | Green | Green |
| Septum | Green | Cream | Cream |
| Ligule | Colourless | White | Colourless |
| Auricle | Colourless | White | Colourless |
| Panicle | Long drooping | Medium compact | Long compact |
| Husk colour | Straw | Straw | Straw |
| Rice colour | White | White | White |
| Abdominal white | Absent | Absent | Absent |
| **Grain size (mm)** | | | |
| Length | 9.3 | 8.2 | 9.6 |
| Breadth | 3 | 2.7 | 2.03 |

| | | | |
|---|---|---|---|
| Thickness | 1.8 | 2 | 1.5 |

## II. PARTICULARS OF RICE VARIETIES (CONTD . . . )

| PARTICULARS | ADT 36 | IR 50 | IR 64 |
|---|---|---|---|
| Parentage | Triveni x IR 20 | IR 2153-14 X IR 28 X IR 36 | IR 5657-33-2-1/ IR 2061-465-1-5-3 |
| Duration (Days) | 110 | 105(Summer) 130 (Winter) | 115 – 120 |
| Average Yield (kg/ha) | 4000 | 6000 | 6146 |
| 1000 grain wt (g) | 20.6 | 20.35 | 23.1 |
| Grain L/B ratio | 3.1 | 3.9 | 3.25 |
| Grain type | Medium | Long Slender | Long slender |
| **Morphological Characters** | | | |
| Habit | Erect | Erect | Semi dwarf |
| Leaf sheath | Green | Green | Green |
| Septum | Green | Green | White |
| Ligule | Colourless | Colourless | Light green |
| Auricle | Colourless | Colourless | Light green |
| Panicle | Long compact | Long & drooping | Intermediate, well exerted |
| Husk colour | Straw | Straw | Straw |
| Rice colour | White | White | White |
| Abdominal white | Absent | Absent | Absent |
| **Grain size (mm)** | | | |
| Length | 7.8 | 8.9 | 10.1 |
| Breadth | 2.5 | 2.3 | 2.9 |
| Thickness | 2 | 1.8 | 2.2 |

## II. PARTICULARS OF RICE VARIETIES (CONTD . . . )

| PARTICULARS | CO 43 | Ponmani | White Ponni |
|---|---|---|---|
| Parentage | Dasal x IR 20 | Pankaj x Jagannath | Taichung 65/2 Mayang Ebos-80 |
| Duration (Days) | 135 - 140 | 155 - 160 | 135 - 140 |
| Average Yield (kg/ha) | 5200 | 5300 | 4500 |
| 1000 grain wt (g) | 20 | 23.5 | 16.4 |
| Grain L/B ratio | 3.5 | 2.2 | 3.22 |
| Grain type | Medium slender | Short bold | Medium slender |
| **Morphological Characters** | | | |
| Habit | Erect | Erect | Medium tall |
| Leaf sheath | Green | Green | Green |
| Septum | Green | Green | Green |
| Ligule | White, longer | White | White |
| Auricle | Colourless | Colourless | Colourless |
| Panicle | Long drooping | Medium drooping | Long drooping |
| Husk colour | Straw | Straw | Straw |
| Rice colour | White | White | White |
| Abdominal white | Absent | Absent | Absent |
| **Grain size (mm)** | | | |
| Length | 8.1 | 6.9 | 8 |

| | | | |
|---|---|---|---|
| Breadth | 2.3 | 3.1 | 3 |
| Thickness | 1.8 | 2.1 | 2 |

## II. PARTICULARS OF RICE VARIETIES (CONTD . . . )

| PARTICULARS | IR 36 | MDU 3 | CO 47 |
|---|---|---|---|
| Parentage | IR 1561-228/1 IR 244/O.nivara./ CR 94-13 | IR 8/W 1263 | IR 50 / CO 43 |
| Duration (Days) | 120 | 120 - 125 | 110-115 |
| Average Yield (kg/ha) | 5000 | 4970 | 5832 |
| 1000 grain wt (g) | 21 | 23.1 | 20.6 |
| Grain L/B ratio | 2.88 | 3.92 | 2.7 |
| Grain type | Medium | Long Slender | Medium slender |
| **Morphological Characters** | | | |
| Habit | Dwarf Erect | Semi dwarf | Erect |
| Leaf sheath | Green | Green | Green |
| Septum | Cream | Cream | Yellow |
| Ligule | Colourless | White | Acute to acuminate, white |
| Auricle | Colourless | White, 2-Clefted | Colourless |
| Panicle | Compact | Medium Compact | Compact |
| Husk colour | Straw | Straw | Straw |
| Rice colour | White | White | White |
| Abdominal white | Absent | Absent | Absent |
| **Grain size (mm)** | | | |
| Length | 8.85 | 9.4 | 7.7 |
| Breadth | 3.07 | 2.4 | 2.3 |
| Thickness | 2.18 | 1.6 | 1.7 |

## II. PARTICULARS OF RICE VARIETIES (CONTD . . . )

| PARTICULARS | ASD 16 | ASD 17 | ADT 37 |
|---|---|---|---|
| Parentage | ADT 31/CO 39 | ADT 31/RATNA/ / ASD 8/IR 8 | BG 280-1 2/ PTB 33 |
| Duration (Days) | 110 - 115 | 101 | 105 |
| Average Yield (kg/ha) | 5600 | 5422 | 6200 |
| 1000 grain wt (g) | 24.2 | 23.8 | 23.4 |
| Grain L/B ratio | 2.6 | 2.24 | 1.79 |
| Grain type | Short Bold | Short bold | Short bold |
| **Morphological Characters** | | | |
| Habit | Semi dwarf erect | Semi dwarf slightly open | Semi dwarf semi erect |
| Leaf sheath | Green | Green | Green |
| Septum | Cream | Cream | White |
| Ligule | White | White, 2-clefted | White |
| Auricle | Colourless | Pale green | White |
| Panicle | Long Compact | Long, Intermediate densely drooping, just exerted | Compact |
| Husk colour | Straw | Straw | Straw |
| Rice colour | White | White | White |
| Abdominal white | Present | Present | Present |
| **Grain size (mm)** | | | |
| Length | 7.86 | 7.9 | 5 |
| Breadth | 3.02 | 2.8 | 2.8 |
| Thickness | 1.96 | 1.88 | 1.88 |

## II. PARTICULARS OF RICE VARIETIES (CONTD . . . )

| PARTICULARS | ADT 38 | TPS 2 | ADT 39 |
|---|---|---|---|
| Parentage | IR 1529-680-3-2/ IR 4432-52-6-4/ IR 7963-30-2 | IR 26/CO 40 | IR 8/IR 20 |
| Duration (Days) | 130 - 135 | 125 - 130 | 120 - 125 |
| Average Yield (kg/ha) | 6200 | 4615 | 5000 |
| 1000 grain wt (g) | 21 | 23.5 | 18 |
| Grain L/B ratio | 3.2 | 2.89 | 2.9 |
| Grain type | Long Slender | Short Bold | Medium slender |
| **Morphological Characters** | | | |
| Habit | Semi dwarf, erect | Semi dwarf | Semi dwarf |
| Leaf sheath | Green | Green | Green |
| Septum | White | Cream | Light Cream |
| Ligule | White Non-prominent | White | Papery white |
| Auricle | White | White | Non-pigmented |
| Panicle | Long moderately dense | Medium | Medium, Modera tely dense |
| Husk colour | Straw | White | Straw |
| Rice colour | White | White | White |
| Abdominal white | Absent | Present | Absent |
| **Grain size (mm)** | | | |
| Length | 6.9 | 8.1 | 7.6 |
| Breadth | 2.4 | 2.8 | 2.3 |
| Thickness | 2 | 2 | 1.9 |

## II. PARTICULARS OF RICE VARIETIES (CONTD . . . )

| PARTICULARS | ASD 18 | MDU 4 | PMK 2 |
|---|---|---|---|
| Parentage | ADT 31/IR 50 | AC 2836/Jagannath | IR13564-149-3/ASD 4 |
| Duration (Days) | 105 - 110 | 120 - 125 | 110-115 |
| Average Yield (kg/ha) | 5900 | 5900 | 3200 |
| 1000 grain wt (g) | 21.8 | 22.9 | 22.1 |
| Grain L/B ratio | 3.2 | 4 | 2.53 |
| Grain type | Medium slender | Long slender | Medium bold |
| **Morphological Characters** | | | |
| Habit | Semi dwarf(90cm) | Erect, tall | Erect |
| Leaf sheath | Pale Green | Green | Green |
| Septum | Light green | Green | Cream |
| Ligule | White clefted | Colourless | Palegreen |
| Auricle | Pale green | Colourless | Palegreen |
| Panicle | Medium, compact exerted | Compact, Medium | Medium, compact |
| Husk colour | Straw | Yellow | straw |
| Rice colour | White | White | Dull white |
| Abdominal white | Slightly present | Absent | Present |
| **Grain size (mm)** | | | |
| Length | 8.64 | 9.12 | 7.8 |
| Breadth | 2.7 | 2.26 | 3 |
| Thickness | 2.2 | — | 2 |

## II. PARTICULARS OF RICE VARIETIES (CONTD . . . )

| PARTICULARS | TKM 10 | TPS 3 | ADT 42 |
|---|---|---|---|
| Parentage | CO 31/C22 | RP 31-492/LMN | AD9246/ ADT29 |
| Duration (Days) | 135 | 135 -140 | 115 |
| Average Yield (kg/ha) | 2563 | 5253 | 5537 |
| 1000 grain wt (g) | 23.2 | 23.2 | 24.8 |
| Grain L/B ratio | 3.6 | 2.06 | 3.6 |
| Grain type | Medium slender | Short Bold | Long, slender |
| **Morphological Characters** | | | |
| Habit | Semi tall | Semi dwarf/erect | Semi-dwarf |
| Leaf sheath | Green | Green | Green |
| Septum | Green | Cream | Cream |
| Ligule | Colourless | — | White |
| Auricle | Colourless | — | Palegreen |
| Panicle | Compact | Long | Intermediate |
| Husk colour | Light brown | Straw | straw |
| Rice colour | White | White | White |
| Abdominal white | Present | Present | Occasionally present |
| **Grain size (mm)** | | | |
| Length | 9 | 7.96 | 9.32 |
| Breadth | 2.53 | 3 | 2.58 |
| Thickness | 1.75 | 2 | 1.89 |

## II. PARTICULARS OF RICE VARIETIES (CONTD . . . )

| PARTICULARS | ASD 19 | TRY 1 | MDU 5 |
|---|---|---|---|
| Parentage | Lalnakanda/ IR 30 | IR578-172-2-2/ BR-1-2-B-1 | *O.glaberrima* x Pokkali |
| Duration(Days) | 127 (120-132) | 135-140 | 95 - 100 |
| Average Yield (kg/ha) | 5800 | 5255 | 4500 |
| 1000 grain wt(g) | 18.39 | 24 | 21.1 |
| Grain/L/B ratio | 3.06 | 2.6 | 3.12 |
| Grain type | Short, slender | Medium | Medium slender |
| **Morphological characters** | | | |
| Habit | Semi-dwarf, erect | Erect | Erect |
| Leaf sheath | Light green | Green | Green |
| Septum | Cream | White | - |
| Ligule | White | White | Colourless |
| Auricle | Palegreen | White | Colourless |
| Panicle | Compact, dense drooping & well exerted | Long, moderately compact | Intermediate |
| Husk colour | Straw | Straw | Straw |
| Rice colour | White | White | White |
| Abdominal white | Absent | Absent | - |
| **Grain size(mm)** | | | |
| Length | 8.28 | 6.2 | 8.45 |
| Breadth | 2.32 | 2.4 | 2.7 |
| Thickness | 1.72 | 1.8 | - |

## II. PARTICULARS OF RICE VARIETIES (CONTD . . . )

| PARTICULARS | ASD 20 | CO 46 | ADT 43 |
|---|---|---|---|
| Parentage | IR 18348/IR25863/ IR 58 | T7/IR 20 | IR 50/White ponni |
| Duration(Days) | 110 | 125 | 110 |
| Average Yield (kg/ha) | 6000 | 6000 | 5900 |
| 1000 grain wt(g) | 22.08 | 23.5 | 15.5 |
| Grain/L/B ratio | 3.12 | 3.14 | 2.81 |
| Grain type | Long Slender | Long Slender | Medium slender |
| **Morphological characters** | | | |
| Habit | Erect | Tall erect | Semi dwarf |
| Leaf sheath | Pale green | Green | Light green |
| Septum | Cream | - | Cream |
| Ligule | Pale white | - | White |
| Auricle | Pale green | Colourless | - |
| Panicle | Medium compact | Long, heavy & Compact | Moderately long, Intermediate type, drooping |
| Husk colour | Straw | Straw | Straw |
| Rice colour | White | White | White |
| Abdominal white | Absent | Absent | Very occasionally present |
| **Grain size(mm)** | | | |
| Length | 9.38 | 6.6 | 5.46 |
| Breadth | 2.18 | 2.1 | 1.94 |
| Thickness | 1.46 | 1.7 | 1.63 |

## II. PARTICULARS OF RICE VARIETIES (CONTD . . . )

| PARTICULARS | TKM 11 | ADTRH 1 | CORH 2 |
|---|---|---|---|
| Parentage | C22/BJ1 | IR 58025 A/IR 66 R | IR 58025 A/C 20R |
| Duration(Days) | 110 - 120 | 115 | 125 |
| Average Yield (kg/ha) | 3000 | 6400 | 6100 |
| 1000 grain wt(g) | 21.4 | 23.8 | 23.77 |
| Grain/L/B ratio | 3.2 | 3.46 | 2.62 |
| Grain type | Long slender | Long slender | Medium |
| **Morphological characters** | | | |
| Habit | Erect | Semi dwarf, erect | Semi dwarf |
| Leaf sheath | Green | Green | Green |
| Septum | Cream | Cream | Cream |
| Ligule | Colourless | White | White |
| Auricle | Light green | - | Absent |
| Panicle | Long, compact drooping | Long | Compact |
| Husk colour | - | - | Straw |
| Rice colour | White | White milky, scented | White |
| Abdominal white | - | Very occasionally present | Occasionally present |
| **Grain size(mm)** | | | |
| Length | 9.3 | 6.96 | 6.11 |
| Breadth | 2.3 | 2.01 | 2.33 |
| Thickness | 1.6 | 1.72 | 1.86 |

## II. PARTICULARS OF RICE VARIETIES (CONTD . . . )

| PARTICULARS | ADT (R) 44 | ADT (R) 45 | ADT (R) 46 |
|---|---|---|---|
| Parentage | IET 14099-IR56/ OR142-99 | IR50 / ADT 37 | ADT38 / CO 45 |
| Duration(Days) | 148 | 110 | 135 |
| Average Yield (Kg / ha) | 6214 | 5400 | 6656 |
| 1000 grain wt(g) | 23.9 | 17.5 | 23.8 |
| Grain/L/B ratio | 2.21 | 2.98 | 3.12 |
| Grain type | Short Bold | Medium slender | Long Slender |
| **Morphological characters** | | | |
| Habit | Medium Tall | Semi dwarf, erect | Erect, semi-dwarf |
| Leaf sheath | Light Green | Green | Green |
| Septum | Cream | Cream | Cream |
| Ligule | White | White | Long white |
| Auricle | - | - | Pale green |
| Panicle | Long Compact | Compact | Intermediate |
| Husk colour | Straw | Straw | Straw |
| Rice colour | White | White | White |
| Abdominal white | Present | Absent | Absent |
| **Grain size(mm)** | | | |
| Length | 8.07 | 8.00 | 9.58 |
| Breadth | 2.95 | 2.16 | 2.46 |
| Thickness | 2.06 | 1.97 | 1.95 |

## II. PARTICULARS OF RICE VARIETIES (CONTD . . . )

| PARTICULARS | TKM (R)12 | TRY (R)2 | PMK (R)3 |
|---|---|---|---|
| Parentage | TKM 9/ TKM 11 | IET 6238 / IR 36 | UPLRi 7/ CO 43 |
| Duration(Days) | 115-120 | 115-120 | 110-115 |
| Average Yield (Kg/ ha) | 3043 | 5362 | 3025 |
| 1000 grain wt(g) | 18.3 | 22.8 | 26.10 |
| Grain/L/B ratio | 2.42 | 3.5 | 2.84 |
| Grain type | Medium slender | Long slender | Long Bold |
| **Morphological characters** | | | |
| Habit | Erect | Semi-dwarf erect | Erect |
| Leaf sheath | Green | Green | Green |
| Septum | Cream | Light green | - |
| Ligule | White | distinct | Pale green |
| Auricle | Dull white | Hairy light brown | - |
| Panicle | Medium compact | Compact | Intermediate |
| Husk colour | Straw | Straw | Golden yellow with brown streaks |
| Rice colour | White | White | White |
| Abdominal white | Present | Absent | - |
| **Grain size(mm)** | | | |
| Length | 7.5 | 9.1 | 6.75 |
| Breadth | 3.1 | 2.6 | 2.38 |
| Thickness | 2.3 | 1.7 | 2.08 |

## II. PARTICULARS OF RICE VARIETIES (CONTD . . . )

| PARTICULARS | ADT (R) 47 | ADT (R) 48 |
|---|---|---|
| Parentage | ADT 43/ Jeeragasamba | IET 11412/ IR 64 |
| Duration(Days) | 118 | 94-99 |
| Average Yield (Kg/ ha) | 6200 | 4800 |
| 1000 grain wt(g) | 13.5 | 22.0 |
| Grain/L/B ratio | 2.72 | 3.25 |
| Grain type | Medium slender | Long slender |
| **Morphological characters** | | |
| Habit | Semidwarf erect | Semidwarf erect |
| Leaf sheath | Green | Green |
| Septum | - | Cream |
| Ligule | - | Acute, prominent |
| Auricle | Light green | - |
| Panicle | Long com droopy | Intermediate |
| Husk colour | straw | straw |
| Rice colour | White | white |
| Abdominal white | Occassionally present | Occassionally present |
| **Grain size(mm)** | | |
| Length | 7.20 | 9.15 |
| Breadth | 2.20 | 2.54 |
| Thickness | 1.80 | 1.90 |

## III RICE SEASONS OF TAMIL NADU

| Month of sowing | Season | Duration (Days) | Districts |
|---|---|---|---|
| Dec - Jan | Navarai | < 120 | Tiruvallur, Vellore, Tiruvannamalai, Cuddalore, Villupuram, Tiruchirapalli, Perambalur, Karur, Nagapattinam, Madurai, Theni, Salem, Namakkal, Dindigul, Dharmapuri, Coimbatore,Erode and Pudukkottai. |
| Apr - May | Sornavari | <120 | Tiruvallur, Vellore, Tiruvannamalai, Cuddalore, Villupuram, Namakkal, Dharmapuri |
| Apr - May<br>May - June | Early Kar<br>Kar | <120 | Tirunelveli, Kanyakumari, Thoothukudi, Erode, Coimbatore, Madurai, Theni, Dindigul, Salem, Namakkal, Dharmapuri. |
| June - July | Kuruvai | <120 | Tiruchirapalli, Perambalur, Karur, Thanjavur, Nagapattinam, Tiruvarur, Pudukottai, Erode |
| July - Aug. | Early Samba | 130 to 135 | Tiruvallur, Vellore, Tiruvannamalai, Salem, Namakkal, Cuddalore, Villupuram, Madurai, Theni, Ramanathapuram, Dharmapuri, Coimbatore, Erode, Pudukkottai, The Nilgiris |
| August | Samba | 130 - 135 and >150 | All districts |
| Sep - Oct | Late Samba Thaladi/Pishanam | 130 - 135 | Tiruvallur, Madurai, Theni, Coimbatore, Erode |
| Sep -Oct | Late Pishanam | 130 - 135 | Madurai, Theni, Dindigul, Kanyakumari Tirunelveli , Thoothukudi |
| Oct - Nov | Late Thaladi | 115 -120 | Thanjavur, Nagapattinam, Tiruvarur, Tiruchirapalli, Perambalur, Karur, |

# CROP MANAGEMENT

## SYSTEMS OF RICE CULTIVATION IN TAMIL NADU

Rice is cultivated under **puddled** and **un-puddled lowland** situations in Tamil Nadu. 'Transplanting' and 'direct wet seeding' are the two environments under puddled lowland. Whereas, un-puddled lowland cultivation undergoes different environments like, dry seeding exclusively with rainfall, locally called as 'rainfed rice', with supplemental irrigation during peak vegetative and reproductive phases by the rain water collected / harvested in tanks ('semi-dry rice') and also assured irrigation from canal after 30-45 days of dry situation (also called semi-dry rice) . They are grouped as follows:

1. **Transplanted puddled lowland rice**
2. **Direct seeded lowland rice**
   a. Wet seeded rice in puddled soil
   b. Dry seeded rice in un-puddled soil
      i) Rainfed
      ii) Semi dry - supplemental irrigation
      iii) Semi dry - canal irrigation* (contingent crop)
3. **Dry seeded upland rice** – This system of rice cultivation is there in areas with high rainfall (like Assam and NE frontiers of India) where the land is sloping and terraced and there is no possibility for bunding to stagnate the water. Grain yield is poor due to loss of nutrients and soil mainly caused by water erosion. Moisture availability is mostly at saturation or at wet range. There is very limited area Dharmapuri district in Tamil Nadu.
4. **Deep water rice** cultivation exits in certain pockets of Nagapattinam and Tiruvarur idstricts. These areas are not always seen with deep water situation. Varieties suitable for that situation can yield better grain yield.

## 1. TRANSPLANTED PUDDLED LOWLAND RICE

**TRANSPLANTED RICE**

**1.1. Nursery management**

**1.1.1. Wet nursery**

**Nursery area**

Select 20 cents (800 $m^2$) of land area near to water source for raising seedlings for one hectare.

**Seed rate**

30 kg for long duration
40 kg for medium duration
60 kg for short duration varieties and
20 kg for hybrids

**Seed treatment**

a. Treat the seeds in Carbendazim or Pyroquilon or Tricyclozole solution at 2 g/l of water for 1 kg of seeds. Soak the seeds in water for 10 hrs and drain excess water.
b. This wet seed treatment gives protection to the seedlings up to 40 days from seedling disease such as blast and this method is better than dry seed treatment.
c. If the seeds are required for sowing immediately, keep the soaked seed in gunny in dark and cover with extra gunnies and leave for 24hrs for sprouting.
d. **Seed treatment with *Pseudomonas fluorescens***: Treat the seeds with talc based formulation of *Pseudomonas fluorescens* 10g/kg of seed and soak in 1lit of water overnight. Decant the excess water and allow the seeds to sprout for 24hrs and then sow.
**e.** **Seed treatment with *Azospirillum***: Three packets (600 g/ha) of Azospirillum and 3 packets (600g/ha) of Phosphobacteria or 6 packets (1200g/ha)of Azophos. In bioinoculants mixed with sufficient water wherein the seeds are soaked overnight before sowing in the nursery bed (The bacterial suspension after decanting may be poured over the nursery area itself).
   - Biocontrol agents are compatible with biofertilizers
   - Biofertilizers and biocontrol agents can be mixed together for seed soaking
   - Fungicides and biocontrol agents are incompatible

**Forming Seedbeds**

- Mark plots of 2.5m breadth with channels 30cm wide all around the seedbeds.
- Length of the seed bed may vary from 8 to 10m according to soil and slope of the land.
- Collect the puddled soil from the channel and spread on the seedbeds or drag a heavy stone along the channel to lower it, so that the seed bed is at a higher level.
- Level the surface of the seedbed, so that the water drains into the channel.

**Sowing**

Sow the sprouted seeds uniformly on the seedbed, having thin film of water in the nursery.

**Water Management**

- Drain the water 18 to 24hrs after sowing
- Care must be taken to avoid stagnation of water in any part of the seedbed.
- Allow enough water to saturate the soil from 3$^{rd}$ to 5$^{th}$ day. From 5$^{th}$ day onwards, increase the water depth to 1.5cm depending on the height of the seedlings.
- Thereafter maintain 2.5cm depth of water.

**Weed Management**

- Apply any one of the pre-emergence herbicides viz., Pretilachlor + safener 0.3kg/ha, on 3$^{rd}$ or 4$^{th}$ day after sowing to control weeds in the lowland nursery. Keep a thin film of water and allow it to disappear. Avoid drainage of water. This will control germinating weeds.

**Nutrient management**

- Apply 1tonne of fully decomposed FYM or compost to 20cents nursery and spread the manure uniformly on dry soil.
- Basal application of DAP is recommended when the seedlings are to be pulled out in 20-25 days after sowing in less fertile nursery soils.
- For that situation, before the last puddling, apply 40kg of DAP and if not readily available, apply straight fertilizers 16kg of urea and 120kg of super phosphate.
- If seedlings are to be pulled out after 25 days, application of DAP is to be done 10 days prior to pulling out.
- For clayey soils where root snapping is a problem, 4kg of gypsum and 1kg of DAP/cent can be applied at 10 days after sowing.

**1.1.2. Dry nursery**

- Dry ploughed field with fine tilth is required.
- Nursery area with sand and loamy soil status is more suitable for this type of nursery.
- Area 20cents.
- Plots of 1 to 1.5 m width of beds and channels may be formed. Length may be according to the slope and soil. Raised beds are more ideal if the soil is clayey in nature.
- Seed rate and seed treatment as that of wet nursery.
- Sowing may be dry seeding. Seeds may be covered with sand and finely powdered well decomposed farm yard manure.
- Irrigation may be done to wet the soil to saturation.
- Optimum age for transplanting – 4$^{th}$ leaf stage
- This type of nursery is handy in times of delayed receipt of canal water.

**1.2 Main Field Management**

**1.2.1. Land preparation**

- Plough the land during summer to economize the water requirement for initial preparation of land.
- Flood the field 1 or 2days before ploughing and allow water to soak in. Keep the surface of the field covered with water.
- Keep water to a depth of 2.5cm at the time of puddling.
- Special technologies for problem soils:
    a) For fluffy paddy soils: compact the soil by passing 400kg stone roller or oil-drum with stones inside, eight times at proper moisture level (moisture level at friable condition of soil which is approximately 13 to18%) once in three years, to prevent the sinking of draught animals and workers during puddling.
    b) For sodic soils with pH values of more than 8.5, plough at optimum moisture regime, apply gypsum at 50% gypsum requirement uniformly, impound water, provide drainage for leaching out soluble salts and apply green leaf manure at 5 t/ha, 10 to 15 days before transplanting. Mix 37.5kg of Zinc sulphate per ha with sand to make a total quantity of 75kg and spread the

mixture uniformly on the leveled field. Do not incorporate the mixture in the soil. Rice under sodic soil responds well to these practices.

c) For saline soils with EC values of more than 4 dS/m, provide lateral and main drainage channels (60cm deep and 45cm wide), apply green leaf manure at 5 t/ha at 10 to 15 days before transplanting and 25% extra dose of nitrogen in addition to recommended P and K and $ZnSo_4$ at 37.5 kg/ha at planting

d) For acid soils apply lime based on the soil analysis for obtaining normal rice yields. Lime is applied 2.5t/ha before last ploughing. Apply lime at this rate to each crop up to the 5th crop.

**1.2.2. Stand Establishment**

**Optimum age of seedlings for quick establishment**

- Optimum age of the seedlings is 18-22 days for short, 25-30 days for medium and 35-40 days for long duration varieties.

**Pulling out the seedlings**

- Pull out the seedlings at the appropriate time (4th leaf stage).
- Pulling at 3rd leaf stage is also possible. These seedlings can produce more tillers, provided enough care taken during the establishment phase (See section 1.8 Integrated Crop Management (ICM) - Rice-SRI) through thin film of water management and perfect leveling of main field.
- Transplanting after 5th and higher order leaf numbers will affect the performance of the crop and grain yield. Then they are called as 'aged seedlings'. Special package is needed to minimize the grain yield loss while planting those aged seedlings.

**Root dipping**

- Prepare the slurry with 5 packets (1000 g)/ha of Azospirillum and 5 packets (1000g/ha) of Phosphobacteria or 10 packets of (2000g/ha) of Azophos inoculant in 40 lit. of water and dip the root portion of the seedlings for 15 - 30 minutes in bacterial suspension and transplant.

**Planting seedlings in the main field**

| Soil | Medium and low fertility | | | High fertility | | |
|---|---|---|---|---|---|---|
| Duration | Short | Medium | Long | Short | Medium | Long |
| Spacing (cm) | 15x10 | 20x10 | 20x15 | 20x10 | 20x15 | 20x20 |
| Hills / $m^2$ | 66 | 50 | 33 | 50 | 33 | 25 |

- Transplant 2-3 seedlings/hill for short duration and 2 seedlings/hill for medium and long duration varieties
- Shallow planting (3 cm) ensures quick establishment and more tillers.
- Deeper planting (> 5cm) leads to delayed establishment and reduced tillers.
- Line planting permits rotary weeding and its associated benefits.
- Allow a minimum row spacing of 20 cm to use rotary weeder.
- Fill up the gaps between 7th and 10th DAT.

**Management of Aged seedlings***

*** Which developed tillers / underwent node elongation in the nursery itself and**
*** About half of its leaf producing capacity may be already over.**

- Follow the spacing recommended to medium and low fertility soil
- Plant one or two seedlings per hill
- Avoid cluster planting of aged seedlings, which are hindering the formation of new tillers.
- New tillers alone are capable of producing normal harvestable panicle. Weak panicle may appear in the mother culm within three weeks after transplanting and vanishes well before harvest.
- To encourage the tiller production, enhance the basal N application by 50% from the recommended and thereafter follow the normal schedule recommended for other stages.

**Gap filling**

- Fill the gaps if any within 7 - 10 days after planting.

**1.2.3. Nutrient management**

**Application of organic manures**

- Apply 12.5 t of FYM or compost or green leaf manure @ 6.25 t/ha.
- If green manure is raised @ 20 kg /ha *in situ,* incorporate it to a depth of 15 cm using a green manure trampler or tractor.
- In the place of green manure, press-mud / composted coir-pith can also be used.

**Stubble incorporation**

- Apply 22 kg urea / ha at the time of first puddling while incorporating the stubbles of previous crop to compensate immobilization of N by the stubbles.
- This may be done at least 10 days prior to planting of subsequent crop. This recommendation is more suitable for double crop wetlands, wherein, the second crop is transplanted in succession with short turn around period.

**Biofertilizer application**

- Broadcast 10 kg of soil based powdered BGA flakes at 10 DAT for the dry season crop. Maintain a thin film of water for multiplication.
- Raise azolla as a dual crop by inoculating 250 kg/ha 3 to 5 DAT and then incorporate during weeding for the wet season crop.
- Mix 10 packets (2000 g)/ha of Azospirillum and 10 packets (2000g/ha)of Phosphobacteria or 20 packets (4000g/ha) of Azophos inoculants with 25 kg FYM and 25 kg of soil and broadcast the mixture uniformly in the main field before transplanting and
- *Pseudomonas fluorescens* (Pf 1) at 2.5 kg/ha mixed with 50 kg FYM and 25 kg of soil and broadcast the mixture uniformly before transplanting.

**Application of inorganic fertilizers**

- Apply fertilizer nutrients as per soil test recommendations
- N dose may be through Leaf Color Chart (**LCC**)
- P & K may be through Site Specific Nutrition Management by Omission plot technique (Ref. Appendix II)
- If the above recommendation are not able to be followed, adopt blanket recommendation as follows:

| Nutrients | N | $P_2O_5$ (kg/ha) | $K_2O$ |
|---|---|---|---|
| Short duration varieties (**dry season**) | | | |
| a) Cauvery delta & Coimbatore tract | 150 | 50 | 50 |
| b) For other tracts | 120 | 40 | 40 |
| Medium and long duration varieties (**wet season**) | 150 | 50 | 50 |
| Hybrid rice | 175 | 60 | 60 |
| Low N responsive cultivars (like Improved White Ponni) | 75* | 50 | 50 |

*For Ponni, N should be applied in three splits at AT, PI and H stages** in addition to GLM or FYM application.

****Phenological stages of rice (days after sowing)**

| Stages | Short (105) | Medium (135) | Long (150) |
|---|---|---|---|
| Active Tillering (AT) | 35-40 | 50-55 | 55-60 |
| Panicle Initiation (PI) | 45-50 | 70-75 | 85-90 |
| Heading (H) | 70-75 | 100-105 | 115-120 |

**Split application of N and K**

- Apply N and K in four equal splits viz., basal, tillering, panicle initiation and heading stages.
- Tillering and Panicle initiation periods are crucial and should not be reduced with the recommended quantity.
- N management through **LCC** may be adopted wherever chart is available as given below

**N management through LCC**

- Time of application is decided by LCC score
- Take observations from 14 DAT in transplanted rice or 21 DAS in direct seeded rice.
- Repeat the observations at weekly intervals up to heading
- Observe the leaf colour in the fully opened third leaf from the top as index leaf.
- Match the leaf color with the colours in the chart during morning hours (8-10 am).
- Take observation in 10 places.
- LCC critical value is 3.0 in low N response cultures like White Ponni and 4.0 in other cultivars and hybrids
- When 6/10 observations show less than the critical colour value, N can be applied @ 35kg N/ha in dry season and 30kg N/ha in wet season per application per ha.

**Application of P fertilizer**

- P may be applied as basal and incorporated.
- When the green manure is applied, rock phosphate can be used as a cheap source of P fertilizer. If rock phosphate is applied, the succeeding rice crop need not be supplied with P. Application of rock phosphate + single super phosphate or DAP mixed in different proportions (75:25 or 50:50) is equally effective as SSP or DAP alone.

**Application of zinc sulphate**

- Apply 25 kg of zinc sulphate mixed with 50 kg dry sand just before transplanting.
- It is enough to apply 12.5 kg zinc sulphate /ha, if green manure (6.25 t/ha) or enriched FYM, is applied.
- If deficiency symptom appears, foliar application of 0.5% Zinc sulphate + 1.0% urea can be given at 15 days interval until the Zn deficiency symptoms disappear.

**Application of gypsum**

- Apply 500 kg of gypsum/ha (as source of Ca and S nutrients) at last ploughing.

**Foliar nutrition**

- Foliar spray of 1% urea + 2% DAP + 1% KCl at PI and 10 days later for all varieties.

**Nutrient deficiency / toxicity symptoms**

- **Nitrogen deficiency**: Plants become stunted and yellow in appearance first on lower leaves. In case of severe deficiency the leaves will turn brown and die. Deficiency symptoms first appear at the leaf-tip and progress along the midrib until the entire leaf is dead.
- **Potassium deficiency**: Bluish green leaves - when young, older leaves irregular. Chlorotic and necrotic areas - grain formation is poor - weakening of the straw which results in lodging.
- **Magnesium deficiency:** Leaves are chlorotic with white tips.
- **Iron toxicity:** Brown spots on the lower leaves starting from tips and proceeding to the leaf base and turns into green or orange purple leaves and spreading to the next above leaves.
- **Zinc deficiency:** Lower leaves have chlorotic particularly towards the base. Deficient plants give a brown rusty appearance.

**Neem treated urea and coal-tar treated urea**

- Blend the urea with crushed neem seed or neem cake 20% by weight. Powder neem cake to pass through 2mm sieve before mixing with urea. Keep it overnight before use (or) urea can be mixed with gypsum in 1:3 ratios, or urea can be mixed with gypsum and neem cake at 5:4:1 ratio to increase the nitrogen use efficiency. For treating 100 kg urea, take one kg coal-tar and 1.5 litres of kerosene. Melt coal-tar over a low flame and dissolve it in kerosene. Mix urea with the solution thoroughly in a plastic container, using a stick. Allow it to dry in shade on a polythene sheet. This can be stored for a month and applied basally.

**1.2.4. Weed management**

- Use of rotary weeder from 15 DAT at 10 days interval. It saves labour for weeding, aerates the soil and root zone, prolongs the root activity, and improves the grain filling though efficient translocation and ultimately the grain yield.
- Cultural practices like dual cropping of rice-azolla, and rice-green manure (described in wet seeded rice section 2.5 & 2.6 of this chapter) reduces the weed infestation to a greater extent.
- Summer ploughing and cultivation of irrigated dry crops during post-rainy periods reduces the weed infestation.

**Pre-emergence herbicides**

- Use Butachlor 1.25kg/ha or Anilophos 0.4kg/ha as pre-emergence application. Alternatively, pre-emergence application of herbicide mixture viz., Butachlor 0.6kg + 2,4 DEE 0.75kg/ha, or Anilophos + 2, 4 DEE 'ready-mix' at 0.4kg/ha followed by one hand weeding on 30 - 35 DAT will have a broad spectrum of weed control.
- Any herbicide has to be mixed with 50kg of dry sand on the day of application (3 - 4 DAT) and applied uniformly to the field with thin film water on the 3$^{rd}$ DAT. Water should not be drained for next 2 days from the field (or) fresh irrigation should not be given.

**Post - emergence herbicides**

- If pre-emergence herbicide application is not done, hand weeding has to be done on 15$^{th}$ DAT.
- 2,4-D sodium salt (Fernoxone 80% WP) 1.25 kg/ha dissolved in 625 litres with a high volume sprayer, three weeks after transplanting or when the weeds are in 3 - 4 leaf stage.

**1.2.5. Water management**

- Puddling and leveling minimizes the water requirement
- Plough with tractor drawn cage wheel to reduce percolation losses and to save water requirement up to 20%.
- Maintain 2.5cm of water over the puddle and allow the green manure to decompose for a minimum of 7 days in the case of less fibrous plants like sunnhemp and 15 days for more fibrous green manure plants like Kolinchi (*Tephrosia purpurea*).
- At the time of transplanting, a shallow depth of 2cm of water is adequate since high depth of water will lead to deep planting resulting in reduction of tillering.
- Maintain 2 cm of water up to seven days of transplanting.
- After the establishment stage, cyclic submergence of water (as in table) is the best practice for rice crop. This cyclic 5cm submergence has to be continued throughout the crop period.

Days after disappearance of ponded water at which irrigation is to be given

| Soil type | Summer | Winter |
|---|---|---|
| Loamy | 1 day | 3 days |
| Clay | Just before/immediately after disappearance | 1 - 2 days |

- Moisture stress due to inadequate water at rooting and tillering stage causes poor root growth leading to reduction in tillering, poor stand and low yield.
- Critical stages of water requirement in rice are a) panicle initiation, b) booting, c) heading and d) flowering. During these stages, the irrigation interval should not exceed the stipulated time so as to cause the depletion of moisture below the saturation level.
- During booting and maturity stages continuous inundation of 5cm and above leads to advancement in root decay and leaf senescence, delay in heading and reduction in the number of filled grains per panicle and poor harvest index.
- Provide adequate drainage facilities to drain excess water or strictly follow irrigation schedule of one day after disappearance of ponded water. Last irrigation may be 15 days ahead of harvest.

**Precautions for irrigation**

- The field plot can be 25 to 50 cents depending on the source of irrigation.
- Field to field irrigation should be avoided. Field should be irrigated individually from a channel.
- Small bund may be formed parallel to the main bund of the field at a distance of 30 to 45cm within the field to avoid leakages of water through main bund crevices.
- To minimize percolation loss, the depth of stagnated water should be 5cm or less.
- In water logged condition, form open drains, about 60cm in depth and 45cm width across the field.
- Care should be taken not to allow development of cracks.
- In canal command area, conjunctive use of surface and ground water may be resorted to for judicious use of water.
- In double cropped wetland of command area, raise groundnut / pulse in the place of *Kuruvai* rice if water is a constraint or go for rice cultivation as described in *5. Dry Seeded Irrigated Un-Puddled Lowland Rice*.

**1.3. Insect management:** See Crop Protection Chapter

**1.4. Disease management:** See Crop Protection Chapter

**1.5. Harvesting**

- Taking the average duration of the crop as an indication, drain the water from the field 7 to 10 days before the expected harvest date as draining hastens maturity and improves harvesting conditions.
- When 80% of the panicles turn straw colour, the crop is ready for harvest. Even at this stage, the leaves of some of the varieties may remain green.
- Confirm maturity by selecting the most mature tiller and dehusk a few grains. If the rice is clear and firm, it is in hard dough stage.
- When most of the grains at the base of the panicle in the selected tiller are in a hard dough stage, the crop is ready for harvest. At this stage harvest the crop, thresh and winnow the grains.
- Dry the grains to 12% moisture level for storage. Grain yield in rice is estimated only at 14% moisture for any comparison.
- Maturity may be hastened by 3-4 days by spraying 20% NaCl a week before harvest to escape

monsoon rains.

### 1.6. SEEDLING THROWING METHOD OF STAND ESTABLISHMENT

- 20 days old seedlings of short duration rice varieties
- Requirement of seedlings will be approximately 20% more than the line planting or equal to random planting.
- The seedlings are thrown into the puddled leveled field by labour without using force.
- Suitable for all seasons except *Thaladi* or heavy rain season.
- 50% labour shaving as compared to line planting and 35% to random planting.
- Up to 7-10 days of seedling throwing care should be taken to maintain thin film of water (similar to wet seeded rice).
- Other cultural operations are same as in transplanted rice
- Grain yield will be equal to line planted crop and 10-12% higher than random planted crop.

### 1.7. TRANSPLANTED HYBRID RICE

| | |
|---|---|
| Seed rate | 20 kg per hectare |
| Nursery | Basal application of DAP at 2 kg/cent of nursery area. Sparse sowing of seeds at one kg/cent of nursery area will give robust seedlings with 1-2 tillers per seedling at the time of planting. If the soil is heavy, apply 4 kg gypsum/cent of nursery area, 10 days before pulling of seedlings. |
| Age of seedling | 20 to 25 days |
| Spacing (cm) | 20 x 10 (50 hills/$m^2$) or 25 x 10 (40 hills/$m^2$) according to soil fertility |
| Seedlings/ hill | One (along with tillers if already produced) |
| Fertilizer | 175:60:60 kg N, $P_2O_5$ and $K_2O$/ha |

**Other package of practices:** same as in transplanted rice varieties.

## 1.8. INTEGRATED CROP MANAGEMENT (ICM) - RICE (SRI)

### 1.8.1. Season

- Dry season with assured irrigation is more suitable.
- Difficulty in crop establishment may be seen in areas with heavy downpour (NE Monsoon periods of Tamil Nadu)

### 1.8.2. Varieties

- Hybrids and varieties with heavy tillering

### 1.8.3. Nursery

#### 1.8.3.1. Seed rate

- 7- 8 kg for single seedling per hill
- 12 -15 kg for two seedlings per hill wherever difficulty in establishment of rice is seen

#### 1.8.3.2. Mat nursery preparation

- *Preparation of nursery area:* Prepare 100 $m^2$ nurseries to plant 1 ha. Select a level area near the water source. Spread a plastic sheet or used polythene gunny bags on the shallow raised bed to prevent roots growing deep into soil.
- *Preparation of soil mixture:* Four (4) $m^3$ of soil mix is needed for each 100 $m^2$ of nursery. Mix 70% soil + 20% well-decomposed pressmud / bio-gas slurry / FYM + 10% rice hull. Incorporate in the soil mixture 1.5 kg of powdered di -ammonium phosphate or 2 kg 17-17-17 NPK fertilizer.
- *Filling in soil mixture:* Place a wooden frame of 0.5 m long, 1 m wide and 4 cm deep divided into 4 equal segments on the plastic sheet or banana leaves. Fill the frame almost to the top with the soil mixture.
- *Pre-germinating the seeds 2 days before sowing:* Soak the seeds for 24 h, drain and incubate the soaked seeds for 24 h, sow when the seeds sprout and radical (seed root) grows to 2-3 mm long.
- *Sowing:* Sow the pre-germinated seeds weighing 90 -100 g / $m^{-2}$ (100g dry seed may weigh 130g after sprouting) uniformly and cover them with dry soil to a thickness of 5mm. Sprinkle water immediately using rose can to soak the bed and remove the wooden frame and continue the process until the required area is completed.

- *Watering:* Water the nursery with rose can as and when needed (twice or thrice a day) to keep the soil moist. Protect the nursery from heavy rains for the first 5 DAS. At 6 DAS, maintain thin film of water all around the seedling mats. Drain the water 2 days before removing the seedling mats for transplanting.
- *Spraying fertilizer solution* (optional): If seedling growth is slow, sprinkle 0.5% urea + 0.5% zinc sulfate solution at 8-10 DAS.
- *Lifting seedling mats:* Seedlings reach sufficient height for planting at 15 days. Lift the seedling mats and transport them to main field.

**1.8.4. Main field preparation**

- Puddled lowland prepared as described in transplanted section
- Perfect leveling is a pre-requisite for the water management proposed hereunder

**1.8.5. Transplanting**

- 1-2 seedlings of 15 days old
- Square planting of 22.5 x 22.5 cm (9 x 9 inch)
- Fill up the gaps between $7^{th}$ and $10^{th}$ DAT.
- Transplant within 30 minutes of pulling out of seedlings.
- There may be difficulty in crop establishment in areas with heavy downpour (North East Monsoon periods of Tamil Nadu)

**1.8.6. Irrigation management**

- Irrigation only to moist the soil in the early period of 10 days
- Restoring irrigation to a maximum depth of 2.5cm after development of hairline cracks in the soil until panicle initiation
- Increasing irrigation depth to 5.0cm after PI one day after disappearance of ponded water

**1.8.7. Weed management**

- Using rotary weeder / Cono weeder
- Moving the weeder with forward and backward motion to bury the weeds and as well to aerate the soil at 7-10 days interval from 10-15 days after planting on either direction of the rows and column.
- Manual weeding is also essential to remove the weeds closer to rice root zone.

**1.8.8. Nutrient management**

- As per transplanted rice.
- Use of LCC has more advantage in N management.
- Green manure and farm yard manure application will enhance the growth and yield of rice in this system approach.

**1.8.9. Other package of practices as recommended to transplanted rice**

**STCR based fertilizer recommendation for transplanted rice** (for some selected districts) is given in the **Appendix I.**

## 2. WET SEEDED PUDDLED LOWLAND RICE

**WET SEEDED RICE**

**2.1. Area**

- Direct wet seeding can be followed in all the areas wherein transplanting is in vogue.

**2.2. Season**

- As that of translated rice

**2.3. Field preparation**

- On receipt of showers during the months of May - July repeated ploughing should be carried out so as to conserve the moisture, destroy the weeds and break the clods.
- After inundation puddling is to be done as per transplanting. More care should be taken to level the field to zero level.
- Stagnation of water in patches during germination and early establishment of the crop leads to uneven crop stand.
- Land leveling has say over efficient weed and water management practices.
- Provision of shallow trenches (15cm width) at an interval of 3m all along the field will facilitate the draining of excess water at the early growth stage.

**2.4. Varieties**

All the varieties recommended for transplanting can do well under direct wet seeded conditions also. However, the following varieties are more suited.

| Varieties | Duration (days) | Time of sowing |
|---|---|---|
| Ponmani | 160 to 165 | 1st to 30th August |
| Co 43, IR20, ADT 38 ADT 39, Ponni, Improved White Ponni | 125 to135 | 1st to 30th September |
| ADT 36, ADT 37 | 105 to 110 | 1st to 10th October |

**2.5. Sowing**

- Follow a seed rate of 60 kg / ha
- Pre-germinate the seeds as for wet nursery
- Seed treatments as adopted for transplanted rice
- Sow the seeds by **drum seeder** or broadcast uniformly with thin film of water.
- Dual cropping of rice-green manure is economic for nutrient budget and efficient for grain production. For this method use 'TNAU Rice-Green manure seeder'.

**2.6. After cultivation**

- Thinning and gap filling should be done 14 - 21 days after sowing, taking advantage of the immediate rain.
- If dual cropped with green manure, incorporate the green manure when grown to 40cm height or at 30 days after sowing, whichever is earlier, using Cono-weeder.
- Green manure incorporated fields may be operated again with rotary weeder a week later in order to aerate the soil and to exploit organic acids formed if any.

**2.7. Manures and fertilizer application**

- For direct wet seeded lowland rice, the recommendation is same at that of transplanted rice.
- Apply N and K as 25% each at 21 DAS, at active tillering, PI and heading stages.
- If N applied through LCC, use the critical value 3 for broadcasted and 4 for line sown drill seeded rice.
- Entire P as basal applied in the last plough or at the time of incorporation of green manure/ compost.
- Biofertilizers as recommended to transplanted rice may be followed wherever feasible and moisture available.
- Micro nutrient, foliar application and biofertlizers as recommended to transplanted rice.

**2.8. Weed management**

- In wet seeded rice, pre-emergence application of pretilachlor 0.75kg/ha on 8 DAS or pretilachlor + safener (Sofit) at 0.45kg/ha on 3-4 DAS followed by one hand weeding on 40 DAS.

**2.9. Water management**

- During first one week just wet the soil by thin film of water.
- Depth of irrigation may be increased to 2.5cm progressively along the crop age.
- Afterwards follow the schedule as given to transplanted rice.

**2.10. Insect management:** See Crop Protection Chapter

**2.11. Disease management:** See Crop Protection Chapter

**Other package of practices**

- As recommended in transplanted rice

## 3. DRY SEEDED RAINFED UN-PUDDLED LOWLAND RICE

**RAINFED RICE**

The crop establishment, growth and maturity depend up on the rainfall received. There will be standing water after crop establishment for a minimum period of few days to a maximum up to grain filling, depending up on the rainfall. This type of cultivation in Tamil Nadu is called as **'rainfed rice'**, with the assumption that the soil moisture will be under unsaturated (dry) condition during establishment or entire growth period, with reference to tropical climate.

**3.1. Area**

- Coastal districts of Tamil Nadu like Kanchipuram, Tiruvallur, Pudukottai, Ramanathapuram, Virudhunagar, Sivagangai and Kanyakumari.

**3.2. Season**

- June – July – (Coastal northern districts)
- September – October (Coastal southern districts)

**3.3. Field preparation**

- Dry plough to get fine tilth taking advantage of rains and soil moisture availability.
- Apply gypsum at 1 t/ha basally wherever soil crusting and soil hardening problem exist.
- Perfect land leveling for efficient weed and water management.
- Provide shallow trenches (15 cm width) at an interval of 3m all along the field to facilitate draining excess water at the early growth stage.

**3.4. Varieties**

- Short duration varieties as mentioned in season and varieties including local land races suitable for those tracts.

**3.5. Sowing**

- Seed rate: 75kg/ha dry seed for any recommended variety.
- Seed hardening with 1% KCl for 16 hours (seed and KCl solution 1:1) and shade dried to bring to storable moisture. This will enable the crop to withstand early moisture stress.
- On the day of sowing, treat the hardened seeds first with *Pseudomonas fluorescens* 10g/kg of seed and then with *Azophos* 2000g or Azospirillum *and Phosphobacteria* @ 600g each per ha seed, whichever is available.
- Drill sow with 20 cm inter row spacing using seed drill.
- The seeds can also be sown behind the country plough
- Depth of sowing should be 3 - 5 cm and the top soil can be made compact with leveling board.
- Pre-monsoon sowing is advocated for uniform germination.

**3.6. After cultivation**

- Azospirillum inoculants 10 packets (2000g/ha) and Phosphobacteria 10 packets (2000g/ha) or 20 packets (4000g/ha) of Azophos mixed with 25 kg of FYM may be broadcasted uniformly over the field just after the receipt soaking rain / moisture.
- Thinning and gap filling should be done 14 - 21days after sowing, taking advantage of the immediate rain
- Spray Cycocel 1000 ppm (1 ml of commercial product in one lit. of water) under water deficit situations to mitigate ill-effects.
- Foliar spray of Kaolin 3% or KCl 1% to overcome moisture stress at different physiological stages of rice.

**3.7. Manures and fertilizer application**

- Blanket recommendation : 50:25:25 kg N:$P_2O_5$:$K_2O$ /ha
- Apply a basal dose of 750 kg of FYM enriched with fertilizer phosphorus (P at 25 kg/ha)
- Apply N and K in two equal splits at 20 - 25 and 40 - 45 days after germination.
- If the moisture availability from the tillering phase is substantial, three splits (25 kg N and 12.5 kg K at 20-25, 40-45 and 60-65 DAG) can be adopted.
- N at PI may be enhanced to 40 kg, if the tiller production is high (may be when the estimated LAI is greater than 5.0) and moisture availability ensured by standing water for 10 days.
- Basal application of $FeSO_4$ at 50 kg/ha is desirable for iron deficient soil.
- Foliar spray of 1% urea + 2% DAP + 1%KCl at PI and 10 days later may be taken up for enhancing the rice yield if sufficient soil moisture is ensured

**3.8. Weed management**

- First weeding can be done between 15 and 21 days after germination.
- Second weeding may be done 30 - 45 days after first weeding.
- Apply pendimethalin 1.0kg/ha on 5 days after sowing or Pretilachlor + safener (Sofit) 0.45kg/ha on the day of receipt of soaking rain followed by one hand weeding on 30 to 35 days after sowing.

**3.9. Insect management:** See Crop Protection Chapter

**3.10. Disease management:** See Crop Protection Chapter

**3.11. Harvesting**

- Same as that for wet rice cultivation

# 4. DRY SEEDED RAINFED UN-PUDDLED LOWLAND RICE WITH SUPPLEMENTAL IRRIGATION

**Semi dry rice**

It is called as **semi-dry rice**. Crop establishment is as that of rainfed rice but the rain water collected in village tank (Kanmai) is supplemented to protect the crop during peak vegetative and reproductive phases. Interaction between applied nutrients and crop is positive here due to better moisture availability than rainfed rice and hence varieties may be improved ones and nutrient levels may be higher than the previous system.

**4.1. Area**

- Kanchipuram/Tiruvallur, Ramanathapuram, Sivaganga, Kanyakumari, Nagapattinam/ Tiruvarur and Pudukottai.

**4.2. Seasons**

- July to August - Kanchipuram/Tiruvallur, Kanyakumari
- August - Nagapattinam/Tiruvarur, Pudukottai
- September to October - Ramanathapuram, Sivaganga

**4.3. Field preparation**

- Dry plough to get fine tilth taking advantage of rains and soil moisture availability.
- Apply gypsum at 1 t/ha basally wherever soil crusting and soil hardening problem exist.
- Perfect land leveling for efficient weed and water management.
- Provide shallow trenches (15 cm width) at an interval of 3m all along the field to facilitate draining excess water at the early growth stage.

**4.4. Varieties**

- Short duration varieties as mentioned in season and vanities including local land races suitable for those tracts.
- Since there is supplemental irrigation high yielding improved short duration varieties can yield more yield than the land races.

**4.5. Sowing**

- Seed rate: 75kg/ha dry seed for any recommended variety.
- Seed hardening with 1% KCl for 16 hours (seed and KCl solution 1:1) and shade dried to bring to storable moisture. This will enable the crop to withstand early moisture stress.
- On the day of sowing, treat the hardened seeds first with *Pseudomonas fluorescens* 10g/kg of seed and then with *Azophos* 2000g or Azospirillum *and Phosphobacteria* @ 600g each per ha seed, whichever is available.
- Drill sow with 20 cm inter row spacing using seed drill.
- The seeds can also be sown behind the country plough
- Depth of sowing should be 3 - 5 cm and the top soil can be made compact with leveling board.
- Pre-monsoon sowing is advocated for uniform germination.

**4.6. After cultivation**

- Azospirillum inoculants 10 packets (2000g/ha) and Phosphobacteria 10 packets (2000g/ha) or 20 packets (4000g/ha) of Azophos mixed with 25 kg of FYM may be broadcasted uniformly over the field just after the receipt soaking rain / moisture.
- Thinning and gap filling should be done 14-21days after sowing, taking advantage of the immediate rain
- Spray Cycocel 1000 ppm (1 ml of commercial product in one lit. of water) under water deficit situations to mitigate ill-effects.
- Foliar spray of Kaolin 3% or KCl 1% to overcome moisture stress at different physiological stages of rice. .

**4.7. Manures and fertilizer application**

- Blanket recommendation : 75:25:37.5 kg N:$P_2O_5$:$K_2O$ /ha
- Apply a basal dose of 750 kg of FYM enriched with fertilizer phosphorus (P at 25 kg/ha)
- Apply N & K in three splits at 20-25, 40-45 and 60-65 days after germination.
- Each split may follow 25kg N and 12.5 kg $K_2O$.
- If the moisture availability is substantial, split application at panicle initiation may be done with 40 kg N and 12.5 kg $K_2O$.

- If the moisture availability is substantial, the split at 40-45 DAS (panicle initiation) may be applied up to 40kg N and 12.5kg $K_2O$ to enhance the growth and the grain yield.
- Basal application of $ZnSO_4$ at 25kg/ha and $FeSO_4$ at 50/ha is desirable wherever Zinc and iron deficiency were noted.
- Biofertilizers as recommended to transplanted rice may be followed wherever feasible and moisture available.
- Foliar spray of 1% urea + 2% DAP + 1%KCl at PI and 10 days later may be taken up for enhancing the rice yield if sufficient soil moisture is ensured

**4.8. Weed management**

- First weeding should be done between 15 and 21 days after germination.
- Second weeding may be done 30 - 45 days after first weeding.
- Apply pendimethalin 1.0kg/ha on 5 days after sowing or Pretilachlor + safener (Sofit) 0.45kg/ha on the day of receipt of soaking rain followed by one hand weeding on 30 to 35 days after sowing.

**4.9. Water management**

- The crop is irrigated from 30-35 days onwards, utilizing water impounded in the tanks.
- Irrigation may be to a depth of 2.5 -5.0cm only. Follow the schedule of one day after disappearance of ponded water in order to save water and to bring additional area under this type of rice cultivation.

**4.10. Insect management:** See Crop Protection Chapter

**4.11. Disease management:** See Crop Protection Chapter

**4.12. Harvest**

- It is same as that of transplanted rice.
- These areas are more suitable for combine-harvester

## 5. DRY SEEDED IRRIGATED UN-PUDDLED LOWLAND RICE

**Also be called 'semi-dry rice'**

It is a contingent plan to command areas, anticipating the release of water; rice crop can be established under rainfed condition up to a maximum of 45 days as that of previous two situations. Filed is converted to wet condition on receipt of canal water. Conversion depends up on receipt of canal water and nutrient management is decided according to the period of irrigation.

**5.1. Area**

- Tiruvarur and Nagapattinam districts

**5.2. Season**

- Samba / Thaladi seasons command areas.

**5.3. Field preparation**

- Dry plough to get fine tilth taking advantage of rains and soil moisture availability.
- Apply gypsum at 1 t/ha basally wherever soil crusting and soil hardening problem exist.
- Perfect land leveling for efficient weed and water management.
- Provide shallow trenches (15 cm width) at an interval of 3m all along the field to facilitate draining excess water at the early growth stage.

**5.4. Varieties**

- Medium duration varieties, if sown in August and short duration varieties beyond September, as mentioned in season and vanities.
- Since there is assured irrigation from canal, high yielding improved short or medium duration varieties can be cultivated depending up on the situation (month of sowing, nearness to canal, depth of standing water during NEM etc).

**5.5. Sowing**

- Seed rate: 75kg/ha dry seed for any recommended variety.
- Seed hardening with 1% KCl for 16 hours (seed and KCl solution 1:1) and shade dried to bring to storable moisture. This will enable the crop to withstand early moisture stress.
- On the day of sowing, treat the hardened seeds first with *Pseudomonas fluorescens* 10g/kg of seed and then with *Azophos* 2000g or Azospirillum *and Phosphobacteria* @ 600g each per ha seed, whichever is available.

- Drill sow with 20 cm inter row spacing using seed drill.
- The seeds can also be sown behind the country plough
- Depth of sowing should be 3 - 5 cm and the top soil can be made compact with leveling board.
- Pre-monsoon sowing is advocated for uniform germination.
- Pre-monsoon sowing with medium duration variety is an advantage for higher grain yield and as well to manage the heavy rainy season.

**5.6. After cultivation**

- Azospirillum inoculants 10 packets (2000g/ha) and Phosphobacteria 10 packets (2000g/ha) or 20 packets (4000g/ha) of Azophos mixed with 25 kg of FYM may be broadcasted uniformly over the field just after the receipt soaking rain / moisture.
- Thinning and gap filling should be done 14 - 21 days after sowing, taking advantage of the immediate rain.

**5.7. Manures and fertilizer application**

- Apply FYM/compost at 12.5 t/ha or 750 kg of FYM enriched with 50 kg $P_2O_5$ as basal dose in clay soils of Nagapattinam / Tiruvarur district.
- N and K in three splits at around 20-25, 40-45 and 60-65 days for short duration varieties or four splits for medium duration varieties at around 20-25, 40-45, 60-65 and 80-85 days after germination is suitable.
- Each split may follow 25kg N and 12.5 kg $K_2O$.
- If the moisture availability is substantial and canal water received from tillering phases itself, the split at panicle initiation (40-45 DAS in short duration and 60-65 DAS in medium duration) may be applied up to 40kg N and 12.5kg $K_2O$ to enhance the growth and the grain yield.
- To induce tolerance under short and prolonged drought situation in Kuruvai season, apart from seed treatment, foliar spray with 1% KCl + CCC at 500ppm during vegetative stage is effective in mitigating the drought and in increasing the yield.
- Basal application of $ZnSO_4$ at 25kg/ha and $FeSO_4$ at 50 kg/ha is desirable wherever Zinc and iron deficiency were noted.
- Biofertilizers as recommended to transplanted rice may be followed wherever feasible and moisture available.
- Foliar spray of 1% urea + 2% DAP + 1%KCl at PI and 10 days later may be taken up for enhancing the rice yield if sufficient soil moisture is ensured

**5.8. Weed management**

- First weeding should be done between 15 and 21 days after germination.
- Second weeding may be done 30 - 45 days after first weeding.
- Apply pendimethalin 1.0kg/ha on 5 days after sowing or Pretilachlor + safener (Sofit) 0.45kg/ha on the day of receipt of soaking rain followed by one hand weeding on 30 to 35 days after sowing.

**5.9. Other special cultural practices**

- Spray Cycocel 1000 ppm (1 ml of commercial product in one lit. of water) under water deficit situations to mitigate ill-effects.
- Foliar spray of Kaolin 3% or KCl 1% to overcome moisture stress at different physiological stages of rice.
- For delayed water release in LBP area, irrigating rice to 5cm depth three days after disappearance of pounded water and growing ADT 38 rice can be resorted to if the release of water is delayed up to September.
- The first top dressing should be applied immediately after the receipt of sufficient rain or canal water.
- Hand weeding, thinning and gap filling should be done before N-fertilizer application.
- Subsequent top dressings in two or three splits should be done before heading.

**5.10. Water management**

- As that of irrigated rice when canal water is used for irrigation
- Possibility of subsequent conversion towards deep water situation is seen in this tract, then variety should be specific for those areas.

**5.11. Insect management:** See Crop Protection Chapter

**5.12. Disease management:** See Crop Protection Chapter

**5.13. Harvest**

- As that of transplanted rice. This area is more suitable to combine harvester.

**5.14. DEEP WATER RICE**

- Cultivation is like the methods described in this section except the harvest. Harvest may some times restricted only to panicle because of the standing water even after maturity.

## DRY SEEDED UPLAND RICE

**Establishment**

- As that of section 3 to 5.

**Area**

- There are small batches in and around Dharmapuri district. Rainfall availability in these tract is better than the rainfed rice cultivated in other parts of Tamil Nadu. There is no bund to stagnate the water. Moisture availability is there but crop growth depends on the nutrient status.

**Other Cultural practices**

- As recommended to semi-dry rice (sec. 4)
- Nutrient may be split applied depending up on the growth.
- LCC based N application is more suitable for this tract.

**Intercropping**

- Blackgram for every four rows of rice.

**Grain Yield**

- Grain yield depends up on the moisture availability and nutrient status.

## Appendix – I

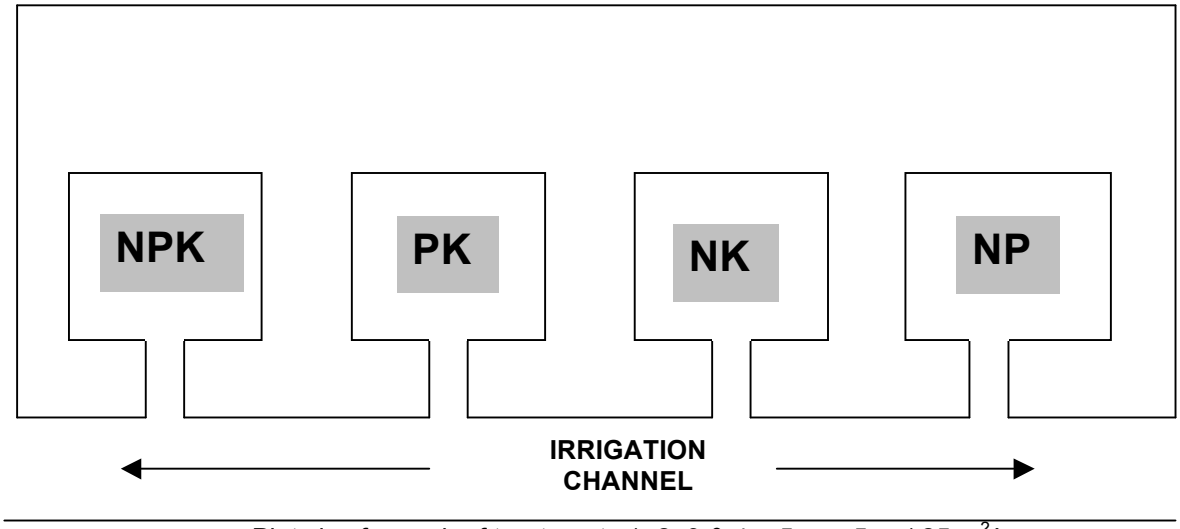
**Site Specific Nutrient Management through Omission Plot Techniques**

An omission plot receives all nutrients except the omitted nutrient (0 P or 0 K). The objective of P and K omission plots is to develop a site-specific P and K fertilizer recommendation. The difference in grain yields between a fully fertilized plot and a P or K omission plot illustrates the deficit between the crop demand for P or K and indigenous supply of P or K, from soil, rain water etc., the deficit has to be met by fertilizer application.

**Guidelines for establishing and managing omission plots**

- Establish omission plots both in dry season as well as wet season (high yielding season of any region is more preferable)
- Measure four plots of 5 m × 5 m size each within a farmer's field. Do not place the plots in a corner of the field. Separate each plot by about 3 m distance from the next plot to minimize risk of fertilizer movement between the plots.
- Construct bunds of sufficient height surrounding each plot to prevent water flow between the plots and the adjacent field. Maintain the bunds nicely throughout the crop season to avoid fertilizer and water contamination from one plot to another or from outside the plot. Provide guard channel all around the plot.
- All the crop management practices including pest management should be under optimal conditions and uniform in the treatment plots.
- Apply fertilizers to the plots strictly as per the treatment with the involvement of the farmers.

**Fig. 1. Field layout of omission plot trial**




Plot size for each of treatments 1, 2, 3 & 4 = 5 m × 5 m ( 25 $m^2$ **)**

***a. For heavy textured soil***

$T_1$ = + NPK - 193 g Urea, 940 g SP, 350 g MOP
$T_2$ = + PK - 940 g SP and 350 g MOP
$T_3$ = + NK - 193 g Urea and 350 g MOP
$T_4$ = + NP - 193 g Urea and 940 g SP

**b. For light textured soil**

$T_1$ = + NPK - 193 g Urea, 940 g SP, 400 g MOP
$T_2$ = + PK - 940 g SP and 400 g MOP
$T_3$ = + NK - 193 g Urea and 400 g MOP
$T_4$ = + NP - 193 g Urea and 940 g SP

SP – Super phosphate full basal,
MOP – Muriate of Potash in two splits 50 % basal and 50 % at PI
Urea - Applied in 4 splits at basal, AT, PI and FF stages each time 193 g
As the Omission Plot technique is only an experimentation for fertilizer calibration we should apply N on blanket basis without using LCC in order to avoid risk in N supply based on color combination
Apply Zinc Sulphate @ 65 g/25 $m^2$ uniformly to all the plots

**Determination of grain yield**

- Harvest all the hills of a central 5-$m^2$ area of each treatment plot separately and properly label each harvested sample.
- Thresh and clean the harvested grain. Record the grain yield at 14% moisture level.
- Convert the sample weight to yield in t $ha^{-1.}$
- Determine the P and K response
- P response = Yield in NPK plot – Yield in P omission plot
- K response = Yield in NPK plot – Yield in K omission plot
- From the P and K response, P & K requirement of the field will be calculated based on the ready reckonar table as suggested by Fairthurst and Witt (2002), as in the table below.

**Note:** The N omission plot yield (0 N Plot) is only to understand the indigenous N supplying capacity of the soil and to decide on the need for basal N application. If the N omission plot yield is < 3.0 t/ha, apply 20-30 kg N/ha basally within 14 DAT, or if the yield is > 3.0 t, there is no need to give basal N application.

**Recommended $P_2O_5$ rates according to target yield and P limited plot**

| Target yield (t/ha) | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|
| **Yield in '0' P plot** | Fertilizer $P_2O_5$ (kg/ha) | | | | |
| 3 | 20 | 40 | 60 | | |
| 4 | 15 | 25 | 40 | 60 | |
| 5 | 0 | 20 | 30 | 40 | 60 |
| 6 | 0 | 0 | 25 | 35 | 45 |
| 7 | 0 | 0 | 0 | 30 | 40 |
| 8 | 0 | 0 | 0 | 0 | 35 |

Example:

| Experiment | Yield (t/ka) |
|---|---|
| **$P_2O_5$** fertilized | 6 |
| P omitted | 5 |
| Calculated $P_2O_5$ (kg/ha) to get an yield of 6 t/ha | 30 |

**Recommended $K_2O$ rates according to target yield and K limited plot**

| Target yield (t/ha) | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|
| **Yield in '0' K plot** | Fertilizer $K_2O$ (kg/ha) | | | | |
| 3 | 30 | 60 | 90 | | |
| 4 | 0 | 35 | 65 | 95 | |
| 5 | 0 | 20 | 50 | 80 | 110 |
| 6 | 0 | 0 | 35 | 65 | 95 |
| 7 | 0 | 0 | 0 | 50 | 80 |
| 8 | 0 | 0 | 0 | 0 | 65 |

Example:

| Experiment | Yield (t/ka) |
|---|---|
| **$K_2O$** fertilized | 6 |
| K omitted | 5 |
| Calculated $K_2O$ (kg/ha) to get an yield of 6 t/ha | 50 |

# Appendix II

**STCR based fertilizer recommendation for transplanted rice**
**Coimbatore STL Jurisdiction**

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Soil type** | : | Alfisol (Noyyal series) | **Season** | : | Kharif | **Yield target** | : | 6.0 t $ha^{-1}$ |

Basic data for Fertiliser Prescription Equations

| | Basic Data | | | | Fertiliser Prescription Equations |
|---|---|---|---|---|---|
| | NR (kg/q) | CS (%) | CF (%) | CO (%) | |
| N | 1.76 | 20.76 | 40.12 | 32.10 | FN = 4.39 T - 0.52 SN - 0.80 ON |
| P | 0.41 | 29.35 | 18.50 | 18.13 | $FP_2O_5$ = 2.22 T - 3.63 SP - 0.98 OP |
| K | 1.50 | 19.83 | 61.50 | 44.65 | $FK_2O$ = 2.44 T - 0.39 SK - 0.72 OK |

Ready reckoner of fertiliser doses at varying soil test values for specific yield target

| S. No. | Initial soil tests (kg $ha^{-1}$) | | | Nutrients required (kg $ha^{-1}$) - target of 6.0 t $ha^{-1}$ | | |
|---|---|---|---|---|---|---|
| | N | P | K | N | $P_2O_5$ | $K_2O$ |
| 1 | 180 | 16 | 180 | 170 | 75 | 76 |
| 2 | 190 | 18 | 190 | 165 | 68 | 72 |
| 3 | 200 | 20 | 200 | 159 | 61 | 68 |
| 4 | 210 | 22 | 210 | 154 | 53 | 64 |
| 5 | 220 | 24 | 220 | 149 | 46 | 60 |
| 6 | 230 | 26 | 230 | 143 | 39 | 56 |

| | | | | | | |
|---|---|---|---|---|---|---|
| 7 | 240 | 28 | 240 | 138 | 32 | 52 |
| 8 | 250 | 30 | 250 | 133 | 24 | 49 |
| 9 | 260 | 32 | 260 | 128 | 17 | 45 |
| 10 | 270 | 34 | 270 | 123 | 10 | 41 |
| 11 | 280 | 36 | 280 | 118 | 3 | 37 |

| **Soil type** | : | Alfisol (Noyyal series) | **Season** | : | Rabi | **Yield target** | : | 6.0 t $ha^{-1}$ |
|---|---|---|---|---|---|---|---|---|

Basic data for Fertiliser Prescription Equations

| | Basic Data | | | | Fertiliser Prescription Equations |
|---|---|---|---|---|---|
| | NR (kg/q) | CS (%) | CF (%) | CO (%) | |
| N | 1.71 | 20.69 | 36.95 | 33.20 | FN = 4.63 T - 0.56 SN - 0.90 ON |
| P | 0.47 | 32.87 | 23.67 | 10.20 | $FP_2O_5$ = 1.98 T - 3.18 SP - 0.99 OP |
| K | 1.94 | 26.18 | 75.42 | 50.17 | $FK_2O$ = 2.57 T - 0.42 SK - 0.67 OK |

Ready reckoner of fertiliser doses at varying soil test values for specific yield target

| S. No. | Initial soil tests (kg $ha^{-1}$) | | | Nutrients required (kg $ha^{-1}$) for an yield target of 6.0 t $ha^{-1}$ | | |
|---|---|---|---|---|---|---|
| | N | P | K | N | $P_2O_5$ | $K_2O$ |
| 1 | 180 | 16 | 180 | 177 | 68 | 78 |
| 2 | 190 | 18 | 190 | 172 | 62 | 74 |
| 3 | 200 | 20 | 200 | 166 | 55 | 70 |
| 4 | 210 | 22 | 210 | 160 | 49 | 66 |
| 5 | 220 | 24 | 220 | 155 | 43 | 62 |
| 6 | 230 | 26 | 230 | 149 | 36 | 57 |
| 7 | 240 | 28 | 240 | 144 | 30 | 53 |
| 8 | 250 | 30 | 250 | 138 | 24 | 49 |
| 9 | 260 | 32 | 260 | 132 | 17 | 45 |
| 10 | 270 | 34 | 270 | 127 | 11 | 41 |

**Erode, Coimbatore, Salem & Trichy STL Jurisdiction**

| **Soil type** | : | Inceptisol (Irugur series) | **Season** | : | Kharif | **Yield target** | : | 6.0 t $ha^{-1}$ |
|---|---|---|---|---|---|---|---|---|

Basic data for Fertiliser Prescription Equations

| | Basic Data | | | | Fertiliser Prescription Equations |
|---|---|---|---|---|---|
| | NR (kg/q) | CS (%) | CF (%) | CO (%) | |
| N | 1.82 | 31.27 | 35.13 | 34.43 | FN = 5.19 T - 0.89 SN - 0.98 ON |
| P | 0.84 | 72.92 | 37.11 | 17.81 | $FP_2O_5$ = 2.27 T - 4.50 SP - 1.09 OP |
| K | 2.14 | 33.55 | 68.81 | 58.00 | $FK_2O$ = 3.11 T - 0.59 SK - 1.02 OK |

Ready reckoner of fertiliser doses at varying soil test values for specific yield target

| S.No. | Initial soil tests (kg $ha^{-1}$) | | | Nutrients required (kg $ha^{-1}$)- target of 6.0 t $ha^{-1}$ | | |
|---|---|---|---|---|---|---|
| | N | P | K | N | $P_2O_5$ | $K_2O$ |
| 1 | 150 | 8 | 150 | 178 | 100 | 98 |
| 2 | 160 | 10 | 160 | 169 | 91 | 93 |
| 3 | 170 | 12 | 170 | 160 | 82 | 87 |
| 4 | 180 | 14 | 180 | 151 | 73 | 81 |
| 5 | 190 | 16 | 190 | 142 | 64 | 75 |
| 6 | 200 | 18 | 200 | 133 | 55 | 69 |
| 7 | 210 | 20 | 210 | 124 | 46 | 63 |
| 8 | 220 | 22 | 220 | 115 | 37 | 57 |
| 9 | 230 | 24 | 230 | 106 | 28 | 51 |
| 10 | 240 | 26 | 240 | 97 | 19 | 45 |

| **Soil type** | : | Inceptisol (Irugur series) | **Season** | : | Rabi | **Yield target** | : | 6.0 t $ha^{-1}$ |
|---|---|---|---|---|---|---|---|---|

Basic data for Fertiliser Prescription Equations

| | Basic Data | | | | Fertiliser Prescription Equations |
|---|---|---|---|---|---|
| | NR kg/ q) | CS (%) | CF (%) | CO (%) | |
| N | 1.75 | 24.31 | 35.75 | 25.74 | FN = 4.88 T - 0.68 SN - 0.72 ON |
| P | 0.92 | 56.98 | 44.87 | 44.42 | $FP_2O_5$ = 2.06 T - 2.91 SP - 2.27 OP |
| K | 1.92 | 26.12 | 66.42 | 32.55 | $FK_2O$ = 2.89 T - 0.47 SK - 0.59 OK |

Ready reckoner of fertiliser doses at varying soil test values for specific yield target

| S.No. | Initial soil tests (kg $ha^{-1}$) | | | Nutrients required (kg $ha^{-1}$)- target of 6.0 t $ha^{-1}$ | | |
|---|---|---|---|---|---|---|
| | N | P | K | N | $P_2O_5$ | $K_2O$ |
| 1 | 150 | 8 | 150 | 191 | 100 | 103 |
| 2 | 160 | 10 | 160 | 184 | 94 | 98 |
| 3 | 170 | 12 | 170 | 177 | 89 | 94 |
| 4 | 180 | 14 | 180 | 171 | 83 | 89 |
| 5 | 190 | 16 | 190 | 164 | 77 | 84 |
| 6 | 200 | 18 | 200 | 157 | 71 | 79 |
| 7 | 210 | 20 | 210 | 150 | 65 | 75 |
| 8 | 220 | 22 | 220 | 143 | 56 | 70 |
| 9 | 230 | 24 | 230 | 137 | 54 | 65 |
| 10 | 240 | 26 | 240 | 130 | 48 | 61 |

**Kovilpatti STL Jurisdiction**

| **Soil type** | : | **Inceptisol** (Manakkarai series) | **Season** | : | Kharif | **Yield target** | : | 6.0 t $ha^{-1}$ |
|---|---|---|---|---|---|---|---|---|

Basic data for Fertiliser Prescription Equations

| | Basic Data | | | | Fertiliser Prescription Equations |
|---|---|---|---|---|---|
| | NR (kg/q) | CS (%) | CF (%) | CO (%) | |
| N | 1.54 | 21.90 | 36.20 | 28.5 | FN = 4.25 T - 0.60 SN - 0.79 ON |
| P | 0.53 | 37.60 | 19.60 | 7.6 | $FP_2O_5$ = 2.71 T - 4.39 SP - 0.89 OP |
| K | 1.94 | 25.04 | 50.90 | 34.5 | $FK_2O$ = 3.83 T - 0.60 SK - 0.82 OK |

Ready reckoner of fertiliser doses at varying soil test values for specific yield target

| **S.No** | Initial soil tests (kg $ha^{-1}$) | | | Nutrients required (kg $ha^{-1}$) - target of 6.0 t $ha^{-1}$ | | |
|---|---|---|---|---|---|---|
| | N | P | K | N | $P_2O_5$ | $K_2O$ |
| 1 | 180 | 12 | 180 | 147 | 110 | 122 |
| 2 | 190 | 14 | 190 | 141 | 102 | 116 |
| 3 | 200 | 16 | 200 | 135 | 93 | 110 |
| 4 | 210 | 18 | 210 | 129 | 84 | 104 |
| 5 | 220 | 20 | 220 | 123 | 75 | 98 |
| 6 | 230 | 22 | 230 | 117 | 66 | 92 |
| 7 | 240 | 24 | 240 | 111 | 58 | 86 |
| 8 | 250 | 26 | 250 | 105 | 49 | 80 |
| 9 | 260 | 28 | 260 | 99 | 40 | 74 |
| 10 | 270 | 30 | 270 | 93 | 31 | 68 |
| 11 | 280 | 32 | 280 | 87 | 23 | 62 |

| **Soil type** | : | **Inceptisol** (Manakkarai series) | **Season** | : | Rabi | **Yield target** | : | 6.0 t $ha^{-1}$ |
|---|---|---|---|---|---|---|---|---|

Basic data for Fertiliser Prescription Equations

| | Basic Data | | | | Fertiliser Prescription Equations |
|---|---|---|---|---|---|
| | NR (kg/q) | CS (%) | CF (%) | CO (%) | |
| N | 1.62 | 21.02 | 36.24 | 28.63 | FN = 4.47 T - 0.58 SN - 0.79 ON |
| P | 0.46 | 27.84 | 17.29 | 6.74 | $FP_2O_5$ = 2.66 T - 3.68 SP - 0.89 OP |
| K | 2.06 | 27.26 | 50.49 | 34.22 | $FK_2O$ = 4.08 T - 0.65 SK - 0.82 OK |

Ready reckoner of fertiliser doses at varying soil test values for specific yield target

| S.No | Initial soil tests (kg $ha^{-1}$) | | | Nutrients required (kg $ha^{-1}$)- target of 6.0 t $ha^{-1}$ | | |
|---|---|---|---|---|---|---|
| | N | P | K | N | $P_2O_5$ | $K_2O$ |
| 1 | 180 | 12 | 180 | 164 | 115 | 128 |
| 2 | 190 | 14 | 190 | 158 | 108 | 121 |
| 3 | 200 | 16 | 200 | 152 | 101 | 115 |
| 4 | 210 | 18 | 210 | 146 | 93 | 108 |
| 5 | 220 | 20 | 220 | 140 | 86 | 102 |
| 6 | 230 | 22 | 230 | 135 | 79 | 95 |
| 7 | 240 | 24 | 240 | 129 | 71 | 89 |
| 8 | 250 | 26 | 250 | 123 | 64 | 82 |
| 9 | 260 | 28 | 260 | 117 | 57 | 76 |
| 10 | 270 | 30 | 270 | 111 | 50 | 69 |
| 11 | 280 | 32 | 280 | 106 | 42 | 63 |

**Aduthurai STL Jurisdiction**

| **Soil type** | : | **Vertisol (Kalathur series)** | **Season** | : | Kharif | **Yield target** | : | 6.0 t $ha^{-1}$ |
|---|---|---|---|---|---|---|---|---|

Basic data for Fertiliser Prescription Equations

| | Basic Data | | | | Fertiliser Prescription Equations |
|---|---|---|---|---|---|
| | NR (kg/ q) | CS (%) | CF (%) | CO (%) | |
| N | 1.68 | 23.80 | 31.75 | 28.26 | FN = 5.29 T - 0.75 SN - 0.89 ON |
| P | 0.87 | 40.40 | 52.70 | 17.92 | $FP_2O_5$ = 1.65 T - 1.76 SP - 0.78 OP |
| K | 1.65 | 18.70 | 60.50 | 41.14 | $FK_2O$ = 2.73 T - 0.37 SK - 0.82 OK |

Ready reckoner of fertiliser doses at varying soil test values for specific yield target

| **S.No** | Initial soil tests (kg $ha^{-1}$) | | | Nutrients required (kg $ha^{-1}$)- target of 6.0 t $ha^{-1}$ | | |
|---|---|---|---|---|---|---|
| | N | P | K | N | $P_2O_5$ | $K_2O$ |
| 1 | 180 | 16 | 180 | 182 | 71 | 97 |
| 2 | 190 | 18 | 190 | 174 | 67 | 93 |
| 3 | 200 | 20 | 200 | 167 | 64 | 90 |
| 4 | 210 | 22 | 210 | 159 | 60 | 86 |
| 5 | 220 | 24 | 220 | 152 | 57 | 82 |
| 6 | 230 | 26 | 230 | 144 | 53 | 79 |
| 7 | 240 | 28 | 240 | 137 | 50 | 75 |
| 8 | 250 | 30 | 250 | 129 | 46 | 71 |
| 9 | 260 | 32 | 260 | 122 | 43 | 68 |
| 10 | 270 | 34 | 270 | 114 | 39 | 64 |
| 11 | 280 | 36 | 280 | 107 | 36 | 60 |

| **Soil type** | : | **Vertisol (Kalathur series)** | **Season** | : | Rabi | **Yield target** | : | 6.0 t $ha^{-1}$ |
|---|---|---|---|---|---|---|---|---|

Basic data for Fertiliser Prescription Equations

| | Basic Data | | | | Fertiliser Prescription Equations |
|---|---|---|---|---|---|
| | NR (Kg/ q) | CS (%) | CF (%) | CO (%) | |
| N | 1.72 | 21.42 | 32.18 | 23.49 | FN = 5.34 T - 0.67 SN - 0.73 ON |
| P | 0.92 | 39.10 | 48.20 | 14.73 | $FP_2O_5$ = 1.90 T - 1.86 SP - 0.70 OP |
| K | 1.84 | 17.80 | 65.42 | 43.25 | $FK_2O$ = 2.81 T - 0.33 SK - 0.80 OK |

Ready reckoner of fertiliser doses at varying soil test values for specific yield target

| S.No | Initial soil tests (kg $ha^{-1}$) | | | Nutrients required (kg $ha^{-1}$) - target of 6.0 t $ha^{-1}$ | | |
|---|---|---|---|---|---|---|
| | N | P | K | N | $P_2O_5$ | $K_2O$ |
| 1 | 180 | 16 | 180 | 199 | 84 | 110 |
| 2 | 190 | 18 | 190 | 193 | 81 | 106 |
| 3 | 200 | 20 | 200 | 186 | 77 | 103 |
| 4 | 210 | 22 | 210 | 179 | 73 | 100 |
| 5 | 220 | 24 | 220 | 173 | 69 | 96 |
| 6 | 230 | 26 | 230 | 166 | 66 | 93 |
| 7 | 240 | 28 | 240 | 159 | 62 | 90 |
| 8 | 250 | 30 | 250 | 152 | 58 | 86 |
| 9 | 260 | 32 | 260 | 146 | 55 | 83 |
| 10 | 270 | 34 | 270 | 139 | 51 | 80 |
| 11 | 280 | 36 | 280 | 132 | 47 | 77 |

**Aduthurai STL Juristiction**

| **Soil type** | : | **Vertisol (Adanur series)** | **Season** | : | Kharif | **Yield target** | : | 7.0 t $ha^{-1}$ |
|---|---|---|---|---|---|---|---|---|

Basic data for Fertiliser Prescription Equations

| | Basic Data | | | | Fertiliser Prescription Equations |
|---|---|---|---|---|---|
| | NR (kg/ q) | CS (%) | CF (%) | CO (%) | |
| N | 1.56 | 16.3 | 56.0 | 49.84 | FN = 2.80 T - 0.29 SN - 0.89 ON |
| P | 0.68 | 28.2 | 50.4 | 39.30 | $FP_2O_5$ = 1.35 T - 1.28 SP - 1.78 OP |
| K | 1.64 | 22.6 | 65.6 | 61.99 | $FK_2O$ = 2.50 T - 0.42 SK - 1.14 OK |

Ready reckoner of fertiliser doses at varying soil test values for specific yield target

| S.No | Initial soil tests (kg $ha^{-1}$) | | | Nutrients required (kg $ha^{-1}$)- target of 7.0 t $ha^{-1}$ | | |
|---|---|---|---|---|---|---|
| | N | P | K | N | $P_2O_5$ | $K_2O$ |
| 1 | 180 | 16 | 180 | 144 | 74 | 99 |
| 2 | 190 | 18 | 190 | 141 | 72 | 95 |
| 3 | 200 | 20 | 200 | 138 | 69 | 91 |
| 4 | 210 | 22 | 210 | 135 | 66 | 87 |
| 5 | 220 | 24 | 220 | 132 | 64 | 83 |
| 6 | 230 | 26 | 230 | 129 | 61 | 78 |
| 7 | 240 | 28 | 240 | 126 | 59 | 74 |
| 8 | 250 | 30 | 250 | 123 | 56 | 70 |
| 9 | 260 | 32 | 260 | 121 | 54 | 66 |
| 10 | 270 | 34 | 270 | 118 | 51 | 62 |
| 11 | 280 | 36 | 280 | 115 | 48 | 57 |

**Note:** Wherever GLM is applied @ 6.25 t $ha^{-1}$ 38, 13 and 33 kg of N, $P_2O_5$ and $K_2O$ and for the addition of azospirillum & phosphobacteria each @ 2 kg $ha^{-1}$ 15 and 10 kg of N & $P_2O_5$ respectively could be reduced from the recommended fertiliser nutrient doses.

**References:**

Selvakumari , G., Santhi,R., Natesan,R and Mathan, K. K. 2000. Compendium of research on soil test crop response studies and rationalized fertilizer recommendations for crops in Tamil Nadu. Dept of SS&AC, TNAU, Coimbatore-3: pp 93

# CROP PROTECTION

**A) Pest management**

**i) Nursery**

An area of 800 sq.m. (20 cents) nursery is required for planting one ha of main field. Forty litres of spray fluid is necessary for spraying the nursery area.

| Pests | Management strategies |
|---|---|
| **Army worm**<br>*Spodoptera mauritia* | ▪ Drain water from the nursery<br>▪ **Spray any one of the following:**<br>Chlorpyriphos 20 EC 80 ml<br>Endosulfan 35 EC 80 ml during late evening |
| **Thrips**<br>*Stenchaetothrips biformis* | ▪ Sampling: Wet your palm with water and pass over the foliage in 12 places in the nursery. If thrips population exceeds 60 numbers in 12 passes or if rolling of 1/2 area of first and 2nd leaves in 10% of seedlings is noticed,<br>▪ **Spray any one of the following:**<br>Phosphamidon 40 SL 50 ml<br>Monocrotophos 36 SL 40 ml<br>Endosulfan 35 EC 80 ml |
| **Green leafhopper**<br>*Nephotettix nigropictus*<br>*N. cincticeps*<br>*N. virescens* | ▪ Sampling: Take 25 net sweepings in the nursery area. If the population exceeds 60 for 25 sweepings or 20/$m^2$ by actual counting,<br>▪ **Spray any one of the following:**<br>Fenitrothion 50 EC 80 ml<br>Phosphamidon 40 SL 50 ml<br>Fenthion 100 EC 40 ml<br>Quinalphos 25 EC 80 ml<br>Phosalone 35 EC 120 ml<br>Endosulfan 35 EC 80 ml<br>Monocrotophos 36 SL 40 ml<br>▪ **Maintain 2.5 cm of water in the nursery and broadcast anyone of the following in 20 cents**<br>Carbofuran 3 G 3.5 kg<br>Phorate 10 G 1.0 kg<br>Quinalphos 5 G 2.0 kg |
| **Caseworm**<br>*Parapoynx stagnalis* | ▪ Mix 250 ml of kerosene to the standing water<br>▪ Dislodge the cases by passing a rope and drain water<br>▪ Collect the cases and destroy<br>▪ **Spray any one of the following:**<br>Monocrotophos 36 SL 40 ml<br>Quinalphos 25 EC 80 ml. |

**ii) Main field**

- Remove/destroy stubbles after harvest
- Keep the fields free from weeds
- Trim field bunds
- Provide effective drainage, if required
- Avoid use of excessive 'N' fertilizers.
- Avoid close planting, especially in BPH and leaffolder prone areas and seasons.
- Leave 30 cm space at every 2.5 M
- Use irrigation water judiciously
- Use light traps to monitor pest incidence
- Remove egg masses of stem borer
- In BPH prone areas/seasons, avoid use of synthetic pyrethroids, methyl parathion and quinalphos and use recommended chemical at recommended doses.
- Use insecticides based on ETLs.

**Economic threshold level for important pests**

| Pests | ETL |
|---|---|
| Stem borer | 2 egg masses/$m^2$ or 10% dead hearts |
| Leaffolder | 10% leaf damage at vegetative phase and 5% of flag leaf damage at flowering |
| Gall midge | 10% silver shoots |
| Whorl maggot | 25% damaged leaves |
| Thrips | 60 numbers in 12 passes or rolling of the first and second leaves in 10% of seedlings. |
| Brown planthopper | 1 hopper/ tiller in the absence of predatory spider and 2 hoppers /tiller when spider is present at 1/hill. |
| Green leafhopper | 60/25 net sweeps or 5/hill at vegetative stage or 10/hill at flowering or 2/hill in tungro endemic area |
| Earhead bug | 5 bugs/100 earheads at flowering and 16 bugs/100 earheads from milky stage to grain maturity |

| Pests | Management strategies |
|---|---|
| **Stem borer**<br>*Scirpophaga incertulas* | ▪ 25% of existing ETL for insecticides (2 egg masses/$m^2$) is the action threshold level (ATL) for release of the egg parasitoid, *Trichogramma japonicum* for the management of the rice yellow stem borer. 25% of ETL adopted for application of insecticide is the ATL for spraying Neem seed kernel extract to control stem borer.<br>▪ **Spray any one of the following insecticides :**<br>Fenthion 100 EC 500 ml/ha<br>Monocrotophos 36 SL 1000 ml/ha<br>Endosulfan 35 EC 1000 ml/ha<br>Fenitrothion 50 EC 1000 ml/ha<br>Phosalone 35 EC 1500 ml/ha<br>Quinalphos 25 EC 1000 ml/ha<br>Phosphamidon 40 SL 600 ml/ha<br>Profenophos 50 EC 1000 ml/ha |
| **Leaffolder**<br>*Cnaphalocrocis medinalis* | ▪ Release both *Trichogramma chilonis* (for leaffolder) and *T. japonicum* (for stem borer) thrice @ 1,00,000/ha each (if moth activity is noticed) and spray *Bacillus thuringiensis* @ 1.0 kg/ha when the stem borer / leaffolder crosses ETL.<br>▪ **Spray any one of the following:**<br>Fenitrothion 50 EC 1000 ml/ha<br>Monocrotophos 36 SL 1000 ml/ha<br>Phosalone 35 EC 1500 ml/ha<br>Quinalphos 25 EC 1000 ml/ha<br>Dichlorvos 76 WSC 250 ml/ha<br>Phosphamidon 40 SL 600 ml/ha<br>Chlorpyriphos 20 EC 1250 ml/ha<br>Carbaryl 50 WP 1.0 kg/ha<br>Fenthion 100 EC 500 ml/ha<br>Profenophos 50 EC 1000 ml/ha<br>Neem seed kernel extract 5% 25 kg/ha |
| **Gall midge**<br>*Orseolia oryzae* | ▪ Release *Platygaster oryzae* parasitised galls at 1 per 10 $m^2$ on 10 days after transplanting (DAT)<br>▪ **Spray any one of the following :**<br>Fenthion 100 EC 500 ml/ha<br>Endosulfan 35 EC 1000 ml/ha<br>Fenitrothion 50 EC 1000 ml/ha |

| | |
|---|---|
| | Phosalone 35 EC 1500 ml/ha<br>Quinalphos 25 EC 1000 ml/ha<br>Phosphamidon 40 SL 600 ml/ha |
| **Thrips**<br>*Stenchaetothrips biformis* | ▪ **Spray any one of the following :**<br>Phosphamidon 40 SL 600 ml /ha<br>Monocrotophos 36 SL 500 ml /ha<br>Endosulfan 35 EC 1000 ml / ha |
| **Brown planthopper**<br>*Nilaparvata lugens* | ▪ Avoid excessive use of nitrogen<br>▪ Control irrigation by intermittent draining<br>▪ Set up light traps during night or yellow pan traps during day time<br>▪ Drain the water before use of insecticides and direct the spray towards the base of the plants.<br>▪ **Apply any one of the following :**<br>Phosphamidon 40 SL 1000 ml/ha<br>Monocrotophos 36 SL 1250 ml/ha<br>Phosalone 35 EC 1500 ml/ha<br>Carbaryl 10 D 25 kg/ha<br>Methyl demeton 25 EC 1000 ml/ha<br>Acephate 75 SP 625 gm/ha<br>Chlorpyriphos 20 EC 1250 ml/ha<br>Carbofuran 3 G 17.5 kg/ha<br>Dichlorvos 76 WSC 350 ml/ ha<br>Neem oil 3% 15 lit/ha<br>Iluppai oil 6% 30 lit/ha<br>. Neem seed kernel extract 5% 25 kg/ha |
| **White backed planthopper**<br>*Sogatella furcifera* | ▪ Phosphamidon 40 SL 1000 ml/ha. |
| **Green leafhopper**<br>*Nephotettix nigropictus*<br>*N. cincticeps*<br>*N. virescens* | ▪ Apply insecticides twice, 15 and 30 days after transplanting.<br>Monocrotophos 36 SL 1000 ml/ha<br>Phosphamidon 40 SL 1000 ml /ha<br>Fenthion 100 EC 500 ml/ha<br>Profenophos 50 EC 1000 ml/ha<br>▪ The vegetation on the bunds should also be sprayed with the insecticides<br>▪ Set up light traps to attract and control the leafhopper vectors as well as to monitor the population.<br>▪ Kill the leafhoppers attracted to light trap by spraying any one of the insecticides every morning. |
| **Mealy bug**<br>*Brevennia rehi* | ▪ **Spray any one of the following:**<br>Phosphamidon 40 SL 600 ml/ha<br>Fenitrothion 50 EC 1000 ml/ha<br>Phosalone 35 EC 1500 ml/ha<br>Dimethoate 30 EC 500 ml/ha |
| **Black bug**<br>*Scotinophora lurida* | ▪ **Spray any one of the following :**<br>Monocrotophos 36 SL 1000 ml/ha<br>Acephate 75 SP 625 g/ha<br>Neem seed kernel extract 5% 25 kg/ha |
| **Earhead bug**<br>*Leptocorisa acuta* | ▪ **Dust any one of the following at 25 kg/ha twice, the first during flowering and second a week later :**<br>Quinalphos 1.5 D<br>Fenitrothion 2 D<br>Carbaryl 10 D<br>Malathion 5 D<br>KKM 10 D |

| | |
|---|---|
| | The new KKM dust formulation consists of 10% of *Acorus calamus* rhizome powder and 90% of flyash which is a waste product from Thermal Power Station. This dust formulation repels the rice earhead bug.<br>▪ **Spray any one of the following twice as above**<br>Monocrotophos 36 SL 500 ml/ha<br>Fenitrothion 50 EC 1000 ml/ha<br>Fenthion 100 EC 500 ml/ha<br>Malathion 50 EC 500 ml/ha<br>Neem seed kernel extract 5% 25 kg/ha<br>Notchi or *Ipomoea* or *Prosopis* leaf extract 10% |
| **Termite**<br>*Anacanthotermus viarum* | ▪ Apply chopped paddy straw treated with quinalphos 1.5 D 50 kg/ha |
| **Mite**<br>*Oligonychus oryzae* | ▪ Apply dicofol 18.5 EC 1250 ml/ha. |
| **Rat**<br>*Rattus rattus rufuscens, Rattus meltada* | ▪ Poison bait at 1 part zinc phosphide with 49 parts popped corn/rice/dry fish or warfarin 0.5% 1 part with 19 parts of popped corn/rice/dry fish or bromodialone 0.25 w/w (1:49) at 0.005%. Mix one part of bromodialone + 49 parts of bait and keep inside the field.<br>▪ Setting up of owl perches |

**IPM module**

- *Pseudomonas fluorescens* – Seed treatment (10 g/kg), seedling dip (2.5 kg/ha), main field application (2.5 kg/ha)
- Pest and disease management in nursery (preferably neem seed kernel extract 5% or Neem oil 2%)
- Integrated Nutrient Management
    - Use of neem cake coated urea (5 : 1)
    - Inclusion of green manures / biofertilizers
    - 'N' management by Leaf Colour Chart (LCC)
    - 'K' application – basal (50%) + one top dressing (50%)
- Adoption of cultural practices
    - Variety selection
    - Spacing based on season, variety and location (endemic / hot spot)
    - Rogueing space (1' for every 8')
- Water management – alternate wetting and drying and submergence of recommended level during critical periods only
- Release of biocontrol agents, when the moth activity is noticed
    - *Trichogramma japonicum* for stem borer @ 1,00,000 (5 cc) / ha at weekly interval for 3 times
    - *Trichogramma chilonis* for leaffolder @ 1,00,000 (5 cc) /ha at weekly interval for 3 times

Set up bird (owl) perches at 40 to 50 /ha

- Application of botanicals especially Neem seed kernel extract 5% against leaffolder
- ETL based insecticide / fungicide application (No synthetic pyrethroids)
- Integrated rodent management
    - Narrow bund maintenance (45 x 30 cms)
    - Zinc phosphide baiting (49 : 1)
    - Trapping with Thanjavur bow trap (100 nos./ha)
    - Baiting with bromodialone

**Insecticide Resistance**

In case of control failures monitor the insecticide resistance with the following discriminating dose screen.

**Rice leaffolder**- *Cnaphalocrocis medinalis* Guenee

(IV instar larvae of 20-30 mg weight and 15-18 mm)

1. Monocrotophos topical 0.3375 µg
2. Quinalphos topical 0.5112 µg
3. Chlorpyriphos topical 1.1405 µg
4. Phosphamidon topical 5.3979 µg

**Resurgence**

**Repeated application of the following insecticides can cause resurgence of insect pests**

- ***Nilaparvata lugens :*** acephate, azinphosmethyl, BPMC, carbofuran, chlorpyriphos, cypermethrin, deltamethrin, diazinon, ethopenprox, fenitrothion, fenthion, fenvalerate, methomyl, methylparathion, monocrotophos, permethrin, perthane, phorate, phosalone, phosphamidon, quinalphos, thiometon, triazophos, vamidothion
- ***Nephotettix virescens :*** deltamethrin, phorate
- ***Sogatella furcifera :*** cypermethrin, deltamethrin, fenvalerate
- ***Cnaphalocrocis medinalis :*** carbofuran, phorate

**RTD Management**

- To control the vector, green leafhopper in the main field, spray two rounds of any one of the insecticides viz., monocrotophos 36 SL (1000 ml/ha), phosphamidon 40 SL (1000 ml/ha), fenthion 100 EC (500 ml/ha) on 15 and 30 days after transplanting. The vegetation on the bunds should also be sprayed with the insecticides.
- Set up light traps to attract and control the leafhopper vectors as well as to monitor the population. In the early morning, the population of leafhopper alighting near the light trap should be killed by spraying/dusting the insecticides. This should be practiced every day.

**B) Disease management**

| **Disease management in nursery** | |
|---|---|
| **Dry seed treatment** | ▪ Thiram or captan or carboxin or carbendazim at 2 g/kg of seeds.<br>▪ Treat the seeds at least 24 hours prior to soaking for sprouting. The treated seeds can be stored for 30 days without any loss in viability. |
| **Wet seed treatment** | ▪ Carbendazim or Tricyclozole at 2 g/lit of water for 1 kg of seed.<br>▪ Soak the seeds in the solution for 2h<br>▪ Drain the solution, sprout the seeds and sow in the nursery bed.<br>▪ This wet seed treatment gives protection to the seedlings up to 40 days from seedling disease such as blast and this method is better than dry seed treatment<br>**or**<br>▪ Treat the seeds with talc based formulation of *P. Fluorescens* (Pf1)@ 10g/kg of seed and soak in 1lit of water for over night. Decant the excess water and allow to sprout the seeds for 24 h and then sow. |
| **Seedling dip with *Pseudomonas fluorescens*** | ▪ Stagnate water to a depth of 2.5cm over an area of $25m^2$ in the main field.<br>▪ Sprinkle 2.5 kg of the talc based formulation of *Pseudomonas fluorescens* ( Pf1) and mix with stagnated water. The seedlings pulled out from the nursery are to be soaked for 30 min. in the stagnated water and then transplanted. |

**Biocontrol agents are compatible with biofertilizers.**
**Biofertilizers and biocontrol agents can be mixed together for seed soaking.**
**Fungicides and biocontrol agents are incompatible.**

| **i.)NURSERY DISEASES** | |
|---|---|
| **Name of the Disease** | **Management stratergies** |
| Blast<br>*Pyricularia grisea*<br>***(Magnaporthe grisea)*** | ➢ **Spray any one of the following: (for 20 cents)**<br>▪ Edifenphos 50 EC 25 ml<br>▪ Carbendazim 50 W 25g |
| Brown spot<br>*Drechslera oryzae*<br>**(*Cochliobolus miyabeanus*)** | ➢ **Spray any one of the following: ( for 20 cents)**<br>▪ Edifenphos 50 EC 40 ml<br>▪ Mancozeb WP 80 g |
| Tungro<br>***Rice tungro Bacilliform virus***<br>***Rice tungro Spherical virus***<br>Vector<br>*Nephotettix virescens*<br>*N. nigropictus*<br>*N. parvus*<br>*N. malayanus*<br>*Recilia dorsalis* | ➢ Apply Carbofuran 3G @ 3.5kg at 10 DAS or<br>➢ **Spray 2 rounds (10 and 20 DAS) with any one of the following insecticides to control the vector (*Nephotettix virescens*) in 20 cents area**<br>▪ Monocrophos 36 WSC 40ml<br>▪ Phosphomidan 85 WSC 40ml<br>▪ Fenthion 100 EC 40 ml |
| **ii.) Main Field** | |
| Name of the Disease | **Management** |
| Blast<br>*Pyricularia grisea*<br>***(Magnaporthe grisea)*** | ❖ **Cultural method**<br>▪ Remove collateral weed hosts from bunds and channels<br>▪ Use only disease free seedlings.<br>▪ Avoid excess nitrogen<br>▪ Apply N in three split doses (50% basal, 25% in tillering phase and 25% N in panicle initiation stage)<br>▪ Use resistant variety CO 47.<br>❖ **Biological**<br>▪ Spray *P. fluorescens* ( Pf1) formulations @ 0.2% (1 kg) dissolved in 500 litres of water for one hectare, commencing from 45 days after transplanting at 10 days interval for 3 times depending on the disease severity.<br>❖ **Chemical**<br>▪ Spray after observing initial infection of the disease,<br>▪ Edifenphos 50 EC 500 ml or Carbendazim WP 250 g or Tricyclozole 75 WP 500 g or Iprobenphos (IBP) 500ml/ha. |
| Brown spot<br>*Drechslera oryzae*<br>**(Cochliobolus miyabeanus)** | ➢ **Spray any one of the following:**<br>▪ Edifenphos 50 EC 500 ml<br>▪ Mancozeb 1000 g/ha when grade reaches 3. If necessary, repeat 15 days later. |

| | |
|---|---|
| Sheath rot<br>***Saroclodium oryzae*** | ❖ **Cultural**<br>▪ Apply Gypsum @ 500 kg/ha at two equal splits once basally and another at active tillering stage.<br>❖ **Botanicals**<br>▪ NSKE (5%)<br>▪ Neem oil 3%<br>▪ Ipomoea leaf powder extract (25 kg/ha)<br>▪ Prosopis leaf powder extract (25 kg/ha). First spray at boot leaf stage and second 15 days later<br>❖ **Chemical**<br>➢ **Spray any one of the following:**<br>▪ Carbendazim 250 g /ha<br>▪ Edifenphos 500 ml /ha<br>▪ Mancozeb 1000g/ha |

| | |
|---|---|
| Sheath blight<br>*Rhizoctonia solani*<br>***(Thanatophorus cucumaris)*** | ❖ **Cutlural**<br>▪ Apply Neem cake at 150 kg/ha<br>❖ **Botanical**<br>▪ Foliar spray with Neem oil at 3% (15 lit /ha) starting from disease appearance.<br>❖ **Biological Control**<br>▪ Soil application of *P. fluorescens* talc based formulation at 30 DAT @ 2.5 Kg/ha and foliar spray (0.2%) at boot leaf and 10 days later @ 1 Kg/ha.<br>❖ **Chemcial**<br>▪ Carbendazim 250 g /ha<br>▪ Iprobenphos ( IBP) 500 ml/ha<br>▪ Edifenphos 500 ml / ha |
| **Rice grain discoloration**<br>1. *Helminthosporium oryzae*<br>2. *Alternaria tenuis*<br>3. *Fusarium moniliforme*<br>4. *Sarocladium oryzae*<br>* | ❖ ***Chemcial***<br>➢ ***Spray any one of the following***<br>▪ Carbendazim + Thiram + Mancozeb (1:1:1) 0.2% at the time of 50% flowering stage. |

** In addition to the above the following are also found to be associated with the grain discoloration viz.,Cladosporium herbarum,Curvularia lunata,C. pallescens,Cephalosporium oryzae,F. semitectum,F. solani,Gailarchia oryzae,H. rostratum,H. tetramera,Nigrospora oryzae,Periconia* sp.,*Pryenochaeta* sp.,*Rhizoctonia solani,Trichoconis padwickii*

| | |
|---|---|
| Bacterial leaf blight<br>*Xanthomonas oryzae* pv. *oryzae* | ❖ **Botanical / others**<br>▪ Spray neem oil 3% or NSKE 5%<br>▪ Spray fresh cowdung extract 20% twice (starting from initial appearance of the disease and another at fortnightly interval)<br>❖ **Chemical / antibiotics**<br>▪ Spray Streptomycin sulphate + Tetracycline combination 300 g + Copper oxychloride 1250 g/ha. If necessary repeat 15 days later. |
| Bacterial leaf streak<br>***Xanthomonas oryzae* pv. *oryzicola*** | ❖ **Botanical / others**<br>▪ Spray fresh cowdung water extract at 20% |
| *Rice tungro disease*<br><br>**Vector**<br>*Nephotettix virescens*<br>*N. nigropictus*<br>*N. parvus*<br>*N. malayanus*<br>*Recilia dorsalis* | ❖ **Physical methods**<br>▪ Light traps are to be set up to attract and control the leaf hopper vectors as well as to monitor the population.<br>▪ In the early morning, the population of leafhopper alighting near the light trap should be killed by spraying/dusting the insecticides. This should be practiced every day.<br>➢ **Spray Two rounds of any one of the following insecticides**<br>▪ Monocrotophos 36 WSC (1000 ml/ha)<br>▪ Fenthion 100 EC (500 ml/ha) may be sprayed 15 and 30 days after transplanting. The vegetation on the bunds should also be sprayed with the insecticides.<br>❖ Special detection technique<br>▪ Collect leaf samples at 6 a.m.<br>▪ The top 10 cm portion of the leaf is immersed in a solution containing 2 g of iodine and 6 g of potassium iodide in 100 ml of water for 15 minutes or 10 ml of tincture of iodine + 140 ml of water for one hour.<br>▪ Washed in water and when examined. Tungro infected leaves develop dark blue streaks. |

| | |
|---|---|
| Rice yellow dwarf<br>***Phytoplasma***<br>Vector<br>*Nephotettix virescens*<br>*N. nigropictus* | ❖ **Cultural method**<br>▪ Plough the stubbles as soon as the crop is harvested to prevent the survival of yellow dwarf pathogen during off-season. |

***C) Nematode management***

| Sl.No | Nematode pest | Control measures |
|---|---|---|
| 1 | **Rice root nematode,** *Hirschmanniella oryzae* | Treat *Pseudomonas fluorescens* at 10 g/kg seed or Apply Carbofuran 3G 33 kg/ha in 2 cm standing water or Carbofuran 3G @ 3.5 kg/20 cent nursery required for 1 ha. |
| 2 | **White tip nematode,** *Aphelenchoides besseyi* | Spray Chlorpyriphos 20 EC 1250 ml or Phosphamidon 85 WSC 300 ml/ha or Monocrotophos 36 WSC at 1 l/ha immediately after the emergence of boot leaf |
| 3 | **Rice root and White tip nematodes** | Seed treatment with *Pseudomonas fluorescens* (10 g/kg seed) followed by foliar spraying of the same @ 1 kg/ha thrice at 45, 55 and 65 DAT. |

## RICE SEED PRODUCTION

**Land requirement**

- Previous crop should not be the different variety of paddy, if same variety it should be passed production of certified procedure.

**Isolation**

- Adopt 3m all around the field

**Pre-sowing seed management**

- For rainfed rice or direct sowing harden the seeds by soaking the seeds in equal volume of 1% KCl solution for 20 h and dry back the seeds to original moisture content.
- In dormant cultivars, break the dormancy by soaking the seeds in equal volume 0.1 N conc.$HNO_3$ or in 0.5% $KNO_3$ for a duration of 12 -16h. The seeds are to be dried to original moisture content.
- Grade the seeds using specific gravity adopting salt water (egg flotation grading) to remove ill filled and immatured seed.

**Blanket fertilizer recommendation**

- Short duration : NPK @ 120:40:40 kg $ha^{-1}$
- Medium duration: NPK @ 150:50:60 kg $ha^{-1}$
- Long duration: NPK @ 150:50:80 kg $ha^{-1}$

**Zinc deficient soils**

- Apply $ZnSO_4$ @ 30 kg $ha^{-1}$

**Roguing space**

- Leave a roguing space of 30 cm between the beds size of 150 cm

**Foliar application**

- DAP 2% at panicle emergence stage.

**Harvesting**

- When 90% of the panicle are in straw colour with the moisture content of 20% for short and medium duration varieties and 17% moisture for long duration varieties harvest the crop as once over harvest.

**Threshing**

- Thresh the ear-heads at 16-17% moisture content either manually or using mechanical threshers.

**Drying**

- Dry the seeds to 12-13% moisture content for short term storage and 8-9% moisture for long term storage.

**Seed Treatment**

- Treat the seeds with carbendazim / Thiram @ 2g $kg^{-1}$ of seed using 5 ml of water $kg^{-1}$ of seed.

**Storage**

- For short term storage (9-12 month) store the seeds with 12-13% moisture content in gunny bag.
- For medium term storage (12-36 months) store the seed in HDPE bag or polylined gunny bag with 10-12% seed moisture

**Mid storage correction**

- Adopt hydration – dehydration treatment with disodium hydrogen phosphate (3.59g dissolved in 100 liters of water) to improve the viability and vigour when the germination of seed reduce to 5-10% lesser than MSCS level (Minimum Seed Certification Standard).

**Other management practices**

- As in crop production technique

**Hybrid seed production Techniques (ADTRH 1 and CORH 2)**

**Land Requirement**

- Select fertile land with good drainage and irrigation.
- Previous crop should not be the different varieties / hybrids of paddy

**Isolation**

- Space isolation : 100 mt
- Time isolation : 25 days (later)
- Barrier isolation : Either a distance of 30 m with vegetative barrier or Plastic sheet with 2mt height

**Staggered sowing**

**Sowing Time (Western zone)**

**Dec – Jan**

**CORH 2**

- Restorer (R) line, the male parent should be sown 3, 6 and 9 days later than sowing of male sterile (A) line, the female parent

**ADTRH 1**

- 'R' line should be sown 14,17and20 days after the sowing of'A' line

**June – July**

**CORH 2**

- Sowing of R line 0,3 and 6 days after 'A' line sowing

**ADTRH – 1**

- 'R' line should be sown 12, 15 and 18 days after 'A' line sowing

**Planting Ratio :** 8:2 to 10:2 (Female : Male)

**Border rows:** 4 rows all around the field

**Main field Management**

**Spacing**

Between 'A' lines 10 cm : between 'R' lines 30 cm ; between A and 'R' line 20 cm : within rows 15 cm.

**Planting Design**

- Two paired row @ 2-3 seedling / hill

**Fertilizer application**

- 150 : 60 : 60 kg NPK / ha. N and K applied in 3 split doses viz., basal, active tillering and panicle initiation

**Foliar spray**

- 2% DAP at panicle differentiation stage.

**Panicle exertion**

- Spray $GA_3$ @ 45 – 60g/ha.
- Spray 40% of $GA_3$ at 5-10% panicle emergence stage (based on tillers)
- Spray the remaining 60% of $GA_3$ at 24 h after first spray

**Note:** $GA_3$ should be dissolved in 70% alcohol

**Supplementary pollination**

- Rope pulling or shaking the pollen parent with the help of two bamboo sticks at 30-40% of spikelets opening stage. The process is repeated for 3 – 4 times during the day time (10 am to 1 pm) at an interval of 30 min. This has to be repeated for 7 – 10 days during flowering period.

**Harvesting**

- 'R' line should be harvested first and removed from the field

**Grading**

For getting better seed quality, the seeds should be size graded using 1.3 mm x 19 mm oblong sieve.

- The size graded seeds may be upgraded by density grading using gravity separator. Heavy and medium fractions with (90 – 92% recovery) could be selected for seed purpose

**Drying**

Moisture content should be reduced to 12 – 13%

**Seed Treatment**

- Treat with carbendazim @ 2g/kg or halogen mixture ($CaOCl_2$ + $CaCO_3$ mixture at 1:1 ratio) @ 3g /kg of seed.
- Sun dry the seeds to reduce the moisture content to 12-13% with adequate stirring

**Storage**

- As that of varieties

# MILLETS

## SORGHUM (*Sorghum bicolor*)

## CROP IMPROVEMENT

### I. SEASON AND VARIETIES

| DISTRICTS/SEASON | VARIETIES/HYBRIDS |
|---|---|
| **Kanchipuram, Tiruvallur** | |
| 1. Thaipattam | CO 26, CO (S) 28, BSR 1, COH 4 |
| 2. Chithiraipattam | CO 26, CO (S) 28, BSR 1, COH 4 |
| 3. Adipattam | K Tall, CO 26, CO (S) 28, BSR 1, COH 4, |
| 4. Puratasipattam | K Tall, CO 26, CO (S) 28, BSR 1 |
| **Vellore, Tiruvannamalai** | |
| 1. Thaipattam | CO 26, CO (S) 28, COH 4, BSR 1 |
| 2. Chithiraipattam | CO 26, CO (S) 28, COH 4, BSR 1 |
| 3. Adipattam | K Tall, CO 26, CO (S) 28, BSR 1, COH 4, Paiyur 1 |
| 4. Puratasipattam | K Tall, CO 26, CO (S) 28, BSR 1, Paiyur 2 |
| **Cuddalore, Villupuram** | |
| 1. Thaipattam | CO 26, CO (S) 28, COH 4, BSR 1 |
| 2. Chithiraipattam | CO 26, CO (S) 28, COH 4, BSR 1 |
| 3. Adipattam | K Tall, CO 26, CO (S) 28, BSR 1, COH 4 |
| 4. Puratasipattam | K Tall, CO 26, CO(S) 28, BSR 1 |
| **Tiruchirapalli, Karur, Perambalur** | |
| 1. Thaipattam | COH 4, BSR 1 |
| 2. Chithiraipattam | BSR 1, COH 4 |
| 3. Adipattam | K Tall, CO 26, CO (S) 28, BSR 1 |
| 4. Puratasipattam | K Tall, CO 26, CO (S) 28, BSR 1 |
| **Kanyakumari** | |
| 1. Thaipattam | CO (S) 28 |
| 2. Chithiraipattam | CO (S) 28 |
| **Salem, Namakkal** | |
| 1. Thaipattam | CO 26, CO (S) 28, BSR 1 |
| 2. Chithiraipattam | CO 26, CO (S) 28, BSR 1 |
| 3. Adipattam | CO (S) 28, Paiyur 1, Paiyur 2, BSR 1 |
| 4. Puratasipattam | CO 26, CO (S) 28, Paiyur 2, BSR 1 |
| **Dharmapuri** | |
| 1. Thaipattam | CO 26, CO (S) 28, BSR 1 |
| 2. Chithiraipattam | CO 26, CO (S) 28, BSR 1 |
| 3. Adipattam | CO 26, CO (S) 28, Paiyur 1, Paiyur 2 |
| 4. Puratasipattam | CO 26, CO (S) 28, Paiyur 2 |
| **Coimbatore** | |
| 1. Thaipattam | CO 26, CO (S) 28, COH 4, BSR 1 |
| 2. Chithiraipattam | CO 26, CO (S) 28, BSR 1, COH 4 |
| 3. Adipattam | CO 26, CO (S) 28, BSR 1 |
| 4. Puratasipattam | CO 26, CO (S) 28 |
| **Erode** | |
| 1. Thaipattam | CO 26, CO (S) 28, COH 4, BSR 1 |
| 2. Chithiraipattam | CO 26, CO (S) 28, BSR 1, COH 4 |
| 3. Adipattam | CO 26, CO (S) 28, BSR 1, Paiyur 2 |
| 4. Puratasipattam | CO 26, CO (S) 28, BSR 1, Paiyur 2 |
| **Pudukottai** | |
| 1. Thaipattam | BSR 1, COH 4 |
| 2. Chithiraipattam | BSR 1, COH 4 |
| 3. Adipattam | CO 26, CO (S) 28, K Tall, BSR 1, COH 4 |
| 4. Puratasipattam | CO 26, CO (S) 28, K Tall, BSR 1 |

**Madurai, Dindigul, Theni**

| | |
|---|---|
| 1. Thaipattam | CO 26, CO (S) 28, COH 4, BSR 1 |
| 2. Chithiraipattam | CO 26, CO (S) 28, BSR 1, COH 4 |
| 3. Adipattam | CO 26, CO(S) 28, COH 4, K 11, BSR 1, COH 4, APK 1 |
| 4. Puratasipattam | CO 26, CO(S) 28, K Tall, K 11, BSR 1, APK 1 |

**Ramanathapuram, Sivaganga, Virudhunagar**

| | |
|---|---|
| 1. Thaipattam | CO 26, CO (S) 28, COH 4 |
| 2. Chithiraipattam | CO 26, CO (S) 28, BSR 1, COH 4 |
| 3. Adipattam | BSR 1, COH 4, APK 1 |
| 4. Puratasipattam | CO 26, K 11, K Tall, BSR 1, APK 1 |

**Tirunelveli, Thoothukudi**

| | |
|---|---|
| 1. Thaipattam | CO 26, CO (S) 28, COH 4, BSR 1 |
| 2. Chithiraipattam | CO 26, CO (S) 28, COH 4, BSR 1 |
| 3. Adipattam | K Tall, K 11, CO 26, CO(S) 28, BSR 1, APK 1 |
| 4. Puratasipattam | K Tall, CO 26, CO (S) 28, K 11, BSR 1, APK 1 |

**Note**: Thaipattam: January - February; Chithiraipattam: April - May; Adipattam: June - July; Puratasipattam: September - October.

## II. PARTICULARS OF SORGHUM VARIETIES

| PARTICULARS | CO 26 | CO (S) 28 | COH 4 |
|---|---|---|---|
| Parentage | Derivative of MS 8271x IS 3691 | Derivative of CO 25 x SPV 942 | 296A X TNS30 |
| Duration (days) | 105 - 110 | 100-105 | 105 - 110 |
| Area (Districts) | All districts | All districts | All districts |
| Season (Pattam) | | | |
| Rainfed | Adi, Puratasi | Adi, Puratasi | … |
| Irrigated | Thai, Chithirai | Thai, Chithirai | Thai, Chithirai |
| **Grain yield kg/ha** | | | |
| Rainfed | 4500 | 2493 | ... |
| Irrigated | 6000 | 4568 | 6500 |
| **Fodder yield kg/ha** | | | |
| Rainfed | 14000 | 12600 | ... |
| Irrigated | 19000 | 17700 | 20000 |
| Stalk | Juicy, Sweet | Juice | Juicy |
| Height (cm) | 160 - 190 | 220-240 | 200 - 215 |
| Sheath colour | Green | Tan | Green |
| Node | Green | Green | Green |
| Midrib | Dull white | Dull white | White |
| Earhead shape | Long cylindrical | Cylindrical | Elliptic |
| Compactness | Semi compact | Semi compact | Semi Compact |
| Grain colour | Pearly white | White | Pearly white |
| Special features | -- | -- | Less incidence of leaf diseases, grain mould and sugary diseases |

## II. PARTICULARS OF SORGHUM VARIETIES (CONTD...)

| PARTICULARS | PAIYUR 1 | K Tall | K 11 |
|---|---|---|---|
| Parentage | CO 19 X CO 24 | 2219A x IS3541 | K 7 x A 6552 |
| Duration (days) | 145 - 150 | 90 | 110-115 |
| Area (Districts) | North east and north | Southern districts and | Southern districts |

| | | | |
|---|---|---|---|
| | western districts | other districts | |
| Season (Pattam) | | | |
| Rainfed | Adi, Puratasi | Adi, Puratasi | Puratasi |
| Irrigated | ... | Thai, Chithirai | … |
| **Grain yield kg/ha** | | | |
| Rainfed | 1000 | 3750 | 1560 |
| Irrigated | ... | 4250 | -- |
| **Straw yield kg/ha** | | | |
| Rainfed | 9000 | 11250 | 10360 |
| Irrigated | ... | 13250 | -- |
| Stalk | Juicy | Juicy | Thin, Juicy and Sweet |
| Height (cm) | 300 | 254 | 220-260 |
| Sheath colour | Green | Brown | Reddish purple at maturity |
| Node | Green | Green | Green, glabrous |
| Midrib | White | Dull white | Dull white |
| Earhead shape | Lax panicle | … | Erect, loose panicle, |
| Compactness | Open | Lanceolate Semi open | Semiopen |
| Grain colour | Pearly white | Cream pearly | Red colour, partially covering the grain |
| Special features | Tolerant to drought, non lodging, photosensitive | -- | Tolerant to drought Resistant to lodging, non shattering |

## II. PARTICULARS OF SORGHUM VARIETIES (CONTD...)

| Particulars | BSR 1 | Paiyur 2 | APK 1 |
|---|---|---|---|
| Parentage | (Multiple cross derivative (SC 108 - 3 x ICSV 4) 16-3-1 x (MR-801 x R 2751) 4-1-1 | Pureline selection from IS 15845 | Hybrid derivative of TNS 30 x CO 26 |
| Duration (days) | 105-110 | 90-95 | 105-110 |
| Area (Districts) | Western Zone (Coimbatore, Erode, parts of Salem, Trichirappalli Perambalur, Karur and Dindigul) | Salem, Namakkal | Southern districts of Tamil Nadu |
| Season (Pattam) | | | |
| Rainfed | Adi-puratasi | Puratasi | Adi, Puratasi |
| Irrigated | Thai-Chitirai | | |
| **Grain yield (Kg/ha)** | | | |
| Rainfed | 2500 - 3500 | 2113 | 2619 |
| Irrigated | 6000 – 6500 | … | … |
| **Fodder Yield(kg/ha)** | | | |
| Rainfed | 8600 | 8789 | 8090 |
| Irrigated | 9600 | … | … |
| Stalk | Juicy, sweet | Juicy, sweet | Pithy |
| Height (cm) | 150 – 180 | 200-215 | 175 |
| Sheath colour | Reddish purple | Green | Tan |
| Node | Green | Green | Green |
| Mid rib | Dull white | White | Dull white |
| Earhead shape | Long, cylindrical | Elliptic | Medium cylindrical |
| Compactness | Semi-compact | Semi-compact | Semi-compact |
| Grain colour | Pearly white | Red | White |

| | | | |
|---|---|---|---|
| Special features | Fertilizer responsive, moderate resistance to earhead bug, shoot fly and stem borer. | Dual purpose red grain sorghum suited to rainfed tract of Salem Namakkal districts. Tolerant to downy mildew and charcoal rot diseases. | Non-lodging |

## CROP MANAGEMENT

### I. SELECTION OF SEEDS

Good quality seeds are collected from disease and pest-free fields.

**Quantity of seed required**

Irrigated Transplanted - 7.5 kg/ha; Direct sown - 10 kg/ha
Rainfed Direct sown - 15 kg/ha
Sorghum under irrigated condition is raised both as a direct sown and transplanted crop.

Transplanted crop has the following advantages:

a. Main field duration is reduced by 10 days.
b. Shoot fly, which attacks direct sown crops during the first 3 weeks and which is difficult to control, can be effectively and economically controlled in the nursery itself.
c. Seedlings which show chlorotic and downy mildew symptoms can be eliminated, thereby incidence of downy mildew in the main field can be minimised.
d. Optimum population can be maintained as only healthy seedlings are used for transplanting.
e. Seed rate can also be reduced by 2.5 kg/ha.

**Population**

150 plants for 10 $m^2$, leaving only one healthy plant per hill.

### II. NURSERY PRACTICES

#### 1. NURSERY PREPARATION

For raising seedlings to plant one hectare, select 7.5 cents (300 $m^2$) near a water source where water will not stagnate.

#### 2. APPLICATION OF FYM TO THE NURSERY

i. Apply 750 kg of FYM or compost for 7.5 cents nursery and apply another 500 kg of compost or FYM for covering the seeds after sowing.
ii. Spread the manure evenly on the unploughed soil and incorporate by ploughing or apply just before last ploughing.

#### 3. LAYING THE NURSERY

i. Provide three separate units of size 2 m x 1.5 m with 30 cm space in between the plots and all around the unit for irrigation.
ii. Excavate the soil from the inter-space and all around to a depth of 15 cm to form channels and spread the soil removed on the bed and level.

#### 4. PRE-TREATMENT OF SEEDS

i. Treat the seeds 24 hours prior to sowing with Carbendazim or Captan or Thiram at 2g/kg of seed.
ii. Treat the seeds with three packets (600 g) / ha of Azospirillum and 3 packets (600g) of phosphobacteria or 6 packets of Azophos (1200g) using rice kanji as binder.

#### 5. SOWING AND COVERING THE SEEDS

1. Make shallow rills, not deeper than 1 cm on the bed by passing the fingers vertically over it.
2. Broadcast 7.5 kg of treated seeds evenly on the beds.
3. Cover by leveling the rills by passing the hand lightly over the soil.

**6. WATER MANAGEMENT**

i. Provide one inlet to each nursery unit.
ii. Allow water to enter through the inlet and cover all the channels till the raised beds are wet and then cut off water and
iii. Adjust the frequency of irrigation according to the soil types as follows:

| Number of irrigations | Red soils | Heavy soils |
|---|---|---|
| First irrigation | Immediately after sowing | Immediately after sowing |
| Second irrigation | 3rd day after sowing | 4th day after sowing |
| Third irrigation | 7th day after sowing | 9th day after sowing |
| Fourth irrigation | 12th day after sowing | 16th day after sowing |

**NOTE: Do not keep the seedlings in the nursery for more than 18 days. If older seedlings are used, establishment and yield are adversely affected. Do not allow cracks to develop in the nursery by properly adjusting the quantity of irrigation water.**

## III. MAIN FIELD PREPARATION FOR IRRIGATED CROP

**1. PLOUGHING**

Plough the field with an iron plough once (or) twice. Sorghum does not require fine tilth since it adversely affects germination and yield in the case of direct sown crop.

To overcome the subsoil hard pan in Alfisols (deep red soils) chiselling the field at 0.5 m intervals to a depth of 40 cm on both the directions of the field followed by disc ploughing once and cultivator ploughing twice help to increase the yield of sorghum and the succeeding blackgram also. This was true with Sorghum followed by Groundnut also.

Application of FYM and 100% of recommended N can also be followed. In soils with sub-soil hard pan, chiselling should be done every year at the start of the cropping sequence to create a favourable physical environment.

**2. APPLICATION OF FYM**

Spread 12.5 t/ha FYM or composted coir pith along with 10 packets of Azospirillum (2000g/ha) and 10 packets (2000 g/ha) of phosphobacteria or 20 packets of Azophos (4000g/ha) on the unploughed field and incorporate the manure in the soil. Apply well decomposed poultry manure @ 5 t/ha to improve the grain yield as well as physical properties of soils.

**3. FORMATION OF RIDGES AND FURROWS**

i. Form ridges and furrows using a ridger at 6 m long and 45 cm apart
ii. Form irrigation channels across the furrows
iii. Alternatively, form beds of size 10 $m^2$ and 20 $m^2$ depending on the availability of water.

**4. APPLICATION OF FERTILIZERS**

**Transplanted crop**

i. Apply NPK fertilizers as per soil test recommendations. If soil test recommendations are not available, adopt a blanket recommendation of 90 N, 45 $P_2O_5$, 45 $K_2O$ kg/ha.
ii. Apply N @ 50:25:25 % at 0, 15 and 30 DAS and full dose of $P_2O_5$ and $K_2O$ basally before planting
iii. In the case of ridge planted crop, open a furrow 5 cm deep on the side of the ridge at two thirds the distance from the top of the ridge and place the fertilizer mixture along the furrow and cover with soil upto 2 cm.
iv. Soil application of Azospirillum at 10 packets (2 kg/ha) and 10 packets (2000g/ha) of

phosphobacteria or 20 packets of Azophos (4000g/ha) after mixing with 25 kg of FYM + 25 kg of soil may be carried out before sowing/planting.

**Direct sown crop**

i. Apply NPK fertilizers as per soil test recommendations as far as possible. If soil test recommendations are not available, adopt a blanket recommendation of 90 N, 45 $P_2O_5$, 45 $K_2O$ kg/ha.
ii. Apply N @ 50:25:25 % at 0, 15 and 30 DAS and full dose of $P_2O$ and $K_2O_5$ basally before sowing and if basal application is not possible the same could be top dressed within 24 hours.
iii. In the case of bed planted crop, mark lines to a depth of 5 cm and 45 cm apart. Place the fertilizer mixture at the depth of 5 cm along the lines. Cover the lines upto 2 cm from the top before sowing.
iv. In the case of sorghum raised as a mixed crop with a pulse crop (Blackgram, Greengram or Cowpea) open furrows 30 cm apart to a depth of 5 cm.
v. Apply fertilizer mixture in two lines in which sorghum is to be raised and cover upto 2 cm.
vi. Skip the third row in which the pulse crop is to be raised and place fertilizer mixture in the next two rows and cover upto 2 cm with soil.
vii. Application of bio-fertilizers: When Azospirillum is used apply only 75% of recommended N for irrigated sorghum.
viii. Soil test based fertilizer recommendation may be adopted in Western and North Western Zone *viz.,* Alfisol, Inceptisol and Vertisol for prescribing fertilizer doses for specified yield targets.

**5. APPLICATION OF MICRONUTRIENT MIXTURE**

**Transplanted Crop**

i. Mix 12.5 kg/ha of micronutrient mixture formulated by the Department of Agriculture, Tamil Nadu with enough sand to make a total quantity of 50 kg and apply the mixture over the furrows and on top one third of the ridges.
ii. If micronutrient mixture is not available, mix 25 kg of zinc sulphate with sand to make a total quantity of 50 kg and apply on the furrows and on the top one third of the ridges.

**Direct Sown Crop**

i. Mix 12.5 kg of micronutrient mixture formulated by the Department of Agriculture, Tamil Nadu with enough sand to make a total quantity of 50 kg.
ii. Spread the mixture evenly on the beds.
iii. Basal application of 25 kg $ZnSO_4$/ha for the deficient soils.
iv. Basal application of $FeSO_4$, 50 kg/ha along with 12.5 t/ha FYM for iron deficient soils.

## IV. MANAGEMENT OF MAIN FIELD

**1. TRANSPLANTED CROP**

i. Pull out the seedlings when they are 15 to 18 days old.
ii. Prepare slurry with 5 packets of Azospirillum (1000g/ha) and 5 packets (1000g/ha) of Phosphobacteria or 10 packets of Azophos (2000 g/ha) in 40 lit. of water and dip the root portion of the seedlings in the solution for 15-30 minutes and transplant.
iii. Let in water through the furrows
iv. Plant one seedling per hill
v. Plant the seedlings at a depth of 3 to 5 cm.
vi. Plant the seedlings on the side of the ridge, half the distance from the top of the ridge and the bottom.
vii. Maintain a spacing of 15 cm between plants in the row which are 45 cm apart ($15/m^2$).

**2. DIRECT SOWN CROP**

i. In the case of pure crop of sorghum, maintain the seed rate at 10kg/ha.
ii. In the case of inter crop of sorghum with pulse crop, maintain the seed rate of sorghum at 10 kg/ha and pulse crop at 10 kg/ha.
iii. In the case of pure crop of sorghum, sow the seeds with a spacing of 15 cm between seeds in the rows which are 45 cm apart.

iv. Maintain one plant per hill.
v. If shootfly attack is there, remove the side shots and retain one healthy shoot.
vi. Sow the seeds over the lines where fertilizers are placed.
vii. Sow the seeds at a depth of 2 cm and cover with soil.
viii. In the case of sorghum intercropped with pulses sow one paired row of sorghum alternated with a single row of pulses. The spacing between the row of sorghum and pulse crop is 30 cm.

Forage cowpea CO 1 can be inter-cropped in sorghum at two rows of fodder cowpea in between paired rows of sorghum.

**3. WEED MANAGEMENT**

i. Apply the pre-emergence herbicide Atrazine 50 WP - 500 g/ha on 3 days after sowing as spray on the soil surface, using Backpack / Knapsack / Rocker sprayer fitted with a flat fan nozzle using 900 l of water/ha.
ii. Sorghum is slow growing in early stages and is adversely affected by weed competition. Therefore keep the field free of weeds upto 45 days. For this, after pre-emergence herbicide application, one hand weeding on 30 - 35 days after sowing may be given
iii. If pulse crop is to be raised as an inter-crop in sorghum do not use Atrazine.
iv. Hoe and hand weed on the 10th day of transplanting if herbicides are not used. Hoe and weed between 30 - 35 days after transplanting and between 35 - 40 days for a direct sown crop, if necessary.

**4. THINNING OF THE SEEDLINGS AND GAP FILLING**

**Direct sown crop**

Thin the seedlings and gap fill with the seedlings thinned out. Maintain a spacing of 15 cm between plants after the first hand weeding on the 23rd day of sowing. Thin the pulse crop to a spacing of 10 cm between plants for all pulse crop except cowpea, for which spacing is maintained at 20 cm between plants.

**5. DEFICIENCY SYMPTOMS**

Zinc: Deficiency symptoms first appear in the newly formed leaves at 20 to 30 days age. Older leaves have yellow streaks or chlorotic striping between veins.

Iron:Interveinal chlorosis will be observed. If the deficiency continues the entire leaf including the veins may exhibit chlorotic symptoms. Newly formed leaves exhibit chlorotic symptoms. The entire crop may exhibit bleached appearance, dry and may die.

**Direct sown crop**

i. Spray only if micronutrient mixture is not applied.
ii. Apply in case of iron deficiency.
iii. If soil is calcareous

## V. WATER MANAGEMENT

**Regulate irrigation according to the following growth phase of the crop.**

| | Transplanted crop | Direct sown crop |
|---|---|---|
| Growth phase | 1 to 40 days | 1 to 33 days |
| Flowering phase | 41 to 70 days | 34 to 65 days |
| Maturity phase | 71 to 95 days | 66 to 95 days |

| STAGES | No. of Irrigation | Days of Transplanting/Sowing of Crop | |
|---|---|---|---|
| | | Transplanted | Direct sown |
| Light soils | | | |
| i. Irrigate for germination | 1 | 1st day | 1st day |
| or establishment | 2 | 4th day | 4th day |
| ii. Regulate irrigation | 1 | 15th day | 15th day |
| during vegetative phase | 2 | 28th day | 28th day |
| iii. Flowering phase (copious irrigation) | 1 | 40th day | 40th day |
| | 2 | 52nd day | 52nd day |
| | 3 | .. | 64th day |
| iv. Maturity phase (Control irrigation) | 1 | 65th day | 76th day |
| | 2 | .. | 88th day |
| v. Stop irrigation thereafter | .. | .. | |
| Heavy soils | | | |
| i. Irrigate for germination | 1 | 1st day | 1st day |
| or establishment | 2 | 4th day | 4th day |
| ii. Regulate irrigation | 1 | 17th day | 17th day |
| during vegetative phase | 2 | 30th day | 30th day |
| iii. Flowering phase (give | 1 | 40th day | 45th day |
| copious irrigation) | 2 | 52nd day | 60th day |
| | 3 | .. | 75th day |
| iv. Maturity phase (Control irrigation) | 1 | 72nd day | 90th day |
| v. Stop irrigation thereafter | | | |

**NOTE :** Adjust irrigation schedule according to the weather conditions and depending upon the receipt of rains. Contingent Plan: This should be done before 75% of soil moisture is lost from available water. Spraying 3% Kaolin (30 g in one litre of water) during periods of stress will mitigate the ill effects.

### V. HARVESTING AND PROCESSING

i. Consider the average duration of the crop and observe the crop. When the crop matures the leaves turn yellow and present a dried up appearance.

ii. The grains are hard and firm.

iii. At this stage, harvest the crop by cutting the earheads separately.

iv. Cut the straw after a week, allow it to dry and then stack.

v. In the case of tall varieties, cut the stem at 10 to 15 cm above ground level and afterwards separate the earheads and stack the straw.

vi. Dry the earheads.

vii. Thresh using a mechanical thresher or by drawing a stone roller over the earheads or by using cattle and dry the produce and store.

## RATOON SORGHUM CROP

### 1. RATOONING TECHNIQUE

i. Harvest the main crop leaving 15 cm stubbles.

ii. Remove the first formed two sprouts from the main crop and allow only the later formed two sprouts to grow. Allow two tillers per hill.

**2. HOEING AND WEEDING**

i. Remove the weeds immediately after harvest of the main crop.
ii Hoe and weed twice on 15$^{th}$ and 30$^{th}$ day after cutting.

**3. APPLICATION OF FERTILIZERS**

i. Apply 100 kg N/ha in two split doses.
ii. Apply the first dose on 15$^{th}$ day after cutting and the second on 45$^{th}$ day after cutting.
iii. Apply 50 kg $P_2O_5$/ha along with the application of N on 45$^{th}$ day.

**4. WATER MANAGEMENT**

i. Irrigate immediately after cutting the main crop.
ii. Irrigation should not be delayed for more than 24 hours after cutting.
iii. Irrigate on 3rd or 4th day after cutting.
iv. Subsequently irrigate once in 7 - 10 days.
v. Stop irrigation on 70 - 80 days after ratooning.

**5. HARVEST**

Harvest the crop when the grains turn yellow.

NOTE: The duration of the ratoon crop is about 15 days less than the main crop.

## RAINFED SORGHUM

**1. RAINFALL**

Average and well distributed rainfall of 250-300 mm is optimum for rainfed sorghum.

**2. DISTRIBUTION**

Madurai, Dindigul, Theni, Ramanathapuram, Tirunelveli, Thoothukudi, Virudhunagar, Sivagangai, Tiruchirapalli, Erode, Salem, Namakkal, Coimbatore and Dharmapuri Districts.

**3. SEASON**

The crop can be grown in South West and North East monsoon seasons provided the rainfall is evenly distributed.

**4. FIELD PREPARATION**

Test the soil and apply fertilizers based on soil test recommendations.

i Field has to be prepared well in advance taking advantage of early showers.
FYM application should be done @ 12.5 t / ha and well incorporated at the time of ploughing.

ii. Chiseling for soils with hard pan

Chisel the soils having hard pan formation at shallow depths with chisel plough at 0.5 M interval, first in one direction and then in the direction perpendicular to the previous one once in three years. Apply 12.5 t FYM or composted Coir pith/ha besides chiseling to get an additional yield of about 30% over control.

iii. To conserve the soil moisture sow the seeds in flat beds and form furrows between crop rows during inter cultivation or during third week after sowing.

**5. SEED RATE**

15 kg/ha

**6. SEED TREATMENT**

**Direct sown crop**

Seed hardening ensures high germination. The seeds are pre-soaked in 2% potassium dihydrogen phosphate solution for 6 hours in equal volum and then dried back to its original moisture content is shade and are used for sowing.

i) Harden the seeds with 1% aquous fresh leaf extract of *Prosopis juliflora* and pungam, (*Pongamia pinnata*) mixed in 1:1 for 16 hrs at 1:0.6 ratio (Seed and solution) followed by drying and subsequently pelleting the seeds with Pungam leaf powder @300 g/kg with gruel.
ii) Halogenise the seeds containing CaOCl, $CaCO_3$ and arappu leaf powder @ 5:4:1 ratio or iodine based (containing 2 mg of Iodine in 3 g of $CaCO_3$) formulation @ 3g/kg packed in polylined cloth bag to maintain seed viability for more than 10 month.
iii) Soak the seeds in 2% (20 g in one litre of water) potassium dihydrogen phosphate or 500 ppm of CCC (500 mg in one litre of water) for six hours and shade dry the seeds for 5 hours. Use 350 ml of solution for soaking one kg of seed.
iv) Treat the seeds with three packets of azospirillum (600 g) and 3 packets of phosphobacteria or 6 packets of Azophos (1200 g/ha). In the main field, apply 10 packets of azospirillum 2000 g/ha and 10 packets (2000g/ha) of phosphobacteria or 20 packets of Azophos (4000 g/ha) with phosphobacteria 2 kg with 25 kg FYM + 25 kg soil.

**7. SOWING**

Sow the seeds well before the onset of monsoon at 5 cm depth (by seed drill or by country plough). The seed is pelletised with 15 g of Chloropyriphos in 150 ml of gum and shade dried.

**Pre-monsoon sowing**

Sow the hardened seeds at 5 cm depth with seed cum fertilizer drill to ensure uniform depth of sowing and fertilizer application before the onset of monsoon as detailed below:

| District | Optimum period |
|---|---|
| 1. Coimbatore | 37-38th week (II to III week of September) |
| 2. Erode | 38th week (III week of September) |
| 3. Sivaganga | 40th week (I week of October) |
| 4. Ramanathapuram | 40th week (I week of October) |
| 5. Thoothukudi | 39-40th week (Last week of September to I week of October) |
| 6.Vellore, Tiruvannamalai | 37th-38th week (September II week to September III week) |

i. Sow the sorghum seeds over the line where the fertilizers are placed.
ii. Sow the seeds at a depth of 5 cm and cover with the soil.
iii. Sow the seeds with the spacings of 15 cm in the paired rows spaced 60 cm apart.
iv. Sow the pulse seeds to fall 10 cm apart in the furrows between the paired rows of sorghum.

**8. SPACING**

45 x 15 cm or 45 x 10 cm.

**9. FERTILIZER**

Apply 12.5 t/ha of Composted Coir pith + NPK at 40:20:0; Apply enriched FYM 750 kg/ha. The recommended dose of 40 kg N and 20 kg $P_2O_5$/ha for rainfed sorghum can be halved if FYM @ 5 t/ ha is applied.

**10. WEED MANAGEMENT**

Keep sorghum field free of weeds from second week after germination till $5^{th}$ week. If sufficient moisture is available spray Atrazine @ 500 g/ha as pre-emergence application within 3 days after the receipt of the soaking rainfall for sole sorghum while for sorghum based intercropping system with pulses, use Pendimethalin at 3.0 l/ha.

**11. CROPPING SYSTEM**

- The most profitable and remunerative sorghum based cropping system adopted is sorghum with cowpea, redgram, lab-lab, blackgram.
- In rainfed Vertisol, adopt paired row planting in sorghum and sow one row of blackgram/ cowpea in between paired rows of sorghum to have 100% population of sorghum plus 33% population of blackgram/cowpea.

- Intercropping of sunflower CO 1, with the main crop of sorghum CO 26 in 4:2 ratio is recommended under rainfed conditions during North-East monsoon for black soils of Coimbatore.
- Intercropping of soyabean with sorghum in the ratio 4:2 is recommended for kharif seasons.
- For sorghum - blackgram intercropping system as well as sole cropping, application of 20 kg N and 20 kg $P_2O_5$/ha through enriched FYM and treating the seeds with Azospirillum is recommended for Aruppukottai region.
- For sorghum (CO 25) + Fodder cowpea (CO 1) intercropping system, application of 20 kg N and 20 kg $P_2O_5$/ha with enriched FYM is recommended for Coimbatore region.
- The intercropping system, fodder sorghum (K 7) + Fodder cowpea (CO 5) at 3:2 ratio is found profitable for rainfed Vertisols of Aruppukottai.
- Tamarind and Neem trees upto 3-4 years from date of planting form an ideal tree component for agroforestry in black cotton soils of Kovilpatti. Sorghum and blackgram gave higher yield even at 50 per cent of the recommended level of fertilizer application.

## CROP PROTECTION

### A. Pest management

**Pre-treatment of seeds**

- Dissolve 0.5 g of gum in 20 ml of water. Add 4 ml of chlorpyriphos 20 EC or monocrotophos 36 WSC or phosalone 35 EC. To this add one kg of seed, pellet and shade dry.

**Economic threshold level for important pests**

| Pests | ETL |
|---|---|
| Shoot fly | 1 egg/plant in 10% of plants in the first two weeks of sowing or 10 % dead hearts |
| Mite | 5 mites/$cm^2$ of leaf area |
| Stemborer | 10% damage |
| Grain midge | 5 / earhead |
| Earhead bug | 10 / earhead |
| Earhead caterpillar | 2 / earhead |

**Pest management strategies**

| Pests | Management strategies |
|---|---|
| **Shoot fly**<br>*Atherigona soccata* | ▪ Take up early sowing of sorghum immediately after the receipt of South West or North East monsoon to minimise the shoot fly incidence.<br>▪ Use seeds pelleted with insecticides<br>▪ Seed treatment with imidacloprid 70 WS @ 10 g/kg of seeds<br>▪ In case of direct seeding, use increased seed rate upto 12.5 kg/per hectare and remove the shoot fly damaged seedlings at the time of thinning or raise nursery and transplant only healthy seedlings.<br>▪ Plough soon after harvest, remove and destroy the stubbles.<br>▪ Set up the TNAU low cost fish meal trap @ 12/ha till the crop is 30 days old.<br>**Spray one of the following for an area of 120 $m^2$ nursery :**<br>Endosulfan 35 EC 18 ml/ha<br>Methyl demeton 25 EC 12 ml/ha<br>Dimethoate 30 EC 12 ml/ha<br>**In main field for direct sown crop spray any one of the following :**<br>Endosulfan 35 EC 500 ml/ha<br>Methyl demeton 25 EC 500 ml/ha<br>Dimethoate 30 EC 500 ml/ha<br>Neem Seed Kernal extract 5%<br>Azadirachtin 1% |

| | |
|---|---|
| | ▪ Soil application of phorate 10 G 10 kg/ha at the time of sowing |
| **Mite**<br>*Oligonychus indicus* | ▪ **Spray any one of the following :**<br>Wettable sulphur 3.75 kg/ha<br>Dicofol 18.5 EC 1500 ml/ha.<br>▪ Direct the spray fluid towards the under surface of the leaves. |
| **Aphids**<br>*Rhopalosiphum maidis*<br>*Melanaphis sacchari* | ▪ **Spray any one of the following :**<br>Methyl demeton 25 EC 500 ml/ha<br>Dimethoate 30 EC 500 ml/ha |
| **Stemborer**<br>*Chilo partellus sesamia inferens* | ▪ Sowing the lab lab / cowpea as an intercrop to minimise stemborer damage (Sorghum: Lab lab /cowpea 4:1).<br>▪ Set up of light traps till mid night to monitor, attract and kill adults of stem borer, grain midge and earhead caterpillars<br>▪ **Mix any one of the following insecticides with sand to make up a total quantity of 50 kg/ha and apply in the leaf whorls :**<br>Quinalphos 5 G 15 kg/ha<br>Endosulfan 4 G 15 kg/ha<br>Phorate 10 G 8 kg/ha<br>Carbofuron 3 G 17 kg/ha<br>Carbaryl 4 G 20 kg/ha<br>Endosulfan 4 D 10 kg/ha<br>Phosalone 4 D 10 kg/ha<br>Phenthoate 2 D 5 kg/ha<br>▪ **Spray any one of the following insecticides :**<br>Endosulfan 35 EC 750 ml/ha<br>Carbaryl 50 WP 1.00 kg/ha (500 l spray fluid/ ha). |
| **Grain midge**<br>*Contarinia sorgicola* | ▪ Sett up of light traps till mid night to monitor, attract and kill adults of stem borer, grain midge and earhead caterpillars.<br>▪ **Apply any one of the following on 3$^{rd}$ and 18$^{th}$ day after panicle emergence :**<br>Carbaryl 10 D 25 kg/ha<br>Malathion 5 D 25 kg/ha<br>Phosalone 4 D 25 kg/ha<br>Neem seed kernel extract 5%<br>Azadirachtin 1%<br>▪ The sowing of sorghum should be completed in as short a time as possible to avoid continuous flowering which favours grain midge and earhead bug multiplication in an area. |
| **Earhead bug**<br>*Calocoris angustatus* | ▪ **Apply any one of the following on 3$^{rd}$ and 18$^{th}$ day after panicle emergence :**<br>Carbaryl 10 D 25 kg/ha<br>Malathion 5 D 25 kg/ha<br>Phosalone 4 D 25 kg/ha<br>Neem seed kernel extract 5%<br>Azadirachtin 1%<br>▪ For the earhead bug and sugary disease predominance, spray twice with Malathion 50 EC 500 ml/ha. plus ziram 1.25 kg/ha in 500 lit of water at 10% heading and 9days after. |
| **Earhead caterpillar**<br>*Helicoverpa armigera* | ▪ Set up of light traps till mid night to monitor, attract and kill adults of stem borer, grain midge and earhead caterpillars.<br>▪ Set up sex pheromone traps at 12/ha to attract male moths of *Helicoverpa armigera* from flowering to grain hardening. Two applications of NPV at 10 days interval at 1.5 X10$^{12}$ POB along with crude sugar 2.5 kg + cotton seed kernel powder 250 g on the earheads is effective in reducing the larval population of *Helicoverpa*.<br>▪ **Apply any one of the following on 3$^{rd}$ and 18$^{th}$ day after** |

| | |
|---|---|
| | **panicle emergence :**<br>Carbaryl 10 D 25 kg/ha<br>Malathion 5 D 25 kg/ha<br>Phosalone 4 D 25 kg/ha |
| **Storage**<br>**Rice weevil**<br>*Sitophilus oryzae* | ▪ Treat the seeds with<br>Monocrotophos 36 WSC 4 ml/kg of seed.<br>Chlorpyriphos 20 EC 4 ml/kg of seed. |

**Resurgence**

Repeated application of the insecticide fenvalerate can cause resurgence of the aphid, *Rhopalosiphum maidis.*

## RATOON SORGHUM CROP

## CROP PROTECTION

**A. Pest management**

- Since the ratoon crop invites pests and diseases in early stages, plant protection measures have to be resorted to. Spray chemicals immediately after application of fertiliser before irrigation.
- Fallow the plant protection measures as recommended for the control of stemborer, earhead midge and earhead bug for the main crop.

**B. Disease Management**

| **Nursery practices** | |
|---|---|
| **Seed treatment** | Treat the seeds 24h prior to sowing with Carbendazim or Captan or Thiram 2g/kg of seed or Metalaxyl+Mancozeb 4g/ kg of seed |
| **Main field** | |
| Name of the Disease | **Management** (Dosage / ha) |
| **Rust**<br>Puccinia purpurea | Spray Mancozeb at 1kg/ha. Repeat fungicidal application after 10 days |
| ***Ergot or Sugary disease***<br>*Sphacelia sorghi*<br>*Claviceps purpurea* | ❖ Cultural method<br>▪ Sowing period to be adjusted so as to prevent heading during rainy season and severe winter.<br>▪ Chemcial<br>➢ Spray any one of the following fungicide at emergence of earhead (5 - 10% flowering stage) followed by a spray at 50% flowering and repeat the spray after a week if necessary<br>▪ Mancozeb – 1000g/ha<br>▪ Propiconazole 500ml/ha |
| ***Head Mould***<br>Fungal complex *Fusarium, Curvularia, Alternaria, Aspergillus* and *Phoma* sp. | ❖ Spray any one of the fungicides like Mancozeb or Captan @ 1000g+ Aureofungin sol. 100 g/ha in case of intermittent rainfall during earhead emergence and repeat if necessary a week later. |
| ***Downy Mildew***<br>*Peronosclerospora sorghi* | ➢ Cultural method<br>▪ Rogue infected plants upto 45 days after sowing<br>▪ Spray any one of the fungicides like<br>Metalaxyl 500 g or Mancozeb 1000g/ha afternoticing the symptoms of foliar diseases, for both transplanted and direct sown crops. |
| **Charcoal Rot**<br>*Macrophomina phaseolina* | ▪ Treat the seeds with *Trichoderma viride* @ 4g/Kg of seed. |

# CUMBU (*Pennisetum glaucum* (L) R. Br. )

## CROP IMPROVEMENT

## I. SEASON AND VARIETIES

| Districts/Season | Varieties/Hybrid |
|---|---|
| **1. IRRIGATED** | |
| i. Chithiraipattam (March-April) | |
| All pearl millet growing districts except Nilgiris | CO 7, CO (Cu) 9, X 7, ICMV 221 |
| ii. Masipattam (January-February) | |
| All pearl millet growing districts except the Nilgiris, | CO 7, CO (Cu) 9, X 7, ICMV 221 |
| **2. RAINFED** | |
| i. Adipattam (June-July) | |
| | CO 7, CO (Cu) 9, X 7, ICMV 221 |
| ii. Purattasipattam (September -October) | CO 7, CO (Cu) 9, X 7, ICMV 221 |

## II. PARTICULARS OF CUMBU HYBRIDS AND VARIETIES

| PARTICULARS | CO 7 | X 7 | CO (Cu) 9 | ICMV 221 |
|---|---|---|---|---|
| Parentage | (CO 6 x PK 560) PT 1921 | L111 A x PT 1890 | Selection from ICMV 93752 | ICRISAT Composite |
| Season-irrigated/ rainfed | Both | Both | Both | Both |
| Duration (Days) | 90 - 100 | 90 | 80-85 | 75-80 |
| Grain yield (kg/ha) | | | | |
| Rainfed | 2500 - 2800 | 2513 | 2354 | 13% >ICTP 8203 |
| Irrigated | 3000 - 3500 | 3295 | 2865 | -- |
| Plant height (cm) | 130 - 145 | 155 - 180 | 186-222 | 140-200 |
| Tillers (No.) | 6 - 10 | 4 - 7 | 3-6 | More tillering |
| Pigmentation | Green | Non- pigmented | -- | -- |
| Hairiness | Glabrous | Glabrous | Glabrous | Absent |
| Days to 50% bloom | 65 - 70 | 50 - 55 | 50-55 | 50-55 |
| Shape of earhead | Conical/ cylindrical/ spindle | Candle | Candle to cylindrical | Semicompact to compact lanceolate to oblanceolate |
| Bristles | Nil | — | Absent | Usually non bristled |
| Length of earhead | 22 - 26 | 25 - 35 | 33-39 | -- |
| (cm) | | | | |
| Breadth of earhead | 3 - 4 | — | 8-10 | Wide in girth |
| (cm) | | | | |
| Grain colour | Slate colour | Slate | Grey seed with yellow base | Dark grey |
| 1000 grains weight (gm) | 6.8 - 7.2 | 8.0 - 9.0 | 9-11 | 10-15 |
| Special features | Resistant to downy mildew | Resistant to downy mildew | Resistant to downy mildew | Resistant to downy mildew |

## CROP MANAGEMENT

### II. NURSERY

**1. PREPARATION OF LAND**

i. For raising seedlings to plant one ha select 7.5 cents near a water source. Water should not stagnate.
ii. Plough the land and bring it to the fine tilth.

**2. APPLICATION OF FYM**

Apply 750 kg of FYM or compost and incorporate by ploughing. Cover the seeds with 500 kg of FYM.

**3. FORMING RAISED BED**

i. In each cent mark 6 plots of the size 3 m x 1.5 m with 30 cm channel in between the plots and all around.
ii. Form the channel to a depth of 15 cm.
iii. Spread the earth excavated from the channel on the beds and level.

NOTE: The Unit of 6 plots in one cent will form one unit for irrigation.

**4. REMOVAL OF ERGOT AFFECTED SEEDS AND SCLEROTIA TO PREVENT PRIMARY INFECTION**

i. Dissolve one kg of common salt in 10 litres of water.
ii. Drop the seeds into the salt solution
iii. Remove the ergot and sclerotia affected seeds which will float.
iv. Wash seeds in fresh water 2 or 3 times to remove the salt on the seeds.
v. Dry the seeds in shade.
vi. Treat the seeds with three packets (600g) of the Azospirillum inoculant and 3 packets (600g) of phosphobacteria or 6 packets (1200g) of azophos.

**5. TREATMENT OF THE NURSERY BED WITH INSECTICIDES**

Apply phorate 10 G 180 g or Carbofuran 3 G 600 g mixed with 2 kg of moist sand, spread on the beds and work into the top 2 cm of soil to protect the seedlings from shootfly infestation.

**6. SOWING AND COVERING THE SEEDS**

i. Open small rills not deeper than 1 cm on the bed by passing the fingers over it.
ii. Sow 3.75 kg of seeds in 7.5 cents (0.5 kg / cent) and use increased seed rate upto 12.5 kg per ha in shootfly endemic area and transplant only healthy seedlings.
iii. Cover the seeds by smoothening out the rills with hand. Sprinkle 500 kg of FYM or compost evenly and cover the seeds completely with hands.

NOTE: Do not sow the seeds deep as germination will be affected.

**7. IRRIGATION TO THE SEED BED**

i. Provide one inlet to each unit of 6 plots to allow water in the channels.
ii. Allow water to enter the channel and rise up in it. Turn off the water when the raised bed is wet.
iii. Irrigate as per the following schedule.

| Light Soil | | Heavy Soil |
|---|---|---|
| 1st | immediately after sowing | Immediately after sowing |
| 2nd | on 3$^{rd}$ day after sowing | On 3$^{rd}$ day after sowing |
| 3rd | on 7$^{th}$ day after sowing | On 9$^{th}$ day after sowing |
| 4th | on 12$^{th}$ day after sowing | On 16$^{th}$ day after sowing |
| 5th | on 17$^{th}$ day after sowing | |

**8. PROTECTION OF SEEDLINGS IN THE NURSERY FROM PEST ATTACK**

If seed bed is not treated before sowing, protect the nursery by applying any one of the insecticides given below on the $7^{th}$ and $14^{th}$ day of sowing by mixing in 6 litres of water. Endosulfan 35 EC 12ml ; Methyl demeton 25 EC 12 ml, Dimethoat 30 EC 12 ml.

Note: 1. The seedlings should not be kept in nursery for more than 18 days. Otherwise the establishment and yield will be affected adversely.
2. Ensure that cracks should not develop in the nursery. This can be avoided by properly adjusting the quantity of irrigation water.

## III. PREPARATION OF MAIN FIELD

**1. FIELD PREPARATION**

i. Plough with an iron plough twice and with country plough twice. Bring the soil into fine tilth.
ii. CHISELING FOR SOILS WITH HARD PAN: Chisel the soils having hard pan formation at shallow depths with chisel plough at 0.5m interval, first in one direction then in the direction perpendicular to the previous one, once in three years.

**2. APPLICATION OF FYM OR COMPOST**

Spread 12.5 t/ha of FYM or compost or composted coir pith uniformly on unploughed soil. Incorporate the manure by working the country plough and apply Azospirillum to the soil @ 10 packets per ha (2000 g) and 10 packets (2000g) of phosphobacteria (or) 20 packets (4000g) of azophos with 25kg of soil and 25 kg of FYM.

**3. FORMING RIDGES AND FURROWS/BEDS**

i. Form ridges and furrows (using 3 ridges) 6 m long and 45 cm apart. If pulses is intercropped, form ridges and furrows 6 m long and 30 cm apart.
ii. If ridge planting is not followed, form beds of the size 10 $m^2$ or 30 $m^2$ depending upon water availability.
iii. Form irrigation channels.
iv. To conserve soil moisture under rainfed condition, sow the seeds in flat and form furrows between crop rows during intercultivation on third week after sowing.

**4. APPLICATION OF FERTILIZERS**

Apply NPK fertilizers as per soil test recommendations as far as possible. If soil test recommendation is not available follow the blanket recommendation of 70:35:35 kg N, $P_2O_5$, $K_2O$/ ha for all varieties. For hybrids, apply 80 kg N, 40 kg $P_2O_5$ and 40 kg $K_2O$ per ha. Apply the recommended N in three splits as 25:50:25 per cent at 0,15 and 30 DAS and full dose of phosphorus and potassium basally. Combined application of azospirillum and phosphobacteria or azophos along with 75 per cent of the recommended level of N and P is recommended for rainfed conditions.

**Method of application:** For transplanted crop, open a furrow more than 5 cm deep on the side of the ridge (1/3 distance from the bottom), place the fertilizer and cover. For the direct sown crop, mark the lines more than 5 cm deep 45 cm apart in the beds. Place the fertilizer below 5 cm depth and cover upto 2 cm from the top before sowing. In the case of intercropping with pulses, mark lines more than 5 cm deep 30 cm apart in the beds. Apply fertilizer only in the rows in which cumbu is to be sown and cover upto 2 cm. When azospirillum inoculant is used for seeds, seedlings use only 50 kg N/ha for variety, 60 kg N/ha for hybrid, as soil application in other words, reduce 25% N of soil test recommendations.

**5. APPLICATION OF MICRONUTRIENT MIXTURE**

Apply 12.5 kg/ha of micronutrient mixture formulated by the Department of Agriculture. Mix the mixture with enough sand to make 50 kg and apply on the surface just before planting/after sowing and cover the seeds. Broadcast the mixture on the surface to seed line. If micronutrient mixture is not

available apply 25 kg of zinc sulphate per ha. Mix the chemical with enough sand to make 50 kg and apply as above.

## IV. MANAGEMENT OF MAIN FIELD

**1. TRANSPLANTING SEEDLINGS OR SOWING PRE-TREATED SEEDS Transplanted Crop**

i. Pull out the seedlings when they are 15 to 18 days old.
ii. Adopt the spacing 45 x 15 cm for all the varieties.
iii. Plant seedlings on the side of ridge, half way from the bottom. Depth of planting should be 3 to 5 cm.
iv. Root dipping with bio-fertilizers: Prepare the slurry with 5 packets (1000 g)/ha of Azospirillum inoculant and 5 packets (1000g/ha) of phosphobacteria or 10 packets of azophos (2000g/ha) in 40 lit. of water and dip the roots of the seedlings 15 - 30 minutes before planting.

**Direct sown crop**

Soaking of cumbu seeds either in 2% Potassium chloride (KCl) or 3% Sodium Chloride (NaCl) for 16 hours followed by 5 hours shade drying improves germination and stand.

i. Adopt the spacing of 45 x 15 cm for all varieties. If pulse is intercropped, adopt a spacing of 30 x 15 cm for cumbu and 30 x 10 cm for pulses. One pair row of cumbu is alternated with a single row of pulse crop.
ii. In the furrows in which fertilizers have been applied, place 5 kg of seed, allowing them to fall 4 - 5 cm apart (Use higher seed rate of 5 kg to offset mortality). The optimum population should be 1,45,000 per ha. Use increased seed rate upto 12.5 kg per hectare in shoot fly endemic area and remove the shootfly damaged seedlings at the time of thinning.
iii. Where pulse seeds are to be sown, drop pulse seeds to fall 5 cm apart and cover.

**2. WEED MANAGEMENT**

**Transplanted crop**

Spray Atrazine 50 WP 500 g/ha on the 3rd day. Then, one hand weeding on 30 - 35 days after transplanting may be given. If herbicide is not used hand weed on 15th day and again between 30 and 35 days after transplanting.

**Direct Sown crop**

i. Apply the pre-emergence herbicide Atrazine 50 WP at 500 g/ha, 3 days after sowing as spray on the soil surface using Back-pack/Knapsack/Rocker sprayer fitted with flat type nozzle using 900 l of water/ha.
ii. Apply herbicide when there is sufficient moisture in the soil.
iii. Hand weed on 30 - 35 days after sowing if pre-emergence herbicide is applied.
iv. If pre-emergence herbicide is not applied hand weed twice on 15 and 30 days after sowing.

**3. THINNING AND GAP FILLING**

In direct sown crop after 1st weeding at the time of irrigation, gap fill and thin the crop to a spacing of 15 cm between plants; cowpea crop to 20 cm between plants and other pulses crops to 10 cm between plants.

**4. TOP DRESSING OF FERTILIZERS**

i. Top dress the nitrogen at 15 and 30 days after transplanting or direct sowing.
ii. In transplanted crop, open a furrow 5 cm deep with a stick or hoe at the bottom of the furrow, place the fertilizer and cover.
iii. In the case of direct sown crop apply the fertilizer in band. If intercropped with pulses apply the fertilizer to cumbu crop only.
iv. After the application of fertilizer, irrigate the crop.

## V. WATER MANAGEMENT

| STAGES | Days after transplantation/sowing<br>Transplanted Crop | Direct Sown Crop |
|---|---|---|
| Light Soils | | |
| i. Germination | 1st day after transplanting | 1st day after sowing |
| | 4th day | 4th day |
| ii. Vegetative phase | 15th Day | 17th day |
| | 28th day | 30th day |
| iii. Flowering phase | 40th day | 42nd day |
| | 52nd day | 55th day |
| | 65th day | 68th day |
| iv. Maturity phase | 77th day | 79th day |
| Total | 8 irrigations | 8 irrigations |
| Heavy Soils | | |
| i. Germination | 1st day after planting | 1st day after sowing |
| | 4th day | 5th day |
| ii. Vegetative phase | 15th day | 15th day |
| | 28th day | 30th day |
| iii. Flowering phase | 42nd day | 45th day |
| | 54th day | 57th day |
| iv. Maturity Phase | 66th day | 70th day |
| Total | 7 irrigations | 7 irrigations |

NOTE: This is only a guideline and the irrigation schedule is to be adjusted depending upon the prevailing weather conditions.

## VI. HARVESTING THE CROP

**1. SYMPTOMS OF MATURITY**

i. Leaves will turn yellow and present a dried apprearance.
ii. Grains will be hardened.

**2. HARVESTING**

i. Cut the earheads separately.
ii. Cut the straw after a week, allowing it to dry and stack it in the field till it can be transported.

**3. THRESHING, CLEANING, DRYING AND STORING**

i. Dry the earheads
ii. Thresh in a mechanical thresher or
iii. Spread it and drag a stone roller over it or
iv. Cattle thresh.
v. Dry the seeds below 10 per cent and mix 100 kg of grains with 1kg of activated kaolin to reduce the rice weevil and rice moth incidence.
vi. Spray Malathion 50EC 10 ml/ lit @ 3 lit of spray fluid/100 $m^2$ over the bags during storage godowns,
vii. For grain purpose the grain should be dried well below 10% moisture and stored in gunny bags.

## CROP PROTECTION

### A. PEST MANAGEMENT

**Pest management strategies**

| Pest | Management strategies |
|---|---|
| **Shoot fly**<br>*Atherigona approximate* | ▪ Use seeds pelleted with insecticides (see sorghum)<br>▪ Seed treatment with imidacloprid 70 WS 10 g/kg of seeds<br>▪ Plough soon after harvest, remove and destroy the stubbles.<br>▪ Set up the TNAU low cost fish meal trap 12/ha till the crop is 30 days old.<br>▪ **Spray any one of the following :**<br>Endosulfan 35 EC 500 ml/ha<br>Methyl demeton 25 EC 500 ml/ha<br>Dimethoate 30 EC 500 ml/ha<br>Neem seed kernel extract 5%<br>Neem Azal 1%<br>▪ Soil application of phorate 10 G 40 kg/ha at the time of sowing |
| **Ear midge**<br>*Geromyia pennisetti* | ▪ **Apply any one of the following at 50 % flowering :**<br>Carbaryl 10 D 25 kg/ha<br>Malathion 5 D 25 kg/ha<br>Carbaryl 50 WP 750 g/ha<br>Endosulfan 35 EC 750 ml/ha (500 l of spray fluid/ ha). |

### B. DISEASE MANAGEMENT

| Nursery | |
|---|---|
| **Seed treatment** | ❖ **Cultural method**<br>**<u>Removal of ergot / sclerotia to prevent primary infection</u>**<br>▪ Dissolve 1 kg of common salt in 10 litres of water.<br>▪ Drop the seeds into the salt solution.<br>▪ Remove the ergot and sclerotia affected seeds which will float.<br>▪ Wash seeds in fresh water 2 or 3 times to remove the salt on the seeds.<br>▪ Dry the seeds in shade.<br>❖ **Chemical**<br>▪ Thiram 75%WDP @ 2g + 5ml of water/kg of seed or Metalaxyl 6 g/kg for the control of downy mildew in the endemic area |
| **Main field** | |
| **Name of the Disease** | **Management** |
| ***Sugary or Ergot disease***<br>*Claviceps fusiformis* | ▪ Spray any one of the fungicides like Carbendazim 500g or Mancozeb 1000g when 5 - 10% flowers have opened and again at 50% flowering stage. |
| ***Rust***<br>*Puccinia penniseti* | ❖ Cultural method<br>▪ Sowing during December - May result in less incidence .<br>▪ Adopt control measures when there is rust incidence in the early stages as spread of infection to top leaves results in poor grain filling.<br>❖ Chemcial<br>➢ Spray any one of the following fungicides when the initial symptoms of the diseases are noticed.<br>▪ Wettable sulphur 2500g / ha.<br>▪ Mancozeb 1000g/ha.<br>▪ Repeat application 10 days after if necessary. |

| | |
|---|---|
| ***Downy Mildew***<br>*Sclerospora graminicola* | ❖ **Cultural method**<br>▪ Grow downy mildew resistant varieties CO7, WCC 75<br>▪ Transplanting reduces disease incidence. At the time of planting infected seedlings should be removed.<br>▪ In the direct sown crop, infested plants should be removed up to 45 days after sowing as and when the symptoms are noticed.<br>➢ Spray any one of the fungicides<br>▪ Metalaxyl 500 g or<br>▪ Mancozeb 1000g/ha |

# RAGI (*Eleusine coracana*)
# IRRIGATED

## CROP IMPROVEMENT

## 1. SEASONS AND VARIETIES

| DISTRICTS/SEASON | VARIETIES |
|---|---|
| **Irrigated** | |
| **i. Marghazipattam (Dec - Jan)** | |
| All districts except Kanyakumari & Nilgiris | CO 9, CO 13, CO (Ra) 14 , TRY 1 |
| **ii. Chithiraipattam (April - May)** | |
| All districts except Kanyakumari & Nilgiris | CO 9, CO 13, CO (Ra) 14 |
| **Rainfed** | |
| **i. Adipattam (June - July)** | |
| All districts except Kanyakumari & Nilgiris | Paiyur 1, CO 13, CO (Ra) 14 |
| **ii. Puratasipattam (September - October)** | |
| All districts except Kanyakumari & Nilgiris | Paiyur 1, CO 13, CO (Ra) 14 |

## II. PARTICULARS OF RAGI STRAINS

| PARTICULARS | CO 9 | CO 13 | CO (Ra) 14 |
|---|---|---|---|
| Parentage | EX 4336 x PLR 1 | CO 7 X TAH 107 | Malawi 1305 x CO 13 |
| Duration (days) | 100-105 | 95 - 100 | 105-110 |
| Season | | | |
| Rainfed/ Irrigated | Both | Both | Both |
| Grain yield kg/ha | | | |
| Irrigated | 4500 | 3600 | 2892 |
| Rainfed | 3100 | 2300 | 2794 |
| Straw yield kg/ha | | | |
| Irrigated | 8000 | 10000 | 8113 |
| Rainfed | 6500 | 7500 | 8503 |
| Stem | Erect | Erect | Erect |
| Height (cm) | 75-80 | 85 - 90 | 115-120 |
| Tillers | 5-8 | 3 - 5 | 8-9 |
| Days to 50% flowering | 65-70 | 55 - 60 | 72 |
| Ear size and shape | incurved fingers | Open | top curved |
| Fingers | 8-9 | 10 –17 | 9-12 |
| Ear length (cm) | 8 | 8-10 | 10-12 |
| Grain colour | White | Light Brown | Brown |
| 1000 grain wt (g) | 2.7 | 1.7 | 3.1 |

**Contd..**

**Contd..**

| PARTICULARS | Paiyur 1 | TRY 1 |
|---|---|---|
| Parentage | Pureline selection from PR 722 | Selection from HR 374 |
| Duration (days) | 115-120 | 102 |
| Season | | |
| Rainfed / Irrigated | Rainfed | Kharif irrigated Sodic/ saline soils |
| Grain yield kg/ha | | |
| Irrigated | .. | 4011 |
| Rainfed | 3125 | .. |
| Straw yield kg/ha | | |
| Irrigated | .. | 6800 |
| Rainfed | 5750 | .. |
| Stem | Erect | Erect |
| Height (cm) | 110 | 100 |
| Tillers | 1 - 3 | 5 - 7 |
| Days to 50% flowering | 80 | 78 |
| Ear size and shape | Open | Incurved |
| Fingers | 6 - 8 | 5-8 |
| Ear length (cm) | 8 | 7.6 |
| Grain colour | Brown | Brown |
| 1000 grain wt (g) | 2.7 | 2.74 |

## CROP MANAGEMENT

### I. PREPARATION OF NURSERY (IRRIGATED TRANSPLANTED CROP)

**1. PREPARATION OF LAND**

i. For raising seedlings to plant one ha of main field, select 12.5 cents (500 $m^2$) of nursery area near a water source, where water does not stagnate.
ii. Mix 37.5 kg of super phosphate with 500 kg of FYM or compost and spread the mixture evenly on the nursery area.
iii. Plough two or three times with a mould board plough or five times with a country plough.

**2. FORMING RAISED BED**

i. Mark units of 6 plots each of size 3 m x 1.5 m. Provide 30 cm space between plots for irrigation.
ii. Excavate the soil from the interspace and all around to a depth of 15 cm to form channels and spread the soil removed from the channels on the bed and level.

**3. PRE-TREATMENT OF THE SEEDS WITH FUNGICIDES**

i. Seed treatment with Azospirillum may be done @ 3 packets/ha (600 g/ha) and 3 packets (600 g/ha) of Phosphobacteria or 6 packets of Azophos (1200 g/ha).
ii. Mix the seeds in a polythene bag to ensure a uniform coating of seeds with Thiram 4 g/ kg or Captan 4 g/kg or Carbendazim 2 g/kg of seeds.

**4. SOWING AND COVERING THE SEEDS**

i. Make shallow rills not deeper than one cm on the beds by passing the fingers vertically over them.
ii. Broadcast 5 kg of treated seeds evenly on the beds.
iii. Cover the seeds by levelling out the hand lightly over the soil.
iv. Sprinkle 500 kg of powdered FYM over the beds evenly to cover the seeds which are exposed and compact the surface lightly.
NOTE: Do not sow the seeds deep as germination will be adversely affected.

**5. WATER MANAGEMENT**

i. Provide one inlet to each nursery unit.

ii. Allow water to enter through the inlet and cover all the channels around the beds. Allow the water in the channels to raise till the raised beds are wet and then cut off water.

iii. Adjust the frequency of irrigation according to the soil type.

| No. of irrigations | RED SOILS | HEAVY SOILS |
|---|---|---|
| 1st | Immediately after sowing | Immediately after sowing |
| 2nd | 3rd day after sowing | 4th day after sowing |
| 3rd | 7th day after sowing | 9th day after sowing |
| 4th | $12^{th}$ day after sowing | 16th day after sowing |
| 5th | $17^{th}$ day after sowing | .. |

NOTE: 1. One irrigation is given on the 3rd day in the case of red soil to soften the hard crust formed on the soil surface and also to facilitate seedlings to emerge out.

2. Do not allow cracks to develop in the nursery bed by properly adjusting the quantity of irrigation water.

**6. PULLING OUT THE SEEDLINGS FOR PLANTING**

Pull out seedlings on the 17th to 20th day of sowing for planting.

## II. PREPARATION OF MAIN FIELD

**1. PLOUGHING THE FIELD**

Plough twice with mould board plough or thrice with wooden plough till a good tilth is obtained.

**2. APPLICATION OF FYM OR COMPOST**

Spread 12.5 t/ha of FYM or compost or composted coir pith evenly on the unploughed field and then plough and incorporate in the soil. NOTE: Do not spread and leave the manure uncovered in the field as nutrients will be lost.

**3. APPLICATION OF FERTILIZERS**

i. Soil test based fertilizer recommendations may be adopted in Western and North Western Zones in soils of Vertisols, Alfisols, Inceptisols and Entisols for prescribing the fertilizer doses for specified yield targets. In soils having high intensive cropping system viz., Ragi-Maize-Cowpea, having high soil available K (310 kg/ha) potassium need not be applied. Apply NPK fertilizers as per soil test recommendation as far as possible. If soil test recommendation is not available, adopt a blanket recommendation of 60 kg N, 30 kg $PO_5$ and 30 kg $K_2O_5$ per ha.

ii. Apply half the dose of N and full dose of N and full dose of $P_2O_5$ basally before planting.

iii. Broadcast the fertilizer mixture over the field before the last ploughing and incorporate into the soil by working a country plough.

iv. Apply 10 packets/ha (2000 g) of azospirillum and 10 packets (2000 g/ha) of Phosphorous solubilizing bacteria or 20 packets of Azophos (4000 g/ha) after mixing with 25 kg of soil and 25 kg FYM before transplanting.

**4. FORMING BEDS AND CHANNELS**

i. Form beds of size 10 $m^2$ to 20 $m^2$ according to topography of the field.

ii. Provide suitable irrigation channels.

**5. APPLICATION OF MICRONUTRIENT MIXTURE**

i. Mix 12.5 kg of micronutrient mixture formulated by the Department of Agriculture, Tamil Nadu with enough sand to make a total quantity of 50 kg/ha.

ii. Apply the mixture evenly on the beds.

iii. Do not incorporate the mixture in the soil.

## III. MANAGEMENT OF MAIN FIELD

**1. TRANSPLANTING THE SEEDLINGS**

i. Let water into the bed, level the bed, if it is not levelled.
ii. Plant 2 seedlings per hill.
iii. Plant the seedlings at a depth of 3 cm.
iv. Plant 18 to 20 days old seedlings.
v. Adopt a spacing of 30x10 cm for planting.
vi. Adopt 22.5 x 10 cm spacing for direct sowing.
vii. Root dipping with Azospirillum prepare slurry with 5 packets (1000 g)/ha of Azospirillum and 5 packets (1000g/ha) of Phosphobacteria or 10 packets of Azophos (2000 g/ha) in 40 litres of water and dip the root portion of the seedlings in the solution for 15-30 minutes and transplant.

**2. WEED MANAGEMENT**

i. Apply Butachlor 2.5 l/ha or Fluchloralin 2 l/ha or Pendimethalin 2.5 l/ha, using Backpack Knapsack/Rocker sprayer fitted with flat fan type of nozzle with 900 litre of water/ha.
ii. Apply the herbicides when there is sufficient moisture in the soil or irrigate immediately after application of herbicide.
iii. If pre-emergence herbicide is not applied, hand weed twice on 10th and 20th day after transplanting.
iv. For rainfed direct seeded crop, apply post emergence herbicide; 2,4-DEE or 2,4-D Na salt at 0.5 kg/ha on 10th day after sowing depending on the moisture availability.

**3. HOEING AND HAND WEEDING**

i. Hoe and hand weed on the 15th day of planting in light soils and 17th day of planting in heavy soils and subsequently on 30th and 32nd days, respectively.
ii. Allow the weeds to dry for 2 or 3 days after hand weeding before giving irrigation. NOTE:

Do not adopt hoeing and hand weeding if herbicide is applied.

## IV. WATER MANAGEMENT

**Regulate irrigation according to the following growth phases of the crop**

| Stages | No. of irrigations | Crop duration 80 days | 100 days | 120 days |
|---|---|---|---|---|
| Vegetative phase(Nursery) | | 1 to 16 | 1 to 18 | 1 to 20 |
| Vegetative phase (in main field) | | 1 to18 | 1 to 20 | 1 to 22 |
| Flowering phase | | 19 to 40 | 21 to 55 | 23 to 69 |
| Maturity phase | | Beyond 40 days | Beyond 55 days | Beyond 69 days |
| **Heavy soils** | | | | |
| Establishment | 1 | 1st day | 1st day | 1st day |
| (1-7 days) | 2 | 5th day | 5th day | 5th day |
| Vegetative phase | 1 | 18th day | 20th day | 20th day |
| (8-20 days) | 2 | 31st day | 33rd day | 30th day |
| Flowering phase | 1 | 41st day | 42nd day | 37thd ay |
| (21-55 days) | 2 | 51st day | 52nd day | 44th day |
| | 3 | -- | -- | 63rd day |
| Maturity phase | 1 | 61st day | 62nd day | 78th day |
| (56-120 days) | 2 | -- | -- | 93rd day |
| Stop irrigation thereafter | | | | |
| **Light soils** | | | | |
| Establishment | 1 | 1st day | 1st day | 1st day |

| | | | | |
|---|---|---|---|---|
| (1 – 7 days) | 2 | 5th day | 5th day | 5th day |
| Vegetative phase | 1 | 15th day | 16th day | 16th day |
| (8 - 20 days) | 2 | 26th day | 28th day | 28th day |
| Flowering phase | 1 | 36th day | 36th day | 36th day |
| (21 - 55 days) | 2 | 45th day | 45th day | 45th day |
| | 3 | .. | 54th day | 54th day |
| Maturity phase | 1 | 58th day | 69th day | 78th day |
| (56 - 120 days) | 2 | 70th day | 85th day | 93rd day |
| Stop irrigation thereafter | | | | |

**NOTE:** The irrigation schedule is given only as a general guideline. Regulate irrigation depending upon the prevailing weather conditions and receipt of rain.

## V. HARVESTING

**1. DECIDE WHEN TO HARVEST**

i. Ragi crop does not mature uniformly and hence the harvest is to be taken up in two stages.
ii. When the earhead on the main shoot and 50% of the earheads on the crop turn brown, the crop is ready for the first harvest.

**2. HARVEST OF THE CROP**

**First harvest**

i. Cut all earheads which have turned brown.
ii. Dry, thresh and clean the grains by winnowing.

**Second Harvest**

i. Seven days after the first harvest, cut all the earheads including the green ones.
ii. Cure the grains to obtain maturity by heaping the harvested earheads in shade for one day without drying, so that the humidity and temperature increase and the grains get cured.
iii. Dry, thresh and clean the grains by winnowing and store the grains in gunnies.

**i. Threshing**

Green earheads if harvested will contaminate the seeds with immature seeds and interfere cleaning, drying and grading. Dry earheads until seed moisture content is 15% and separate manually by threshing with bamboo stick or machine thresher.

**ii. Precleaning and drying**

Threshed seeds should be precleaned before sundrying, seeds must be dried to 12% before grading.

**Protection from storage pests**

1. Grain purpose: Dry the seeds adequately to reduce the moisture level to 10%.
2. Seed purpose: Admix one kg of Activated kaolin or Malathion 5% D for every 100 kg of seed. Pack in gunny or polythene lined gunny bags for storage.

**Special problems**

i. Root Aphids: Mix Dimethoate 3 ml in one litre of water and drench the rhizosphere of the infested and surrounding plants with the insecticidal solution.
ii. Rainfed ragi: Azospirillum mixed with FYM and applied to field saves the cost of nitrogen by 50% with a comparable yield obtained with 40 kg N/ha.
iii. Management of aged seedlings of ragi under rainfed conditions: When planting **ragi** seedlings beyond 21 days, increase the number of seedlings to 3/hill and increase N level by 25% to minimise yield loss.
iv. Apply VAM culture (*Glomus fasciculatum*) at 100 g/$m^2$ in the nursery and also treat with Azospirillum and Phosphobacterium as seed treatment, seedling dip and field application to reduce the reniform nematode population in ragi.

# RAGI : RAINFED

**Rainfall**

Average and well distributed rainfall of 450-500 mm is optimum for rainfed ragi

**Season**

Finger millet is grown in different seasons in different parts of the country. As a rainfed crop, it is normally sown in June- July in Tamil Nadu. It also grown in winter season (rabi) by planting in September – October in Tamil Nadu and as a summer irrigated crop by planting January – February.

**Tillage**

Fall ploughing is advantageous for moisture conservation. In the month of April or May, one deep ploughing with mould board plough followed by ploughing with wooden plough twice is necessary. Before sowing secondary tillage with cultivator and multiple tooth hoe to prepare smooth seed bed is necessary.

**Seed rate and planting**

A plant population of 4 – 5 lakhs per ha is optimum for getting higher yields and higher or lower population than the optimum will reduce the yield. Line sowing is ideal and seed drills giving spacing of 22.5 – 30 cm between rows should be used. Finger millet seeds are very small (400 seeds/g) and the recommended seed rate of 15-20 kg per hectare will contain about 4 million seeds. Therefore, even when seed drill is used thinning within the row leaving a spacing of 7.5 – 10 cm between plants, must be followed.

Sowing by seed-cum-fertilizer drill is advantageous for line sowing besides efficient utilization of applied nutrients.

Maintenance of optimum plant population is an important prerequisite for getting higher yield under rainfed conditions. Poor germination, often, is the result of inadequate moisture after sowing in low rainfall areas. Under these conditions, the adoption of a simple technique like seed hardening will not only improve germination and subsequent plant stand but also impart early seedling vigour and tolerance to drought.

The procedure of seed hardening technique is as follows.

1. Sole seeds in water for 6 hours. Use one litre water for every kg seed for soaking.
2. Drain the water and keep the seeds in wet cloth bag tightly tied for two days.
3. At this stage, the seeds will show initial signs of germination.
4. Remove seeds from the wet cloth bag and dry them in shade on a dry cloth for 2 days.
5. Use the above hardened seeds for sowing.

**Manuring and fertilization**

Finger millet responds well to fertilizer application especially to N and P. The recommended doses of fertilizers vary from state to state for rainfed crop. Recommended dose of 40:20:20 kg/ha N:P:K was applied. With judicious application of farmyard manure inorganic fertilizer efficiency is enhanced. Entire $P_2O_5$ and $K_2O$ are to be applied at sowing, whereas nitrogen is to be applied in two or three split doses depending upon moisture availability. In areas of good rainfall and moisture availability, 50% of recommended nitrogen is to be applied at sowing and the remaining 50% in two equal splits at 25-30 and 40-45 days after sowing. In areas of uncertain rainfall, 50% at sowing and the remaining 50% around 35 days after sowing is recommended.

**Bio-fertilizers**

Treating seeds with *Azospirillum brasilense* (N fixing bacterium) and *Aspergillus awamori* (P solubilizing fungs) @ 25 g/kg seed is beneficial. In case seeds are to be treated with seed dressing chemicals, treat the seeds first with seed dressing chemicals and then with bio-fertilizers at the time of sowing.

**Procedures for inoculating seeds with biofertilizers**

1. Bio-fertilizer culture specific to the crop is to be used @ 25 g per kg of seed.
2. Sticker solution is necessary for effective seed inoculation. This can be prepared by dissolving 25 g jaggery or sugar in 250 ml water and boiling for 5 minutes. The solution thus prepared is cooled.
3. Smear the seeds well using the required quantity of sticker solution. Then add culture to the seeds and mix thoroughly so as to get a fine coating of culture on the seed.
4. The culture-coated seeds is to be dried well in shade to avoid clumping of seeds.
5. Use the inoculated seeds for sowing.

**Weed control**

In line sown crop 2-3 inter-cultivations are necessary. In assured rainfall and irrigated areas spraying 2,4-D sodium salt @ 0.75 kg.a.i./ha as post-emergent spray around 20-25 days after sowing effectively controls weeds. Isoproturon @ 0.5 a.i/ha as pre-emergence spray is also effective in control of weeds. In broadcast crop two effective hand weedings will minimize weeds as inter cultivations is not possible.

**Cropping systems**

**Crop rotation**

Rotation with legumes like green / black gram / field gram / soybean / horse gram or ground nut in southern state will minimize inorganic fertilizer application and also sustain higher yields.

**Intercropping**

Finger millet + Pigeon pea combinations in 8 : 2 or 6 : 2 proportion brings higher returns in Tamil Nadu. Finger millet + field bean in 8 : 1 proportion or finger millet + blackgram or green gram in 8 : 2 proportion are the other profitable crop combinations.

## CROP PROTECTION

### A. PEST MANAGEMENT

| Pests | Management strategies |
|---|---|
| **Aphids**<br>*Schizaphis graminum*<br>*Rhopalosiphum maidis* | ▪ Spray any one of the following insecticides mixed in 10 litres of water using a high volume sprayer if dusting is not done :<br>Methyl demeton 25 EC 20 ml/ha<br>Dimethoate 30 EC 20 ml/ha |
| **Stem borer**<br>*Sesamia inferens* | ▪ Spray any one of the following :<br>Endosulfan 35 EC 1000 ml/ha<br>Carbaryl 50 WP 1 kg/ha (500 l spray fluid/ha). |
| **Root aphid**<br>*Tetraneura nigriabdominalis* | ▪ Spray any one of the following :<br>Endosulfan 35 EC 1000 ml/ha<br>Carbaryl 50 WP 1 kg/ha (500 l spray fluid/ha). |
| **Ear head bug** | ▪ Spray carbaryl 50 WP 1.0 kg/ha at milky stage |
| **Earhead caterpillar** | ▪ Spray carbaryl 50 WP 1.0 kg/ha at milky stage |

**B. DISEASE MANAGEMENT**

| **i)Nursery** | |
|---|---|
| **Seed treatment** | ▪ Thiram 4 g or Captan 4 g or Carbendazim 2 g/kg of seed.<br>▪ Treat the seeds 24 hr prior to sowing. |
| **ii)Main field** | |
| **Name of the Disease** | **Management** |
| **Blast**<br>*Pyricularia grisea* | ➢ Spray any one of the fungicides<br>▪ Edifenphos 500 ml or Carbendazim 250 g or IBP 500 ml/ha. First spray immediately after noticing the symptoms. Second and third sprays at flowering stage at 15 days interval to control neck and finger infection.<br>▪ Treat the seeds with *P. fluorescens* 10 g/kg and spray the extracts of *Prosopis juliflora* leaf extract (10%), *Ipomoea cornea* leaf extract (10%).<br>▪ Foliar spray with Aureofungin sol 100 ppm at 50% earhead emergence followed by a second spray with Mancozeb 1000g/ha 10 days later. |
| **Mosaic**<br>(*Potyvirus*)<br>*Vector*<br>*Aphis* sp.<br>**Ragi mottle streak**<br>*(Nucleo rhabdovirus)*<br>Vector<br>*Cicadulina bipunctella*<br>*C. chinai* | ❖ **Cultural method**<br>▪ Rogue out the affected plants.<br>▪ Spray any one of the insecticide like Monocrotophos 36 WSC 700 ml/ha or Methyl demeton 25EC 500 ml/ha on noticing symptoms and repeat twice if necessary at 20 days intervals. |

# MAIZE (*Zea mays* L.)

## I. SEASON AND VARIETIES

| SEASON | STRAIN |
|---|---|
| 1. Adipattam (July - August) | CO 1, COH (M) 4, COBC 1 |
| 2. Purattasipattam (September -October) | CO 1, COH (M) 4, COBC 1 |
| 3. Thaipattam (January -February) | CO 1, COH (M) 4, COBC 1 |

**MAIZE - MORPHOLOGICAL DESCRIPTION**

| PARTICULARS | CO 1 | COH(M) 4 | COBC 1 |
|---|---|---|---|
| Parentage | Unit Selection from Indonesian popu lation Suwan 1 | (UMI 90 x xUMI 285) x UMI 112 | Composite involving UMI 836 and UMI 836-1-2 |
| Duration (days) | 105 - 110 | 94 (90 – 95 days) | 55 - 65<br>(fresh cobs harvest) |
| Area of Adoption | Coimbatore, Erode, Tirunelveli, Tanjore and Pudukottai | Coimbatore, Dindugul, Erode, Salem, Virudhunagar, Perambalur, Tuticorin, Madurai and Theni districts | All maize growing areas |
| Rainfed/Irrigated | Both | Both | Both |
| **Grain yield (kg/ha)**<br>Irrigated | 5200 | 5694 | 6.7 tonnes of tender cobs & 32.3 tonnes of green fodder / ha |
| Rainfed | 3300 | 4521 | 3 tonnes of tender cobs per ha. |
| Special features | Resistant to downy mildew<br>Orange flint grains | Shorter duration, suitable for rainfed areas<br>Moderate resistance to downy mildew and stemborer | 2 to 3 cobs / plant, possessing delightful sweet taste and flavour<br>Suitable for inland and export market<br>Green fodder has good forage value |
| Sheath colour | Green | Green | Green |
| Node colour | Green | Green | Green |
| Mid rib | White | White | White |
| Cob size | Big | Big | Small |
| Husk coverage | Full | Full | Nil |
| Grain colour | Deep orange | Yellow | Yellow |
| Nature of kernels | Flint | Flint | Dent |

## CROP MANAGEMENT

## I. IRRIGATED MAIZE

**1. APPLICATION OF FYM OR COMPOST**

Spread 12.5 t/ha of FYM or compost or composted coir pith evenly on the unploughed field along with 10 packets of Azospirillum (2000 g/ha) and incorporate in the soil.

**2. FIELD PREPARATION**

Plough the field with disc plough once followed by cultivator ploughing twice, after spreading FYM or compost till a fine tilth is obtained.

**3. FORMING RIDGES AND FURROWS OR BEDS**

i. Form ridges and furrows providing sufficient irrigation channels. The ridges should be 6 m long and 60 cm apart.
ii. If ridges and furrows are not made, form beds of size 10 $m^2$ or 20 $m^2$ depending on the availability of water.
iii. Use a bund former or ridge plough to economise cost of production.

**4. APPLICATION OF FERTILIZERS**

i. Apply NPK fertilizers as per soil test recommendation as far as possible. If soil test recommendation is not available adopt a blanket recommendation of 135:62.5:50 NPK kg/ha.
ii. Apply quarter of the dose of N; full dose of $P_2O$ and $K_2O$ basally before sowing.
iii. In the case of ridge planted crop, open a furrow 6 cm deep on the side of the ridge, at two thirds the distance from the top of the ridge.
iv. Apply the fertilizer mixture along the furrows evenly and cover to a depth of 4 cm with soil.
v. If bed system of planting is followed, open furrows 6 cm deep at a distance of 60 cm apart.
vi. Place the fertilizer mixture along the furrows evenly and cover to a depth of 4 cm with soil.
vii. When Azospirillum is used as seed and soil application, apply 100 kg of N/ha (25% reduction on the total N recommended by soil test).

**Defieciency symptoms**

| | | |
|---|---|---|
| Nitrogen deficiency | : | Leaves become yellow, older leaves show drying at the tips which progress along mid veins, stalks become slender. |
| Phosphorus deficiency | : | Leaves are purplish green during early growth. Growth spindly, slow maturity, irregular ear formation. |
| Potassium deficiency | : | Leaves show yellow or yellowish green streaks, become corrugated. Tips and marginal scorch. Tips end in ears are poorly filled. Stalks have short internode. Plants become weak and may fall down. |
| Magnesium deficiency | : | Older leaves are the first to become chlorotic at margins and between veins. Streaked appearance of leaves. Necrotic or chlorotic spots seen in leaves. |
| Zinc deficiency | : | Older leaves have yellow streaks or chlorotic striping between veins. In several cases, unfolding of young leaves, which may be white or yellow. |
| Iron deficiency | : | Interveinal chlorosis. The entire crop may exhibit bleached appearance. |

**5. APPLICATION OF MICRONUTRIENT**

i. 12.5 kg of micronutrient mixture formulated by the Department of Agriculture, Tamil Nadu, mixed with sand to make a total quantity of 50 kg/ha is to be applied.
ii. Apply the mixture over the furrows and two thirds in the top of ridges, if ridge planting is followed.
iii. If bed system of sowing is followed, apply the micronutrient mixture over the furrows.
iv. Do not incorporate the micronutrient mixture in the soil.

**6. SEED RATE**

Select good quality seeds and adopt the seed rate of 20 kg/ha for CO 1 and COH(M) 4 and 25 kg /ha for COBC 1.

**7. SPACING**

Adopt a spacing of 25 cm between plants in the rows which are 60 cm apart.

Population : For varieties and hybrids 6 – 7 plants / sq. m. and
For baby corn, 8 – 9 plants / sq. m.

**8. SEED TREATMENT**

**Step 1:** Use pelleted seeds with insecticides (treat one kg of seeds with Chlorpyriphos 20EC or Monocrotophos 36 WSC or Phosalone 35 EC @ 4 ml + 0.5 gram gum in 20 ml of water) for the control of stem borer or seed treatment with imidacloprid 70 WS 10 g/kg of seeds.

**Step 2:** Seed treatment with Metalaxyl or Thiram @ 2 g/kg of seed for the control of downy mildew and crazy top

**Step 3**: Seeds treated with fungicides may be treated with three packets (600 g/ha) of Azospirillum before sowing.

**9. SOWING**

i. Dibble the seeds at a depth of 4 cm along the furrow in which fertilizers are placed and cover with soil.
ii. Put one seed per hole if the germination is assured otherwise put two seeds per hole

**10. WEED MANAGEMENT**

i. Apply the pre-emergence herbicide, Atrazine 50 WP at 500 g/ha (900 lit of water) at 3 days after sowing as spray on the soil surface followed by one hand weeding on 40-45 days after sowing.
ii. Apply herbicide when there is sufficient moisture in the soil.
iii. Do not disturb the soil after herbicide application.
iv. Hoe and hand weed on the 17th or 18th day of sowing, if herbicide is not applied. NOTE: If pulse crop is to be raised as intercrop, do not use Atrazine.

**11. THINNING AND GAP FILLING**

i. If two seeds were sown, leave only one healthy and vigorous seedling per hole and remove the other on the 12-15 days after sowing.
ii. Where seedlings have not germinated, dibble presoaked seeds at the rate of 2 seeds per hole and immediately irrigate.

**12. HOEING, HAND-WEEDING AND EARTHING UP**

**i. Hoe and hand-weed on the 30th day of sowing.**
ii. Earth up and form new ridges so that the plants come directly on the top of the ridges. This will provide additional anchorage to the plants.

**13. TOP DRESSING WITH N**

i. Place half of the dose of N on the 25$^{th}$ day of sowing along the furrows evenly and cover it with soil.
ii. Place the remaining quarter of N on the 45th day of sowing

**14. WATER MANAGEMENT**

Maize crop is sensitive to both moisture stress and excessive moisture, hence regulate irrigation according to the requirement. Ensure optimum moisture availability during the most critical phase (45 to 65 days after sowing); otherwise yield will be reduced by a considerable extent.

Regulate irrigation according to the following growth phase of the crop.

| | |
|---|---|
| Germination & establishment phase | 1 to 14 days |
| Vegetative phase | 15 to 39 days |
| Flowering phase | 40 to 65 days |
| Maturity phase | 66 to 95 days |

| **Heavy soils** | | |
|---|---|---|
| **Stage** | **No. of irrigation** | **Days after sowing** |
| Germination & establishment | 3 | After sowing, Life irrigation -4$^{th}$,12$^{th}$ day |
| Vegetative | 2 | 25$^{th}$, 36$^{th}$ day |
| Flowering(Irrigate copiously) | 2 | 48$^{th,}$ 60$^{th}$ day |
| Maturity phase (Control irrigation) | 2 | 72$^{nd}$, 85$^{th}$ day |

| Light soils | | |
|---|---|---|
| Germination & establishment | 3 | After sowing, Life irrigation -4$^{th}$ ,12$^{th}$day |
| Vegetative Phase | 3 | 22$^{nd}$ ,32$^{nd}$ & 40$^{th}$ day |
| Flowering phase (Irrigate copiously) | 3 | 50$^{th}$ ,60$^{th}$ & 72$^{nd}$ day |
| Maturity phase (Controlled irrigation) | 2 | 85$^{th}$ , 95$^{th}$ day |

**15. HARVESTING**

**STAGE OF HARVEST**

Observe the following symptoms, taking into consideration the average duration of the crop.

i. The sheath covering the cob will turn yellow and dry at maturity.
ii. The seeds become fairly hard and dry. At this stage the crop is ready for harvest.

**HARVESTING THE CROP**

i. Tear off the cob sheath by using the gunny needle and remove the cobs from the plant.
ii. Carry out harvest operations at a single stage for easy transportation.

**THRESHING THE COBS**

i. Dry the cobs under the sun till the grains are dry.
ii. Use mechanical threshers or by running the tractor over dried cobs to separate the grains from the shank.
iii. Clean the seeds by winnowing
iv. Collect and store the dry grains in gunnies.

**STACKING THE STRAW FOR FEEDING CATTLE**

i. Maize straw can also be used as a good cattle feed when it is green.
ii. Harvest the crop and cut the green straw into bits with a chaff cutter or chopping knife and feed the cattle.

## II. RAINFED MAIZE

**1. FIELD PREPARATION**

Chisel the soil having hard pan formation at shallow depths with chisel plough at 0.5 M interval first in one direction and then in the direction perpendicular to the previous one once in three years. Apply 12.5 t/ha of FYM or compost or composted coir pith besides chiselling, to get an additional yield of about 30% over control.

**2. APPLICATION OF FYM OR COMPOST**

Spread 12.5 t/ha of FYM or compost or composted coir pith evenly on the unploughed field along with 10 packets of Azospirillum (2000 g/ha) and incorporate in the soil.

**3. APPLICATION OF FERTILIZER**

i. Apply NPK as per soil test recommendation as far as possible. If soil test recommendation is not available, adopt a blanket recommendation of 60 : 30 : 30 NPK kg/ha for Alfisols and 40 : 20 : 0 NPK kg/ha for Vertisols.
ii. Apply half of N and full dose of $P_2O$ and $K_2O$ with enriched FYM as basal along with Azospirillum (10 packets/ha).
iii. Top dress remaining half of N at tasseling.

**4. SEED RATE**

Select good quality seeds. Adopt the seed rate @ 20 kg/ha for hybrids and 25 kg/ha. for varieties

**5. SPACING**

Adopt a spacing of 45 cm between rows and 20 cm between plants in the row.
Population : 10 - 11plants/$m^2$

**6. PRE-TREATMENT OF SEEDS WITH BIOFERTILIZER**

Seeds treated with fungicides may be treated with three packets (600 g/ha) of Azospirillum

**7. SOWING**

Dibble or drill the seeds at a depth of 4 cm.

**8. CROPPING SYSTEMS**

i. Intercropping system of maize + cowpea or maize + blackgram is recommended for higher net returns in the red lateritic soils of Southern districts.
ii. For Vertisols of Southern district, maize + redgram intercropping systems is ideal.

## CROP PROTECTION

**A. PEST MANAGEMENT**

| Pests | Management strategies |
|---|---|
| **Shoot fly**<br>*Atherigona orientalis* | ▪ Use seeds pelleted with insecticides (see sorghum)<br>▪ Seed treatment with imidacloprid 70 WS 10 g/kg of seeds<br>▪ Plough soon after harvest, remove and destroy the stubbles.<br>▪ Set up the TNAU low cost fish meal trap 12/ha till the crop is 30 days old.<br>▪ Spray any one of the following :<br>Endosulfan 35 EC 500 ml/ha<br>Methyl demeton 25 EC 500 ml/ha<br>Dimethoate 30 EC 500 ml/ha<br>Neem seed kernel extract 5%<br>Neem azal 1%<br>▪ Soil application of phorate 10 G 40 kg/ha at the time of sowing |
| **Stem borer**<br>*Chilo partellus*<br>**Ash weevil**<br>*Myllocerus* spp<br>**Aphids**<br>*Rhopalosiphum maidis* | ▪ Mix any of the following granular insecticides with sand to make up a total quantity of 50 kg and apply in the leaf whorls on the 20th day of sowing :<br>Quinalphos 5 G 15 kg/ha<br>Carbaryl 4 G 20 kg/ha.<br>▪ If granular insecticides are not used, spray any one of the following :<br>Quinalphos 25 EC 1 lit/ha<br>Carbaryl 50 WP 1 kg/ha on the 20th day of sowing (500 l of spray fluid/ha). |
| **Cob borer**<br>*Helicoverpa armigera* | ▪ Apply any one of the following insecticides at silk drying stage :<br>Carbaryl 10 % D 25 kg/ha<br>Carbaryl 50 WP 1.0 kg/ha.<br>▪ Repeat the insecticidal application 15 days later (500 l spray fluid/ha). |

**B. DISEASE MANAGEMENT**

| **Seed treatment** | Metalaxyl or Carbendazim or Thiram @ 2 g/kg of seed. |
|---|---|
| **Name of the Disease** | **Management** |
| Downy mildew or Crazy top<br>***Peronosclerospora sorghi*** | ▪ Use resistant hybrids / varieties<br>▪ Rogue out affected plants.<br>▪ Spray Metalaxyl 72 WP @ 1000g, Mancozeb 1000g/ha 20 days after sowing. |
| **Leaf spot**<br>*Helminthosporium turcicum* | ▪ Spray Mancozeb or Captan 1000g/ha when the disease intensity reaches grade 3 |

# SMALL MILLETS

## CROP IMPROVEMENT

### I. SEASON AND VARIETIES

| Crop | Districts grown |
|---|---|
| Tenai | Dharmapuri, Krishnagiri, Vellore, Tiruvannamalai, Cuddalore, Villupuram, Salem, Namakkal, Erode, Coimbatore, Tiruchirapalli, Perambalur, Karur, Madurai, Theni, Tirunelveli and Thoothukudi. |
| Samai | Dharmapuri, Vellore, Tiruvannamalai, Erode, Salem, Namakkal, Coimbatore, Madurai, Dindigul, Theni, Tirunelveli and Thoothukudi. |
| Varagu | Vellore, Thiruvannamalai, Kanchipuram, Tiruvallur, Dharmapuri, Krishnagiri, Salem, Namakkal, Cuddalore, Villupuram, Tiruchirapalli, Perambalur, Karur, Thanjavur, Tiruvarur, Nagapattinam, Madurai, Dindigul, Theni and Ramanathapuram |
| Panivaragu | Vellore, Tiruvannamalai, Salem, Namakkal, Dharmapuri, Krishnagiri, Madurai, Dindigul, Theni, Tirunelveli and Thoothukudi |
| Kudiraivali | Salem, Namakkal, Dharmapuri, Krishnagiri, Coimbatore, Tiruchirapalli, Perambalur,Karur, Pudukottai, Madurai, Dindigul, Theni, Ramanathanpuram, Tirunelveli and Thoothukudi. |

| District/Season | Variety |
|---|---|
| **1. TENAI** | |
| **Rainfed** | |
| a) June-July (Hill slopes of Coimbatore and Erode Districts) | CO 6, CO (Te) 7 |
| b) September-October (Coimbatore and Southerndistricts) | CO 6, CO (Te) 7 |
| **Irrigated** | |
| a) February-March | |
| b) September-October | CO 6, CO (Te) 7 |
| **2. SAMAI** | |
| a) June-July (Hill slopes of Coimbatore and Erode districts) | CO 3 |
| b) July-August (Dharmapuri) | Paiyur 1, Paiyur 2, CO 3 |
| c) September-October | CO 3 |
| **3. VARAGU** | |
| **Rainfed** | |
| a) June-July | CO 3 |
| b) July-August | |
| **4.PANIVARAGU** | |
| **Rainfed** | |
| a) September-October (Madurai, Dindigul, Theni, Tirunelveli & Thoothukudi) | CO 4<br>K 2 |
| **Irrigated** | |
| a) February-March | K 2 , CO 4 |
| **5. KUDIRAIVALI** | |
| **Rainfed** | |
| a) September-October | CO 1 |
| **Irrigated** | |
| a) February-March | CO 1 |

## SMALL MILLETS - MORPHOLOGICAL DESCRIPTION

**TENAI**

| PARTICULARS | CO 6 | CO (Te) 7 |
|---|---|---|
| Parentage | Hybrid derivative of CO 5 x ISE 301 | CO 5 x ISE 248 |
| Duration (days) | 85-90 | 80-85 |
| Pigmentation | green | Greenish purple |
| Tillering ability | High | high |
| Panicles | Long and drooping | Long, compact |
| Grain Character | Bold & Yellow | yellow |
| Grain Yield (kg/ha)<br>Straw | 1565<br>3.7 t/ha | 1855<br>5.1 t/ha |
| Special features | Suitable for irrigated & rainfed conditions, good grain quality, rich in protein, fat and minerals than CO 5 | Non lodging<br>High yielding |

**SAMAI**

| PARTICULARS | Paiyur 1 | Paiyur 2 | CO 3 |
|---|---|---|---|
| Parentage | Pureline selection from Santhur local | Pure line selection from PM 295 | Selection from germplasm bank |
| Duration (days) | 105 -110 | 85 | 80 - 85 |
| Pigmentation | Dark green | Green | Green |
| Tillering ability | Moderate | Moderate | High |
| Panicles | Semi compact long | long, loose panicle | long, loose panicle |
| Grain Character | Brown | Brown | Brown |
| Grain Yield (kg/ha)<br>Rainfed | 870 | 850 | 1066 |
| Special features | Long duration | Short duration, suitable for little millet – Horse gram cropping sequence | Bold grain, Non-lodging suits for early & late sowing |

**VARAGU**

| PARTICULARS | CO 3 |
|---|---|
| Parentage | Selection from Georgia variety |
| Duration (days) | 120 |
| Pigmentation | Purple stem |
| Tillering ability | High |
| Panicles | Well exposed clusters and spikelets |
| Grain Character | Brown & Bold with hard seed coat |
| Grain Yield (kg/ha)<br>Rainfed | 1500 - 1800 |
| Special features | Tolerant to smut, short duration |

**PANIVARAGU**

| PARTICULARS | K 2 | CO 4 |
|---|---|---|
| Parentage | Selection from PV1685 | Pureline selection from Sengathur local |
| Duration (days) | 75 | 75 |
| Pigmentation | Green | Green |
| Tillering ability | High | High |
| Panicles | Loose | Loose density subscent |

| | | |
|---|---|---|
| Grain character | Grey | Golden yellow |
| Grain yield (kg/ha)<br>Rainfed | 3184 | 1500 |
| Special features | Non-lodging drought tolerant<br>Non-shattering and responsive to fertilizers | High tillering,wider adaptability |

**KUDIRAIVALI**

| **PARTICULARS** | **CO 1** |
|---|---|
| Parentage | Pureline selection from Coimbatore local |
| Duration (days) | 75 |
| Pigmentation | Green |
| Tillering ability | High |
| Panicles | Loose |
| Grain Character | Yellow |
| **Grain Yield (kg/ha)**<br>Rainfed | 1750 |
| Special features | No major diseases noted |

## CROP MANAGEMENT

**Package of practices for Tenai**

**Seeds and sowing :**

| | | |
|---|---|---|
| For line planting | : | 10kg/ha |
| For sowing | : | 12.5kg/ha for use of Gorru or seed drill is recommended. |
| Seed treatment | : | Treat 1 kg of seeds with 2 g Thiram or Carbendazim. |
| Field preparation | : | Plough the field thoroughly using a small iron plough or country plough to fine tilth. |
| Fertilizer application | : | Apply basally<br>FYM/Compost 12.5 t/ha<br>Nitrogen 44 kg/ha<br>Phosphorus 22 kg/ha |
| Spacing | : | For line planting 22 x 10 cm, for sowing 10 cm in between plants |
| Weeding | : | First weeding on 15 th DAS and the second on 40$^{th}$ DAS |
| Thinning | : | Before 20 DAS |
| Plant protection | : | Generally no major problem of pests and diseases |

**Package of practices for Samai**

| | | |
|---|---|---|
| Seeds and sowing | : | For line planting 10 kg/ha<br>For sowing 12.5 kg/ha for use of Gorru or seed drill is recommended. |
| Seed treatment | : | Treat 1 kg of seeds with 2 g Thiram or Carbendazim. |
| Field preparation | : | Plough the field thoroughly 2 or 3 times using a small iron plough or country plough to fine tilth. |
| Fertilizer application | : | Apply basally<br>FYM/COMPOST: 12.5 t/ha<br>Nitrogen : 44 kg/ha<br>Phosphorus : 22 kg/ha |
| Spacing | : | For line planting 25 x 10 cm for sowing 10 cm in between plants |

| | | |
|---|---|---|
| Weeding | : | First weeding is done on the 15th DAS and the second weeding on 40th DAS |
| Thinning | : | Thinning is done soon after weeding or before 20 DAS |
| Plant Protection | : | Usually no major problem of pests and diseases |

**Package of practices for Varagu**

| | | |
|---|---|---|
| Seeds and sowing | : | For line planting 10 kg/ha; For sowing 12.5 kg/ha Use of Gorru or seed drill is recommended. |
| Seed treatment | : | Treat 1 kg of seeds with 2 g Thiram or Carbendazim. |
| Field preparation | : | Plough the field thoroughly using a small iron plough or country plough to fine tilth. |
| Fertilizer application | : | Apply basally FYM/Compost 12.5 t/ha; Nitrogen 44 kg/ha Phosphorus 22 kg/ha |
| Spacing | : | For line planting 45 x 10 cm for sowing 10 cm in between plants |
| Weeding | : | First weeding is done on the 15th DAS and the second weeding on 40th DAS |
| Thinning | : | Thinning is done soon after weeding or before 20 DAS |
| Plant protection | : | Generally no major problem of pests and diseases |

# WHEAT (*Triticum aestivum.*)

## CROP IMPROVEMENT

## I. SEASON AND VARIETY

**Suitable districts**

Plains & adjoining areas near to hills and hills in Theni, Dindigul, Karur, Coimbatore, Erode, Salem, Dharmapuri, Vellore, Thiruvannamalai and Kancheepuram Districts

**Season**

Ideal sowing time is 15$^{th}$ October to 1$^{st}$ week of November. Sowing must be completed within the first fortnight of November.

**Variety** : COW(W)1

**2. Morphological Description of COW(W) 1**

| Particulars | | |
|---|---|---|
| Parentage | : | HD2646/HW2002A/CPAN3057 |
| Duration (days) | : | 85-90 |
| Grain yield (Kg /ha) | : | 2364 |
| Stem | : | Erect |
| Height (cm) | : | 73 – 78 |
| Tillers | : | 5-6 |
| Days to 50% flowering | : | 50 days |
| Ear size and shape | : | Fusiform ears |
| Grain colour | : | Amber |
| 1000 grains weight (g) | : | 37 |
| Special features | : | Non lodging, non shattering; tolerance to stem and leaf rust ; suitable for chappathi and bread making. |

**3. SEED RATE:** 100 kg/ha

## CROP MANAGEMENT

**1. FIELD PREPARATION**

Plough twice with an iron plough and two to three times with cultivator and prepare the land to a fine tilth.

**2. APPLICATION OF FYM OR COMPOST**

Spread 12.5 t/ha of FYM or compost on the unploughed field.

**3. SEED TREATMENT WITH FUNGICIDES**

Treat the seeds with Carbendazim or Thiram at 2 g/kg of seeds 24 hours before sowing.

**4. FORMING BEDS AND CHANNEL**

Form beds of size 10 $m^2$ or 20 $m^2$. The irrigation channels are to be provided sufficiently.

**5. APPLICATION OF FERTILIZERS**

i. Apply NPK fertilizer as per soil test recommendation as far as possible. If soil test recommendation is not available, adopt a blanket recommendation of 80:40:40 NPK kg/ha.

ii. Apply half of N and full dose of $P_2O$ and $K_2O_5$ basally before sowing and incorporate in the sowing line.

**6. SOWING**

Draw the lines 20 cm apart and sow the seeds continuously after application of fertilizers to a depth of 5 cm. Avoid deep sowing.

**7. WEED MANAGEMENT**

i. Spray Isoproturon 800 g/ha as pre-emergence spraying 3 days after sowing followed by one hand weeding on 35th day after sowing.
ii. If herbicide is not applied, give two hand weedings on 20th and 35th day after sowing.

**8. WATER MANAGEMENT**

The crop requires 4 - 6 irrigations depending on the soil type and rainfall. Wheat crop requires minimum of 5 irrigations at the following critical stages.

I – Immediately after sowing
II – Crown root intiation : 15-20 DAS
III- Active tillering stage : 35-40 DAS
IV- Flowering stage : 50-55 DAS
V- Grain filling stage : 70-75 DAS

Crown root initiation and flowering are the most critical stages. Water stagnation should be voided at the time of germination.

**9. TOP DRESSING**

Apply remaining half of N at crown root initiation stage (15-20 DAS).

**10. HARVESTING**

Harvest the crop when the grains become hard and straw becomes dry and brittle. Trash and winnow the grains. Use mechanical threshers to reduce the cost of threshing and winnowing.

## CROP PROTECTION

**Seed treatment** Treat the seed with any one of the following fungicides
Carbendazim @ 2 g/kg of seed, Thiram @ 2 g/kg of seed
Vitavax @ 2 g/kg of seed

# PULSES

## REDGRAM ( *Cajanus cajan (L.) Millsp.* )

### CROP IMPROVEMENT

### I. SEASON AND VARIETIES

| District/season | Varieties |
|---|---|
| **Adipattam** (June - August) | |
| Kanchipuram, Tiruvallur, Dharmapuri, Erode | SA 1, CO 6, CO(RG) 7, COPH 2, Vamban 2, VBN (RG) 3 |
| Coimbatore, Vellore, Tiruvannamalai, | |
| Cuddalore, Villupuram, Salem, Namakkal, Pudukottai | CO 6*, COPH 2, CO(RG) 7, Vamban 1, VBN (RG) 3 |
| * For all districts | |
| Thanjavur, Tiruvarur, Nagapattinam, Tiruchirapalli, Perambalur, Karur, Pudukottai, Madurai, Dindigul, Theni, Ramanathapuram, Virudhunagar, Tirunelveli, Thoothukudi, Sivagangai | CO(RG) 7 |
| **Purattasipattam (September – November)** | |
| Kanchipuram, Tiruvallur, Vellore, Tiruvannamalai, Dharmapuri, Salem, Namakkal, Erode, Coimbatore, Madurai, Dindigul, Theni | COPH 2, CO(RG) 7, APK 1, VBN (RG) 3 |
| **Summer** (February - March) | |
| All districts except The Nilgiris and Kanyakumari | COPH 2, CO(RG) 7, VBN (RG) 3 |
| Pudukottai | Vamban 1, VBN (RG) 3 |
| Wetland bunds | BSR 1 , Vamban 2 |

### II DESCRIPTION OF REDGRAM VARIETIES

| PARTICULARS | BSR 1 | CO 6 | VBN 1 |
|---|---|---|---|
| Parentage | Pureline selection from Mayiladumparai | Mutant of SA 1 (25 Kr gamma rays) | (Prabhat x HY 3A) x (T 21 x 102) |
| Year of release | 1986 | 1991 | 1992 |
| 50% flowering (days) | 100-110 | 120 - 130 | 70 |
| Duration days) | Perennial | 170 - 180 | 95 – 100 |
| Grain Yield (kg/ha) | | | |
| Rainfed | 0.75 - 1.0 kg of green pods/plant | 893 | 840 |
| Irrigated | .. | .. | 1200 |
| Height (cm) | 150 - 200 | 166 | 92 – 100 |
| Branches | 7-10 | 8-12 | 4-6 |
| Plant spread | Semi spreading | Semi spreading | Erect |
| Colour of standard petal | Red at dorsal side | Yellow with light purple streaks at the base | Yellow with faint Red veins at the base |
| Colour of pod | Red with diagnal | Green with purple streaks | Green with purple |

| | | | |
|---|---|---|---|
| | constriction | | streaks |
| Colour of grain | Reddish brown | Reddish brown | Brown |
| 100 seed weight (g) | 12 .0 | 8.8 | 6.8-7.5 |
| Pattern of growth | NDT | NDT | DT |

**NDT: Non-Determinate** **DT : Determinate**

| **PARTICULARS** | **COPH 2** | **APK 1** | **VBN 2** |
|---|---|---|---|
| Parentage | ms CO 5 x ICPL 83027 | Pure line selection from ICPL 87101 | ICPL 341 x Bhavani-sagar local |
| Year of release | 1997 | 1999 | 1999 |
| 50% flowering (days) | 60 - 75 | 70 | 70 |
| Duration days) | 120 - 130 | 95 – 105 | 172 – 180 |
| Grain Yield (kg/ha) | | | |
| Rainfed | - | 900 | 1050 |
| Irrigated | 1050 | 1250 | - |
| Height (cm) | 100 - 120 | 91 – 128.2 | 200 - 250 |
| Branches | 4 - 6 | 4 – 5 | 8-12 |
| Plant spread | Erect | Erect | Semi spreading |
| Colour of standard petal | Yellow with faint Red veins at the base | Deep red in back of standard petal | Yellow with faint Red veins at the base |
| Colour of pod | Green with purple streaks | Green with purple streaks | Green with purple streaks |
| Colour of grain | Tan brown | Reddish brown | Reddish brown |
| 100 seed weight (g) | 9.0 to 9.4 | 10.9 – 11.0 | 7.52 8-0 |
| Pattern of growth | NDT | DT | NDT |

| **PARTICULARS** | **CO(Rg) 7** | **VBN(Rg) 3** |
|---|---|---|
| Parentage | Selection from PB 9825 | Vamban 1 x Gulbarga |
| Year of release | 2004 | 2005 |
| 50% flowering (days) | 70 - 90 | 65 - 70 |
| Duration days) | 120 - 130 | 100 - 105 |
| Grain Yield (kg/ha) | | |
| Rainfed | 950 | 885 |
| Irrigated | 1168 | - |
| Height (cm) | 120 – 130 | 100 - 120 |
| Branches | 7 – 9 | 3 - 10 |
| Plant spread | Semi spreading | Erect, determinate and open type |
| Colour of standard petal | Yellow with light red vein at the base | Yellow |
| Colour of pod | Green with purple streaks | Green with purple streaks |
| Colour of grain | Reddish brown | Reddish brown |
| 100 seed weight (g) | 8.5– 11.0 | 7.5 – 8.0 |
| Pattern of growth | NDT | DT |

**NDT: Non-Determinate** **DT : Determinate**

## CROP MANAGEMENT

## III. SEED RATE

**Quantity of seed required kg/ha**

| Varieties | CO 6 | CO(Rg) 7 | Vamban 1 | APK 1 | Vamban 2 | COPH 2 | VBN (Rg) 3 |
|---|---|---|---|---|---|---|---|
| Sole Crop | 10 | 25 | 25 | 25 | 10 | 25 | 25 |
| Mixed Crop | 5 | 10 | 10 | 10 | 5 | -- | 10 |

[BSR 1 (Bund planting) 50 g/100 metre]

Select good seeds from pest and disease free plants.

## IV. MANAGEMENT OF FIELD OPERATION

**1. PREPARATION OF THE LAND**

Prepare the land to fine tilth and apply 12.5 t FYM/ha

**2. SEED TREATMENT**

Treat the seeds with Carbendazim or Thiram @ 2 g/kg of seed 24 hours before sowing (or) with talc formulation of *Trichoderma viride* @ 4g/kg of seed (or) Pseudomonas fluorescens @ 10 g/kg seed. Bio control agents are compatible with biofertilizers. First treat the seeds with biocontrol agents and then with rhizobium. Fungicides and biocontrol agents are incompatible.

**3. TREATMENT OF THE SEEDS WITH BIOFERTILIZER**

a) Fungicide treated seeds should be again treated with a bacterial culture. Treat with Rhizobial culture CRR6. There should be an interval of atleast 24 hours after fungicidal treatment for giving the bacterial culture treatment. For red lateritic soil Rhizobial culture VPR 1 is effective.

b) Treat the seeds with 3 packets (600 g/ha) of Rhizobial culture CRR6 and 3 packets (600 g/ha) of Phosphobacteria developed at TNAU using rice kanji as binder. If the seed treatment is not carried out apply 10 packets of Rhizobium (2000 g/ha) and 10 packets(2000 g) of Phosphobacteria with 25 kg of FYM and 25 kg of soil before sowing.

**4. APPLICATION OF FERTILIZERS**

a) Apply fertilizers basally before sowing.
Rainfed : 12.5 kg N + 25 kg $P_2O_5$ + 12.5 kg $K_2O$ +10 kg S*/ha
Irrigated : 25 kg N + 50 kg $P_2O_5$ + 25 kg $K_2O$ + 20 kg S*/ha
*Note : Applied in the form of gypsum if Single Super Phospate is not applied as a source of phosphorus

d) Soil application of 25 kg $ZnSo_4$/ha under irrigated condition

**5. SOWING THE SEEDS**

Dibble the seeds adopting the following spacing.

| Variety | Pure crop | Mixed crop |
|---|---|---|
| CO(Rg) 7 | 45 cm x 30 cm | 120 cm x 30 cm |
| COPH 2 | 45 cm x 15 cm | …….. |
| Vamban 1, APK 1, Vamban (Rg) 3 | 45 cm x 20 cm | 120 cm x 30 cm |
| CO 6, Vamban 2 | 90 cm x 30 cm | 240 cm x 30 cm |
| Bund Crop | 60 cm for BSR 1 and 30 cm for others. | |

**6. WEED MANAGEMENT**

i) Pre emergence application of Pendimethalin 2 litres /ha 3 days after sowing mixed with 500 l of water using Backpack/Knapsack/Rocker sprayer using flat fan type of nozzle. Then irrigate the field. Following this, one hand weeding may be given on 30-35days after sowing.

ii) If herbicide is not given, give two hand weedings on 20 and 35days after sowing.

**7. WATER MANAGEMENT**

Irrigate immediately after sowing, 3rd day after sowing, bud initiation, 50 % flowering and pod development stages. Water stagnation should be avoided.

**8. SPRAYING OF DIAMMONIUM PHOSPHATE OR UREA, NAA AND SALICYLIC ACID**

a) Foliar Spray of NAA 40 mg/l and Salycylic acid 100 mg/l once at Pre-flowering and another at 15 days thereafter

b) Foliar Spray of DAP 20 g/l or urea 20 g/l once at flowering and another at 15 days thereafter

**9. HARVESTING THE CROP**

1) Picking the matured pods and drying and processing
2) Harvesting the whole plants heaping ,drying and processing.

**10. INTER-CROPPING**

a) Raising one row of long duration redgram varieties as inter crop for every six rows of groundnut (6:1) is recommended for rainfed crops.

b) Raising one row of short and medium duration redgram as inter crop for every four rows of groundnut (4:1) is recommended for rainfed as well as for irrigated crops.

c) **Multistoreyed cropping**: For rainfed Vertisols of Virudhunagar, Tirunelveli, Thoothukudi districts recording more than 300 mm of rainfall during the crop growth period, multistoreyed cropping system Agathi + Redgram (CO 5) + Cotton (MCU 10) + Blackgram (CO 5) is highly profitable. (Agathi in I tier with 1 x 1 m spacing - Redgram in II tier with a spacing of 45 x 20 cm - Cotton in the III tier with a spacing of 45 x 15 cm - Blackgram in the IV tier with the spacing of 30 x 10 cm).

For rainfed Vertisols receiving less than 300 mm of rainfall, Agathi + Sorghum (CO 26) + Cotton (MCU 10) + Blackgram (CO 5) system is ideal. For both systems, apply 40 kg N and 20 kg $P_2O_5$/ha. (Agathi in I tier with a spacing of 1 x 1 m - sorghum in II tier with a spacing of 45 x 15 cm - cotton in III tier with the spacing of 45 x 15 cm and Blackgram in IV tier with 30 x 10 cm).

**11. NUTRITIONAL DISORDERS**

Redgram / Greengram/Blackgram/Cowpea

Zinc: Symptom appears within a month of sowing. The plants are stripped with yellow or pale green foliage. Veins and mid ribs of the leaves are green although tissue around them becomes yellow and bronzed.

Iron: Reduced concentration of Chlorophyll in leaves - pale leaf colour may be indistinguishable from deficiency of nitrogen or other elements.

## CROP PROTECTION

## A. Pest management

**Economic threshold level for important pests**

| Pests | ETL |
|---|---|
| Aphids | 20/2.5 cm shoot length |
| Pod borers | 10% of affected pods |
| Plume moth | 5/plant |
| Spotted pod borer | 3/plant |

| Pests | Management strategies |
|---|---|
| **Aphids**<br>*Aphis cracivora*<br>**Red spider mite**<br>*Schizotetranychus cajani* | ▪ **Spray any one of the following :**<br>Methyl demeton 25 EC 500 ml/ha<br>Dimethoate 30 EC 500 ml/ha<br>Phosphamidon 40 SL 500 ml/ha (250 l spray fluid/ha). |
| **Blister beetle**<br>*Mylabris* spp. | ▪ **Spray any one of the following :**<br>Dichlorvos 76 EC 500 ml/ha<br>Carbaryl 50 WP 500 g/ha |
| **Pod borers**<br>Spotted pod borer<br>*Maruca testulalis*<br>**Plume moth**<br>*Exelastis atomosa*<br>**Grampod borer**<br>*Helicoverpa armigera*<br>**Pod fly**<br>*Melanagromyza obtuse*<br>**Pod bug**<br>*Clavigralla gibbesa* | ▪ Pheromone traps for *Helicoverpa armigera* 12/ha<br>▪ Bird perches 50/ha<br>▪ Mechanical collection of grown up larva and blister beetle<br>▪ Ha NPV 3 x$10^{12}$ POB/ha in 0.1% teepol<br>▪ **Apply any one of the following :**<br>Dichlorvos 0.07%<br>Endosulfan 4% D 25 kg/ha<br>Quinalphos 4% D 25 kg/ha<br>Carbaryl 5% D 25 kg/ha<br>Endosulfan 35 EC 1.25 lit./ha<br>Monocrotophos 36 WSC 625 ml/ha.<br>NSKE 5% twice followed by triazophos 0.05%<br>Neem oil 2%<br>Phosalone 0.07%<br>(Spray fluid 625 ml/ha)<br>(Note : Insecticide / Ha NPV spray should be made when the larvae are upto third instar) |

## B. Disease Management

- **Seed treatment**
  - Talc formulation of *Trichoderma viride* @ 4g or *P. fluorescens* @ 10 g/kg seed (or) Carbendazim or Thiram @ 2 g/kg

| Name of the Disease | Management |
|---|---|
| **Wilt**<br>*Fusarium oxysporum f.sp.udum* | ▪ Spot drench Carbendazim – 1 gm/ litre (or)<br>▪ *P. fluorescens* (or) *T. viride*– 2.5 Kg / ha + 50 Kg of well decomposed FYM or sand at 30 days after sowing. |
| **Root rot**<br>*Rhizoctonia bataticola*<br>*Macrophomina phaseolina* | ▪ Spot drench Carbendazim – 1 gm/ litre (or)<br>▪ *P. fluorescens* or *T. viride*– 2.5 Kg / ha + 50 Kg of well decomposed FYM or sand at 30 days after sowing. |
| **Sterility Mosaic**<br>*Pigeonpea sterility mosaic virus*<br>*Vector : Aceria cajani* | ▪ Rogue out the infected plants in the early stages of growth.<br>▪ Spray monocrotophos 500 ml/ha on noticing the initial symptoms and repeat after a fortnight. |

**C. Nematode management**

| Nematode pest | Control measures |
|---|---|
| **Cyst nematode-**<br>*Heterodera cajani* | Soil application of *Pseudomonas fluorescens* or *Trichoderma viride* @ 2.5 kg/ha at the time of sowing. |

# SEED PRODUCTION

**Variety Seed Production**

**Land requirement**
- Land should be free of volunteer plants.

**Isolation**
- Adopt 100 m of certified seed production

**Pre-sowing seed treatment**
- Soak the seeds before sowing for 3 hrs in aqueous solution of $ZnSO_4$, 100 ppm (10g/100 lit of water) at 1/3 volume of seeds and quickly air dry in shade to their original moisture content.
- Slurry treat the seeds with Thiram 75% WP @ 2 g dissolved in 5 ml of water per kg of seeds. After air-drying, treat the seeds with the Rhizobial culture before sowing, following the recommended procedures.

**Foliar Application**
- Spray 2% DAP at the time of first appearance of flowering and give a second spray 15 days after first spray.
- Spray NAA 40 ppm first round at 1$^{st}$ flowering and a second round after fortnight.
- NAA can be mixed with fungicide and pesticide

**Pre-harvesting sanitation spray**
- Apply of Endosulfan 35 EC @ 0.07% or malathion 50 EC @ 0.05% before harvesting (3 - 5 days before) to minimise bruchid infestation in seed crop.

**Harvesting**
- Pods should be harvested 40 days after 50% flowering stage for production of quality seed.

**Processing**
- Dry the pods immediately to about 12 to 13 per cent moisture content.
- Dry the seeds to 10 % moisture content.
- Size grade using wire mesh sieve B.S.S. 5 x 5 (width of square aperture 3.35 mm) for large seeded varieties and 6 x 6 (2.8 mm) for small seeded varieties and rejected discoloured and broken seeds.

**Pre-storage Seed treatment**
- Dry the seed to 7 - 8 % moisture content
- Treat the seed with Carbendazim @ 2g dissolved in 5 ml of water kg$^{-1}$ of seed.
- Dry dressing of seeds with halogen mixture (Pure $CaOCl_2$ + $CaCO_3$ + arappu leaf powder at (5:4:1) ratio @ 3g/kg of seed
- For grain cum seed storage, treat the seed with activated clay, neem oil, groundnut oil and leaf powder of tobacco, nochi, neem, *Albizzia amara* (arappu) and fruit rind powder of *Sapindus laurifolius* (Poochi kottai) and *Acacia concinna* (Soapnut powder).

**Storage**
- Use gunny or cloth bags for short term storage with seed moisture content of 8 - 9%
- Use polylined gunny bag for medium term storage with seed moisture content of 8 - 9%
- Use 700 gauge polythene bag for long term storage with seed moisture content of less than 8%.

**Other management techniques**
- As in crop management technique

## HYBRID SEED PRODUCTION (COPH 2)

**Isolation**

- Adopt 200 m for parental line seed production (foundation) and 100 m for hybrid (Certified) seed production

**Planting ratio**

- Adopt 4:2 ratio of female : male

**Border rows**

- Plant 2 rows of male all around the field

**Spacing**

- Adopt 45 x 15 cm

**Fertilizer**

- Apply NPK @ 25:50:25 kg $ha^{-1}$

**Roguing**

- Pull out all fertile plants in female rows

**Seed Treatment**

- Treat the seed with halogen mixture @ 3g/kg of seed.
- Treat with turmeric rhizome powder (or) Neem leaf powder @ 1:50 powder to seed ratio for ecofriendly seed treatment with nil bruchid infestation

**Other management practices**

- As in varietal seed production

## PERENNIAL REDGRAM

| | | |
|---|---|---|
| Variety | : | BSR 1 |
| Economic uses | : | Tender beans are pinkish green in colour and can be cooked as curry or added to Kurma or Sabji. When the beans mature they can be used as Dhal. Recommended for growing in kitchen gardens, backyards, farm road sides, as border crop in sugarcane, banana and betelvine and as a shade crop in turmeric and as a bund crop in paddy double cropped wetlands. |
| Season | : | June – July |
| Height of the plant | : | 150 - 200 cm |
| Number of branches | | 7 - 10 |
| Flowering | : | Five months from date of sowing |
| Pit Size | : | Small pits are dug 90 cm apart and the pits are filled with a mixture of well decomposed manure or compost and soil. |
| Fertilizer application | : | Urea 15 g and superphosphate 30 g / pit. |
| Planting methods | : | Two to three seeds are dibbled per pit and watered. When they grow six inches height one plant may be retained in each pit. |
| Irrigation | : | Need based |
| Harvesting | : | If harvested when the pods are tender the beans will be fit for making curry. Each plant will yield two to three kg of green pods at an average seed yield of 750 g to one kg per plant. After the first harvest the branches are pruned and allowed to grow further. In another 45 - 60 days the plants produce the second flush. For pure crop, about 3 kg of seeds may be required. |

# BLACKGRAM ( *Vigna mungo L.*)

## CROP IMROVEMENT

### 1. SEASON AND VARIETIES

| District/Season | Varieties |
|---|---|
| **Adipattam (June-August)** | |
| Kanchipuram, Tiruvallur, Trichirapalli, Perambalur, Karur | T 9, VBN 1, VBN 2, VBN 3, VBN(Bg) 4 |
| Vellore, Tiruvannamalai, Dharmapuri, Pudukottai Cuddalore, Villupuram, Salem and Namakkal | T 9, VBN 1, VBN 2, VBN 3, VBN(BG) 4 |
| Ramanathapuram, Virudhunagar, Sivagangai Tirunelveli and Thoothukudi | CO 5, T 9, VBN 2, VBN 3, VBN(BG) 4 |
| Madurai, Dindigul, Theni | CO 5, VBN 1, VBN 2, VBN 3, VBN(BG) 4 |
| Erode, Coimbatore | CO 5, VBN 1, VBN 2, VBN 3, VBN (BG) 4 |
| **Puratasipattam (September-November)** | |
| Kanchipuram, Tiruvallur, Vellore, Tiruvannamalai Cuddalore, Villupuram, Dharmapuri, Salem, Namakkal | VBN 1, K 1*, VBN 2, VBN 3, VBN(Bg) 4 |
| Pudukottai, Madurai, Dindigul, Theni, Ramanathapuram, Sivagangai, Virudhunagar, Thoothukudi, Tirunelveli. | VBN 1, VBN 2, VBN 3, VBN(BG) 4 |
| Erode,Coimbatore,Tiruchirapalli,Perambalur,Karur | |
| **Rice fallows (January)** | |
| Cuddalore, Villupuram, Thanjavur, Tiruvarur, Nagapattinam, Trichirapalli, Perambalur, Karur | ADT 3 |
| Tirunelveli, Thoothukudi | ADT 3 |
| **Summer (February-March)** | |
| Kanchipuram, Tiruvallur, Vellore, Tiruvannamalai, Cuddalore, Villupuram, Dharmapuri Thanjavur, Tiruvarur, Nagapattinam, Tiruchirapalli, Perambalur, Karur, | ADT 5 |
| Pudukottai, Madurai, Dindigul, Theni, Ramanathapuram, Sivagangai, Virudhunagar | T 9, VBN 3, VBN(Bg) 4 |
| Tirunelveli, Thothukudi, Salem, Namakkal | VBN 3, VBN(Bg) 4 |
| Erode, Coimbatore | VBN 3, VBN(BG) 4 |

*** suitable for intercropping in cotton.**

## II. DESCRIPTION OF BLACKGRAM VARIETIES

| Particulars | T 9 | CO 5 | ADT 3 | VBN 1 |
|---|---|---|---|---|
| Parentage | Selection from Bareilly, U.P | Pureline selec tion from Musiri - | Pureline selection from Tirunelveli local | KM 1xH 76-1 |
| Year of release | 1972 | 1981 | 1981 | 1987 |
| 50% flowering (days) | 30 - 35 | 35 - 40 | 30 - 35 | 30 - 35 |
| Maturity duration (days) | 65 - 70 | 70 - 75 | 70 - 75 | 60 - 65 |
| Grain yield (kg/ha) | | | | |
| Rainfed | - | 740 | .. | 700 |
| Irrigated | 1000 | 1270 | - | 850 |
| Rice fallows | | | 720 | |
| Height (cm) | 35 - 40 | 30 - 35 | 50 | 30 - 35 |
| Clusters | 10 - 12 | 10 to 12 | 10 - 15 | 12 – 13 |
| Hairiness of pods | Glabrous | Hairy | Hairy | Hairy |
| Colour of grain | Black & Dull | Black & Dull | Black & dull | Black |
| 100 grain wt (g) | 4.0 | 5.7 | 3.6 | 5.1 |

| Particulars | ADT 5 | K 1 | VBN 2 | VBN 3 | VBN(Bg) 4 |
|---|---|---|---|---|---|
| Parentage | Selection from Kanpur variety | CO 3 x US 131 | Spontaneous mutant | LBG 402 x LBG 17 | CO 4 x PDU 102 |
| Year of release | 1988 | 1994 | 1996 | 2000 | 2003 |
| 50% flowering (days) | 32 | 40 - 45 | 33 - 40 | 35 -40 | 35 – 40 |
| Maturity duration (days) | 62 | 70 - 75 | 60 - 70 | 65 – 70 | 75 – 80 |
| Grain yield (kg/ha) | | | | | |
| Rainfed | - | 707 | 750 | 775 | 780 |
| Irrigated | 1323 | - | 1000 | 825 | 900 |
| Rice fallows | | - | - | | |
| Height (cm) | 20 - 25 | 30 - 35 | 25 - 35 | 25 – 35 | 40 – 45 |
| Clusters | 13 - 15 | 10 - 13 | 4 - 18 | 5 - 10 | 8 – 10 |
| Hairiness of pods | Hairy | Hairy | Glabrous | Hairy pods | Glabrous |
| Colour of grain | Black | Dull black | Black with green tinge | Dull black | Black |
| 100 grain wt (g) | 3.6 | 5.3 | 3.83 | 4.4 | 4.8 |

## III. SEED RATE

| STRAIN | Quantity of seed required kg/ha Pure crop | Mixed crop |
|---|---|---|
| T 9, CO 5, TMV 1, VBN 1, VBN 2, VBN 3, VBN (Bg) 4 | 20 | 10 |
| ADT 5, TMV 1 (Rice fallows) | 25 | .. |

Optimum plant population 3,25,000/ha

## CROP MANAGEMENT

# IV. MANAGEMENT OF FIELD OPERATIONS

### 1. FIELD PREPARATION

i. Prepare the land to fine tilth and form beds and channels.

ii. Amendments for soil surface crusting: To tide over the soil surface crusting apply lime at the rate of 2t /ha along with FYM at 12.5 t/ha or composted coirpith at 12.5 t/ha to get an additional yield of about 15 - 20%.

### 2. SEED TREATMENT

Treat the seeds with Carbendazim or Thiram @ 2 g/kg of seed 24 hours before sowing (or) with talc formulation of *Trichoderma viride* @ 4g/kg of seed (or) *Pseudomonas fluorescens* @ 10 g/kg seed. Bio control agents are compatible with biofertilizers. First treat the seeds with Biocontrol agents and then with Rhizobium. Fungicides and biocontrol agents are incompatible.

Note: Seed treatment will protect the seedlings from seed borne pathogens, root-rot and seedlings diseases.

### *3.* SEED TREATMENT WITH BIOFERTILIZER

Treat the seeds with 3 packets (600 g/ha) of Rhizobial culture CRU-7 + 3 packets (600 g/ha) of PGPR and 3 packets(600 g/ha) of Phosphobacteria developed at TNAU using rice kanji as binder. If the seed treatment is not carried out apply 10packets of Rhizobium (2000 g/ha) + 10 packets of PGPR (2000 g/ha) and 10 packets (2000 g) of Phosphobacteria with 25 kg of FYM and 25 kg of soil before sowing.

### 4. FERTILIZER APPLICATION

a) Apply fertilizers basally before sowing.
   Rainfed : 12.5 kg N + 25 kg $P_2O_5$ + 12.5 kg $K_2O$ +10 kg S*/ha
   Irrigated : 25 kg N + 50 kg $P_2O_5$ + 25 kg $K_2O$ + 20 kg S*/ha
   *Note : Applied in the form of gypsum if Single Super Phospate is not applied as a source of phosphorus
b) Soil application of 25 kg $ZnSo_4$/ha under irrigated condition

### 5. SOWING OF SEEDS

a) For irrigated crop dibble the seeds adopting 30 x 10 cm cm spacing
b) For rainfed crop dibble the seeds adopting 25 cm x 10 cm spacing

### 6. WATER MANAGEMENT

Irrigate immediately after sowing, followed by life irrigation on the third day. Irrigate at intervals of 7to 10 days depending upon soil and climatic conditions. Flowering and pod formation stages are critical periods when irrigation is a must. Avoid water stagnation at all stages. Apply KCl at 0.5 per cent as foliar spray during vegetative stage if there is moisture stress.

### 7. SPRAYING OF DIAMMONIUM PHOSPHATE OR UREA, NAA AND SALICYLIC ACID

a. Foliar spray of Spray of NAA 40 mg/lt and Salicylic acid 100 mg/lt once at pre-flowering and another at 15 days thereafter
b. i) For rice fallow crops foliar spray of DAP 20 g/lt once at flowering and another at 15 days thereafter
   ii) For irrigated and rainfed crops, foliar spray of DAP 20 g/litre or Urea 20 g/litre once at flowering and another at 15 days thereafter.

**8. WEED MANAGEMENT**

i) Pre emergence application of Pendimethalin 2 litres/ha 3 days after sowing using Backpack/ Knapsack/Rocker sprayer fitted with flat fan nozzle using 500 l of water for spraying one ha. After this, one hand weeding on 30 days after sowing gives weed free environment throughout the crop period.

ii) If herbicides are not applied give two hand weedings on 15 and 30 days after sowing.

**9. Multi bloom technology**

A special technology being practiced in Pattukottai block of Tanjore district for blackgram and greengram. The soil is alluvial and rich in organic matter and nutrients. The crop is sown during early summer (Jan.-Feb.) as normal crop and fertilizer is applied as per the recommendation for irrigated crop. In addition to that, top dressing of Nitrogen is done with an extra dose of 25 to 30 kg through urea. Since pulses are indeterminate growth habit and continue to produce new flashes the top dressing will be done on 40-45 days after sowing. The crop complete its first flesh of matured pods during 60-65$^{th}$ day, further their second new flesh within 20-25 days. Therefore two fleshes of pods can be harvested at a time within the duration of 100 days.

## CROP PROTECTION

**A. Pest management**

**Economic threshold level for important pests**

| Pests | ETL |
|---|---|
| Aphids | 20/2.5 cm shoot length |
| Pod borers | 10% of affected pods |
| Spotted pod borer | 3/plant |
| Stem fly | 10% of affected plants |
| Tobacco cut worm | 8 egg masses/100 m |

**Pest management strategies**

| Pests | Management strategies |
|---|---|
| **Stem fly**<br>*Ophiomyia phaseoli* | ▪ Seed treatment with dimethoate 30 EC 5 ml/kg of seed<br>▪ **Spray any one of the following** (Spray fluid 250 l /ha)<br>Methyl demeton 25 EC 500 ml/ha<br>Dimethoate 30 EC 500 ml/ha |
| **Aphids**<br>*Aphis craccivora* | |
| **Whitefly**<br>*Bemisia tabaci* | ▪ **Spray any one of the following** (Spray fluid 250 l /ha)<br>Methyl demeton 25 EC 500 ml/ha<br>Dimethoate 30 EC 500 ml/ha<br>Wettable sulphur 1.5 kg/ha |
| **Mite**<br>*Tetranychus urticae* | |
| **Tobacco cut worm**<br>*Spodoptera litura* | ▪ Use of light trap to monitor and kill the attracted adult moths. Set up the sex pheromone traps at 12/ha to monitor the activity of the pest and to synchronise the pesticide application, if need be, at the maximum activity stage.<br>▪ Growing castor along borders.<br>▪ Removal and destruction of egg masses in castor and cotton crops.<br>▪ Removal and destruction of early stage larvae found in clusters which can be located easily even from a distance.<br>▪ Hand picking and destruction of grown up caterpillars.<br>▪ **Spray any one of the following insecticides using, a high volume sprayer covering the foliage and soil surface :**<br>Dichlorvos 76 WSC 1.0 l/ha<br>Phenthoate 50 EC 2.0 l/ha |

| | |
|---|---|
| | Chlorpyriphos 20 EC 1.25 l/ha<br>Fenitrothion 50 EC 625 ml/ha.<br>▪ Spraying NPV at 1.5 x $10^{12}$ POB/ha with teepol (1ml/ lit.)<br>▪ Spraying of insecticide should be done either in the early morning or in the evening and virus in the evening.<br>▪ Use of poison bait pellets prepared with rice bran 12.5 kg, jaggery 1.25 kg, carbaryl 50 WP 1.25 kg and water 7.5 litres. This bait can be spread in the fields in the evening hours so that the caterpillars coming out of the soil, feed and get killed. |
| **Blue butterflies**<br>*Lampides boeticus*<br>*Euchrysops cnejus*<br>**Apids**<br>*Aphis craccivora*<br>**Spotted pod borer**<br>*Maruca vitrata* | ▪ **Apply any one of the following :**<br>Endosulfan 4 D 25 kg/ha<br>Quinalphos 1.5 D 25 kg/ha<br>Phosalone 4 D 25 kg/ha<br>Carbaryl 5 D 25 kg/ha<br>▪ **Spray any one of the following** (Spray fluid 500 l/ha)<br>Endosulfan 35 EC 1.0 l/ha<br>Monocrotophos 36 SL 500 ml/ha<br><br>Note : When the activity of coccinellid predator (both grubs and adults) is seen, insecticide application should be avoided. |
| **Storage pests**<br>Bruchid-<br>*Callosobruchus chinensis*<br>*C. maculatus* | ▪ Dry the seeds adequately to reduce moisture level to 10 %.<br>▪ Use pitfall traps or two in one model trap to assess the time of emergence of field carried over pulse beetle in storage and accordingly sun-dry the produce.<br>▪ Seed: **Mix any one of the following for every 100 kg :**<br>Activated kaolin 1 kg<br>Malathion 5 D 1 kg<br>TNAU Neem oil 60 EC (C) 1 lit<br>Pungam oil 1 lit.<br>Monocrotophos 36 SL 400 ml<br>▪ Pack in polythene lined gunny bags for storage |

**B.Disease management**

| | |
|---|---|
| **Seed treatment** | Talc formulation of *T. viride* @ 4g or *P. fluorescens* @ 10 g/kg seed (or) Carbendazim or Thiram @ 2 g/kg |

| Name of the Disease | Management |
|---|---|
| ***Powdery Mildew***<br>*Erysiphe polygoni* | ▪ Spray NSKE 5% or Neem oil 3% twice at 10 days interval from initial disease appearance.<br>▪ Spray Carbendazim 250 g or Wettable sulphur 2500g/ha.<br>**Botanical**<br>▪ Spray Eucalyptus leaf extract 10% at initiation of the disease and 10 days later. |
| **Rust**<br>*Uromyces appendiculatus* | ▪ Spray Mancozeb – 1000g or Wettable sulphur–2500g |
| **Leaf spot**<br>*Cercospora canescens* | ▪ Spray Carbendazim – 250 g |

| | |
|---|---|
| **Root rot**<br>*Macrophomina phaseolina (Rhizoctonia bataticola)* | ▪ Neem cake @ 150 Kg/ha<br>▪ Soil application *P. fluorescens* or *T. viride*– 2.5 Kg / ha + 50 Kg of well decomposed FYM or sand at 30 days after sowing.<br>▪ Spot drench Carbendazim – 1 gm/ litre<br>▪ Basal application of zinc sulphate 25 kg/ha |
| **Yellow mosaic** (*Geminivirus*) and **Leaf Crinkle**<br>Vector:<br>*Bemisia tabaci*<br><br>**Leaf Curl** (*Tospovirus*)<br>Vector:<br>*Frankliniella schultzii*<br>*Thrips tabaci*<br>*Scirtothrips dorsalis* | ▪ Rogue out the infected plants up to 30 days<br><br>▪ Spray Monocrotophos 500 ml or Methyl demeton 500 ml/ha and repeat after 15 days, if necessary.<br><br>▪ For seed crop, the plants affected by leaf crinkle should be periodically removed upto 45 days after sowing since the leaf crinkle virus is seed borne. |

**C. Nematode management**

| Nematode pest | Control measures |
|---|---|
| **Cyst nematode,** *Heterodera cajani* | Soil application of *Pseudomonas fluorescens* or *Trichoderma viride* @ 2.5 kg/ha at the time of sowing. |

## RICE-FALLOWS

**VARIETIES AND SEED RATE**

| VARIETIES | Quantity of seed required kg/ha<br>**Sole crop** | <br>**Mixed crop** |
|---|---|---|
| CO 4, ADT 2, ADT 3, ADT 4, ADT 5, TMV 1 (Rice fallows) | 30 | .. |

**1.TIME OF SOWING**

Third week of January –Second week of February

**2.SOWING OF SEEDS**

a) For relay cropping broadcast the seeds in the standing crop 5 to 10 days before the harvest of the paddy crop uniformly under optimum soil moisture conditions so that the seeds should get embedded in the waxy mire.
b) For combined harvesting areas, broadcast the seeds before harvesting the paddy crop with machinerie

**3. SPRAYING OF DIAMMONIUM PHOSPHATE , NAA AND SALICYLIC ACID**

a. Foliar Spray of NAA 40 mg/lt and Salicylic acid 100 mg/lt once at pre-flowering and another at 15 days thereafter
b. Foliar spray of DAP 20 g/lt once at flowering and another at 15 days thereafter

**4. HARVESTING**

i) Picking the matured pods, drying and processing
ii) Uprooting or cutting the whole plants ,heaping ,drying and processing

# SEED PRODUCTION

**Variety Seed Production**

**Land Requirement**

- Land should be free of volunteer plants.

**Isolation**

- Adopt 5 m certified seed production

**Presowing seed treatment**

- Remove all discoloured seeds and use only normal coloured seeds (black coloured in blackgram and olive green in greengram).
- Avoid seeds with bruchid infestation for sowing.
- If the hard seed percentage exceeds more than 10 per cent scarify with commercial $H_2SO_4$ for 2 min.
- For rainfed sowing harden the greengram seeds for 3 h in aqueous solution of manganese sulphate @ 100 ppm / (0.1 g/lit) at 1/3 volume of seeds and quickly air-dry in shade to their original moisture content. For blackgram, zinc sulphate @ 100 ppm may be used for hardening.
- Fortify the blackgram seed with $ZnSO_4$ 0.2% $MnSO_4$ 0.2% and $Na_2MO_4$ 0.1% in 1/3 volume to enhance the field establishment under irrigated conditions.

**Foliar application**

- Spray 2% DAP at the time of first appearance of flowers.
- Give a second spray 15 days after first spraying
- Spray NAA 40 PPM at first flowering and a second spray after a fortnight.
- NAA can be mixed with pesticides and fungicides.

**Preharvest sanitation spray**

- Three to 5 days before harvesting spray endosulphan 35 EC @ 0.07% or malathion 50 EC @ 0.05% for seed crop to minimise the bruchid infestation in storage.

**Harvesting**

- Harvest the pods 30 days after the 50 per cent flowering for blackgram and greengram.
- At this stage the colour of the majority of the pods (80%) will be black in blackgram and brown in greengram.
- The pod moisture content will be about 17 – 18%.
- Harvest as picking if the flowering period is long
- Dry the pods to 13 to 15 per cent moisture content

**Threshing**

- Thresh the seed with pliable bamboo stick or pulses thresher.

**Drying**

- Dry the seeds to 8 - 9 per cent moisture content

**Processing**

- Grade the seeds using B.S.S.7 x 7 wire mesh sieve for large seeded varieties
- Reject the discoloured and broken seeds.

**Seed treatment**

- Treat the seed with Carbendazim 75% WDP at 2g dissolved in 5 ml of water $kg^{-1}$ of seed (or)
- Dry dress the seeds with halogen mixture (Pure $CaOCl_2$ + $CaCO_3$ + arappu leaf powder at (5:4:1) ratio @ 3g/kg of seed

- For grain cum seed storage, treat the seed with activated clay, neem oil, groundnut oil and leaf powder of tobacco, nochi, neem, *Albizzia amara* (arappu) and fruit rind powder of *Sapindus laurifolius* (Poochi kottai) and *Acacia concinna* (Soapnut powder).

**Storage**

- Use gunny or cloth bags for short term storage with seed moisture content of 8 - 9%
- Use polylined gunny bag for medium term storage with seed moisture content of 8 - 9%
- Use 700 gauge polythene bag for long term storage with seed moisture content of Less than 8%.

**Others management practices**

- As in crop management

# GREENGRAM ( *Vigna radiata* L. )

## CROP IMPROVEMENT

### I. SEASON AND VARIETIES

| District/Season | Varieties |
|---|---|
| **Adipattam (June - July)** | |
| Kanchipuram, Tiruvallur, Vellore, Tiruvannamalai, Cuddalore, Villupuram, Pudukkottai, Madurai, Dindigul, Theni | KM 2, CO 4, CO 6, VBN 1, VBN(Gg) 2, VRM(Gg) 1 |
| Thanjavur, Tiruvarur, Nagapattinam, Tiruchirapalli, Perambalur, Karur | CO 4, CO 6, KM 2, Paiyur 1, VBN 1, VBN(Gg) 2 |
| Salem, Namakkal | CO 4, CO 6, KM 2, Paiyur 1 |
| Dharmapuri | CO 4, CO 6, KM 2, Paiyur 1, VBN 1, VBN(Gg) 2 |
| Erode, Coimbatore, Kanyakumari | CO 4, CO 6, VBN 1, VBN(Gg) 2 |
| Ramanathapuram, Sivagangai Virudhunagar, Tirunelveli, Thoothukudi | CO 4, CO 6, KM 2, VBN 1, VBN(Gg) 2 |
| **Puratasipattam (September - October)** | |
| Kanchipuram, Tiruvallur, Vellore, Tiruvannamalai, Cuddalore, Villupuram, Pudukottai | CO 4, CO 6, KM 2, VBN 1, VBN(Gg) 2 |
| Dharmapuri, Salem, Namakkal, Tirunelveli, Thoothukudi | CO 4, CO 6, Paiyur 1, VBN 1, VBN(Gg) 2, K1 |
| Erode, Coimbatore | CO 4, CO 6, VBN 1, VBN(Gg) 2 |
| Ramanathapuram, Virudhunagar, Sivagangai | CO 4, CO 6, KM 2, Paiyur 1, K1 |
| Madurai, Dindigul, Theni | CO 4, CO 6, Paiyur 1 |
| Tiruchirapalli, Perambalur, Karur | Paiyur 1, CO 6, VBN 1, VBN (Gg) 2 |
| **Rice fallows (January - February)** | |
| Thanjavur, Tiruvarur, Nagapattinam, Tiruchirapalli, Perambalur, Karur, Pudukottai, Tirunelveli, Thoothukudi | ADT 3 |
| **Summer (February - March)** | |
| Kanchipuram, Tiruvallore, Vellore, Tiruvannamalai, Cuddalore, Villupuram, Pudukkottai | CO 4, CO 6, KM 2 |
| Salem, Namakkal,Dharmapuri,Tiruchirapalli, Perambalur, Karur, Tirunelveli, Thoothuku | CO 4, CO 6, Paiyur 1 |
| Erode, Coimbatore, Thanjavur, Tiruvarur, Nagapattinam | CO 4, CO 6, Paiyur 1 |
| Madurai, Dindigul, Theni, Ramanathapuram, Virudhunagar, Sivagangai | CO 4, CO 6, KM 2, Paiyur 1 |

## II. DESCRIPTION OF GREENGRAM VARIETIES

| Particulars | KM 2 | CO 4 | ADT 3 | Paiyur 1 | VBN 1 | K 1 |
|---|---|---|---|---|---|---|
| Parentage | Hybrid derivative of No.127xS.9 | Mutant of CO 1 | Hybrid derivative of H70-16/ Rajendran/ G65 | PS from DPT 703 | Hybrid derivative of S.8xPIMS 3 | Co 4 x ML 65 |
| Year of release | 1978 | 1981 | 1988 | 1988 | 1989 | 1998 |
| 50% flowering (days) | 35 - 40 | 40 – 45 | 36 | 45 - 50 | 30 - 35 | 70 - 75 |
| Duration (days) | 65 - 70 | 85 | 66 | 85 – 90 | 65 | - |
| Grain yield (kg/ha) | | | | | 770 | 670 |
| Rainfed | 767 | 910 | 500 | 742 | | |
| Irrigated | .. | 1550 | 500 (Rice fallows) | - | .. | |
| Plant height (cm) | 35 - 45 | 60 - 70 | 35 – 45 | 5 5- 60 | 55 - 60 | 70 - 75 |
| Stem, branches etc. | Green stem | Green stem | Green stem | Green | Green sparsely hairy | Green, slightly Pubescent |
| Branches | - | 3 – 4 | 2 – 3 | 3 – 4 | 2 – 3 | 3 - 4 |
| Leaves | Medium | Broad tripartate and hairy | Medium broad, Petiole long | Broad | Medium, foliate, ovate | Broad |
| Pod characters | Dark brown and tip covered | Green when tender & black on drying | Dark brown | Dull green | Sparsely hairy , light black pods | Long and slightly pubescent, brown |
| Colour of grain | Shiny green | Dull green | Green | Dull Green | Green | Dull green |
| 100 grain wt (g) | 3.0 | 4.1 | 2 - 3 | 3 - 5 | 3.6 | 3.5 |

| Particulars | CO 6 | VRM(Gg) 1 | VBN(Gg) 2 |
|---|---|---|---|
| Parentage | WGG 37 x CO 5 | Pure line selection from K 851 | Cross derivative of VGG 4 x MH 309 |
| Year of release | 1999 | 2001 | 2001 |
| 50% flowering (days) | 26 – 30 | 31 – 35 | 30 – 35 |
| Duration (days) | 62 – 67 | 56 – 67 | 65 – 70 |
| Grain yield (kg/ha) | | | |
| Rainfed | 900 | 1100 | 750 |
| Irrigated | 1050 | - | 900 |
| Plant height (cm) | 35 - 55 | 45 – 55 | 50 – 60 |
| Leaves | Broad | Broad | Trifoliate with lobes |
| Colour of pod | Pubescent | Pubescent | Sparsly hairy pods |
| Colour of grain | Green with smooth surface | Shiny green bold and smooth surface | Shiny green |
| 100 grain wt (g) | 3 .4 | 3.5 | 3.6 – 3.9 |

## CROP MANAGEMENT

### III. SEED RATE

| Particulars | Quantity of seed required kg/ha | |
|---|---|---|
| | Pure crop | Mixed crop |
| All strains | 20 | 10 |
| Rice fallows - ADT 3 | 25 | -- |

Optimum plant population 3,25,000/ha.

### IV. MANAGEMENT OF FIELD OPERATIONS

#### 1. FIELD PREPARATION

i. Prepare the land to get fine tilth and form beds and channels.

Ii Amendments for soil surface crusting: To tide over the soil surface crusting apply lime at the rate of 2 t/ha along with FYM at 12.5 t/ha or composted coir pith at 12.5 t/ha to get an additional yield of about 15 - 20%.

#### 2. SEED TREATMENT

Treat the seeds with Carbendazim or Thiram @ 2 g/kg of seed 24 hours before sowing (or) with talc formulation of *Trichoderma viride* @ 4g/kg of seed (or) Pseudomonas fluorescens @ 10 g/kg seed. Bio control agents are compatible with biofertilizers. First treat the seeds with Biocontrol agents and then with Rhizobium. Fungicides and biocontrol agents are incompatible.

#### *3.* SEED TREATMENT WITH BIOFERTILIZER

Treat the seeds with 3 packets (600 g/ha) of Rhizobial culture CRM 6 and 3 packets (600 g/ha) of Phosphobacteria developed at TNAU using rice kanji as binder. If the seed treatment is not carried out apply 10 packets of Rhizobium (2000g/ha) and 10 packets (2000 g) of Phosphobacteria with 25 kg of FYM and 25 kg of soil before sowing.

**4. FERTILIZER APPLICATION**

a) Apply fertilizers basally before sowing.

Rainfed : 12.5 kg N + 25 kg $P_2O_5$ + 12.5 kg $K_2O$ +10 kg S*/ha

Irrigated : 25 kg N + 50 kg $P_2O_5$ + 25 kg $K_2O$ + 20 kg S*/ha

*Note : Applied in the form of gypsum if Single Super Phospate is not applied as a source of phosphorus

b) Soil application of 25 kg $ZnSo_4$/ha under irrigated condition

**5. SOWING**

Dibble the seeds adopting a spacing of 30 x 10 cm. For bund crop dibble the seeds at 30 cm spacing.

**6. WATER MANAGEMENT**

Irrigate immediately after sowing, followed by life irrigation on the third day. Irrigate at intervals of 7to 10 days depending upon soil and climatic conditions. Flowering and pod formation stages are critical periods when irrigation is a must. Avoid water stagnation at all stages. . Apply KCl at 0.5 per cent as foliar spray during vegetative stage if there is moisture stress.

**7. SPRAYING OF DIAMMONIUM PHOSPHATE OR UREA, NAA AND SALICYLIC ACID**

a. Foliar spray of Spray of NAA 40 mg/litre and Salicylic acid 100 mg/litre once at pre-flowering and another at 15 days thereafter

b. i) For rice fallow crops, foliar spray of DAP 20 g/litre once at flowering and another at 15 days thereafter

ii) For irrigated and rainfed crops foliar spray of DAP 20 g/litre or urea 20 g/litres once at flowering and another at 15 days thereafter.

**8. WEED MANAGEMENT**

i) Pre emergence application of Pendimethalin 2 litres on 3 days after sowing using Backpack/ Knapsack/Rocker sprayer fitted with flat fan nozzle using 500 litre of water for spraying one ha. After this, one hand weeding on 30 days after sowing gives weed free environment throughout the crop period.

ii) If herbicides are not applied give two hand weedings on 15 and 30 days after sowing.

**9. MULTI BLOOM TECHNOLOGY**

A special technology being practiced in Pattukottai block of Tanjore district for blackgram and greengram. The soil is alluvial and rich in organic matter and nutrients. The crop is sown during early summer (Jan.-Feb.) as normal crop and fertilizer is applied as per the recommendation for irrigated crop. In addition to that, top dressing of Nitrogen is done with an extra dose of 25 to 30 kg through urea. Since pulses are indeterminate growth habit and continue to produced new flushes, the top dressing will be done on 40-45 days after sowing. The crop complete its first flushes of matured pods during 60-65$^{th}$ day and put further second new flush within 20-25 days. Therefore two flushes of pods can be harvested at a time within the duration of 100 days.

## CROP PROTECTION

### A. Pest management

**Economic threshold level for important pests**

| Pests | ETL |
|---|---|
| Aphids | 20/2.5 cm shoot length |
| Pod borers | 10% of affected pods |
| Spotted pod borer | 3/plant |
| Stem fly | 10% of affected plants |
| Tobacco cut worm | 8 egg masses/100 m |

**Pest management strategies**

| Pests | Management strategies |
|---|---|
| **Stem fly**<br>*Ophiomyia phaseoli*<br>**Aphid**<br>*Aphis craccivora* | ▪ Seed treatment with dimethoate 30 EC 5 ml/kg of seed<br>▪ **Spray any one of the following** (Spray fluid 250 l /ha)<br>Methyl demeton 25 EC 500 ml/ha<br>Dimethoate 30 EC 500 ml/ha |
| **Whitefly**<br>*Bemisia tabaci*<br>**Mite**<br>*Tetranychus urticae* | ▪ **Spray any one of the following** (Spray fluid 250 l /ha)<br>Methyl demeton 25 EC 500 ml/ha<br>Dimethoate 30 EC 500 ml/ha<br>Wettable sulphur 1.5 kg/ha |
| **Apid**<br>*Aphis craccivora*<br>**Spotted pod borer**<br>*Maruca vitrata*<br>**Pod bug**<br>*Nezara viridula*<br>*Riptortus pedestris* | ▪ **Apply any one of the following :**<br>Endosulfan 4 D 25 kg/ha<br>Quinalphos 1.5 D 25 kg/ha<br>Phosalone 4 D 25 kg/ha<br>Carbaryl 5 D 25 kg/ha<br>▪ **Spray any one of the following** (Spray fluid 500 l/ha)<br>Endosulfan 35 EC 1.0 l/ha<br>Monocrotophos 36 SL 500 ml/ha<br>▪ Neem seed kernel extract 5% twice, starting from 50% flowering stage followed by one round of endosulfan 0.07%. |
| **Gram caterpillar**<br>*Helicoverpa armigera* | ▪ Pheromone traps for *Helicoverpa armigera* 12/ha<br>▪ Bird perches 50/ha<br>▪ Hand picking of larvae<br>▪ Ha NPV 1.5 x$10^{12}$ POB/ha with teepol (1 ml/lit.)<br>▪ **Apply any one of the following** (Spray fluid 625 l/ha)<br>Dichlorvos 76 WSC 625 ml/ha<br>Endosulfan 4 D 25 kg/ha<br>Quinalphos 4 D 25 kg/ha<br>Carbaryl 5 D 25 kg/ha<br>Endosulfan 35 EC 1.25 lit./ha<br>Monocrotophos 36 SL 625 ml/ha.<br>Neem seed kernel extract 5% twice followed by<br>triazophos 40 EC 780 ml/ha<br>Neem oil 12.5 l/ha<br>Phosalone 35 EC 1.25 l/ha<br>(Note : Insecticide / Ha NPV spray should be applied when the larvae are in early stage) |
| **Storage pests**<br>Bruchid-<br>*Callosobruchus chinensis*<br>*C. maculatus* | ▪ Dry the seeds adequately to reduce moisture level to 10 %.<br>▪ Use pitfall traps or two in one model trap to assess the time of emergence of field carried over pulse beetle in storage and accordingly sun-dry the produce.<br>▪ Seed: **Mix any one of the following for every 100 kg** :<br>Activated kaolin 1 kg<br>Malathion 5 D 1 kg |

| | |
|---|---|
| | TNAU Neem oil 60 EC (C) 1 lit.<br>Pungam oil 1 lit.<br>Monocrotophos 36 SL 400 ml<br>▪ Pack in polythene lined gunny bags for storage |

**B. Disease management**

| **Seed treatment** | Talc formulation of *T. viride* @ 4g or *P. fluorescens* @ 10 g/kg seed (or) Carbendazim or Thiram @ 2 g/kg |
|---|---|
| **Name of the Disease** | **Management** |
| ***Powdery Mildew***<br>*Erysiphe polygoni* | ▪ Spray NSKE 5% or Neem oil 3% twice at 10 days interval from initial disease appearance.<br>▪ Spray Carbendazim 250 g or Wettable sulphur 2500g/ha.<br>**Botanical**<br>▪ Spray Eucalyptus leaf extract 10% at initiation of the disease and 10 days later. |
| ***Rust***<br>*Uromyces appendiculatus* | **Spray any one of the following:**<br>Mancozeb 1000g /ha<br>Wettable sulphur 2500g/ha |
| ***Leaf spot***<br>*Cercospora canescens* | **Spray any one of the following:**<br>▪ Carbendazim 250 g<br>▪ Mancozeb 1000g/ha |
| ***Root rot***<br>*Macrophomina phaseolina (Rhizoctonia bataticola)* | ▪ Basal application of zinc sulphate – 25kg/ha.<br>▪ Neem cake @ 150 Kg/ha<br>▪ Soil application *P. fluorescens* or *T. viride*– 2.5 Kg / ha + 50 Kg of well decomposed FYM or sand at 30 days after sowing.<br>▪ Spot drench Carbendazim – 1 gm/ litre<br>▪ Basal application of zinc sulphate 25 kg/ha |
| **Yellow mosaic** (*Geminivirus*) and Leaf **Crinkle**<br>Vector:<br>*Bemisia tabaci*<br><br>**Leaf Curl** (*Tospovirus*)<br>Vector:<br>*Franklinielle schultzii*<br>*Thrips tabaci*<br>*Scirtothrips dorsalis* | ▪ Rogue out the infected plants up to 30 days<br>**Spray any one of the following:**<br>Monocrotophos 500 ml<br>Methyl demeton 500 ml/ha and repeat after 15 days, if necessary.<br><br>▪ For seed crop, the plants affected by leaf crinkle should be periodically removed upto 45 days after sowing since the leaf crinkle virus is seed borne. |

**Integrated management of viral diseases of blackgram and greengram**

1. Grow resistant variety like Vamban 4 black gram
2. Grow seven rows of sorghum as border crops
3. Treat the seeds with Imidacloprid 70WS@ 5ml/kg
4. Give one foliar spray of insecticide (Dimethoate @750ml/ha) on 30 days after sowing

**C. Nematode management**

| Nematode pest | Control measures |
|---|---|
| Cyst nematode, *Heterodera cajani* | Soil application of *Pseudomonas fluorescens* or *Trichoderma viride* @ 2.5 kg/ha at the time of sowing. |

# RICE-FALLOWS

**VARIETIES AND SEED RATE**

| Varieties | Quantity of seed required kg/ha | |
|---|---|---|
| | Sole crop | Mixed crop |
| All varieties | 30 | - |

**1.TIME OF SOWING**

Third week of January –Second week of February

**2.SOWING OF SEEDS**

a) For relay cropping broadcast the seeds in the standing crop 5 to 10 days before the harvest of the paddy crop uniformly under optimum soil moisture conditions so that the seeds should get embedded in the waxy mire.

b) For combined harvesting areas, broadcast the seeds before harvesting the paddy crop with machineries

**3. SPRAYING OF DIAMMONIUM PHOSPHATE , NAA AND SALICYLIC ACID**

a. Foliar spray of NAA 40 mg/litre and Salicylic acid 100 mg/litre once at pre-flowering and another at 15 days thereafter

b. Foliar spray of DAP 20 g/lt once at flowering and another at 15 days thereafter

**4. HARVESTING**

i) Picking the matured pods, drying and processing

ii) Uprooting or cutting the whole plants ,heaping ,drying and processing

# COWPEA ( *Vigna unguiculata* (L.) Walp.aggreg.)

## CROP IMPROVEMENT

## I. SEASON AND VARIETIES

| DISTRICT/SEASON | VARIETIES |
|---|---|
| **Adipattam (June-August)** | |
| For all districts | CO 6, VBN 1, VBN 2, CO(CP) 7 |
| Kanchipuram, Tiruvallur, | CO 6 , CO(CP) 7 |
| Vellore, Tiruvannamalai, Dharmapuri | Paiyur 1 |
| Cuddalore, Villupuram | CO 6 , CO(CP) 7 |
| Salem, Namakkal | Paiyur 1, CO 6 , CO(CP) 7 |
| Tiruchirapalli, Perambalur, Karur | CO 6 , CO(CP) 7 |
| Pudukottai | CO 2, CO 6 , CO(CP) 7 |
| Erode | Paiyur 1, CO 6 , CO(CP) 7 |
| Coimbatore, Madurai, Dindigul, Theni, Ramanathapuram, Sivagangai, Virudhunagar | CO 2, CO 6 , CO(CP) 7 |
| Tirunelveli, Thoothukudi , Kanyakumari, Thanjavur, Tiruvarur, Nagapattinam | CO 6 , CO(CP) 7 |
| **Purattasipattam (September - November)** | |
| For all districts | CO 6, VBN 1, VBN 2, CO(CP) 7 |
| Kanchipuram, Tiruvallur | CO(CP) 7 |
| Dharmapuri | Paiyur 1 |
| Cuddalore, Villupuram | Paiyur 1, CO (CP) 7 |
| Salem, Namakkal | CO 2, Paiyur 1, CO (CP) 7 |
| Pudukottai | CO 2, VBN 2 |
| Erode | CO 2, CO(CP) 7 |
| Coimbatore | CO 2, Paiyur 1, CO 6 , CO(CP) 7 |
| Madurai, Dindigul, Theni, Ramanathapuram, Virudhunagar | CO, 2, VBN 2, CO(CP) 7 |
| Sivagangai, Tirunelveli, Thoothukudi , | CO 2, VBN 2, CO(CP) 7 |
| Kanyakumari, Vellore, Tiruvannamalai, | Paiyur 1, CO(CP) 7 |
| **Summer irrigated** | |
| Salem, Namakkal, Coimbatore, Erode, | CO 2, CO 6, CO (CP) 7, VBN 2 |
| Madurai, Dindigul, Theni, Tiruchirapalli, Perambalur, Karur and Tirunelveli, Thoothukudi | |

## II. PARTICULARS OF VARIETIES

| Particulars | CO 2 | Paiyur 1 | CO 6 | VBN 1 | VBN 2 | CO(CP) 7 |
|---|---|---|---|---|---|---|
| Parentage | Hybrid derivative (C521 x C49) | Selection from VM 16 | MS 9804 x C 152 | Selection from T 85F 2020 | Selection from IT 81-D-1228-10 | Gamma mutant of CO 4 (20 Kr) |
| Year of release | 1972 | 1985 | 1993 | 1997 | 1998 | 2002 |
| 50% flowering(days) | 45 | 75 | 35 | 35 – 40 | 40-45 | 40 – 45 |
| Duration (days) | 90 | 90 | 65 – 70 | 55 – 65 | 75 - 80 | 70 – 75 |
| Grain yield(kg/ha) | | | | | | |
| Rainfed | | 900 | 671 | 950 | - | 1000 |
| Irrigated | ... 1375 | -- | - | - | - | 1600 |
| Vegetable (Kg/ha) | 9400 | ... | - | | 10580 | - |
| Plant height (cm) | 30 - 40 | 60 - 70 | 34.5 | 25 - 30 | 45 - 60 | 40 – 55 |
| Stem, branches | Green stem purple wash at nodes, 3-4 branches | Erect | Green stem purple wash at nodes, 3-4 branches | Erect green, 2 – 4 branches | Green erect, 2 – 3 branches | Green with purple ring at fruiting nodes, 5 – 8 branches |
| Leaves | Trifoliate, broad and long petioles | Dark green leaflets possessing triangular white spots | Trifoliate entire, green, purple spot at the abse of leaflet | Light green | Dark green trifoliate | Ovote, trifoliate, entire, green and glabrous |
| Colour of pods | Tender greenish white | Green | Dark green pigmented tip | Greenish white | Light green | Green |
| Dry | Greyish white | Brown | Brown at maturity | Cream | Cream | Light brown |
| Colour of grain | Reddish brown with irregular patches | Brick red | Light cream | White | Ivory white, hilum with tan ring surrounded by brown band | Brownish white and squire shape |
| 100 grain wt (g) | 12.5 | 9.9 | 9.9 | 12 - 15 | 13 - 15 | 12 - 14 |

## III. SEED RATE

| STRAIN | Quantity of seed required (kg/ha) | |
|---|---|---|
| | Pure crop | Mixed crop |
| Paiyur 1, VBN 1, VBN 2, CO 6, CO(CP) 7 | 25 | 12.5 |

Optimum plant population 3,50,000/ha.

## CROP MANAGEMENT

## IV. MANAGEMENT OF FIELD OPERATIONS

### 1. FIELD PREPARATION

Prepare the land to fine tilth and form beds and channels.

### 2. SEED TREATMENT

Treat the seeds with Carbendazim or Thiram 2 g/kg of seed 24 hours before sowing (or) with talc formulation of *Trichoderma viride* @ 4g/kg of seed (or) Pseudomonas fluorescens @ 10 g/kg seed.

- Biocontrol agents are compatible with biofertilizers.
- First treat the seeds with biocontrol agents and then with Rhizobium.
- Fungicides and biocontrol agents are incompatible.

### 3. SEED TREATMENT WITH BIOFERTILIZER

a) Fungicide-treated seeds, should be again treated with a bacterial culture. There should be an interval of atleast 24 hours between fungicidal and biofertilizer treatments.

b) The improved rhizobial strain COC 10 is more effective in increasing the yield.
Treat the seeds with 3 packets (600 g/ha) of Rhizobial culture COC 10 and 3 packets (600 g/ha) of Phosphobacteria developed at TNAU using rice kanji as binder. If the seed treatment is not carried out apply 10packets of Rhizobium (2000 g/ha) and 10 packets(2000 g) of Phosphobacteria with 25 kg of FYM and 25 kg of soil before sowing. Dry the biofertilizer treated seeds in shade for 15 minutes before sowing.

### 4. FERTILIZER APPLICATION

a) Apply fertilizers basally before sowing.
Rainfed : 12.5 kg N + 25 kg $P_2O_5$ + 12.5 kg $K_2O$ +10 kg S*/ha
Irrigated : 25 kg N + 50 kg $P_2O_5$ + 25 kg $K_2O$ + 20 kg S*/ha
*Note : Applied in the form of gypsum if Single Super Phospate is not applied as a source of phosphorus

b) Soil application of 25 kg $ZnSo_4$/ha under irrigated condition

### 5. SOWING

Dibble the seeds adopting the following spacing.

| Varieties | Sole crop | Mixed crop |
|---|---|---|
| CO 6, VBN 1 | 30 cm X 15 cm | 200 cm X 15 cm |
| CO (CP) 7, VBN 2 | 45 cm x 15 cm | .. |
| Paiyur 1 | 30 cm x 15 cm | .. |

### 6. WATER MANAGEMENT

Irrigate immediately after sowing followed by life irrigation on the third day. Irrigate at intervals of 7 to 10 days depending upon soil and climatic conditions. Flowering and pod formation stages are critical periods when irrigation is a must. Avoid water stagnation at all stages. . Apply KCl at 0.5 per cent as foliar spray during vegetative stage if there is moisture stress.

### 7. SPRAYING OF DIAMMONIUM PHOSPHATE OR UREA, NAA AND SALICYLIC ACID

a. Foliar spray of Spray of NAA 40 mg/litre and Salicylic acid 100 mg/litre once at pre-flowering and another at 15 days thereafter
b. Foliar spray of DAP 20 g/litre or urea 20 g/litre once at flowering and another at 15 days thereafter

**8. WEED MANAGEMENT**

i) Pre emergence application of Pendimethalin 2 litres on 3 days after sowing using Backpack/ Knapsack/Rocker sprayer fitted with flat fan nozzle using 500 l of water for spraying one ha. After this, one hand weeding on 30 days after sowing gives weed free environment throughout the crop period.
ii) If herbicides are not applied give two hand weedings on 15 and 30 days after sowing.

## CROP PROTECTION

**A. Pest management**

**Economic threshold level for important pests**

| Pests | ETL |
|---|---|
| Aphids | 20/2.5 cm shoot length |
| Spotted pod borer | 3/plant |
| Stem fly | 10% of affected plants |

| Pests | Management strategies |
|---|---|
| **Stem fly**<br>*Ophiomyia phaseoli* | ▪ Soil application of carbofuran 3 G (15 kg/ha) at the time of sowing<br>▪ Spray endosulfan 35 EC 500 ml/ha a week after germination and second round 10 days after first round. |
| **Aphid**<br>*Aphis craccivora* | |
| **Spotted pod borer**<br>*Maruca vitrata* | ▪ Apply any one of the following :<br>Endosulfan 4 D 25 kg/ha<br>Quinalphos 1.5 D 25 kg/ha<br>Phosalone 4 D 25 kg/ha<br>Carbaryl 5 D 25 kg/ha<br>▪ Spray any one of the following (Spray fluid 500 l/ha)<br>Endosulfan 35 EC 1.0 l/ha<br>Monocrotophos 36 SL 500 ml/ha |
| **Blue butterflies**<br>*Lampides boeticus*<br>*Euchrysops cnejus* | |
| **Pod bug**<br>*Nezara viridula*<br>*Riptortus pedestris*<br>*Clavigralla gibbosa* | |
| **Blister beetle**<br>*Mylabris pustulata* | |
| **Storage pests**<br>Bruchid-<br>*Callosobruchus chinensis*<br>*C. maculatus* | ▪ Dry the seeds adequately to reduce moisture level to 10 %.<br>▪ Use pitfall traps or two in one model trap to assess the time of emergence of field carried over pulse beetle in storage and accordingly sun-dry the produce.<br>▪ Seed: Mix any one of the following for every 100 kg :<br>Activated kaolin 1 kg<br>Malathion 5 D 1 kg<br>TNAU Neem oil 60 EC (C) 1 lit.<br>Pungam oil 1 lit.<br>Monocrotophos 36 SL 400 ml<br>▪ Pack in polythene lined gunny bags for storage |

**B. Disease management**

| | |
|---|---|
| **Seed treatment** | Talc formulation of *T. viride* @ 4g or *P. fluorescens* @ 10 g/kg seed (or) Carbendazim or Thiram @ 2 g/kg |

| Name of the Disease | Management |
|---|---|
| **Rust**<br>*Uromyces appendiculatus* | ▪ Two sprays of chlorothalanil 0.1% or one spray with 0.1% chlorothalanil followed by 3% Neem oil after the appearance of rust disease, effectively controls the disease. |
| **Root rot**<br>*Macrophomina phaseolina (Rhizoctonia bataticola)* | ▪ Soil application *P. fluorescens* or *T. viride*– 2.5 Kg / ha + 50 Kg of well decomposed FYM or sand at 30 days after sowing.<br>▪ Spot drench Carbendazim – 1 gm/ litre |
| **Aphid borne Mosaic Virus**<br>(*Potyvirus*)<br>Vector:<br>*Aphis craccivora*<br>*A. fabae*<br>*A. gossypii*<br>*Myzus persicae* | ▪ Roguing out of cowpea mosaic virus diseased plants in the early stage of growth upto 30 days and spraying twice at fortnightly intervals with monocrotophos 500 ml/ ha (or) methyldemeton 25 EC500 ml/ha. |

## SEED PRODUCTION

**Variety Seed Production**

**Land requirement**

- Land should be free of volunteer plants.

**Isolation**

- Adopt 5 m for certified seed production

**Season**

- September to October

**Sowing seed treatment**

- Remove all discoloured seeds and use highly germinable (>75%) seeds.

**Intercultural operation**

- Pinching the tendrils and application of NAA 40 ppm (940 mg/l) may be followed at flower initiation and at peak flowering stage to promote pod setting.
- Pull out and destroy plants exhibiting severe symptoms of mosaic in the early stages of growth.

**Harvesting**

- Seeds attain physiological maturity 27-30 days after anthesis
- Harvest the pods as they turn light straw in colour and the seeds turn brown or mottled in colour.
- At this stage the moisture content of seeds will be about 18 per cent.
- Harvest the pods as picking (2 –3 no) at 10 days interval
- Air dry pods at first for 1-2 days and sun dry until they become brittle
- Beat with pliable bamboo stick or pulse thresher by adjusting the cylinder to avoid splitting and cracking of seeds.
- The seed should be dried to 8-10% moisture content.

**Processing**

- Grade the seeds at 10% moisture content using 12/64" diameter (Aperture width 4.60 mm) round perforated sieve for CO 2 and 10/64" diameter (aperture width 3.96 mm) for small seeded varieties.

**Drying**

- Remove of the broken and immature seeds

- Dry seed to 7 to 8 % moisture content.

***Seed Treatment***

- Treat with thiram / Carbendazim @ 2g $kg^{-1}$ of seed
- Treat with halogen mixture @ 3g $kg^{-1}$ of seed
- Treat with activated clay @ 1 kg/100 kg of seeds. May be dry dressed for grain cum seed storage use.

**Storage**

- Use gunny or cloth bags for short term storage with seed moisture content of 8 - 9%
- Use polylined gunny bag for medium term storage with seed moisture content of 8 - 9%
- Use 700 gauge polythene bag for long term storage with seed moisture content of less than 8%.

**Other management practices**

As in crop management technique.

# HORSEGRAM ( *Macrotyloma uniflorum* )

## CROP IMPROVEMENT

### I. SEASON AND VARIETIES

| DISTRICT/SEASON | VARIETIES |
|---|---|
| **November (Winter season) (Rainfed)** | |
| All districts except The Nilgiris and Kanyakumari | CO 1, Paiyur 1, Paiyur 2 |

### II. DESCRIPTION OF VARIETIES

| Variety | CO 1 | Paiyur 1 | Paiyur 2 |
|---|---|---|---|
| Parentage | PurelineSelection from Mudukulathur | PurelineSelection from Mettur local | Gamma irradiation of CO 1 |
| Year of release | 1953 | 1988 | 1998 |
| 50% flowering (days) | 55 - 60 | 45 - 50 | 45 - 50 |
| Maturity duration(days) | 110 | 110 | 100 - 105 |
| Grain yield (Kg/ha) | | | |
| Rainfed | 560 | 650 | 870 |
| Height (cm) | 30-40 | 35 - 40 | 40 - 45 |
| Branches | 2-3 | 2-3 | 4 - 5 |
| Colour of grain | Buff mottled | Light brown | Pale brown |
| 100 grain weight (g) | 4.6 | 3.4 | 3.56 |

### III. SEED RATE

For a pure crop 20 kg/ha is needed.

## CROP MANAGEMENT

### IV. MANAGEMENT OF FIELD OPERATIONS

**1. FIELD PREPARATION**

Prepare the land to a fine tilth.

**2. SEED TREATMENT WITH FUNGICIDES**

Treat the seeds with any one of the following fungicides. Carbendazim or Thiram at 2 g/kg seed.

**3. FERTILIZER RECOMMENDATION**

Apply basally 12.5 t/ha FYM/Compost, 12.5 kg/ha nitrogen, 25 kg/ha phosphorus , 12.5 kg/ha potassium if soil is deficient in NPK status.

**4. SEED TREATMENT WITH BIOFERTILIZER**

Treat the seeds with 3 packets (600 g/ha) of Rhizobial culture and 3 packets (600 g/ha) of Phosphobacteria developed at TNAU using rice kanji as binder. If the seed treatment is not carried out apply 10packets of Rhizobium (2000 g/ha) and 10 packets(2000 g) of Phosphobacteria with 25 kg of FYM and 25 kg of soil before sowing. Dry the biofertilizer treated seeds in shade for 15 minutes before sowing.

**5. SOWING**

Dibble the seeds with a spacing of 30 x 10 cm.

**6. WEED MANAGEMENT**

Give one weeding and hoeing on 25-30 days after sowing

**7. HARVESTING**

Harvest the matured whole plant, thresh the pods and extract seeds

## SEED PRODUCTION

**Variety Seed Production**

**Land Requirement**

- Should be free from volunteer plants

**Isolation**

- Adopt a isolation of 10mt

**Season**

- October - November

**Pre sowing Seed Treatment**

- Treat the seed with Captan 75% WP at 2 g/kg of seed.

**Harvesting**

- Seeds attain physiological maturity when the pods turn yellowish brown in colour. Harvest pods when 75 - 80% of the pods have matured.

**Processing**

- Grade the seed with a sieve having 8/64" round perforation.

**Storage**

- Store in cloth upto two years
- For long term storage (more than three years) store in 700 gauge polythene bag

**Note**

- Magnesium chloride is to be sprayed against any chlorotic symptom at 6 g/litre with a power sprayer 2 - 3 times at 5-day interval.
- Timely harvest is essential taking care not to expose the pods to rain or very moist weather which may change the seed coat colour from light brown to dark brown or light black.
- The discoloured seeds loose viability much faster than the normal seeds in storage.
- The hard seededness of 50 - 55% noticed immediately after harvest declined to 4 to 6% over a period of 2 months.

**Others**

As in management practices

# BENGALGRAM ( *Cicer arietinum L.*)

## CROP IMPROVEMENT

### I. SEASON AND VARIETIES

| DISTRICT/SEASON | VARIETIES |
|---|---|
| **November (Winter season) Rainfed**<br>Vellore, Tiruvannamalai, Salem, Namakkal, Tiruchirapalli, Perambalur, Karur, Dharmapuri, Pudukottai, Erode, Coimbatore, Madurai, Dindigul, Theni, Virudhunagar, Ramanathapuram, Sivagangai, Tirunelveli, Thoothukudi. | CO 3, CO 4 |

### II. DESCRIPTION OF VARIETIES

| Variety | CO 3 | CO 4 |
|---|---|---|
| Parentage | Pureline selection from Maharashtra collection | Cross derivative of ICC 42 x ICC 12237 |
| Year of release | 1986 | 1998 |
| 50% flowering (days) | 35 – 40 | 40 |
| Duration (days) | 85 | 85 |
| Grain yield (Kg/ha) | | |
| Rainfed | 1000 | 1150 |
| Height (cm) | 25 - 30 | 35 - 40 |
| Branches | 3 - 5 | 3 - 5 |
| Flower colour | Light pink & veined | Light pink & veined |
| Colour of grain | Light brown | Brown |
| 100 seed weight (g) | 30-32 | 30 - 32 |

### III. SEED RATE

- CO 3 - 90 kg/ha.
- CO 4 - 75 kg/ha.

As a pure crop to have an optimum plant population 325000 / ha

## CROP MANAGEMENT

### IV. MANAGEMENT OF FIELD OPERATIONS

**1. FIELD PREPARATION**

Prepare the land to fine tilth and apply 12.5 t FYM/ha

**2. SEED TREATMENT**

Treat the seeds with Carbendezim (or) Thiram @ 2g/kg of seed 24hrs before sowing (or) with talc formulation of *Trichoderma viride* @ 4 g/kg seed (or) *Pseudomonas fluorescens* @ 10 g/kg seed. Biocontrol agents are compatible with biofertilizers. First treat the seeds with biocontrol agents and then with Rhizobium. Fungicides and biocontrol agents are incompatible.

The above seed treatment will protect the seedlings from seed borne pathogens in the early stages.

**3. SEED TREATMENT WITH BIOFERTILIZER**

Treat the seeds with 3 packets (600 g/ha) of Rhizobial culture and 3 packets (600 g/ha) of Phosphobacteria developed at TNAU using rice kanji as binder. If the seed treatment is not carried out apply 10packets of Rhizobium (2000g/ha) and 10 packets(2000 g) of Phosphobacteria with 25 kg of FYM and 25 kg of soil before sowing. Dry the biofertilizer treated seeds in shade for 15 minutes before sowing.

**4. FERTILIZER APPLICATION**

a) Apply fertilizers basally before sowing.
   Rainfed : 12.5 kg N + 25 kg $P_2O_5$ + 12.5 kg $K_2O$ +10 kg S*/ha
   Irrigated : 25 kg N + 50 kg $P_2O_5$ + 25 kg $K_2O$ + 20 kg S*/ha
   *Note : Applied in the form of gypsum, if Single Super Phospate is not applied as a source of phosphorus
b) Soil application of 25 kg $ZnSo_4$/ha under irrigated condition

**5. SOWING**

Dibble the seeds by adopting the spacing of 30 cm x 10 cm.

**6. WEED MANAGEMENT**

i) Pre emergence application of Pendimethalin 2 litres on 3rd day after sowing using Backpack/ Knapsack/Rocker sprayer fitted with flat fan nozzle using 500 l of water for spraying one ha. followed by one hand weeding on 25 - 30 days after sowing.
ii) If herbicide is not applied give two hand weedings on 15th and 30th day after sowing.

**7. INTERCROPPING IN BENGALGRAM**

Bengalgram in paired row planting with one or two rows of Coriander as intercrop would give the highest return. Wheat can also be intercropped in deep black cotton soil in Coimbatore, Erode, Salem, Namakkal and Dharmapuri districts.

## CROP PROTECTION

**A. Pest management**

**Economic threshold level for important pests**

| Pests | ETL |
|---|---|
| Gram caterpillar | 2 early instar larvae/plant<br>5-8 eggs/plant |
| Aphids | 20/2.5 cm shoot length |

**Pest management strategies**

| Pests | Management strategies |
|---|---|
| **Aphid**<br>*Aphis craccivora* | ▪ **Spray any one of the following :**<br>Methyl demeton 25 EC 500 ml/ha<br>Dimethoate 30 EC 500 ml/ha |
| **Gram caterpillar**<br>*Helicoverpa armigera* | ▪ Pheromone traps for *Helicoverpa armigera* 12/ha<br>▪ Bird perches 50/ha<br>▪ Hand picking of grown up larvae and blister beetles<br>▪ Ha NPV 1.5 x$10^{12}$ POB/ha with teepol (1 ml/lit.)<br>▪ **Apply any one of the following** (Spray fluid 625 ml/ha)<br>Endosulfan 4 D 25 kg/ha<br>Quinalphos 4 D 25 kg/ha<br>Carbaryl 5 D 25 kg/ha<br>Dichlorvos 76 WSC 625 ml/ha<br>Endosulfan 35 EC 1.25 lit./ha<br>Monocrotophos 36 SL 625 ml/ha<br>Neem seed kernel extract 5% (31.0 kg/ha) twice followed by |

| | |
|---|---|
| | triazophos 40 EC 780 ml/ha<br>Neem oil 12.5 lit./ha<br>Phosalone 35 EC 1.25 lit./ha<br>(Note : Insecticide / Ha NPV spray should be made when the larvae were upto third instar) |
| **Storage pests** | ▪ Dry the seeds adequately to reduce moisture level to 10 %.<br>▪ Use pitfall traps or two in one model trap to assess the time of emergence of field carried over pulse beetle in storage and accordingly sun-dry the produce.<br>▪ Seed: **Mix any one of the following for every 100 kg** :<br>Activated kaolin 1 kg<br>Malathion 5 D 1 kg<br>TNAU Neem oil 60 EC (C) 1 lit.<br>Pungam oil 1 lit.<br>Monocrotophos 36 SL 400 ml<br>▪ Pack in polythene lined gunny bags for storage |

**B.Disease management**

| | |
|---|---|
| **Seed treatment** | ▪ Talc formulation of *T. viride* @ 4g or *P. fluorescens* @ 10 g/kg seed (or) Carbendazim or Thiram @ 2 g/kg |

| Name of the Disease | Management |
|---|---|
| ***Wilt***<br>*Fusarium oxysporum* f.sp. *ciceri* | ▪ Soil application with P. *fluorescens* @ 2.5 kg/ha. |
| ***Root rot***<br>*Macrophomina phaseolina (Rhizoctonia bataticola)* | **Biological control**<br>▪ Soil application *P. fluorescens* or *T. viride*– 2.5 Kg / ha + 50 Kg of well decomposed FYM or sand at 30 days after sowing.<br>**Chemical**<br>▪ Spot drench Carbendazim – 1 gm/ litre |

## SEED PRODUCTION

**Variety Seed Production**

**Land Requirements**

- Land to be used for seed production of Bengal gram shall be free of volunteer plants.

**Isolation**

- Adopt 5 m for certified seed production

**Pre-Sowing treatment**

- Soak the seeds in 1% aqueous solution of $KH_2PO_4$ for 3-4 h at 1/3$^{rd}$ volume of seeds and quickly air dry in shade.
- Avoid using of bruchid infected seed for sowing

**Harvesting**

- Seeds attain physiological maturity @ 35 – 40 days after anthesis
- Harvest when 70 – 80% of pods are creamy in colour.

**Processing**

- Grade the seeds using 13/64” or 18/64" round perforated metal sieve depending on the variety.
- Dry the seed to 8-10% moisture content

**Seed treatment**

- Slurry treat the seeds with Carbendazim (or) Thiram 75% WP @ 2 g $kg^{-1}$ of seed (or).
- Treat the seed with halogen mixture @ 3g + Carbendazim @ 2g $kg^{-1}$ of seed
- For grain cum seed storage treat the seed with activated clay @ 1:100 (W/W).

**Storage**

- Use gunny or cloth bags for short term storage with seed moisture content of 8 - 9%
- Use polylined gunny bag for medium term storage with seed moisture content of 8 - 9%
- Use 700 gauge polythene bag for long term storage with seed moisture content of Less than 8%.

**Other Management practices**

- As in crop management technique

# GARDEN LAB LAB (AVARAI)
## (*Lab lab purpureus* (L.) var. *typicus*. )

## CROP IMPROVEMENT

### I.SEASON AND VARIETIES

| DISTRICT/SEASON | VARIETIES |
|---|---|
| **Adipattam (Jul - Aug)** | |
| Kanchipuram, Tiruvallur, Dharmapuri, Coimbatore, Madurai, Dindigul, Theni, Vellore, Tiruvannamalai, Ramanathapuram, Virudhunagar, Sivagangai, Tirunelveli, Thoothukudi , Salem, Namakkal, | CO 13 |
| Thanjavur, Tiruvarur, Nagapattinam, Tiruchirapalli, Perambalur, Karur, Pudukottai, Kanyakumari, Erode | CO 12, CO 13 |
| **Purattasipattam (Sept - Nov)** | |
| Kanchipuram, Tiruvallur, Tiruchirapalli, Perambalur, Karur, Vellore, Tiruvannamalai, Cuddalore, Villupuram. | CO 13 |
| Dharmapuri, Salem, Namakkal | CO 13 |
| Pudukottai, Erode, Coimbatore, Madurai, Dindigul, Theni, Ramanathapuram, Sivagangai, Virudhunagar, Tirunelveli, Thoothukudi , Thanjavur, Tiruvarur, Nagapattinam. | CO 12, CO 13 |
| **Summer (April)** | |
| Kanchipuram, Tiruvallur, Vellore, Tiruvannamalai, Cuddalore, Villupuram. | CO 12, CO 13 |
| Dharmapuri, Salem, Namakkal, Thanjavur, Tiruvarur, Nagapattinam. | CO 12, CO 13 |
| Kanyakumari, Pudukottai, Erode, Coimbatore, Madurai, Dindigul, Theni, Ramanathapuram, Virudhunagar, Tirunelveli, Thoothukudi, Sivagangai | CO 12, CO 13 |

### II.DESCRIPTION OF AVARAI VARIETIES

| Particulars | CO 12 | CO 13 |
|---|---|---|
| Parentage | Pedigree selection from CO 9 x CO 4 | Derivative of CO 9 x Florikifield |
| Year of release | 1991 | 1997 |
| 1st flowering (days) | 40 | 40 |
| Duration | 100 - 110 | 110 - 120 |
| Grain yield (kg/ha) | | |
| Irrigated | 9700 | 10000 |
| Habit | Erect, bushy | Dwarf, bushy and tendency to form tendrils |
| Height (cm) | 60 - 70 | 50 - 75 |
| Colour of flowers | Purple | White |
| Colour of pod | Deep purple | Whitish green |
| Shape of pod | Broad, flat | Flat and long |
| Colour of grain | Black | Brown |
| 100 seed wt(g) | 38.4 | 35.2 |

## III. SEED RATE

| Particulars | Quantity of seed required kg/ha | |
|---|---|---|
| | Sole crop | Mixed crop |
| CO 12 | 20 | 10.0 |
| CO 13 | 25 | - |

## CROP MANAGEMENT

## IV. MANAGEMENT OF FIELD OPERATIONS

**1. FIELD PREPARATION**

Prepare the land to fine tilth. Form beds and channels for bushy types.

**2. SEED TREATMENT WITH FUNGICIDES**

Treat the seeds with Carbendezim (or) Thiram @ 2g/kg of seed 24hrs before sowing (or) with talc formulation of Trichoderma viride @ 4 g/kg seed (or) Pseudomonas fluorescens@ 10 g/kg seed. Biocontrol agents are compatible with biofertilizers. First treat the seeds with biocontrol agents and then with Rhizobium. Fungicides and biocontrol agents are incompatible.

**3. SEED TREATMENT WITH BACTERIAL CULTURE**

Fungicide treated seeds should be again treated with bacterial culture. There should be an interval of atleast 24 hours between fungicidal and bacterial culture treatments. Three packets of bacterial culture are sufficient for treating seeds required for one ha. The bacterial culture slurry may be prepared with rice kanji. Dry the inoculated seeds in shade for 15 minutes, before sowing.

**4. FERTILIZER APPLICATION**

a) Apply fertilizers basally before sowing.
   Rainfed : 12.5 kg N + 25 kg $P_2O_5$ + 12.5 kg $K_2O$ +10 kg S*/ha
   Irrigated : 25 kg N + 50 kg $P_2O_5$ + 25 kg $K_2O$ + 20 kg S*/ha
   *Note : Applied in the form of gypsum if Single Super Phospate is not applied as a source of phosphorus
b) Soil application of 25 kg $ZnSo_4$/ha under irrigated condition

**5. SOWING**

Dibble the seeds adopting the following spacing.

| Varieties | |
|---|---|
| CO 12 | : 45 cm X 15 cm |
| CO 13 | : 45 cm X 30 cm |

**6. WEED MANAGEMENT**

i) Pre emergence application of Pendimethalin 2 litres/ha on 3 days after sowing using Backpack/ Knapsack/Rocker sprayer fitted with flat fan nozzle using 500 l of water for spraying one ha. After this, one hand weeding on 40-45 days after sowing gives weed free environment throughout the crop period.
ii) If herbicides are not applied give two hand weedings on 25 and 45days after sowing.

**7. WATER MANAGEMENT**

Irrigate immediately after sowing, followed by life irrigation on the third day. Irrigate at intervals of 7to 10 days depending upon soil and climatic conditions. Flowering and pod formation stages are critical periods when irrigation is a must. Avoid water stagnation at all stages. Apply KCl at 0.5 per cent as foliar spray during vegetative stage if there is moisture stress.

**8. PRUNING TECHNIQUE**

A spacing of about 10 feet between lines and four feet between plants are adopted. Pits are dug and two to three seeds are sown in the middle of the pit. One healthy seedling is allowed to grow and the rest removed. The vine is propped with a stick. When the vine reaches the pandal, the terminal bud is nipped. Allow the branches to trail over the pandal. Each branch may be pruned at three feet length so that the pandal is covered with vines. Branches arising on the main vine below the pandal are removed. When flowering starts, prune the tip of the branches bearing inflorescence having three nodes from the productive axil. Continue this procedure throughout the reproductive phase.

**9. HARVESTING**

Pick the pods when they are completely dry. Thresh the pods and clean the beans. Pick the tender pods once in a week for vegetable purpose.

## CROP PROTECTION

**A. Pest management**

**Economic threshold level for important pests**

| Pests | ETL |
|---|---|
| Aphids | 20/2.5 cm shoot length |
| Spotted pod borer | 3 larvae/plant |
| Gram caterpillar | 10% of affected pods |

**Pest management strategies**

| Pests | Management strategies |
|---|---|
| **Aphid**<br>*Aphis craccivora* | ▪ **Spray any one of the following :**<br>Methyl demeton 25 EC 500 ml/ha<br>Dimethoate 30 EC 500 ml/ha |
| **Spotted pod borer**<br>*Maruca vitrata*<br>**Gram caterpillar**<br>*Helicoverpa armigera*<br>**Blue butterflies**<br>*Lampides boeticus*<br>*Euchrysops cnejus*<br>**Pod bug**<br>*Nezara viridula*<br>*Riptortus pedestris*<br>*Clavigralla gibbosa*<br>**Aphid**<br>*Aphis craccivora*<br>**Blister beetle**<br>*Mylabris pustulata* | ▪ **Apply any one of the following :**<br>Endosulfan 4 D 25 kg/ha<br>Quinalphos 1.5 D 25 kg/ha<br>Phosalone 4 D 25 kg/ha<br>Carbaryl 5 D 25 kg/ha<br>▪ **Spray any one of the following** (Spray fluid 500 l/ha)<br>Endosulfan 35 EC 1.0 l/ha<br>Monocrotophos 36 SL 500 ml/ha |
| **Storage pests**<br>Bruchid-<br>*Callosobruchus chinensis* | ▪ Dry the seeds adequately to reduce moisture level to 10 %.<br>▪ Use pitfall traps or two in one model trap to assess the time of emergence of field carried over pulse beetle in storage and accordingly sun-dry the produce.<br>▪ Seed: **Mix any one of the following for every 100 kg** :<br>Activated kaolin 1 kg<br>Malathion 5 D 1 kg<br>TNAU Neem oil 60 EC (C) 1 lit.<br>Pungam oil 1 lit.<br>Monocrotophos 36 SL 400 ml<br>▪ Pack in polythene lined gunny bags for storage |

**B. Disease management**

| | |
|---|---|
| **Seed treatment** | ▪ Talc formulation of *T. viride* @ 4g or *P. fluorescens* @ 10 g/kg seed (or) Carbendazim or Thiram @ 2 g/kg |

| Name of the Disease | Management |
|---|---|
| **Anthracnose and die-back** *Colletotrichum lindemuthianum* | ▪ Spray Mancozeb 1000g or Carbendazim 250 g/ha soon after the appearance of the disease and if necessary, spray once again a fortnight later. |

## SEED PRODUCTION

**Variety Seed Production**

**Land requirement**

- Land should be free of volunteer plants.

**Isolation**

- Adopt 5 m for certified seed production

**Preharvest sanitation spray**

- Spray with 0.07% Malathion or Endosulfan @ 0.07 % before harvesting the pods for seed crop.

**Harvesting**

- Harvest the pods as they turn straw coloured.
- Discard the terminal pods, as they invariably contain immature and diseased seeds.
- The seed moisture content at the stage will be about 15% and the green colour of the seed coat will turn to chocolate brown colour.
- Dry pods to 15-18% moisture content.

**Drying**

- Dry the seeds to 8 -10% moisture content.

**Grading**

- Grade the seed using 18/64" (aperture width 7.28 mm) round perforated metal sieve.
- Remove the broken and immature seeds
- Dry the seed to 7 to 8 per cent moisture content.

**Seed Treatment**

- Treat with Carbendazim @ 2 g + Carbaryl 50% WP at 200 mg/kg$^{-1}$ of seed
- Activated clay @ 1:100 kg$^{-1}$ of seed

**Storage**

- Use gunny or cloth bags for short term storage with seed moisture content of 8 - 9%
- Use polylined gunny bag for medium term storage with seed moisture content of 8 - 9%
- Use 700 gauge polythene bag for long term storage with seed moisture content of less than 8%.

**Other management practices**

- As in crop management technique

# FIELD LAB-LAB (MOCHAI)

**( *Lab lab purpureus* (L.) var. *lignosus* )**

## CROP IMPROVEMENT

### I. SEASON AND VARIETIES

| DISTRICT/SEASON | VARIETIES |
|---|---|
| All districts except The Nilgiris | |
| June - July | CO 1 |
| All through the year | CO 2 |

### II. DESCRIPTION OF MOCHAI VARIETIES

| Particulars | CO 1 | CO 2 |
|---|---|---|
| Parentage | Pureline selection | Derivative of CO 8 X CO 1 |
| Year of release | 1980 | 1984 |
| 50% flowering (days) | 75 - 90 | 35 - 45 |
| Duration (days) | 140 | 105 |
| **Grain yield (Kg/ha)** | | |
| Rainfed | 1600 | 900 |
| Irrigated | .. | 1400 |
| Habit | Medium Erect and bushy (indeterminate) photo sensitive | Erect and bushy determinate photo insensitive |
| Height (cms) | 60 - 70 | 60 |
| Colour of flowers | Purple | Purple |
| Colour of pod | Green | Green |
| Shape of pod | Flat | Flat |
| Colour of grain | Black | Black |
| 100 seed weight (g) | 24.4 | 20.0 |

#### III. SEED RATE

| Particulars | Quantity of seed required kg/ha | |
|---|---|---|
| | Sole crop | Mixed crop |
| CO 1 | 20 | 10.O |
| CO 2 | 25 | 12.5 |

## CROP MANAGEMENT

### IV. MANAGEMENT OF FIELD OPERATIONS

#### 1. FIELD PREPARATION

Prepare the land to fine tilth.

#### 2. SEED TREATMENT WITH FUNGICIDES

Treat the seeds with Carbendezim (or) Thiram @ 2g/kg of seed 24hrs before sowing (or) with talc formulation of *Trichoderma viride* @ 4 g/kg seed (or) *Pseudomonas fluorescens* @ 10 g/kg seed.

- Biocontrol agents are compatible with biofertilizers.
- First treat the seeds with biocontrol agents and then with Rhizobium.
- Fungicides and biocontrol agents are incompatible.

**3. SEED TREATMENT WITH BACTERIAL CULTURE**

Fungicide treated seeds should be again treated with bacterial culture. There should be an interval of atleast 24 hours between fungicidal and bacterial culture treatments. Three packets of bacterial culture are sufficient for treating seeds required for one hectare. The bacterial culture may be prepared with rice kanji. Dry the inoculated seeds in shade for 15 minutes, before sowing.

**4. FERTILIZER APPLICATION**

a)Apply fertilizers basally before sowing.

Rainfed : 12.5 kg N + 25 kg $P_2O_5$ + 12.5 kg $K_2O$ +10 kg S*/ha

Irrigated : 25 kg N + 50 kg $P_2O_5$ + 25 kg $K_2O$ + 20 kg S*/ha

*Note : Applied in the form of gypsum, if Single Super Phospate is not applied as a source of phosphorus

b) Soil application of 25 kg $ZnSo_4$/ha under irrigated condition

**5. SOWING**

Dibble the seeds, adopting the following spacing.

| Strain | Sole crop | Mixed crop |
|---|---|---|
| CO 1 | 90 cm x 30 cm | 200 cm x 30 cm |
| CO 2 | 45 cm x 15 cm | 200 cm x 15 cm |

**6. WEED MANAGEMENT**

i) Pre emergence application of Pendimethalin 2 litres/ha on 3 days after sowing using Backpack/ Knapsack/Rocker sprayer fitted with flat fan nozzle using 500 l of water for spraying one ha. After this, one hand weeding on 40-45 days after sowing gives weed free environment throughout the crop period.

ii) If herbicides are not applied, give two hand weedings on 25$^{th}$ and 45$^{th}$ days after sowing.

**7. WATER MANAGEMENT**

Irrigate immediately after sowing, followed by life irrigation on the third day. Irrigate at intervals of 7to 10 days depending upon soil and climatic conditions. Flowering and pod formation stages are critical periods when irrigation is a must. Avoid water stagnation at all stages. Apply KCl at 0.5 per cent as foliar spray during vegetative stage if there is moisture stress.

**8. HARVESTING**

Dry pods may be collected for grain purposes. Green mature pods may be collected for vegetable purpose.

## CROP PROTECTION

**A. Pest management**

**Economic threshold level for important pests**

| Pests | ETL |
|---|---|
| Aphids | 20/2.5 cm shoot length |
| Spotted pod borer | 3 larvae/plant |
| Gram caterpillar | 10% of affected pods |

**Pest management strategies**

| Pests | Management practices |
|---|---|
| **Aphid**<br>*Aphis craccivora* | ▪ **Spray any one of the following :**<br>Methyl demeton 25 EC 500 ml/ha<br>Dimethoate 30 EC 500 ml/ha |

| | |
|---|---|
| **Gram caterpillar**<br>*Helicoverpa armigera*<br><br>**Field bean pod borer**<br>*Adisura atkinsoni*<br><br>**Pod bug**<br>*Nezara viridula*<br>*Riptortus pedestris* | ▪ **Apply any one of the following :**<br>Endosulfan 4 D 25 kg/ha<br>Quinalphos 1.5 D 25 kg/ha<br>Phosalone 4 D 25 kg/ha<br>Carbaryl 5 D 25 kg/ha<br>▪ **Spray any one of the following** (Spray fluid 500 l/ha)<br>Endosulfan 35 EC 1.0 l/ha<br>Monocrotophos 36 SL 500 ml/ha |
| **Storage pests**<br>Bruchid-<br>*Callosobruchus chinensis* | ▪ Dry the seeds adequately to reduce moisture level to 10 %.<br>▪ Use pitfall traps or two in one model trap to assess the time of emergence of field carried over pulse beetle in storage and accordingly sun-dry the produce.<br>▪ Seed: **Mix any one of the following for every 100 kg** :<br>Activated kaolin 1 kg<br>Malathion 5 D 1 kg<br>TNAU Neem oil 60 EC (C) 1 lit.<br>Pungam oil 1 lit.<br>Monocrotophos 36 SL 400 ml<br>▪ Pack in polythene lined gunny bags for storage |

**B. Disease management**

| | |
|---|---|
| **Seed treatment** | ▪ Talc formulation of *T. viride* @ 4g or *P. fluorescens* @ 10 g/kg seed or Carbendazim or Thiram @ 2 g/kg |

| Name of the Disease | Management |
|---|---|
| **Anthracnose and die-back**<br>*Colletotrichum lindemuthianum* | ▪ Spray Mancozeb 1000g or Carbendazim 250 g/ha soon after the appearance of the disease and if necessary, spray once again a fortnight later. |

# SOYBEAN ( *Glycine max* (L.) Merr.)

## CROP IMPROVEMENT

### I. SEASON AND VARIETIES

**Adipattam** (June - July)
**Purattasipattam** (Sep. - Oct.) ——— CO 1 (Irrigated), CO 2
**Masipattam** (February - March)
**Rice fallows** CO 1, CO 2*
* Under protected irrigation.

### II. DESCRIPTION OF THE VARIETY

| Variety | CO 1 | CO 2 | CO (Soy) 3 |
|---|---|---|---|
| Parentage | Re-selection from a Thailand variety | Cross derivative of UGM 21 x JS 335 | Cross derivative of UGM 69 x JS 335 |
| Year of release | 1980 | 1995 | 2005 |
| 50% flowering | 37 days | 30-40 days | 39 – 41 days |
| Duration (days) | 85 | 75 - 80 | 85 – 90 |
| **Grain yield (Kg/ha)** | | | |
| Rainfed | 1080 | 1340 | - |
| Irrigated | 1640 | 1650 | 1700 |
| Height (cm) | 58 | 30 - 40 | 53.5 |
| Branches | 6 | - | 5 - 6 |
| Flower colour | Pink | Pink to purple | Pink |
| Colour of grain | Cream | Creamy yellow | Creamy yellow with brown hilum |
| 100 seed weight (g) | 12.5 | 13 - 14 | 10.95 – 11.75 |

### III. SEED RATE

CO 1 - 80 kg/ha. Optimum plant population 6,66,000/ha.
CO 2 (irrigated) Pure crop : 60-70 Kg/ha; Inter crop : 25 Kg/ha
CO(Soy) 3 Pure crop : 50 Kg/ha

## CROP MANAGEMENT

### IV. MANAGEMENT OF FIELD OPERATIONS

#### 1. FIELD PREPARATION

Prepare the land to get fine tilth and form beds and channels.

#### 2. SEED TREATMENT WITH FUNGICIDES

a) Treat the seeds with Carbendezim or Thiram @ 2g/kg of seed 24hrs before sowing or with talc formulation of *Trichoderma viride* @ 4 g/kg seed (or) *Pseudomonas fluorescens* @ 10 g/kg seed.

- Biocontrol agents are compatible with biofertilizers.
- First treat the seeds with biocontrol agents and then with Rhizobium.
- Fungicides and biocontrol agents are incompatible.

b) Coat the seeds with ZnSO4 @ 300 mg/kg using 10% maida solution as adhesive (250 ml/ kg) or

gruel and arappu leaf powder (250 g/kg) as carrier to increase the field stand.

**3. SEED TREATMENT WITH BIOFERTILIZER**

a) Treat the seeds atleast 24 hours before sowing.

Treat the seeds with 3 packets (600 g/ha) of Rhizobial culture (COS-1) and 3 packets (600 g/ha) of Phosphobacteria developed at TNAU using rice kanji as binder. If the seed treatment is not carried out apply 10 packets of Rhizobium (2000 g/ha) and 10 packets (2000 g) of Phosphobacteria with 25 kg of FYM and 25 kg of soil before sowing. Dry the bacterial culture treated seeds in shade for 15 minutes before sowing.

**4. FERTILIZER APPLICATION**

i) Apply 20 kg N and 80 kg $P_2O_5$ and 40 kg $K_2O$ per ha 40 kg of S as gypsum (220 kg/ha) / ha as basal dressing. Soil application of 25 kg $ZnSo_4$/ha under irrigated condition

ii) Foliar spray of NAA 40 mg/litre and Salicylic acid 100 mg/litre once at pre-flowering and another at 15 days thereafter

iii). Foliar spray of DAP 20 g/litre or urea 20 g/litre once at flowering and another at 15 days thereafter

**5. SOWING**

Dibble the seeds at a depth of 2 - 3 cm adopting a spacing of 30 x 5 cm. In Erode district, Soybean + Castor (60 cm apart) cropping system gives high net return.

**6. WATER MANAGEMENT**

Irrigate immediately after sowing. Give life irrigation on the 3$^{rd}$ day. Further irrigations at intervals of 7 - 10 and 10 - 15 days during summer and winter season respectively may be given depending on soil and weather conditions. Soyabean is very sensitive to excess moisture and the crop is affected, if water stagnates in the fields. The crop should not suffer due to water stress from flowering to maturity. To alleviate moisture stress spray of either Kaolin 3% or liquid paraffin at 1% on the foliage. In Erode district Soybean + castor with irrigation at 0.60 IW/CPE ratio (i.e.) once in 10 to 12 days is recommended to realise maximum benefits.

**7. WEED MANAGEMENT**

i) Alachlor may be applied to the irrigated crop at 4 litres/ha or Pendimethalin 3.3 litre ai/ha after sowing followed by one hand weeding on 30 days after sowing.

ii) If herbicide spray is not given two hand weedings on 20 and 35 days after sowing may be given.

**8 HARVESTING**

Yellowing of leaves and shedding, indicate the maturity of the crop. Cut the entire plant when most of the pods have turned yellow, drying and processing.

# SOYABEAN IN RICE FALLOWS

Soyabean can be sown in rice fallows from middle of January to middle of March. Seeds can be dibbled at 75 kg/ha.

**SPECIAL SITUATIONS**

1. Optimum time of sowing Soyabean CO 1 - 2$^{nd}$ fortnight of June in Kharif at Bhavanisagar.
2. Intercropping of Soyabean CO 2 in Sugarcane is recommended for North Western Zone.
3. Intercropping of Soyabean in coconut gardens of more than 10 years is recommended.
4. Soyabean varieties UGM21, UGM 37 and ADT 1 are recommended for sowing in rice fallows of Thanjavur, Tiruvarur, Nagapattinam district.
5. Vermipelleting (50 g/kg) and adopting spacing of 30 x 10 cm and two foliar sprays of 2% DAP

during flowering is recommended to achieve higher yield.

# RAINFED SOYABEAN

**1. VARIETIES**

CO 1, ADT 1

**2. SEASON**

The crop can be grown in South-West and North-East monsoon seasons. The middle of July is the optimum time of sowing for rainfed Soyabean in North Western Zone.

**3. SEED TREATMENT WITH THE FUNGICIDES AND BIOFERTILIZERS**

a) Treat the seeds with Carbendezim or Thiram @ 2g/kg of seed 24hrs before sowing or with talc formulation of *Trichoderma viride* @ 4 g/kg seed or *Pseudomonas fluorescens* @ 10 g/kg seed.

- Biocontrol agents are compatible with biofertilizers.
- First treat the seeds with biocontrol agents and then with Rhizobium.
- Fungicides and biocontrol agents are incompatible.

b*)* Treat the seeds required for ha. with three pockets of Rhizobium and 3 packets of Phosphobacteria

**4.FERTILIZER APPLICATION**

i) Apply NPK as per soil test recommendation as far as possible. If soil test recommendation is not available adopt blanket recommendation of 20:40:20:20 NPKS kg/ha, if adequate moisture is available.

ii) Apply entire dose of N, P ,K and S as basal.

**5. SPACING**

Adopt a spacing of 30 cm between rows and 5 cm between plants in the row.

**6. SOWING**

Dibble or drill the seeds.

**7. WEED MANAGEMENT**

i) If sufficient moisture is available, Alachlor may be applied to the irrigated crop at 4.0 litres/ha or Pendimethalin 3.3 litres/ha after sowing followed by one hand weeding on 30 days after sowing.

ii) If herbicide spray is not given, two hand weeding on 20 and 35th day after sowing.

## CROP PROTECTION

**A. Pest management**

| Pests | Management strategies |
|---|---|
| **Whitefly**<br>*Bemisia tabaci* | ▪ **Spray any one of the following** (Spray fluid 500 l/ha)<br>Methyl demeton 25 EC 500 ml/ha<br>Dimethoate 30 EC 500 ml/ha |
| **Tobacco caterpillar**<br>*Spodoptera litura* | ▪ **Spray Endosulfan 35 EC 1000 ml/ha** |
| **Leaf miner**<br>*Aproaerema modicella* | ▪ Quinalphos 1.5 D 25 kg/ha or Neem seed kernel extract (25 kg/ha) spray on 15$^{th}$ and 25$^{th}$ day after sowing |
| **Galerucid beetle**<br>*Madurasia obscurella* | ▪ Methyl demeton 25 EC 500 ml/ha |

**B. Disease management**

| | |
|---|---|
| **Seed treatment** | ▪ Talc formulation of *T. viride* @ 4g or *P. fluorescens* @ 10 g/kg seed (or) Carbendazim or Thiram @ 2 g/kg |

| Name of the Disease | Management |
|---|---|
| **Rust**<br>*Phakopspora pachyrhizi* | **Spray any one of the following:**<br>▪ Triadimefon - 0.1 % or Propiconazole – 0.1% or<br>▪ Hexaconazole - 0.1% at flowering stage or at the onset of disease. |
| **Yellow Mosaic** (*Geminivirus*)<br>Vector -*Bemisia tabaci*<br>**Bud blight (*Ilarvirus*)**<br>Vector- *Thrips palmi* | **Cultural Method**<br>▪ Rogue out infected plants up to 30 days.<br><br>**Spray twice on 15 and 30 days after sowing with any one of the following:**<br>▪ Monocrotophos 36 WSC 500 ml<br>▪ Methyldemeton 25EC500 ml/ ha. |

## SEED PRODUCTION

**Variety Seed Production**

**Land requirement**

- Land to be used for seed production of soyabean shall be free of volunteer plants.

**Isolation**

- Adopt 3 m for certified seed production

**Presowing seed treatment**

**Mid storage seed treatment**

- Pellet the seeds with $ZnSO_4$ @ 0.25 to 0.3 g $kg^{-1}$ of seed using 2% CMC or 10% maida as adhesive @ 250 ml $kg^{-1}$ of seed and arappu leaf powder / vermicompost as filler @ 300g $kg^{-1}$ for better field establishment.
- Moist conditioning of seeds in 5% sand for 16h and drying back to original moisture content (or)
- Moisture equilibration treatment with water for 24 h followed by dry dressing with thiram @ 2.5 g $kg^{-1}$ of seed resulted in better crop establishment.

**Harvesting**

- Seeds attain physiological maturity 27 –30 days after flowering
- Harvest the plants as such on yellowing of pods

**Threshing**

- Dry the plant and thresh with pliable bamboo stick

**Processing**

- Grade the seeds using 14/64” to 12/64” round perforated metal sieve based on varieties.

**Drying**

- Dry the seeds to 7- 8% moisture content

**Seed Treatment**

- Treat the seeds with Thiram @ 2 g $kg^{-1}$ of seed.

**Storage**

- Use gunny or cloth bags for short term storage with seed moisture content of 10 - 12%
- Use polylined gunny bag for medium term storage with seed moisture content of 8 - 10%
- Use 700 gauge polythene bag for long term storage with seed moisture content of less than 7%.

**Other management practices**

- As in crop management technique

# SWORD BEAN ( *Canavalia gladiata L.*)

## CROP IMPROVEMENT

Sword bean SBS 1 is an introduction and is one of the vegetables with photo-insensitivity. It matures in 110 - 120 days. It can be grown throughout the year and gives good response to irrigation. Tender pods are ready for harvest from 75 days after sowing. As a pure crop it gives an average grain yield of 1356 kg/ha and green pod yield of 7500 kg/ha. This can also be grown as border crop, intercrop and a shade crop.

## I. SEASON

June - July (Rainfed), September - October (Rabi), February - March (Summer).

## II. DESCRIPTION OF VARIETY - SBS 1

| | |
|---|---|
| Year of release | 1990 |
| Plant habit | Dwarf, erect, bushy |
| Pigmentation | Green |
| Branches (No) | 4 - 6 |
| Inflorescence | Axillary raceme |
| Flower | Bold, light purple |
| Pods | Long, pendulous, green, flat and fleshy (for vegetable use). Becomes very hard on maturity. |
| 100 seed weight (g) | 131.6 |
| Seed colour | Milky white |
| Days to 50% bloom | 45 - 50 |
| Salient features | Early duration (110 - 120 days)<br>Vegetable cum grain crop<br>Free from beany odour<br>Highly nutritious and delicious (25.9% protein)<br>No major pests and diseases |

## III. MANAGEMENT OF FIELD OPERATIONS

- Seed rate (kg/ha) : 110-120 (Pure crop)
- Fertilizers (kg/ha) : 25 N 50 $P_2O_5$
- Spacing : 45 x 30 cm (irrigated), 30x20 cm rainfed

### INTEGRATED PEST MANAGEMENT FOR PULSE PESTS

**1. Stem fly**

- It attacks blackgram, greengram and cowpea.
- Adult fly is blackish and lay eggs on the young leaves
- Affected plants get dried
- Immature stage will be inside the stem
- Economic threshold level is 10% damage

**2. Aphids**

- Attacks blackgram, greengram, lab lab, cowpea and redgram.
- Congregated on the growing shoots, leaves, flowers and pods.
- Affected plants will be weak and stunted
- Because of honeydew ant movements will be there

**3. Whiteflies**

- Attacks blackgram, greengram, cowpea and soyabean
- Act as vector for yellow mosaic virus disease

**4. Bugs**

- Desap the flowers and pods
- Affected pods show shriveled grains

**5. Pod borers**

- Gram pod borer, spotted pod borer, blue butterflies, pod fly and blister beetles are the major borers
- Blister beetles feed on flower buds, flowers and young pods
- Spotted pod borers web the flowers and young pods
- Gram pod borer, plume moth and blue butterflies bore into the pods
- Pod fly feed on the seeds of redgram.

**IPM**

- Take up the sowing of blackgram from September to November with increased seed rate (25 kg/ha) in stem fly endemic areas.
- Remove alternate hosts
- Use of pheromone traps @ 12/ha for Gram pod borer
- Spray insecticides like methyl demeton or dimethoate or monocrotophos @ 500ml/ha to reduce the sucking insects
- Spray endosulfan @1.25 l/ha or Neem seed kernel extract (25 kg/ha) against pod borers
- Avoid insecticidal spray when parasitoids and predators activity is high.

**6. Storage pests**

- Dry the seeds adequately to reduce moisture level to 10 %.
- Use pitfall traps or two in one model trap to assess the time of emergence of field carried over pulse beetle in storage and accordingly sun-dry the produce.
- Seed: **Mix any one of the following for every 100 kg** :

  Activated kaolin 1 kg
  Malathion 5 D 1 kg
  TNAU Neem oil 60 EC (C) 1lit.
  Pungam oil 1lit.
  Monocrotophos 36 SL 400 ml
- Pack in polythene lined gunny bags for storage

# OILSEEDS

## GROUNDNUT (*Arachis hypogaea* )

### CROP IMPROVEMENT

### I. SEASON AND VARIETIES

**DISTRICT / SEASON VARIETIES**

**A. Rainfed**

**1. Chithiraipattam (Apr - May)**

Pollachi, Theni and Tenkasi — TMV 7, ALR 3

**2. Early Adipattam (Jun - Jul)**

All districts — Bunch : TMV 7, CO 2, VRI 2, VRIGn 5, Semispreading : TMV 10, COGn 5

**3. Late Adipattam (Jul - Aug)**

All districts — TMV 7, VRI 2, CO 2

**B. Irrigated**

**1. Margazhipattam (Dec - Jan)**

All districts — TMV 7, CO 2, CO 3, COGn 4, VRI 2, VRI 3, ALR 3, VRIGn 5

**2. Masipattam (Feb - Mar)**

(New Delta areas of Thanjavur, Tiruvarur, Nagapattinam) — TMV 7, CO 2, VRI 2, VRI 3,

**3. Chithiraipattam (Apr – May))** — TMV 7, CO 2, CO 3, COGn 4, VRI 2, VRI 3

### II. PARTICULARS OF GROUNDNUT VARIETIES

| Particulars | TMV 7 | TMV 10 | CO 3 | COGn 4 | COGn 5 |
|---|---|---|---|---|---|
| Parentage | Pureline selection from Tennessee white | Spontaneous mutant from Argentina | Derivative of VG 55 x JL 24 | Derivative of TMV 10 x ICGV 82 | Multiple cross derivative |
| Duration (days) | 100 - 105 | 120 - 130 | 115-120 | 115-120 | 125-130 |
| Average yield of pods kg/ha | 1400 | 1650 | 1950 | 2150 | 1850 |
| Shelling % | 74 | 77 | 70 | 70 | 70 |
| 100 seed wt (g) | 36 | 43 | 65 | 60 | 47 |
| Oil content % | 49.6 | 54.4 | 49.2 | 52.7 | 51 |
| Special features | Seed dormancy for 10 days | Yield, oil, shelling high | Bold 1-2 seeded, HPS kernels. Low bud necrosis incidence | Bold pods; 1-2 seeded; high oil content. | Dark green foliage; Tolerant to foliar diseases |
| Growth habit | Bunch | Semi-spreading | Bunch | Bunch | Semispreading |
| Leaf colour | Green | Dark green | Green | Dark green | Dark green |
| Seed colour | Light rose | Red blotched with white | Rose | Rose | Red testa |

| Particulars | ALR 3 | VRI 2 | VRI 3 | VRIGn 5 |
|---|---|---|---|---|
| Parentage | Derivative of (R33-1 x ICGV 68) x (NCAC 17090 x ALR 1) | JL24 x CO2 | J11 x R 33-1 | Derivative of CG 26 x ICGS 44 |
| Duration(days) | 110-115 | 100 - 105 | 90 | 105 -110 |
| Average yield of pods (kg/ha) | 2095 | 2060 | 1882 | 2133 |
| Shelling % | 69 | 74.8 | 73 | 75 |
| 100 Seed wt. (g) | 46 | 49.9 | 35 | 46 |
| Oil content % | 50 | 48 | 48 | 51 |
| Special features | Suitable for rainfed, rust resistant, tolerant to late leaf spot. | suitable for irrigated | Suitable for intercropping | High reproductive efficiency. Dormancy 45 days. |
| Growth habit | Bunch | Bunch | Bunch | Bunch |
| Leaf colour | Dark green | Ashy green | Light green | Dark green |
| Seed colour | Rose | Light rose | Light rose | Red testa. |

## CROP MANAGEMENT

## I. Rainfed

### 1. FIELD PREPARATION

i) Plough with tractor using a disc followed by harrow, once or twice with iron plough or 3 - 4 times with country plough till all the clods are broken and a fine tilth is obtained.

ii) Chiselling for soils with hard pan: Chisel the soils having hard pan formation at shallow depth with chisel plough first at 0.5 m interval in one direction and then in the direction perpendicular to the previous one, once in three years. Apply 12.5 t/ha of FYM or composted coir pith besides chiselling.

iii) Amendments for soil surface crusting: a) To tide over the surface crusting, apply lime @ 2 t/ha along with FYM or composted coir pith @ 12.5 t/ha. b) Coir pith at 12.5 t/ha converted to compost by inoculating with *Pleurotus* and applied serves as a good source of nutrients.

### 2. APPLICATION OF FERTILIZERS

Apply NPK fertilizers as per soil test recommendation. If soil test is not done, follow the blanket recommendation.

| N | P | K |
|---|---|---|
| 10 | 10 | 45 kg/ha |

### 3. FORMING BEDS

i) Form beds of size 10 $m^2$ to 20 $m^2$ depending upon the slope of the land and type of soil.

ii) Wherever tractor is engaged, bed former may be used.

### 4. APPLICATION OF MICRONUTRIENTS

Mix 12.5 kg/ha of micronutrient mixture developed by Department of Agriculture with enough dry sand to make a total quantity of 50 kg/ha. Broadcast evenly on the soil surface immediately after sowing. Do not incorporate micronutrient mixture in to the soil.

### 5. NUTRITIONAL DISORDER

**Zinc deficiency**: Apply 25 kg $ZnSO_4$/ha as basal.

If soil analysis shows less than 1.3 ppm of zinc, soil application of 25 kg $ZnSo_4$ is recommended. For the standing crop, less than 39.4 ppm of zinc in leaves, foliar spray of 0.5% $ZnSo_4$ is recommended.

**Iron deficiency**: spray 1% $FeSo_4$ on 30, 40 and 50 days after sowing.

**Boron deficiency**: Apply Borax 10 kg + Gypsum 400 kg/ha at 45th day after sowing.

**6. SEED RATE**

Use 120 kg/ha of kernels. Increase the seed rate by 15% in the case of bold seeded varieties.

**7. SPACING**

Adopt a spacing of 30 cm between rows and 10 cm between plants. Wherever groundnut ring mosaic (bud necrosis) is prevalent, adopt a spacing of 15cm x 15 cm.

**8. SEED TREATMENT**

i) Treat the seeds with talc formulation of *Trichoderma viride* @ 4 g/kg seed or *Pseudomonas fluorescens* @ 10 g/kg seed.

Biocontrol agents are compatible with biofertilizers.

Treat the seeds with biocontrol agents first and then with *Rhizobium*.

Fungicides and biocontrol agents are incompatible.

ii) Treat the seeds with *Trichoderma* @ 4g/kg. This can be done just before sowing. It is compatible with biofertilizers. SUCH SEEDS SHOULD NOT BE TREATED WITH FUNGICIDES (or)

iii) Treat the seeds with Thiram or Mancozeb @ 4 g/kg of seed or Carboxin or Carbendazim at 2 g/kg of seed.

iv) Treat the seeds with 3 packets (600 g)/ha of Rhizobialculture TNAU14 + 3 packets of Azospirillum (600 g/ha) and 3 packets(600 g/ha) of Phosphobacteria or 6 packets of Azophos(1200 g/ha)developed at TNAU using rice kanji as binder. If the seed treatment is not carried out apply 10packets of Rhizobium (2000g/ha) + 10 packets of Azospirillum (2000 g/ha) and 10 packets(2000 g) of Phosphobacteria with 25 kg of FYM and 25 kg of soil before sowing.

**9. SOWING**

- Use Kovai seed drill/gorru to sow the seeds in lines.
- Put one seed in each hole. Protect the seeds from crows and squirrels.

**10. INTERCROPPING**

i) Raise one row of cowpea for every five rows of groundnut wherever red hairy caterpillar is endemic.
ii) Raise intercrops like redgram, blackgram, sunflower, gingelly or other pulses.
iii) Cumbu can be raised as intercrop.
iv) Groundnut + Gingelly or Groundnut + Blackgram in the ratio of 4:1 or Groundnut + Cowpea at 6:1 ratio and Groundnut + Sunflower at 6:2 ratio may be raised.

**11. WEED MANAGEMENT**

i) **Pre-sowing**: Fluchloralin at 2.0 l/ha soil applied and incorporated.
ii) **Pre-emergence**: Fluchloralin 2.0 l/ha applied through flat fan nozzle with 900 l of water/ha followed by irrigation. After 35 - 40 days one hand weeding may be given.
iii) If no herbicide is applied two hand weeding and hoeing are given on $20^{th}$ and $40^{th}$ day after sowing.

**12. EARTHING UP**

Accomplish earthing up during second hand weeding/late hand weeding (in herbicide application).

NOTE: i) Earthing up provides medium for the peg development ii) Use the improved hoe with long handle which can be worked more efficiently in a standing position. iii) Do not disturb the soil after $45^{th}$ day of sowing as it will affect pod formation adversely.

**13. APPLICATION OF CALCIUM SULPHATE (GYPSUM)**

i) Apply gypsum @ 400 kg/ha by the side of the plants on 40th to 70th day depending upon soil moisture.
ii) Apply gypsum, hoe and incorporate it in the soil and then earth up.

iii) Avoid gypsum in calciferous soils.
iv) Gypsum is effective in soils deficient in calcium and sulphur.

*NOTE: Application of gypsum encourages pod formation and better filling up of the pods.*

Application of gypsum at the rate of 50 % basal both in rainfed and irrigated condition reduces Khadhasty malady and pod scab nematode

**Combined nutrient spray**

Pod filling is a major problem especially in the bold seed varieties. To improve pod filling spraying of nutrient solution is to be given. This can be prepared by soaking DAP 2.5 kg, Ammonium sulphate 1 kg and borax 0.5 kg in 37 lit of water overnight. The next day morning it can be filtered and about 32 litre of mixture can be obtained and it may be diluted with 468 lit of water so as to made up to 500 litre to spray for one ha. Plano fix at the rate of 350 ml. can also be mixed while spraying. This can be sprayed on 25$^{th}$ and 35$^{th}$ day after sowing.

**14. HARVESTING**

i) Observe the crop, considering its average duration. Drying and falling of older leaves and yellowing of the top leaves indicate maturity.
ii) Pull out a few plants at random and shell the pods. If the inner shell is brownish black and not white, then the crop has matured.
iii) Irrigate prior to harvest, if the soil is dry, as this will facilitate easy harvesting. If there is enough moisture in the soil, there is no need for irrigation for harvesting.
iv) If water is not available for irrigating the field prior to harvest, work a mould board plough or work a country plough, so that the plants are uprooted. Engage labour to search pods left out in the soil, if necessary.

*NOTE: Do not keep the pulled out plants in heaps when they are wet, especially the bunch varieties, as the pods will start sprouting.*

v) Strip off the pods from the plants. Groundnut stripper developed by TNAU can be used.
vi) Dry the pods in the sun for 4 or 5 days. Repeat drying for 2 or 3 more days after an interval of 2 or 3 days to ensure complete drying. When temperature is very high, avoid direct sun drying. Collect the pods in gunnies and store on the ground over a layer of sand to avoid any moisture coming in contact with dry pods.

**II. Irrigated**

**1. FIELD PREPARATION**

i) Plough with tractor using a disc followed by harrow, once or twice with iron plough or 3 - 4 times with country plough till all the clods are broken and a fine tilth is obtained.
ii) Chiselling for soils with hard pan: Chisel the soils having hard pan formation at shallow depth with chisel plough first at 0.5 m interval in one direction and then in the direction perpendicular to the previous one, once in three years. Apply 12.5 t/ha FYM or composted coir pith besides chiselling.
iii) Amendments for soil surface crusting: a) To tide over the surface crusting, apply lime @ 2 t/ha along with FYM or composted coir pith @ 12.5 t/ha. b) When coir pith at 12.5 t/ha is converted into compost by inoculating with *Pleurotus* and applied, it serves as a good source of nutrient

**2. APPLICATION OF FERTILIZERS**

Apply NPK fertilizers as per soil test recommendation. If soil test is not done, follow the blanket recommendation.

| N | P | K | Sulphur sludge |
|---|---|---|---|
| 17 | 34 | 54 kg/ha | 60 kg/ha |

**3. FORMING BEDS**

- Form beds of size 10 $m^2$ to 20 $m^2$ depending upon the availability of water, slope of the land and type of soil.
- Wherever tractor is engaged, bed former may be used.

**4. POLYTHENE FILM MULCHING**

Spread black polythene sheet over the soil surface. Seven micron polythene film sheet @50 kg/ha is required. Holes can be made at required spacing before spreading of the sheets.

**5. APPLICATION OF MICRONUTRIENTS**

Mix 12.5 kg/ha of micronutrient mixture developed by Department of Agriculture with enough dry sand to make a total quantity of 50 kg/ha. Broadcast evenly on the soil surface immediately after sowing. Do not incorporate the micronutrient mixture to the soil.

**6. NUTRITIONAL DISORDER**

**Zinc deficiency**: Apply 25 kg $ZnSO_4$/ha as basal.

If soil analysis shows less than 1.3 ppm of zinc, soil application of 25 kg $ZnSo_4$ is recommended. For the standing crop, less than 39.4 ppm of zinc in leaves, foliar spray of 0.5% $ZnSo_4$ is recommended.

**Iron deficiency** : spray 1% $FeSO_4$ on 30, 40 and 50 days after sowing.

**Boron deficiency**: Apply Borax 10 kg + Gypsum 200 kg/ha at 45th day after sowing.

**7. SEED RATE**

Use 125 kg/ha of kernels. Increase the seed rate by 15% in the case of bold seeded varieties.

**8. SPACING**

Adopt a spacing of 30 cm between rows and 10 cm between plants. Wherever groundnut ring mosaic (bud necrosis) is prevalent, adopt a spacing of 15cm x 15 cm.

**9. SEED TREATMENT**

i) Treat the seeds with *Trichoderma viride* @ 4 g/kg seed or *Pseudomonas fluorescens* @ 10 g/kg seed.

**Biocontrol agents are compatible with biofertilizers.**

**First treat the seeds with biocontrol agents and then with *Rhizobium*.**

**Fungicides and biocontrol agents are incompatible.**

ii) Treatment with *Trichoderma* can be done just before sowing. SUCH SEEDS SHOULD NOT BE TREATED WITH FUNGICIDES. (or)

iii) Treat the seeds with Thiram or Mancozeb @ 4 g/kg of seed or Carboxin or Carbendazim at 2 g/kg of seed.

iv) Treat the seeds with 3 packets (600 g)/ha of Rhizobial culture TNAU14 developed at TNAU using rice kanji as binder. If the seed treatment is not carried out, apply 10 packets/ha (2000 g) with 25 kg of FYM and 25 kg of soil before sowing.

*Seed treatment will protect the young seedlings from root-rot and collar rot infection.*

**10. SOWING**

a) Dibble the seeds at 4 cm depth along with fertilizer.

**11. WEED MANAGEMENT**

i) **Pre-sowing**: Fluchloralin at 2.0 l/ha soil applied and incorporatede followed by light irrigation.

ii) **Pre-emergence**: Fluchloralin 2.0 l/ha applied through flat fan nozzle with 900 l of water/ha followed by irrigation. After 35 - 40 days one hand weeding may be given.

iii) If no herbicide is applied two hand hoeing and weeding are given on 20$^{th}$ and 40$^{th}$ day after sowing.

**12. EARTHING UP:**

Accomplish earthing up during second hand weeding/late hand weeding (in herbicide application).

NOTE: i) Earthing up provides medium for the peg development. ii) Use the improved hoe with long handle which can be worked more efficiently in a standing position. iii) Do not disturb the soil after the 45$^{th}$ day of sowing as it will affect pod formation adversely.

**13. APPLICATION OF CALCIUM SULPHATE (GYPSUM)**

- Apply gypsum @ 400 kg/ha by the side of the plants on the 40$^{th}$ to 45$^{th}$ day of sowing. Apply gypsum, hoe and incorporate in the soil and then earth up.
- Avoid gypsum in calciferous soils.
- Gypsum is effective in soils deficient in calcium and sulphur.

*NOTE: Application of gypsum encourages pod formation and better filling up of the pods.*

Application of gypsum at the rate of 50 % basal both in rainfed and irrigated condition reduces Khadhasty malady and pod scab nematode

**Combined nutrient spray**

Pod filling is a major problem especially in the bold seed varieties. To improve pod filling spraying of nutrient solution is to be given. This can be prepared by soaking DAP 2.5 kg, Ammonium sulphate 1 kg and borax 0.5 kg in 37 lit of water overnight. The next day morning it can be filtered and about 32 litre of mixture can be obtained and it may be diluted with 468 lit of water so as to made up to 500 litre to spray for one ha. Plano fix at the rate of 350 ml. can also be mixed while spraying. This can be sprayed on 25$^{th}$ and 35$^{th}$ day after sowing.

**14. WATER MANAGEMENT**

Schedule the irrigation at 0.40 and 0.60 IW/CPE ratio during vegetative and reproductive phases respectively. Regulate irrigation according to the following growth phase of the crop.

Pre-flowering phase : 1 to 25 days
Flowering phase : 26 to 60 days
Maturity phase : 61 to 105 days

Regulate irrigation based on physiological growth phases. Pegging, flowering and pod development phases are critical for irrigation during which period adequate soil moisture is essential. Apply irrigation as follows:

i) Sowing or pre-sowing
ii) Life irrigation, 4 - 5 days after sowing if sowing irrigation given to break the surface crust.
iii) 20 days after sowing
iv) At flowering give two irrigations
v) At pegging stage give one or two irrigations
vi) In pod development stage, 2 - 3 irrigations depending on the soil type

*Note: Spraying 0.5% Potassium chloride during flowering and pod development stages will aid to mitigate the ill effects of water stress. Sprinkler irrigation will save water to the tune of about 30%. Borderstrip irrigation is recommended in command areas in light textured soils. Composted coir pith increases moisture availability and better drainage in heavy textured soil.*

**15. HARVESTING**

i) Observe the crop, considering its average duration. Drying and falling of older leaves and yellowing of the top leaves indicate maturity.
ii) Pull out a few plants at random and shell the pods. If the inner shell is brownish black and not white, then the crop has matured.

iii) Irrigate prior to harvest, if the soil is dry, as this will facilitate easy harvesting. If there is enough moisture in the soil, there is no need for irrigation for harvesting.

iv) If water is not available for irrigating the field prior to harvest, work a mould board plough or work a country plough, so that the plants are uprooted. Engage labour to search pods left out in the soil, if necessary.

*NOTE: Do not keep the pulled out plants in heaps when they are wet, especially the bunch varieties, as the pods will start sprouting.*

v) Strip off the pods from the plants. Groundnut stripper developed by TNAU can be used.

vi) Dry the pods in the sun for 4 or 5 days. Repeat drying for 2 or 3 more days after an interval of 2 or 3 days to ensure complete drying. When temperature is very high, avoid direct sun drying. Collect the pods in gunnies and store on the ground over a layer of sand to avoid any moisture coming in contact with dry pods.

## CROP PROTECTION

### Pest management

**Economic threshold level for important pests**

| Pests | ETL |
|---|---|
| Leaf miner | 1 larvae /meter row |
| Tobacco cutworm | 8 egg masses/100 m row |

| Pests | Management strategies |
|---|---|
| **Red hairy caterpillar**<br>*Amsacta albistriga* | ▪ Dig out and destroy the pupae from the field bunds and shady spots prior to summer rains.<br>▪ Set up 3 to 4 light traps and bonfires immediately after receipt of rains, after sowing in the rainfed season to attract and kill the moths and also to know brood emergence.<br>▪ Collect and destroy gregarious, early instar larvae on lace-like leaves of intercrops such as redgram and cowpea.<br>▪ Collect and destroy egg masses in the cropped area.<br>▪ Avoid migration of larvae by digging a trench 30 cm deep and 25 cm wide with perpendicular sides around the infested fields.<br>▪ **Apply any one of the following insecticides at 25 kg/ha (for young caterpillars) :**<br>Quinalphos 1.5 D<br>Phosalone 4 D<br>Endosulfan 4 D<br>Carbaryl 10 D<br>▪ **Spray any one of the following insecticides :**<br>Endosulfan 35 EC 750 ml/ha<br>Fenitrothion 50 EC 750 ml/ha<br>Quinalphos 25 EC 750 ml/ha<br>Dichlorvos 76 WSC 625 ml/ha<br>Chlorpyriphos 20 EC 1250 ml/ha<br>Ethion 50 EC 500 ml/ha<br>Phosalone 35 EC 750 ml/ha<br>in 375 l of water.<br>▪ Use Nuclear Polyhedrosis Virus (NPV) at 1.5 x $10^{12}$ POBs per ha as detailed below.<br>▪ **Virus multiplication**<br>Collect medium sized larvae of *Amsacta albistriga* from the field and starve them over night. Make a pure suspension of virus with the nucleus culture, in water. Dip *Calotropis* leaves in virus |

| | |
|---|---|
| | suspension, shade dry and feed them to starved larvae for 1 or 2 days. From third day, normal, untreated leaves can be fed to these larvae. From 5th day, the treated larvae will start dying. Virus infected larvae can be diagnosed by their pinkish ventral surface, their head hanging downwards with white body contents oozing out through ruptured body wall in the late stage. Collect the dying larvae, keep in fresh potable water for a few days, grind the larvae and filter through several layers of fine cloth and collect filtrate (Crude virus suspension). Use virus suspension obtained from 750 medium sized larvae for spraying one hectare along with a sticker 250 ml or Triton in 350 l of water. Use potable water for mixing and spray in the evening hours. |
| **Tobacco cut worm**<br>*Spodoptera litura* | ▪ Grow castor as border or intercrop in groundnut fields to serve as indicator or trap crop.<br>▪ Monitor the emergence of adult moths by setting up light and pheromone traps.<br>▪ Collect egg masses and destroy.<br>▪ Collect the gregarious larvae and destroy them as soon as the early symptoms of lace-like leaves appear on castor, cowpea and groundnut.<br>▪ **Apply anyone of the following insecticides to control the early instar (1st to 3rd instar) larvae**<br>Carbaryl 10 D 25 kg/ha<br>Fenitrothion 50 EC 750 ml/ha<br>Carbaryl 50 WP 2.0 kg/ha<br>Quinalphos 25 EC 750 ml/ha<br>Fenthion 100 EC 500 ml/ha<br>Phenthoate 50EC 1250 ml/ha<br>Dichlorvos 76 WSC 750 ml/ha<br>Endosulfan 35 EC 1.0 l/ha<br>▪ **Spray any one of the following insecticides to control the 4th to 6th instar larvae :**<br>Chlorpyriphos 20 EC 2.0 l/ha<br>Dichlorvos 76 WSC 1.0 l/ha<br>Phenthoate 50 EC 2.0 l/ha<br>Profenophos 2.0 l/ha<br>Neem oil (2%) 20 lit /ha<br>▪ Prepare a bait with the following materials to cover one ha. Rice bran 12.5 kg; Molasses or brown sugar 1.25 kg or carbaryl 50 WP 1.25 kg. Mix the ingredients to obtain a homogeneous mixture, sprinkle water (7 lit.) gradually and bring the bait to a dough consistency. Distribute the above bait on the soil, around the field and inside in the evening hours immediately after preparation.<br>▪ Apply Nuclear Polyhedrosis Virus 1.5 x $10^{12}$ POBs/ha with crude sugar 2.5 kg/ha and Teepol 250 ml/ ha. Methods of mass culturing and application are the same as for *Amsacta* NPV. Use castor leaves for larvae |
| **Leaf hopper**<br>*Empoasca kerri* | ▪ Intercrop lab lab with groundnut 1:4 ratio<br>▪ Spray Imidacloprid 200 SL at 100 ml/ha twice, at 30 and 51 days after sowing |
| **Leaf miner**<br>*Aproaerema modicella* | ▪ Set up light traps between 8 and 11 p.m at ground level<br>▪ **Apply anyone of the following insecticides at 25 kg/ha :**<br>Phosalone 4 D<br>Endosulfan 4 D<br>Carbaryl 10 D<br>Fenitrothion 2 D |

| | |
|---|---|
| | ▪ **Spray any one of the following :**<br>Endosulfan 35 EC 750 ml/ha<br>Dichlorvos 76 WSC 625 ml/ha<br>Monocrotophos 36 WSC 750 ml/ha<br>Phosphamidon 40 SL 750 ml/ha<br>Chlorpyriphos 20 EC 1250 ml/ha<br>Phosalone 35 EC 750 ml/ha<br>Quinalphos 25 EC 750 ml/ha<br>Phenthoate 50 EC 750 ml/ha<br>Imidacloprid 200 SL 150 ml/ha (in 375 l of water) |
| **Pod borer (Earwig)** *Anisolabis stali* | ▪ **Apply any one of the following to the soil prior to sowing in endemic areas :**<br>Malathion 5 D 25 kg/ha<br>Endosulfan 4 D 25 kg/ha<br>▪ Repeat soil application of any one of the above dust formulations on the 40th day of sowing and incorporate in the soil during the earthing up. |
| **Millipede**<br>*Spirostreptus* spp. | ▪ Apply methyl parathion 2D 25 kg/ha during gypsum application and earthing up |

## B.Disease management

| | |
|---|---|
| **Seed treatment** | ➢ ***Treat the seeds with any one of the following***<br>▪ *Thiram @ 4g/Kg of seed*<br>▪ *Mancozeb @ 4g/Kg of seed*<br>▪ *Carboxin @ 2g/kg of seed*<br>▪ *Carbendazim @ 2g/kg of seed.*<br>▪ Talc formulation of *T. viride* @ 4g/kg of seed<br>▪ *P. fluorescens* @10g/kg of seed |
| ***Rust***<br>*Puccinia arachidis* | ➢ **Spray any one of the following :**<br>▪ Mancozeb 1000g /ha<br>▪ Chlorothalonil 1000g /ha<br>▪ Wettable sulphur 2500g /ha<br>▪ Tridemorph 500 ml/ha<br>If necessary, repeat the spray 15 days later. |
| ***Early leaf Spot***<br>*Cercopora arachidicola*<br>*Mycosphaerella arachidis*<br>***Late leaf Spot***<br>*Phaeoisariopsis personata*<br>*Mycosphaerella berkeleyii* | ➢ **Spray any one of the following :**<br>▪ Carbendazim 500 g/ha<br>▪ Mancozeb 1000 g/ha<br>▪ Chlorothalonil 1000 g/ha<br>If necessary give the second round 15 days later. |
| Combined infection of rust and Leaf spot | ➢ **Spray any one of the following :**<br>❖ **Botanical /others**<br>▪ Spray 10% *Calotropis* leaf extract<br>❖ **Chemical**<br>▪ Spray Carbendazim 250 g + Mancozeb 1000g/ha<br>▪ Chlorothalonil 1000g/ha. If necessary give the second round 15 days later. |
| ***Root rot***<br>*Macrophomina phaseolina*<br>*Rhizoctonia bataticola* | ❖ **Biological control**<br>▪ Soil application of *P. fluorescens* @ 2.5g /ha mixed with 50 kg of well decomposed FYM / sand at 30 DAS<br>❖ **Chemical**<br>▪ Spot drench with Carbendazim 1 g / l |
| Groundnut Bud Necrosis (*Peanut Bud Nercrosis Virus*) | ❖ **Cultural Method**<br>▪ Adopt a close spacing of 15 x 15 cm. |

| | |
|---|---|
| **vector:**<br>*Thrips tabaci*<br>*Frankliniella schultzeii* | ▪ Remove infected plants up to 6 weeks after sowing and spray Monocrotophos 36 WSC 500 ml/ha, 30 days after sowing either alone or in combination with antiviral principles.<br>**Botanical/others**<br>▪ Antiviral principles from sorghum or coconut leaves. AVP are extracted as follows: Sorghum or coconut leaves collected, dried, cut into small bits and powdered. To one kg of leaf powder two litres of water is added and heated to 60°C for one hour. It is then filtered through muslin cloth and diluted to 10 litres and sprayed. To cover one ha 500 litre of fluid will be required. Two sprays at 10 and 20 days after sowing will be needed. |

## SEED PRODUCTION

**Variety Seed Production**

**Land Requirements**

- A seed crop of groundnut shall not be eligible for certification if planted on land on which the crops grown within previous two seasons were of the same kind unless the crops grown within the previous two seasons were of the same variety and of an equivalent or higher class of certified seed and were certified.
- Boron deficient soil should be avoided as it produces single compartment seeds.
- Calcium deficient soils should be avoided as it causes production of darkened plumules.

**Isolation**

- Adopt 3 m for certified seed production all around the field against other varieties

**Season**

- June – July and December – January.

**Spacing**

- Adopt spacing of 25 x 15 cm

**Pre-sowing seed treatment**

- Obtain seed from healthy pods.
- Shrivelled and disfigured seeds should be discarded.
- Others as in crop management techniques for crop production.

**Pre-sowing seed hardening**

- The graded seeds can be hardened by soaking in 0.5% $CaCl_2$ (50% seed volume) for 6 hrs. After 6 hrs soaking, seeds should be incubated in between moist gunny bags for 12–18 h. The sprouting of radicle should be observed periodically at 2 hr intervel after 12 h of incubation. The seeds with sprouted radicle (just visible expression of radicle) should be separated and dried under shade. All the viable seeds with the expression of radicle emergence should be separated and dried under shade.

**Fertilizers**

- NPK @ 40:40:60 kg $ha^{-1}$ as basal
- Apply borax @ 10 kg $ha^{-1}$ as basal
- Apply gypsum @ 200 kg $ha^{-1}$ at peg formation stage

**Harvesting**

- Harvest the pods as and when the color of the inner side of the shell turns black.
- The pod moisture will be 35 – 40% at harvest.
- Strip the pods and dry to 10 – 12 per cent moisture.
- Mechanically injured pods should be rejected.
- Remove all discolored pods.
- Pods sorting should be practiced to remove genetically impure seed.

**Drying**

- Stake the plants as the pods are exposed to outside for easy drying of pod.
- Dry the pod to 15 – 20 % moisture content under sun.

**Decortication**

- Pods dried to 16 per cent moisture content could be decorticated either manually or using hand operated decorticator with proper adjustment
- The kernels should be dried to 7 to 8 per cent moisture.

**Seed Treatment**

- Treat the pods with Thiram 75% WP @ 2 g $kg^{-1}$ of seed at pods with 6 -7 per cent kernel moisture.

**Seed Storage**

- Store the pods in closed plastic container or gunny bags with Calcium chloride at 250g/30 kg of pods.
- Use gunny or cloth bags for short term storage with seed moisture content of 8 - 9%
- Use polylined gunny bag for medium term storage with seed moisture content of 6- 8%
- Use 700 gauge polythene bag for long term storage with seed moisture content of less than 5%.

**Other management practices**

- As in crop management technique

# SESAME (*Sesamum indicum*)

## CROP IMPROVEMENT

## I. SEASON AND VARIETIES

| DISTRICT/SEASON | VARIETIES |
|---|---|
| **A) Rainfed** | |
| **1. Adipattam (Jun - Jul)** | |
| All districts | CO 1, TMV 3 |
| **2. Karthigaipattam (Oct - Nov)** | |
| All districts | CO 1, TMV 3, TMV 5, SVPR 1, VRI(SV) 2 |
| **B) Irrigated - Masipattam (Feb - Mar)** | |
| Coimbatore, Erode, Tiruchirapalli, Perambalur, Karur, Madurai, Dindigul, Theni, Thanjavur, Tiruvarur,Nagapattinam | TMV 3, TMV 4, TMV 6, CO 1, VRI(SV) 1, SVPR 1, VRI(SV) 2 |
| **C) Rice fallows** | |
| Coastal situations | VRI(SV)1 |

## II. PARTICULARS OF SESAME VARIETIES

| Particulars | CO 1 | TMV 3 | TMV 4 | TMV 5 |
|---|---|---|---|---|
| Parentage | (TMV3 x SI 1878) x SI 1878 | South Arcot local x Malabar | Pureline from Sattur local variety | Pureline from Srivaikuntam variety |
| Duration (Days) | 85 - 90 | 80 - 85 | 85 - 90 | 80 - 85 |
| Oil content(%) | 51 | 51 | 50 | 51 |
| **Yield kg/ha** | | | | |
| Irrigated | 750 - 790 | 625 - 750 | 700 - 850 | .. |
| Rainfed | 450 - 650 | 400 - 650 | .. | 450 - 650 |
| Habit | Erect with branching and short internodes on the main stem | Bushy with profuse branching | Bushy with profuse branching | Erect with moderate branching |
| Capsules | 4 loculed | 4 loculed | 4 loculed | 4 loculed |
| Seeds | Intense dark brown almost black | Darkbrown | Brown | Brown |

| Particulars | TMV 6 | SVPR 1 | VRI (SV)1 |
|---|---|---|---|
| Parentage | Pureline selection from Andhra Pradesh variety | Selection from Western Ghat White | Pureline seletion from Tirukattupalli local |
| Duration (Days) | 85 - 90 | 75 - 80 | 70 - 75 |
| **Yield kg/ha** | | | |
| Irrigated | 700 - 950 | 800 | 650 - 900 |
| Rainfed | .. | 600 | 450 - 650 |
| Oil content(%) | 54 | 53.8 | 51 |
| Habit | Erect with moderate branching | Erect and moderate branching | Moderate branching |
| Capsules | 4 loculed | 4 loculed | 4 loculed |
| Seeds | Brown | White | Brown |

## CROP MANAGEMENT

### 1. FIELD PREPARATION

a) Plough the field with tractor twice or with mould board plough thrice or five times with a country plough.

b) Break the clods in between ploughings and bring the soil to a fine tilth to facilitate quick germination as the seeds are small.

c) Chiselling for soils with hard pan: Chisel the soils having hard pan formation at shallow depth with chisel plough first at 0.5 m interval in one direction and then in the direction perpendicular to the previous one once in three years. Apply 12.5t FYM/composted coir pith besides chiselling.

d) For irrigated gingelly, form beds of size 10 $m^2$ or 20 $m^2$ depending upon the availability, inflow of water and slope of the land. Level the beds perfectly without any depressions to prevent water stagnation, which will affect the germination adversely.

e) In rice fallows, field is ploughed once with optimum moisture, seeds are sown immediately and covered with one more ploughing.

### 2. APPLICATION OF FERTILIZERS

i) Spread FYM or composted coir pith or compost @ 12.5 t/ha evenly on the unploughed field and plough it in.

ii) If the manure is not applied before commencement of ploughing, spread 12.5 t/ha of FYM or compost evenly on the field before the last ploughing and incorporate in the soil.

iii) Apply NPK fertilizers as per soil test recommendation. If soil tests are not available, follow the blanket recommendations. **Rainfed:** Apply 23:13:13 kg NPK/ha or 17:13:13 kg NPK/ ha + 3 packets of Azospirillum (600 g/ha) and 3 packets (600 g/ha) of Phosphobacteria or 6 packets of Azophos(1200 g/ha). **Irrigated:** Apply 35:23:23 kg NPK/ha or 21:23:23 kg NPK/ha + 3 packets of Azospirillum (600 g/ha) and 3 packets(600 g/ha) of Phosphobacteria or 6 packets of Azophos(1200 g/ha)

iv) Apply full dose of N, P and K basally. Add 5 kg of Manganese sulphate per hectare. Apply 50% of the recommended $P_2O_5$ and $K_2O$ with full recommended dose of N to irrigated gingelly raised after groundnut fertilized with 100% of recommended NPK.

v) Open furrows to a depth of 5 cm and 30 cm apart and place the fertilizer mixture along the furrows and cover to a depth of 3 cm with soil before sowing.

vi) If furrow application is not done, broadcast the fertilizer mixture evenly on the beds before sowing.

### 3. APPLICATION OF AZOSPIRILLUM

25% of the N can be substituted by 3 packets of Azospirillum (600 g/ha) and 3 packets (600 g/ha) of Phosphobacteria or 6 packets of Azophos (1200 g/ha) by seed treatment and 10 packets of Azospirillum (2000 g/ha) and 10 packets (2000 g/ha) of Phosphobacteria or 20 packets of Azophos(4000 g/ha)as soil application.

### 4. NUTRITIONAL DISORDERS

a) **Manganese deficiency** : Leaves develop interveinal chlorosis, chlorotic tissue, later develop light brown or husk coloured necrotic lesions.

b) **Zinc deficiency**: Middle leaves develop chlorosis in the interveinal areas and necrosis along the apical leaf margins. Mix 5 kg/ha of Zinc sulphate with 45 kg of soil and broadcast evenly in the beds after sowing.

*Note: Do not incorporate the micronutrient in the soil.*

### 5. SEED RATE

Adopt a seed rate of 5 kg/ha.

### 6. SPACING

a) Give a spacing of 30 cm between rows and 30 cm between plants. b) For rice fallows, seeds are broadcasted and thinned to maintain 11 plants/$m^2$.

**7. QUALITY OF SEEDS**

Select mature, good quality seeds free from pest and fungal damage.

**8. SEED TREATMENT**

Treat the seed with *Trichoderma*@ 4g/kg. This can be done just before sowing. SUCH SEEDS SHOULD NOT BE TREATED WITH FUNGICIDES or treat the seed with Thiram 4 g or Carbendazim at 2 g/kg of seeds before sowing.

**9. SOWING**

a) Sow the seeds preferably in lines.
b) Mix the seeds with four times its volume of dry sand and drop the mixture evenly along the furrows in which fertilizers are applied.
c) Sow the seeds to a depth of 3 cm and cover with soil.
d) The optimum time of sowing for VRI (SV) 1 sesame is second fortnight of February to first fortnight of March under summer irrigated conditions.

**10. WATER MANAGEMENT**

i) Irrigate at sowing and give life irrigation 7 days after sowing depending on the soil and climatic condition and allow excess water to percolate.
ii) Give one pre-flowering irrigation (25 days): One at flowering and one or two at pod setting. An irrigation at flowering period is critical.

*NOTE: The critical stage for moisture requirement is the flowering phase i.e, between 35th to 45th days of sowing. During the maturity phase, moisture status should be low. If more water is given during this phase, maturity of seeds is affected and filling up of the capsules will be poor. Therefore, stop irrigation after 65 days of sowing.*

**11. THINNING**

Thin out the seedlings to a spacing of 15 cm between the plants on the 15th day of sowing and 30 cm on 30th day of sowing. This operation is very important for the crop in order to induce basal branches.

**12. WEED MANAGEMENT**

Weed and hoe on $15^{th}$ and $35^{th}$ day of sowing. Apply Alachlor @ 20 kg/ha on $20^{th}$ day after sowing and irrigate the crop immediately.

**13. HARVESTING**

**a) Decide when to harvest**

i. Observe the crop, considering the average duration of the crop.
ii. Twenty five per cent of the leaves from the bottom are shed and the top leaves loose their colour and turn yellow at maturity.
iii. The colour of the stem turns yellow.
iv. The colour of the capsules turn yellow upto the middle.
v. Harvest before the bottom capsules turn brown.
vi. Examine the $10^{th}$ capsule from the bottom by opening. If the seeds attained the full color of the variety harvest may be taken up.
vii. If harvest is delayed/ the capsules will dehisce resulting in yield reduction.

**b) Harvest**

i. Pull out the plants from the bottom.
ii. Stack in the open, one over the other in a circle with the stems pointing out and the top portion pointing inside.
iii. Cover the top with straw, so that humidity and temperature increases.
iv. Cure like this for 3 days, shake the plants. About 75 per cent of the seeds will fall off.
v. Dry the plants for one more day and again shake the plants. All the mature seeds will fall off.
vi. Winnow the seeds and dry in the sun for 3 days. Stir once in 3 hours to give uniform drying.
vii. Collect the seeds and store in gunnies.

## CROP PROTECTION

### A. Pest management

**Economic threshold level for important pests**

| Pests | ETL |
|---|---|
| Shoot webber - Shoot damage | 2 larvae /$m^2$ or 10% plant damage |
| **Leaf amage** | 10 larvae/$m^2$ in the vegetative stage and 2 larvae/$m^2$ during the reproductive stage |

**Pest management strategies**

| Pests | Management strategies |
|---|---|
| **Shoot and Leaf webber**<br>*Antigastra catalaunalis*<br>**Pod borer**<br>*Elasmolomus (= Aphanus) sordidus*<br>**Gall fly**<br>*Asphondylia ricini* | ▪ Two sprayings with neem formulation 0.03%<br>▪ **Apply any one of the following insecticides 25 kg/ha on the 25th, 35th and 50th day of sowing :**<br>Endosulfan 4 D<br>Phosalone 4 D<br>Quinalphos 1.5 D<br>Malathion 5 D<br>▪ **Spray any one of the following :**<br>Phosalone 35 EC 1000 ml/ha<br>Quinalphos 25 EC 1000 ml/ha<br>Dichlorvos 6W SC 500 ml/ha<br>Monocrotophos 36 WSC 625 ml/ha<br>Endosulfan 35 EC 1000 ml/ha<br>Carbaryl 50 WP 1000 g/ha in 500 litre of water<br>Neem seed kernel extract (5%).<br>Neem oil 2% (two rounds)<br>▪ Use alternate insecticides each time and avoid the usage of same insecticide every time. |
| **Leaf webber**<br>*Antigastra catalaunalis* | |
| **Storage pests**<br>*Triboilum castaneum*<br>*Corcyra cephalonica* | ▪ **Dust any one of the following on gunny :**<br>Malathion 5 D<br>Phosalone 4 D<br>Carbaryl 10 D<br>▪ Mix one kg of activated clay with 100 kg of seeds after adequate drying of seeds |

### B. Disease management

| Seed treatment | ➢ Treat the seeds with any one of the following<br>▪ ***P. fluorescens* @ 10g/kg of seed**<br>▪ *T. viride* @ 4g/kg of seed.<br>▪ Thiram @ 4g/kg of seed<br>▪ Carbendazim @ 2g/kg of seed. |
|---|---|
| **Name of the Disease** | ***Management*** |
| ***Powdery mildew***<br>*Erysiphe cichoracearum* | Apply any one of the following<br>▪ Sulphur dust 25 kg/ha<br>▪ Wettable sulphur 25 kg/ha |
| ***Alternaria blight***<br>*Alternaria sesami* | ▪ Spray Mancozeb 1000g/ha |
| ***Cercospora leaf spot***<br>*Cercospora sesami* | ▪ Spray Mancozeb 1000g/ha |

| | |
|---|---|
| Root rot<br>*Macrophomina phaseolina (Rhizoctonia bataticola)* | **Biological control**<br>▪ Soil application of *P. fluorescens* or *T. viride* – 2.5 Kg / ha + 50 Kg of well decomposed FYM or sand at 30 days after sowing.<br>**Chemical**<br>▪ Spot drench Carbendazim – 1 gm/ litre |
| ***Phyllody***<br>Phytoplasma<br>Vector:<br>*Orosius albicinctus* | ❖ **Cultural Method**<br>▪ Remove and destroy infected plants.<br>❖ **Chemical**<br>▪ To control vector, spray Monocrotophos 36 or Dimethoate 30 EC 500 ml/ha combined with Intercropping of Sesamum + Redgram (6 : 1) |

## SEED PRODUCTION

**Variety Seed Production**

**Land requirement**
- Land should be free of volunteer plants.

**Isolation**
- Adopt 50 m for certified seed production

**Fertilizer**
- NPK @ 50:25:25 kg $ha^{-1}$ as basal
- Apply manganese sulphate @ 5 kg $ha^{-1}$ as basal

**Foliar application**
- Spray 1% DAP at the time of first flowering and again 10 days after first spray.

**Harvest**
- Harvest when 75–80% of the pods start yellowing and bottom 1 or 2 pods have dehisced.
- At this stage, the pod moisture content will be 50–60% and seed moisture content will be 25 – 30%.
- Seeds would have attained chocolate brown colour.
- Stack the plants in inverted position and allow them to dry for 3 – 4 days.

**Threshing**
- Remove the staked plants and beat with pliable bamboo stick

**Processing**
- Use 4/64" (1.6 mm) round perforated metal sieve for grading.

**Drying**
- Dry the seeds to 7- 8% moisture content and treat the seeds with Carbendazim or Thiram @ 2 g $kg^{-1}$ of seed.

**Storage**
- Use gunny or cloth bags for short term storage with seed moisture content of 8 - 9%
- Use polylined gunny bag for medium term storage with seed moisture content of 7 - 8%
- Use 700 gauge polythene bag for long term storage with seed moisture content of less than 5%.

**Mid-storage correction**
- Soak the seeds for two hours in double the volume of Disodium phosphate solution (3.60 g/100 lit of water).
- Remove the floaters
- Dry back the seeds to original moisture content 6 – 7%.

**Other management practices**
- As in crop management technique

# CASTOR (*Ricinus communis*)

## CROP IMPROVEMENT

### I. SEASON AND VARIETIES

**SEASON**

i) Rainfed
Adipattam (June - July)
ii) Gardenland (border)

VARIETIES : TMV 5, TMV 6
HYBRID : TMVCH 1
CO 1 (Perennial)

### II. DESCRIPTION OF CASTOR VARIETIES

| Particulars | CO 1 |
|---|---|
| Parentage | Pureline selection from Anamalai |
| Duration (months) | Perennial |
| Average yield (kg/ha) | |
| Pure crop | 2.5 kg/tree/yr |
| Mixed crop | .. |
| Oil percentage | 57 |
| Special features | |
| Stem colour | Pinkish green |
| Bloom (Waxy coat) | No bloom |
| Receme/ capsule | Bold, sparse settings, non dehiscent |
| Suitability | Bund crop and fit for raising in vacant areas |

| Particulars | TMV 5 | TMV 6 | TMVCH1 |
|---|---|---|---|
| Parentage | SA 2 X S248/2 | VP 1 x RC 962 | LRES 17 x TMV 5 |
| Duration (months) | 4 | 160 | 160 - 170 |
| Average yield(kg/ha) | | | |
| Pure crop | 850 | 950 | 1300 |
| Mixed crop | ... | 500 | 600 |
| Oil percentage | 50 | 51.9 | 51.7 |
| Special features | | | |
| Stem colour | Rose | Red | Red |
| Bloom (Waxy coat) | Trible | Double | Trible |
| Receme/ capsule | Spiny, non-dehiscent, resistant to leafhopper | Medium length, spiny capsules | Semi compact, spiny capsules |
| Suitability | Pure and mixed crop | Pure and mixed crop | Pure and mixed crop |

## CROP MANAGEMENT

**1. PREPARATION OF THE FIELD**

Plough two-three times with country or mould board plough.

**2. APPLICATION OF FERTILIZERS**

Spread 12.5 t/ha of FYM or compost evenly on the main field before last ploughing and incorporate in to soil by working a country plough.

NOTE: Do not leave FYM or compost exposed to sunlight as nutrients will be lost.

**3. SEED RATE**

Adopt a seed rate of 10 kg/ha for varieties and 5 kg/ha for hybrid.

**4. SPACING**

Adopt the following spacing for short and long duration strains.

| | **Spacing** | | **Spacing** |
|---|---|---|---|
| **Long duration** | 90 x 90 cm | **Short duration** | 60 x 45 cm |
| TMV 6, | 90 x 60 cm | TMV 5 | 60 x 30 cm |
| TMVCH 1 | 90 x 60 cm | | |

**5. APPLICATION OF FERTILIZERS**

Apply NPK fertilizers basally as per soil test recommendations as far as possible.

If soil test recommendations are not available, follow the blanket recommendation of 30:15:15 kg/ha NPK.

**6. PRE TREATMENT OF SEEDS**

a) Treat the seeds with Carbendazim @ 2 g/kg of seed. b) Soak the seeds in water for 20 hours.

**7. SOWING**

a) Sow the seeds adopting the recommended spacing. b) Place the seeds at depth 4 - 6 cm. c) Put one seed in each hole.

**8. GAP FILLING**

Gap fill on the 10th day of sowing.

**9. WEED MANAGEMENT**

Hoe and hand weed on 20th day of sowing to remove the weeds and repeat the operation on 40th day of sowing, if necessary.

**10. INTERCROPPING**

Raise one row of castor for every six rows of groundnut. In the case of late receipt of monsoon blackgram + castor at 6:1 ratio is recommended.

**11. HARVESTING THE CROP**

Observe the crop considering the average duration of the variety. i) One or more capsules show sign of drying. ii) Cut the matured racemes without damaging the secondaries. iii) Dry the capsule in the sun without heaping it in the shade. iv) Use castor sheller to separate the seeds or beat the dried capsule with wooden planks, winnow and collect the seeds.

## CROP PROTECTION

**A. Pest management**

| Pests | Management strategies |
|---|---|
| **Capsule borer**<br>*Conogethus punctiferalis* | ▪ **Spray any one of the following insecticides, thrice from flowering at three weeks interval** :<br>Malathion 50 EC 2.0 l /ha<br>Fenthion 100 EC 1.0 l /ha<br>Carbaryl 50 WP 2.0 kg / ha in 1000 l of water<br>▪ Apply endosulfan 4 D 25 kg/ha |
| **Semiloopers**<br>*Achaea janata*<br>*Paralellia algira* | ▪ **Spray any one of the following insecticides thrice from flowering at three weeks interval** :<br>Malathion 50 EC 2.0 l / ha<br>Fenthion 100 EC 1.0 l / ha<br>Carbaryl 50 WP 2.0 kg / ha in 1000 l of water<br>▪ Apply endosulfan 4 D 25 kg/ha |

| | |
|---|---|
| | ▪ Apply neem seed kernel extract 5% + Neem oil 2% |

**B. Disease management**

| Name of the Disease | *Management* |
|---|---|
| Botrytis<br>*Botrytis ricini* | ▪ Removal and destroyal of affected spikes.<br>▪ During cloudy weather and rainy season, prophylactic spray of carbendazim 2g/l of water twice at 15 day interval (or)<br>▪ Prophylactic spray of *P. fluorescens* @ 2g/l and seconds pray after a fortnight. |

**C. Nematode management**

| Nematode pest | Control measure |
|---|---|
| Reniform nematode,<br>*Rotylenchulus reniformis* | Seed treatment with *Pseudomonas fluorescens* @ 20 g/kg seed. |

## SEED PRODUCTION

**Variety Seed Production**

**Land requirement**

- Land to be used for seed production shall be free from volunteer plants.

**Isolation**

- Adopt 200 m all around the field

**Spacing**

- Adopt 60 x 20 m

**Fertilizer**

- Apply NPK @ 60 : 60 : 20 kg NPK ha-1 as basal

**Physiological maturity**

- Seeds attain physiological maturity 35 days after anthesis

**Processing**

- Grade the seeds using BSS 6 x 6 wire mesh sieve

**Seed Treatment**

- Treat the seed with mixture of halogen mixture @ 3g $kg^{-1}$ of seed along with carbendazim @ 2 g $kg^{-1}$ of seed

**Storage**

- Use gunny or cloth bags for short term storage with seed moisture content of 8 - 9%
- Use polylined gunny bag for medium term storage with seed moisture content of 7 - 8%
- Use 700 gauge polythene bag for long term storage with seed moisture content of less than 5%.

**Other management practices**

- As in crop management technique

**Hybrid Seed Production**

**LAND REQUIREMENT**
- Free from volunteer plants

**Isolation**
- Adopt 100 m distance for Certified seed production around the plot for other varieties and hybrids of castor

**Planting ratio**
- Adopt a ratio of 3:1 between female and male

**Season**
- First fortnight of September for production of more pistillate inflorescence in female line

**Fertilizer**
- Apply NPK @ 90:70:70 kg ha $^{-1}$

**Spacing**
- Adopt 90 X 30 cm

**Physiological Maturation**
- Forty five days after 50% flowering
- Seed coat will be mottled
- Well developed caruncle
- Slight Splitting of capsule

**Harvesting**
- Once over harvest after drying of capsules (browning) in 80% of plants
- The seeds from secondary raceme are better than primary and others

**Threshing**
- Use power oprerated thresher for shelling
- Avoid hand operated thresher to avoid mechanical damage

**Grading**
- Grade with two screen cleaner cum grader with top screen of 7.2mm for obtaining higher recovery of quality seeds

**Drying**
- Thin layer drying at $60^0$C for one hour reduce the moisture content to safe level ( 8- 10%) with high seed quality characters

**Storage**
- Slurry treat the seed with carbendazim @ 2g kg$^{-1}$ of seed or dry dress with *Achorus calamus* rhizome powder at the ratio of 1:100 for preservation of seed quality.
- Use 700 gauge polythene bag for long term storage
- The seeds of female parent are poor storer than male and hybrid

# SAFFLOWER (*Carthamus tinctorius*)

## CROP IMPROVEMENT

| **I. Season** | **Varieties** |
|---|---|
| Rainfed crop (November) | K1, CO 1 |

**Particularas of varieties**

| | **K1** | **CO 1** |
|---|---|---|
| Duration (days) | 120 | 125 |
| Yield (kg/ha) Rainfed | 700 | 800 |
| Percentage of oil | 31 | 33 |
| Plant character | Spiny | Non-spiny |

## CROP MANAGEMENT

### III. PREPARATION OF THE FIELD

**1. FIELD PREPARATION**

a) Plough with tractor 2-3 times with a mould board plough or 5 times with a country plough.
b) Break the clods in between the ploughings and bring the soil to a fine tilth.

**2. APPLICATION OF FYM**

a) Spread 12.5 t of FYM or compost or composted coir pith per ha evenly and incorporate in the soil. b) If the manure is not applied before commencement of ploughing, spread the manure evenly before the last ploughing and incorporate in the soil.
*NOTE: Do not leave the organic manure exposed to sunlight as nutrients will be lost.*

**3. APPLICATION OF FERTILIZERS**

Apply N at 20 kg/ha basally.

**4. SEED RATE**

Adopt a seed rate of 10 kg/ha.

**5. SPACING**

Adopt a spacing of 45 cm between rows and 15 cm between plants.

**6. SELECTION OF GOOD QUALITY SEEDS**

Select mature good quality seeds, free from pest damage and fungal attack.

**7. PRE-TREATMENT OF SEEDS WITH FUNGICIDES**

a) Treat with Carbendazim or Thiram at 4 g/kg of seed in a polythene bag and ensure a uniform coating of the fungicide over the seed. b) Treat the seeds 24 hours prior to sowing.
*NOTE: Seed treatment will protect the young seedlings from root rot disease in the early stage.*

**8. SOWING**

a. Sow the seeds in line at a depth of 2 to 3 cm and cover with soil. b. Sow using gorru or country plough.

*NOTE: First week of November is the best sowing time.*

**9. THINNING OUT SEEDLINGS**

Thin out the seedlings to a spacing of 15 cm between plants on the 15$^{th}$ day of sowing.

**10. WEED MANAGEMENT**

Hoe and weed on 25th and 40th day of sowing.

**11. HARVESTING**

i. Observe the crop considering the average duration of the crop.
ii. The leaves and entire plant loose their colour and turn brown at maturity.
iii. Cut the plants at the bottom.
iv. Keep the plants in the threshing floor and beat the plants (heads) with sticks till the mature seeds are separated.
v. Winnow the seed and dry in the sun.
vi. Collect and store the seeds in gunnies.

# SUNFLOWER ( *Helianthus annuus* )

## CROP IMPROVEMENT

### I. SEASON AND VARIETIES

| SEASON/DISTRICT | VARIETIES |
|---|---|
| **A. Rainfed :** | |
| **1. Adipattam (June-July)** | |
| Coimbatore, Erode, Salem, Namakkal, Tirunelveli, Dindigul, Dharmapuri, Tiruchirapalli, Perambalur, Karur | Variety : Morden, CO 4<br>Hybrid : TCSH 1, KBSH 1, KBSH 44, PAC 1091, MSFH 17 |
| **2. Karthigaipattam (Oct - Nov)** | |
| Cuddalore, Villupuram, Virudhunagar, Sivagangai , Ramanathapuram, Madurai, Dindigul, Theni, Tiruchirapalli, Perambalur, Karur, Tirunelveli | Variety : Morden, CO 4<br>Hybrid : TCSH 1, KBSH 1, KBSH 44, PAC 1091, MSFH 17 |
| **B. Irrigated :** | |
| **1. Margazhipattam (December - January)** | |
| Salem, Namakkal, Dharmapuri, Erode Coimbatore, Madurai, Dindigul, Theni, Tirunelveli, Thoothukudi | Variety : Morden , CO 4<br>Hybrid : TCSH 1, KBSH 1, KBSH 44, PAC 1091, MSFH 17 |
| **2. Chithiraipattam (April - May)** | |
| Coimbatore, Erode, Dharmapuri, Salem, Namakkal, Tiruchirapalli, Perambalur, Karur | Variety : Morden, CO 4<br>Hybrid : TCSH 1, KBSH 1, KBSH 44, PAC 1091, MSFH 17 |

### II. DESCRIPTION OF SUNFLOWER VARIETIES

| Particulars | MORDEN | CO 4 | TCSH 1 |
|---|---|---|---|
| Parentage | Selection from Cernianka 66 | Extract from the cross Dwarf x Surya | 234A x R 272 |
| Duration (days) | 75 | 80 - 85 | 85 |
| Yield (kg/ha) | | | |
| Rainfed | 900 | 1500 | 1800 |
| Irrigated | 1000 | 1750 | 2500 |
| Oil percentage | 36 | 39.7 | 41 |

| | | | |
|---|---|---|---|
| Ray floret | Light yellow | Light yellow | Light yellow |
| Height (cm) | 90 | 145 - 175 | 160 |
| Seed size & Seed Colour | Medium, black | Black | Black<br>Very few seeds may have strips |
| 1000 seed weight (g) | 44 | 56 | 60 |

## CROP MANAGEMENT

### 1. FIELD PREPARATION

Plough once with tractor or twice with iron-plough or three to four times with country-plough till all the clods are broken and a fine tilth is obtained.

### 2. APPLICATION OF FERTILIZERS

i) Spread 12.5 t/ha of FYM or compost or composted coir pith evenly on the field before the last ploughing and incorporate in the soil by working a country plough.

ii) Apply NPK fertilizers basally as per soil test recommendations. If soil test recommendations are not available, follow the blanket NPK/ha for both irrigated and rainfed crops.

| | Season | Blanket recommendation of Nutrients (kg/ha) | | |
|---|---|---|---|---|
| | | N | $P_2O_5$ | $K_2O$ |
| **Hybrids** | **IRRI** | 60 | 90 | 60 |
| | **RF** | *40* | *50* | *40* |
| **Varieties** | **IRRI** | 50 | 60 | 40 |
| | **RF** | *40* | *50* | *40* |

iii) Biofertilizer : Soil application - Mix 10 packets (2000 g/ha) of Azospirillum and 10 packets(2000 g/ha) of Phosphobacteria or 20 packets of Azophos(4000 g/ha) with 25 kg FYM and 25 kg soil and apply before sowing.

### 3. APPLICATION OF MICRONUTRIENTS

a) Mix 12.5 kg/ha of micronutrient mixture formulated by the Department of Agriculture, Tamil Nadu with enough sand to make total quantity of 50 kg/ha.
b) Apply the mixture over the furrows and top two thirds of the ridges before sowing.
c) Do not incorporate the mixture in the soil.
   i) To overcome manganese deficiency, foliar spray of 0.5% $MnSO_4$ on 30, 40 and 50$^{th}$ day / after sowing.
   ii) For zinc deficiency, apply 25 kg/ha $ZnSO_4$ as basal, or 0.5% $ZnSO_4$ spray on 30, 40 and 50$^{th}$ day after sowing.

### 4. FORMING RIDGES AND FURROWS

i. Form ridges and furrows 6 m long.
ii. Use bund-former or ridge plough to economise and
iii. Form irrigation channels across and ridges according to the topography of the field.

### 5. SEED RATE

| | Rainfed | Irrigated |
|---|---|---|
| Varieties | 7 kg/ha | 6 kg/ha |
| Hybrids | 5 kg/ha | 4 kg/ha |

### 6. SEED TREATMENT

Soaking seeds in 2% $ZnSO_4$ for 12 hrs and shade drying is recommended for rainfed sowing.

i) Treat the seed with *Trichoderma* @4g/kg. This can be done just before sowing. It is compatible with biofertilizers. Such seeds should not be treated with fungicides.
ii) Treat the seeds with Carbendazim or Thiram at 2 g/kg of seed.

iii) Treat the seeds 24 hours prior to sowing.
iv) Azospirillum : Use 3 packets of Azospirillum (600 g/ha) and 3 packets (600 g/ha) of Phosphobacteria or 6 packets of Azophos (1200 g/ha) for treating seeds using rice kanji as binder. Dry the treated seeds in shade for 15 minutes and sow immediately.
v) Moist hydration for 24 hours in moist gunny bags followed by drying and seed dressing with Thiram @ 2g/kg to enhance field emergence.
vi) Seeds dried to 8 - 9% moisture content, treated with Thiram @ 2g/kg and packed in polylined (300 guage) cloth bag can store upto 9 months with 70% germination.

## 7. SOWING

Spacing : Hybrids : 60 cm x 30cm
Varieties : 45 cm x 30cm

i) Place the seeds at a depth of 3 cm along the furrows in which the fertilizer mixture is placed and cover with soil. Put two seeds per hole.
ii) Irrigate the crop according to the different growth stages. Regulate irrigation according to the following growth phase.
Pre-sowing irrigation; Life irrigation; 20th day after sowing; Early bud development; Flowering-2 irrigations and Seed development-2 irrigations; Flowering period is critical.

## 8. THINNING

Thin out seedlings leaving only one healthy and vigorous seedling in each hole on the 10$^{th}$ day of sowing.

## 9. WEED MANAGEMENT

i) Apply Fluchloralin at 2.0 l/ha before sowing and incorporate or apply as pre-emergence spray on 5 day after sowing followed by irrigation or apply Pendimethalin as pre-emergence spray 3 days after sowing. The spray of these herbicides has to be accomplished with Back Pack/Knapsack/Rocker sprayer fitted with flat fan nozzle using 900 l water/ha as spray fluid.
All the herbicide application is to be followed by one late hand weeding 30 - 35 days after sowing.
ii) Hoe and hand weed on the 15th and 30th day of sowing and remove the weeds. Allow the weeds to dry for 2 - 3 days in the case of irrigated and then give irrigation.

## 10. WATER MANAGEMENT

Irrigate immediately after sowing followed by an irrigation on 4 - 5th day and later at intervals of 7 to 8 days according to soil and climatic conditions, seeding, flowering and seed development stage (ie) two weeks before and after flowering.

## 11. SPRAYING NAA

i) Spray the hormone Napthalene Acetic Acid (NAA) at 20 ppm concentration (280 g NAA in 625 litres of water per ha) on the 30th and 60th day of sowing.
ii) Use a high volume sprayer and give a thorough coverage of the entire plant.
iii) Do not use brackish water.

## 12. SULPHUR FERTILIZATION

Apply sulphur @ 20 kg/ha through ammonium sulphate or single super phosphate.

## 13. Boron application

Spray borax @ 0.2 % (2g/l of water) to capitulum at ray floret opening stage to improve seed set and seed filling.

## 14. IMPROVING SEED SET BY MECHANICAL MEANS

a. During the mid flowering phase, improve pollination by :
i. Mild rubbing of the capitulum with the hand covered with soft cloth or
ii. Rubbing two flowers face to face gently.
b. The mid-flowering phase are: 58 to 60 days of planting for long duration varieties, 45 to 48 days

of planting for short duration varieties.

c. Do this operation in the morning hours between 9.0 and 11.00 am when pollen shedding is high.

d. Keeping bee hives at the rate of 5/ha improves seed setting.

## 15. JUDGE WHEN TO HARVEST

Observe the bracts on the backside of the capitula. When they turn lemon yellow, the heads harden and the crop is ready for harvest.

Bird damage: Use of reflective ribbons scares the birds effectively and thus prevents loss of grain.

## 16. HARVESTING

i. Cut the capitula (flower heads) only

ii. Thresh and clean

a. Immediately after harvest, dry the heads in the sun for 3 days.

b. Spread the heads in thin layer and give turning once in 3 hours.

*NOTE: Do not heap or store the heads before drying properly as mould fungi will develop and spoil the grain quality.*

c. Thresh using a mechanical thresher, or beat with a stick and separate the grains.

d. Winnow and clean the seeds

e. Dry the seeds again in the sun for another two days

f. Store in gunny bags

# CROP PROTECTION

**A. Pest management**

| Pests | Management strategies |
|---|---|
| **Weevil**<br>*Myllocerus* spp. | ▪ Hand pick the *Helicoverpa* larvae and destroy.<br>▪ **Spray any of the following insecticides :**<br>Endosulfan 35 EC 1000 ml/ha<br>Fenthion 100 EC 500 ml/ha<br>Phosalone 35 EC 1000 ml/ha<br>Phenthoate 50 EC 500 ml/ha |
| **Tobacco cut worm**<br>*Spodoptera litura* | ▪ **Dust any one of the following :**<br>Endosulfan 4 D 25 kg/ha<br>Phosalone 4 D 25 kg/ha |
| **Gram pod borer**<br>*Helicoverpa armigera* | ▪ Insecticidal application at the time of bee visit is toxic to honey bees. So, apply the insecticides after 4 pm when the bee activity is minimum.<br>▪ Do not spray insecticides on the same day when NAA is sprayed |
| **Leaf hopper**<br>*Amrasca devastans* | ▪ Treat seed with imidacloprid 70 WS at 7 g/kg protected the sunflower plants from leaf hopper upto 7 weeks.<br>▪ Spray Imidacloprid 200 SL at 100 ml/ha |

**B. Disease management**

| Seed treatment | ➢ **Treat the seeds with any one of the following:**<br>▪ *T. viride* @ 4g/kg of seed.<br>▪ Thiram @ 4g/kg of seed<br>▪ Carbendazim @ 2g/kg of seed. |
|---|---|
| **Name of the Disease** | ***Management*** |
| ***Alternaria leaf spot***<br>*Alternaria helianthi* | ▪ Spray Mancozeb 1000g/ha |
| ***Rust***<br>*Puccinia helianthi* | ▪ Spray Mancozeb 1000g/ha |

| | |
|---|---|
| ***Charcoal rot***<br>*Macrophomina phaseolina (Rhizoctonia bataticola)* | ❖ **Biological control**<br>▪ Soil application *P. fluorescens* or *T. viride* – 2.5 Kg / ha + 50 Kg of well decomposed FYM or sand at 30 days after sowing.<br>❖ **Chemical**<br>▪ Spot drench Carbendazim – 1 gm/ litre |
| ***Head rot***<br>*Rhizopus sp* | ▪ Spray Mancozeb 1000g/ha in case of intermittent rainfall at the head stage, directing the spray to cover the capitulum.<br>▪ Repeat fungicidal application after 10 days if humid weather continues. |
| ***Necrosis virus disease***<br>*Tobacco streak virus (Ilarvirus)*<br>Vector<br>Thrips | ❖ **Cultural Method**<br>▪ Raise sorghum as border crop (One month prior to sunflower sowing)<br>❖ **Chemical**<br>▪ Imidacloprid seed treatment 2g/kg<br>▪ 0.01 % Imidacloprid foliar spray at 30 & 45 DAS. |

## SEED PRODUCTION

**Variety Seed Production**

**Land requirement**

- Same kind of crop should not be grown in the previous year
- It can be same variety if it is certified.

**Isolation**

- Adopt 200 m for certified seed production

**Spacing**

- Adopt a spacing of 45 x 30 cm

***Presowing seed treatment***

- Soaking in water @ 1:1 volume for 16h enhanced the germination and field establishment.

**Fertilizer**

- Apply NPK @ 60:45:45 kg $ha^{-1}$.

**Foliar application**

- Spray 0.5% borax at the stage of button opening for increased seed set.

***Supplementary pollination***

- During flowering rub the heads with muslin cloth between 8–11 AM at alternate days till the completion of flowering.

**Harvesting**

- Harvest when the drooping peduncular receptacle turns lemon or pale yellow in colour
- At this stage the seed moisture content will be 25 % and the seeds will be black in colour.
- Cut and dry the heads immediately until the seeds contain 15–16 per cent moisture.
- Separate the seeds with a mechanical thresher or manual labour and pre-clean.

**Processing**

- Use sieve 9/64" (3.6 mm) round perforated metal sieve or BSS 7x7 wire mesh sieve.
- Remove broken and dehulled seed

**Seed Treatments**

- Treat with Thiram 75% WP @ 2 g in 5 ml of water $kg^{-1}$ of seed.

**Storage**

- Use gunny or cloth bags for short term storage with seed moisture content of 8-9%
- Use polylined gunny bag for medium term storage with seed moisture content of 8-9%
- Use 700 gauge polythene bag for long term storage with seed moisture content of less than 8%.

**Mid storage seed treatment**

- Soak sunflower achenes in dilute solution of sodium dihydrogen phosphate ($10^{-4}$ M) for 2 h followed by dry back to original moisture content (8%), when the germination of seeds falls 5-10% lesser than MSCS level (70%)

***Hybrid Seed Production***

**Land requirement**

- Same kind of crop should not be grown in the previous year
- It can be same variety if it is certified.

**Isolation**

- Adopt 400 m for certified seed production

**Planting ratio :**

- Adopt a planting ratio of 4 : 1 / 3:1 (female : male)

**Border rows**

- Four number of male rows all around the field

**Harvesting**

- Harvest the male line first and remove the produce from the field then harvest the hybrid seed

**Other management practices**

As that of varieties / crop management techniques

# NIGER (Guizotia abyssinica)

## CROP IMPROVEMENT

### I. SEASON AND VARIETIES

| | | |
|---|---|---|
| Season | : | |
| 1. Adipattam | : | June - July |
| 2. Purattasipattam | : | Sept.-Oct. |
| Areas of adoption | : | Hosur and Denkanikotta taluks of Dharmapuri district and hilly regions of Shevroy, Kolli hills, Jawad hills and Thalavadi hills |

### II. DESCRIPTION OF VARIETIES

| | | |
|---|---|---|
| Variety | : | Paiyur 1 |
| Parentage | : | Mass selection from composite II |
| Duration (days) | : | 80 |
| Yield (kg/ha) | : | 260 |
| Oil content (%) | : | 44.6 |
| Plant height (cm) | : | 80-85 |
| Branches | : | Profuse |
| Seed | : | Bold |
| Colour of seed | : | Brown |
| 50% flowering (days) | : | 50 |

## CROP MANAGEMENT

### III. PREPARATION OF THE FIELD

**1. FIELD PREPARATION**

a) Plough with tractor 2-3 times with a mould board plough or 5 times with a country plough.
b) Break the clods in between the ploughings and bring the soil to a fine tilth.

**2. APPLICATION OF FYM**

a) Spread 12.5 t of FYM or compost or composted coir pith per ha evenly and incorporate in the soil.
b) If the manure is not applied before commencement of ploughing, spread the manure evenly before the

last ploughing and incorporate in the soil.
*NOTE: Do not leave the organic manure exposed to sunlight as nutrients will be lost.*

**3. APPLICATION OF FERTILIZERS**

Apply N at 20 kg/ha basally.

**4. SEED RATE**

Adopt a seed rate of 5 kg/ha.

**5. SPACING**

Adopt a spacing of 30 cm between rows and 10 cm between plants.

**6. SELECTION OF GOOD QUALITY SEEDS**

Select mature good quality seeds, free from pest damage and fungal attack.

**7. PRE-TREATMENT OF SEEDS WITH FUNGICIDES**

a) Treat with Carbendazim or Thiram at 4 g/kg of seed in a polythene bag and ensure a uniform coating of the fungicide over the seed. b) Treat the seeds 24 hours prior to sowing.
*NOTE: Seed treatment will protect the young seedlings from root rot disease in the early stage.*

**8. SOWING**

a. Sow the seeds in line at a depth of 2 to 3 cm and cover with soil. b. Sow using gorru or country plough.

**9. THINNING OUT SEEDLINGS**

Thin out the seedlings to a spacing of 10 cm between plants on the 15th day of sowing.

**10. WEED MANAGEMENT**

Hoe and weed on $20^{th}$ and $35^{th}$ day of sowing.

**11. HARVESTING**

i) Observe the crop considering the average duration of the crop.
ii) The leaves and entire plant loose their colour and turn brown at maturity.
iii) Cut the plants at the bottom.
iv) Keep the plants in the threshing floor and beat the plants (heads) with sticks till the mature seeds are separated.
v) Winnow the seed and dry in the sun.
vi) Collect and store the seeds in gunnies.

## SEED PRODUCTION

### Variety seed production

**Land requirement**

- Same kind of crop should not be grown in the previous season.

**Isolation**

- Adopt 200 m all around the plot for certified seed production

**Spacing**

- Adopt 30 x 30 cm

**Fertilizer**

- Apply 40 : 40 : 20 kg of NPK $ha^{-1}$ as basal application

**Physiological maturation**

- Harvest when seeds attain physiological maturation ie 85 days after sowing.

**Processing**

- Grade the seeds using BSS 16 x 16 wire mesh sieve

**Seed Treatment**

- Treat the seed with halogen mixture @ 3g $kg^{-1}$ of seed

**Storage**

- Use gunny or cloth bags for short term storage with seed moisture content of 8 - 9%
- Use polylined gunny bag for medium term storage with seed moisture content of 7 - 8%
- Use 700 gauge polythene bag for long term storage with seed moisture content of less than 5%.

**Mid storage correction**

- Soak the seed in double the volume of disodium phosphate ($10^{-4}$ M) solution for 3 h and dry back to original moisture content.

**Other management practices**

- As in crop management techniques

# COTTON

## CROP IMPROVEMENT

## I. SEASON AND VARIETIES

| District/Season | | Varieties/Hybrids |
|---|---|---|
| **Irrigated (Main)**<br>**Winter Irrigated (Aug – Sep)** | | |
| Coimbatore, Erode, Madurai, Dindigul, Theni | : | MCU 5, MCU 5 VT, Suvin, TCHB 213*, MCU 12, MCU 13, Surabhi, Sumangala, Sruthi* |
| Dharmapuri | : | MCU 5, TCHB 213*, MCU 12, MCU 13, Suvin |
| Salem, Namakkal | : | MCU 5, Suvin, TCHB 213*, MCU 12, MCU 13, Sumangala |
| Cuddalore, Villupuram | : | MCU 5, MCU 12, MCU 13, LRA 5166, TCHB 213*, SVPR 2, Surabhi, |
| **Summer – Irrigated (Feb – Mar)** | | |
| Erode | : | MCU 5, MCU 5 VT, SVPR 2 , Supriya, MCU 12, MCU 13, TCHB 213 |
| Madurai, Dindigul, Theni | : | MCU 5, MCU 5 VT, MCU 12,MCU 13, SVPR 2, Supriya, |
| Ramanathapuram, Virudhunagar, Sivagangai, Tirunelveli, Thoothukudi | : | MCU 5, MCU 5 VT, SVPR 2, MCU 12, MCU 13 |
| **Rainfed (Sep – Oct)** | | |
| Madurai, Dindigul, Theni | : | LRA 5166, K11, KC 2, SVPR 2 |
| Ramanathapuram, Virudhunagar, Sivagangai | : | LRA 5166, K 11, KC 2, SVPR 2 |
| Tirunelveli, Thoothukudi, Dharmapuri | : | LRA 5166, K 11, KC 2, SVPR 2 |
| **Rice Fallow** | | |
| Thanjavur, Tiruvarur, Nagapattinam, Parts of Trichirapalli, Perambalur, Karur, Cuddalore and Villupuram | : | MCU 7, SVPR 3, Anjali |

* Hybrid

## II. PARTICULARS OF COTTON VARIETIES/HYBRIDS

| Varieties/ Hybrids | Parentage | Season | Irrigated/ Rainfed | Mean yield of seed (kg/ha) | Special features |
|---|---|---|---|---|---|
| MCU 5 | Multiple cross | Aug-Oct<br>Feb-Mar | Irrigated | 1850 | Extra long staple (29 mm MHL), Can spun upto 70s, ginning 34% |
| MCU 7 | X ray irradiation of X L 1143 EE | Jan-Feb | Rice fallows | 1330 | Medium staple (23.7 mm MHL), Can spun upto 30s, early maturing with 33.2% ginning outturn. Tolerant to Black arm |
| MCU 12 | Derivative from the cross LRA 5166 x MCU 11 | Aug-Oct | Irrigated | 2000 | Shorter in duration than MCU 5, GOT 34.8%<br>Can spun upto 50s |

| | | | | | |
|---|---|---|---|---|---|
| MCU 13 | It is a multiple cross derivative involving the parents of [(TCH 665 x LS 149) x (TCH 665 x TCH 21)] x (TCH 21 x EECH) x (TCH 92-7 x EECH) | Aug- Oct Jan-Feb | Irrigated | 2200 | Early duration Can spun upto 50s |
| LRA 5166 | Laxmi x Reba B.50 x AC 122 | Aug-Oct Jan –Feb | Irrigated Rainfed | 1800 725 | Medium staple (29 mm), Can spun upto 40s, ginning 36.2% |
| MCU 5 VT | Reselection from MCU 5 | Aug-Oct Jan –Feb | Irrigated | 2000 | Extra long staple, *Verticillium* wilt tolerant |
| Supriya | MCU 5 x C 1998 | Aug-Oct Jan-Feb | Irrigated | 2000 | White fly tolerant |
| Anjali | LRA 5166 x (Khandwa 2 x Reba B 50) BC 2 | Jan –Feb | Irrigated (Rice fallows) | 1800 | Dwarf, semi compact plant type |
| Surabhi | MCU 5 VT (MCU 5 x *G.mexicanum*) | Aug-Oct | Irrigated | 2200 | Exta long staple, *Verticillium* wilt resistant |
| Sumangala | CW 134 x Reba B 50 x Khandwa 2 | Sept-Oct Jan –Feb | Irrigated Rainfed | 2000 1200 | Suitable for rainfed tract |
| Sruthi | 70 E x RSP 4 | Sept-Oct Jan –Feb | Irrigated | 2500 | Early duration |
| K 11 | (0794-1-DX H 876) x (0794-1-DX H 450) Multiple Hybrid derivative | Sept – Oct | Rainfed | 1100 | Better fibre properties with lesser pest incidence than K10 |
| Suvin | Hybrid derivative from the cross Sujatha x St. Vincent | Aug-Oct | Irrigated | 1020 | Extra long staple cotton with 28% ginning outturn and 32 mm MHL, spins 100s |

## II. PARTICULARS OF COTTON VARIETIES/HYBRIDS (CONTD...)

| Varieties/ Hybrids | Parentage | Season | Irrigated/ Rainfed | Mean yield of seed (kg/ha) | Special features |
|---|---|---|---|---|---|
| TCHB 213 | Interspecific Hybrid of TCH 1218 (*G.hirsutum*) and TCB 209 (*G.barbadense*) | Aug-Sep | Irrigated | 2215 | High yielding, early maturing, Tolerant to leaf spot diseases |
| SVPR 2 | TSDT 22 x JR 36 | Feb - Mar Sep-Oct | Irrigated Rainfed | 2000 | High ginning out turn of 36.4%, medium staple (24.3 mm), can spin 30's, suited to summer irrigated, winter rainfed and tankfed rice fallow tracts of Tamil Nadu. |

| | | | | | |
|---|---|---|---|---|---|
| SVPR 3 | Selection from L.H 900 x 1301 D.D | Jan-Feb | Rice fallows | 1800 | Suitable for rice fallow tract. early duration (135-140 days). Tolerant to drought, leafhopper, alternaria spot, black arm disease. |
| KC 2 | MCU 10 x KC 1 | Sep - Oct | Rainfed | 1000 | High ginning out turn of 37.5%, medium staple cotton - 24.4 mm, Suited for rainfed black cotton soil of Tirunelveli, Thoothukudi and Virudhunagar Districts. |

**CROP MANAGEMENT**

## I. PREPARATION OF FIELD FOR IRRIGATED COTTON CROP

### 1. PREPARATION OF THE FIELD

i) Prepare the field to get a fine tilth.
ii) Chiselling for soils with hard pan: Chisel the soils having hard pan formation at shallow depths with chisel plough at 0.5 M interval, first in one direction and then in the direction perpendicular to the previous one, once in three years. Apply 12.5 t farm yard manure or composted coir pith/ha besides chiselling to get increased yield
iii) If intercropping of Greengram/Soyabean is proposed, prepare the main field, so as to provide ridges and furrows to take up sowing 20 days prior to cotton sowing.

### 2. APPLICATION OF FYM OR COMPOST

Spread 12.5 t of FYM or compost or 2.5 t of vermicompost per ha if available, uniformly on the unploughed soil.

### 3. APPLICATION OF BIOFERTILIZER

Seed treatment with 3 packets of Azospirillum (600 g/ha) and 3 packets (600 g/ha) of Phosphobacteria or 6 packets of Azophos(1200 g/ha). In addition apply and 10 packets of Azospirillum (2000 g/ha) and 10 packets(2000 g/ha) of Phosphobacteria or 20 packets of Azophos(4000 g/ha)mixed with 25 kg FYM and 25 kg of soil on the seed line. This saves 25% nitrogen besides increasing yield.

### 4. FORMATION OF RIDGES AND FURROWS

i) Form ridges and furrows 10 m long with appropriate spacing depending upon the variety.
ii) Use ridge plough or bund former to form ridges so as to economise on cost of cultivation.
iii) In fields with ragi stubbles, just dibble cotton seeds at the specified spacings.
v) Adopt the following spacing between ridges for different varieties/hybrids.

| Varieties/Hybrids | Spacing between ridges (cm) |
|---|---|
| MCU 5, SVPR 2, LRA 5166, MCU 12, MCU 13 | 75 |
| TCHB 213 | 120 |
| Suvin | 90 |
| MCU 7 | 60 |

*NOTE: Adopt higher spacing rows in fertile soils by 15 to 30 cm.*

### 5. APPLICATION OF INORGANIC FERTILIZERS

i) Apply NPK fertilizers as per soil test recommendations.
ii) If soil test recommendations are not available, follow the blanket recommendation for the different varieties.

| Varieties / Hybrids | Quantity of fertilizers (Kg/ha) | | |
|---|---|---|---|
| | **N** | **$P_2O_5$** | **$K_20$** |
| MCU 7, SVPR 3 | 60 | 30 | 30 |
| MCU 5, MCU 5 VT, MCU 12, MCU 13,Suvin, SVPR 2 | 80 | 40 | 40 |
| TCHB 213, | 120 | 60 | 60 |

iii) If basal application could not be done, apply on the 25th day after sowing.

iv) Apply 50 per cent of N and K full dose of $P_2O_5$ as basal and remaining ½ N and K at 40 – 45 DAS for varieties. For hybrids apply N in three splits *viz*., basal, 45 and 65 DAS.

v) Foliar application of 2% DAP + 1% KCl will improve kapas yield.

vi) Apply the fertilizers in a band, two-thirds of the distance from the top of the ridge, and incorporate.

**6. APPLICATION OF MICRONUTRIENT MIXTURE**

Mix 12.5 kg of micronutrient mixture formulated by the Department of Agriculture, Tamil Nadu with enough sand to make a total quantity of 50 kg for one ha.

**7. NUTRITIONAL DISORDERS' CORRECTION**

a) In the case of Zinc deficient soils $ZnSO_4$ @ 50 kg/ha as basal or $ZnSO_4$ 0.5% spray thrice at 45, 60 and 75 DAS.

b) When reddening occurs in leaves apply 5% $MgSO_4$ Urea (1.0%) and $ZnSO_4$ (0.10%) as foliar spray on 50th and 80th day to correct this malady. In Mg deficient areas apply $MgSo_4$ @ 20 kg/ha basally.

## II. MANAGEMENT OF MAIN FIELD OPERATIONS

**I. SEED RATE**

Adopt the following seed rates for different varieties/hybrids

| **Varieties / Hybrids** | **Quantity of seed (Kg/ha)** | | |
|---|---|---|---|
| | **With fuzz** | **Delinted** | **Naked** |
| MCU 5, MCU 5 VT, MCU 7, MCU 12, MCU 13 | 15.00 | 7.50 | .. |
| SVPR 2 | 15.00 | .. | .. |
| KC 2 | 20.00 | 15.00 | .. |
| SUVIN | .. | .. | 6.00 |
| TCHB 213 | 2.5 | 2.0 | .. |

**2. SPACING**

In a pure crop of cotton, adopt the spacing as below for the different varieties.

| **Varieties / hybrids** | **Spacing ( cm)** | |
|---|---|---|
| | **Between rows** | **Between plants** |
| MCU 5, MCU 5 VT, MCU 12, MCU 13 | 75 | 30 |
| LRA 5166, SVPR 2 | | |
| KC 2 | 45 | 15 |
| SUVIN | 90 | 45 |
| TCHB 213 | 120 | 60 |
| MCU 7, SVPR 3 | 60 or 75 * | 30 |

* Fertile soils

a. If cotton intercropped with other crops, one paired row of cotton is alternated with three rows of intercrop and the total population of cotton crop is maintained at the same level as in the case of pure crop.
b. For intercropping with Greengram / Soyabean, complete the sowing and irrigation 20 days prior to cotton sowing on one side of the ridge.

| Varieties/hybrid | Spacing for cotton crop ( cm) | | |
|---|---|---|---|
| | Within Paired row | Between Paired rows | Between plants |
| MCU 5, MCU 5 VT, MCU 12,MCU13 | 60 | 90 | 30 |
| SUVIN | 80 | 100 | 45 |
| TCHB 213 | 100 | 140 | 60 |

Plant two rows of intercrop between each paired row of cotton

| Intercrop | Seed rate(kg/ha) | Spacing (cm) | |
|---|---|---|---|
| | | Rows | Plants |
| Blackgram | 12.5 | 30 | 10 |
| Greengram | 12.5 | 30 | 10 |
| Cowpea | 7.5 | 30 | 20 |
| Soyabean | 20.0 | 30 | 10 |

For higher returns, advance sowing of either greengram or soyabean 20 days before sowing of cotton in winter season.

**3. ACID-DELINTING OF COTTON SEEDS**

i) Choose plastic bucket bucket for acid delinting of seeds.
ii) Do not use earthen wares, metal vessels, porcelain wares or wooden drum for acid delinting as concentrated sulphuric acid will corrode them.
iii) Put the required quantity of seeds in the container and add commercial concentrated sulphuric acid at the rate of 100 ml per kg of fuzzy seed.
iv) Stir vigorously and continuously with a wooden stick for 2 to 3 minutes till the fuzz sticking to the seeds is completely digested and the seed coat attains a dark brown colour of coffee powder.
v) Add water to fill the container. Drain the acid water and repeat the washing 4 or 5 times to remove any trace of acid.
vi) Remove the floating, ill-filled and damaged seeds while retaining the healthy and good seeds which remain at the bottom.
vii) Drain the water completely and dry the delinted seeds in shade.

*NOTE: Acid delinting has the following advantages:*

i) Eliminates some externally seed borne pathogenic organisms.
ii) Kills eggs, larvae and pupae of pink boll worm.
iii) Helps to remove immature, ill-filled, cut and damaged seeds.
iv) Makes seed dressing more effective and easy
v) Facilitates easy sowing and good germination.

**4A. PRE-TREATMENT OF ACID DELINTED SEEDS WITH FUNGICIDES**

i) Treat the delinted seeds with talc formulation of Trichoderma viride @ 4g/kg of seed or with Carbendazim (or) Thiram @ 2g/kg of seed.
Biocontrol agents are compatible with biofertilizers.
First treat the seeds with biocontrol agents and then with biofertilizers.
Fungicides and biocontrol agents are incompatible.
ii) Treat the delinted fungicide treated seeds with 3 packets (600 g) of Azospirillum and 3 packets of phosphobacteria 600g (or) 6 packets of Azophos (1200 g) and sow immediately.

**4B. SEED HARDENING**

Soak the seeds in equal volume of Pungam leaf extract (1%) for 8 hours and dry back to original moisture to increase germination and vigour. Dry the seeds in shade.

Seed pelleting: Seeds coated with arappu leaf powder (100 g/kg) along with DAP (40 g/kg), micronutrient mixture (15 g/kg) and Azospirillum (200 g/kg) phosphobacteria (200 g/ha) or Azophos (400 g/ha) using 5% maida solution or gruel as adhesive (300 ml/kg) to increase the germination and vigour.

**5. SOWING**

i) Dibble the seeds at a depth of 3-5 cm on the side of the ridge 2/3 height from the top and above the band where fertilisers and insecticides are applied, maintaining the correct spacing and then cover seeds with soil.
ii) In the case of intercropping, sow the seeds of the intercrop in between the paired rows of cotton in a row of 5 cm apart and cover the seeds.
iii) Sow the required number of seeds in each hole.

| Varieties / hybrids | No. of seeds / hole | |
|---|---|---|
| | Fuzzy seeds | Delinted seeds |
| Hybrids | 2 | 1 |
| Varieties | 3 | 2 |

**6. WEED MANAGEMENT**

i) Apply Pendimethalin @ 3.3 l/ha three days after sowing, using a hand operated sprayer fitted with deflecting or fan type nozzle. Sufficient moisture should be present in the soil at the time of herbicide application. This will ensure weed free condition upto 40 days.
ii) One hand weeding on 45 DAS will keep weed free environment upto 60 DAS.
iii) Hoe and hand weed between 18th to 20th day of sowing, if herbicide is not applied at the time of sowing.

**7. GAP FILLING**

a. Take up gap filling on the 10th day of sowing.
i) In the case of TCHB 213, raise seedlings in polythene bags of size 15 x 10 cm.
ii) Fill the polythene bags with a mixture of FYM and soil in the ratio of 1:3.
iii) Dibble one seed per bag on the same day when sowing is taken up in the field.
iv) Pot water and maintain.
v) On the 10th day of sowing, plant seedlings maintained in the polythene bags, one in each of the gaps in the field by cutting open the polythene bag and planting the seedling along with the soil intact and then pot water.
b. In the case of all other varieties, dibble 3 to 4 seeds in each gap and pot water.

**8. THINNING**

Thin out the seedlings on the 15th day of sowing. In the case of fertile soils, allow only one seedling per hole, whereas in poor soil allow two seedlings per hole.

**9. TOP DRESSING**

i) Top dress 50% of the recommended dose of N and K on 40 – 45 DAS for varieties.
ii) Top dress 1/3rd of recommended dose of N on 40-45 DAS and the remaining 1/3rd on 60-65$^{th}$ DAS for hybrids.

**10. RECTIFICATION OF RIDGES AND FURROWS**

Reform the ridges and furrows after first top dressing in such a way that the plants are on the top of the ridges and well supported by soil.

**11. SPRAYING OF NAPTHALENE ACETIC ACID (NAA)**

Spray 40 ppm NAA at 60 and 90 days after sowing on the crop to prevent early shedding of buds and squares and to increase the yield.

*NOTE: 40 mg of NAA dissolved in one litre of water will give 40 ppm.*

**12. MANAGEMENT STRATEGIES FOR DELAYED UMMER IRRIGATED COTTON SOWING**

KCI 1% spray, twice on 50 and 70 DAS for delayed sowing (first fortnight of March) of summer irrigated cotton in rice-cotton cropping system for Srivilliputhur region.

**13. ARRESTING TERMINAL GROWTH:**

Nip the terminal portion of the main stem as indicated below:

For varieties having less than 160 days duration nip the terminal portion of the main stem beyond the 15th node (75 to 80 DAS) and for varieties and hybrids having more than 160 days duration beyond the 20th node (85 - 90 DAS).

## III. WATER MANAGEMENT

Regulate irrigation according to the following growth phases of the crop.

| Stages | No. of Irrigations | Days after dibbling seeds | |
|---|---|---|---|
| | | Light soil | Heavy soil |
| **Germination Phase(1-15 days)** | | | |
| Irrigate for germination | 1 | Immediately after sowing | Immediately after sowing |
| and establishment | 2 | Give a life irrigation on | Give a life irrigation on |
| | | 5th day of sowing to | 5th day of sowing to |
| | | facilitate the seedlings | facilitate the seedlings |
| | | to emerge out | to emerge out |
| **Vegetative phase (16-44 days)** | | | |
| Regulate | 1 | Irrigate on the 20th or | Irrigate on the 20th or 21st day |
| | | 21st day of sowing, three | of sowing, three days after |
| | | days after hoeing and | hoeing and weeding |
| | | Weeding | |
| | 2 | Irrigate again on | Irrigate again on |
| | | the 35th or 36th | the 40th day of |
| | | day of sowing | Sowing |
| **Flowering phase (45-100daysfor hybrids and 87 days for varieties)** | | | |
| Irrigate copiously | 1 | 48th day | 55th day |
| | 2 | 60th day | 70th day |
| | 3 | 72nd day | 85th day |
| | 4 | 84th day | 100th day |
| | 5 | 96th day | ** |

** For TCHB 213 and Suvin only.

| **Maturity phase(beyond 100 days for hybrids and 88 days for varieties)** | | For all varieties other than Suvin, and TCHB 213 | |
|---|---|---|---|
| Control irriga | 1 | 108th day | 115th day |
| tion during | 2 | 120th day | 130th day |
| maturity phase | 3 | 130th day | |
| | 4 | 144th day | |
| | Stop Irrigation after 150th day | | |
| | For Suvin, TCHB213 | | |
| | 1 | 108th day | 115th day |
| | 2 | 120th day | 130th day |
| | 3 | 132nd day | 145th day |
| | 4 | 144th day | 160th day |
| | 5 | 158th day | ... |
| Stop irrigation after 160th day | | | |

| | |
|---|---|
| *NOTE: i.* | *If irrigation is given on climatological approach, Schedule the irrigation at 0.40 and 0.60 IW/CPE ratio during vegetative and reproductive phases respectively.* |
| *ii)* | *The irrigation schedule given above is only a guideline and regulate the irrigation* |
| | *depending upon the prevailing weather condition and receipt of rains.* |
| *iii)* | *Adopt alternate furrow or skip furrow irrigation to save irrigation water.* |

The features of the methods are furnished below:

**Skip furrow irrigation**

a) Suited to heavy soils like clay and loam
b) Alternate furrows should be skipped and may be converted to ridges having a wide bed formation.
c) Short term crops like pulses may be raised in wider bed without exclusive irrigation.
d) Water saving is 50% when compared to control.

**Alternate furrow irrigation**

a) During any one run of irrigation a particular set of alternate furrows is irrigated.
b) The interval of irrigation should be shortened when compared to the conventional furrows.
c) During the next run, the left over furrows be irrigated.
d) Suited to heavy soils like clay and loam.

## IV. HARVESTING

a) Harvest at frequent intervals, at less than 7 days interval.
b) Harvest in the morning hours upto 10 to 11 a.m only when there is moisture so that dry leaves and bracts do not stick to the kapas and lower the market value.
c) Pick kapas from well burst bolls only.
d) Remove only the kapas from the bolls and leave the bracts on the plants.
e) After kapas is picked, sort out good puffy ones and keep separately.
f) Keep stained, discoloured and insect attacked kapas separately.

*NOTE: Do not mix stained, discoloured and insect damaged kapas with good kapas, as they will spoil the good kapas also and lower the market value of the produce.*

## V. POST HARVEST OPERATIONS

1) Immediately after picking, dry the kapas in shade. If it is not dried immediately the colour will change which will lower the market value.

2) Do not dry the kapas under direct sun as the fibre strength and luster will be lost.
3) Grade the kapas into good and second quality ones, if it is not sorted out at the time of picking.
4) Spread a thin layer of dry sand on the ground and keep the kapas over it.

# RICE FALLOW COTTON

## MANAGEMENT OF FIELD OPERATIONS

**1. PREPARATION OF THE FIELD**

i) If the soil is in waxy condition, instead of Zero tillage, the seed rows may be tilled and the seed dibbled in Virudhunagar district.
ii) If the soil is dry and not in condition to take up sowing, let in water and then allow the soil to dry till soil comes to waxy condition.
iii) At the lower level of the field dig a trench 15 cm wide and connect this trench to the outside channel to drain off the excess water.

**2. PRE-TREATMENT OF ACID DELINTED SEEDS WITH FUNGICIDES**

i) Same as for the irrigated crop.
ii) Treat the acid delinted and fungicide treated seeds with 3 packets (600g) of Azospirillum and sow immediately.

**3. SOWING THE SEEDS**

| Particulars | | MCU 7 | SVPR 3 |
|---|---|---|---|
| a) Seed rate (kg/ha) | | | |
| | i) Fuzzy seed | 15.0 | 15 |
| | ii) Acid delinted | 7.5 | 7.5 |
| b) Spacing (cm) | | | |
| | i) Between rows | 60 | 60 or 75* |
| | ii) Between plants | 30 | 30 |
| c) Number of seeds / hole | | | |
| | i) Fuzzy seeds | 4 | 4 |
| | ii) Acid delinted | 2 | 2 |
| d) Depth of sowing (cm) | | 3 | 3 |

* In fertile soils

**4. FILLING UP GAPS**

i) Fill up gaps on the 10th day of sowing.
ii) Dibble 2 to 3 acid delinted seeds or 4 to 5 fuzzy seeds in the gaps in the case of MCU 7 and SVPR 3

**5. THINNING SEEDLINGS**

i) Thin out seedlings on the 20th day of sowing
ii) Leave only one healthy and vigorous seedling per hill.

**6. WEED MANAGEMENT**

i) Pre-emergence application of Pendimethalin 3.3 l/ha ensures weed free condition for 40 - 45 days. This should be followed by one hand weeding and earthing up during 40 - 45 days.
ii) Take up hoeing and weeding 20 days after sowing.
iii) Take up this operation when the top soil dries up and comes to proper condition.

**7. APPLICATION OF FERTILIZERS**

a) Apply NPK fertilisers as per soil test recommendations. If soil test is not done follow the blanket recommendation of 60:30:30 kg NPK/ha.

b) Apply half the dose of N and K full dose of $P_5O_5$ at 35th day in old delta and balance in 55 days the rows of cotton plants. In the case New delta apply full P and 1/3 of N and K at 20 DAS and 2/3 N and K at 40 DAS.

**8. APPLICATION OF MICRONUTRIENTS**

Apply basally12.5 kg/ha micronutrient mixture prepared by Department of Agriculture. Apply $MgSo_4$ basally @ 20 kg/ha to prevent reddening.

**9. FORMATION OF RIDGES**

**Old delta**

a) If soil is in condition, give a hoeing with mammutti and form ridges and incorporate the fertilizer in the soil around the plants between 30th to 35th day of sowing.
b) If soil is not in condition, give one hoeing and weeding and cover the fertilizers.
c) Form long ridges and furrows from one end of the field to the other without forming any separate channels for carrying water to prevent excessive soaking of water.
d) Form ridges and furrows on alternate rows of plants. Skip furrow method of irrigation to prevent excessive irrigation

**New delta**

a) Give a hoeing with mummutti and form ridges and incorporate the fertiliser in the soil around the plants on the 40th day of sowing.
b) If soil is not in condition give one hoeing and weeding and cover the fertilizers.
c) Form long ridges & furrows on alternate rows of plants to adopt skip furrow irrigation.

*Note: In case of zinc deficient soils, apply 50 kg ZnSo4 /ha*

**10. APPLYING OF NAA**

Spray 40 ppm of NAA (40 mg of NAA dissolved in one litre of water) at 40/45th day using high volume spray. Repeat the same dose after 15 days of first spraying.

**11. TOPPING**

Arrest terminal growth by nipping the terminal 15$^{th}$ node for controlling excessive vegetative growth. (70-75 DAS)

**12. WATER MANAGEMENT**

Regulate irrigation according to the growth phases of the crops.

| Stages | No. of Irrigations | Days after dibbling seeds | |
|---|---|---|---|
| | | Old delta | New delta |
| 1. Vegetative Phase | | | |
| Regulate irrigation during the germination phase | 1 | One wetting on the 30th to 35th day of sowing after the application of fertilisers | One irrigation on the 20th day after the application of fertilisers |
| | 2 | ... | One irrigation on the 40th day after the application of N |
| 2. Flowering Phase | | | |
| Irrigate more frequently | 1 | 45th day of sowing after the application of 2nd dose of N | 45th day |
| | 2 | 55th day | 51st day |
| | 3 | 65th day | 56th day |
| | 4 | 75th day | 61st day |
| | 5 | 85th day | 66th day |
| | 6 | ... | 71st day |
| | 7 | ... | 76th day |
| | 8 | ... | 81st day |
| | 9 | ... | 86th day |

| | | | |
|---|---|---|---|
| | 10 | ... | 91st day |
| 3. Control | 1 | 99th day | 98th day |
| Irrigation during | 2 | 113th day | 105th day |
| maturity phase | 3 | ... | 112th day |

Stop irrigation from the 113th day onwards.

*Note: 1) The irrigation schedule given above is only a guideline and regulate irrigation depending upon the prevailing weather conditions and receipt of rains.*

*2) Observe the crop and if the plants show wilting symptoms in the afternoon and in the evening hours, give an additional irrigation.*

13. Harvesting |
14. Post harvest operation | As that of the irrigated cotton.
15. Pest and disease management |

# RAINFED COTTON

Follow water harvesting techniques and raise a successful crop of cotton.

## I. SEASON AND VARIETIES

For Thirumangalam in Madurai district, Sattur in Virudhunagar district and parts of Kovilpatti in Thoothukudi district, where the seasonal rainfall is 375 mm and most of it is received during September or first week of October, select LRA 5166 (or) SVPR 2 (or) KC 2.

In places where rains are received during October or November, Select K 11 for Ramanathapuram, Virudhunagar, Tirunelveli and Thoothukudi districts.

## II. PREPARATION OF LAND FOR RAINFED COTTON

**1. PREPARATION OF THE FIELD**

i) Start preparation of the land immediately after harvest of the previous crop.
ii) Adopt permanent broad ridges system.

**2. APPLICATION OF FYM OR COMPOST**

i) Spread 12.5 t of FYM or compost or composted coir pith or 2.5t of vermicompost per ha uniformly on the unploughed soil.
ii) Incorporate the manure in the soil by working the multipurpose implement or country plough.

**3. APPLICATION OF INORGANIC FERTILIZERS**

i) Apply NPK fertilizers as per soil test recommendation as far as possible.
ii) If soil tests are not done, follow the blanket recommendations for the different varieties.

| Varieties | Quantity of fertilizers (Kg/ha) | | |
|---|---|---|---|
| | N | $P_2O_5$ | $K_20$ |
| K 11 | 20 | 0 | 0 |
| SVPR 2 | 40 | 20 | 40 |
| KC 2 | 40 | 20 | 40 |

**4. APPLICATION OF MICRONUTRIENT MIXTURE**

i) Mix 12.5 kg of micronutrient mixture formulated by the Department of Agriculture, Tamil Nadu with enough sand to make a total quantity of 50 kg.
ii) Apply uniformly over the furrows after sowing and cover the seeds.
iii) Do not incorporate in the soil.

**5. SEEDS AND SOWING**

i) Adopt the following seed rates for different varieties/hybrids.

| Varieties | Quantity of seeds (kg/ha) | |
|---|---|---|
| | Fuzzy seeds | Delinted seeds |
| K11 | 20 | .. |
| LRA 5166, SVPR 2 | 20 | 15 |

*Note: Delint only LRA 5166 and SVPR2 seeds. Do not delint seeds of K 9, K10 & K 11*

ii) In the case of mixed crop of cotton, maintaining the same seed rates as for a pure crop and adopt the following seed rate for the pulses crop.

| | |
|---|---|
| Blackgram/greengram | 10 kg/ha |
| Cowpea | 7.5 kg/ha |

**6. SPACING**

i) In the case of pure crop of varieties/hybrids, a spacing of 45 cm between rows and 15 cm between plants may be adopted.

ii) In the case of cotton, intercropped with pulses, one paired row of cotton is alternated with two rows of pulses and the total population of cotton crop is maintained at the same line as that for a pure crop of cotton.

| **Varieties** | **Spacing for cotton crop (cm)** | | |
|---|---|---|---|
| | **Within Paired row** | **Between Paired rows** | **Between plants** |
| K 11, LRA 5166, SVPR 2 | 30 | 60 | 15 |

iii) Adopt a spacing of 30 x 10 cm for the pulse crop in between each paired row of cotton. APK 1 Blackgram is best suited for this situation.

**7. ACID DELINTING**

Adopt procedure for acid delinting as for an irrigated crop.

**8. PRETREATMENT OF ACID DELINTED SEEDS WITH FUNGICIDES**

Same as for the irrigated crop.

**9. SOWING**

i) Use the multipurpose farming implement to sow the seeds and to apply basal fertilizers simultaneously.

ii) Fill the hopper in the implement with the fertilizer mixtures and work the implement.

iii) Engage 3 persons for dropping the seeds, 2 for cotton and one for pulses.

In one operation, placement of fertilizer, sowing of seeds and covering will be completed.

*NOTE: Cotton and pulses can be sown at a depth of 5 cm in black cotton soil even before the onset of monsoon rains in dry bed sowing. When light rains are received, the moisture will not penetrate deeper and the seeds will not germinate and die away. Only when good rains are received, the moisture level will be sufficient to penetrate to the level of the seed and facilitate germination and proper establishment.*

**10. WEED MANAGEMENT**

i) Pre-emergence application of Pendimethalin 3.3 l/ha followed by one hand weeding on 40 days after crop emergence. At the time of herbicide application sufficient soil moisture must be there.

ii) If sufficient soil moisture is not available for applying herbicides hand weeding may be given at 10 - 20 days after crop emergence.

**11. GAP FILLING**

Dibble 3 to 4 seeds in each gap if sufficient moisture is available.

**12. THINNING SEEDLINGS**

i) Allow two seedlings per hole and thin out on 15th day of sowing, adopting proper spacing between plants.

ii) Thin the pulse crop on the 20th day of sowing, adopting a spacing of 15 cm between plants for cowpea and 10 cm for other pulse crop.

**13. FOLIAR FERTILIZATION**

Spray 0.5% urea and 1% KCl on the 45th and 65th day of sowing if fufficient moisture is available.

**14. INTERCULTIVATION WITH DHANTHULU/BLADE HARROW**

Work dhanthulu or blade harrow on the 30th and 45th day of sowing.

*NOTE: Other cultivation practices, plant protection measures, harvest etc., are the same as for the irrigated crop.*

# COTTON

## CROP PROTECTION

**A. Pest management**

- Remove the cotton crop and dispose off the crop residues as soon as harvest is over.
- Avoid stacking of stalks in the field.
- Avoid ratoon and double cotton crop.
- Adopt proper crop rotation.
- Use optimum irrigation and fertilizers.
- Grow one variety throughout the village as far as possible.
- Treat the seeds with imidacloprid or use designer seed (Delinted seed + polykote @ 3g/kg + carbendazim @ 2g/kg + imidacloprid @ 7g/kg + *Pseudomonas fluorescens* 10g/kg + Azophos 40g/kg). When the treated seeds are used, it protects against sucking pests upto 45 days after sowing and promotes early vigour of the crop
- Synchronise the sowing time in the villages and complete the sowing within 10 to 15 days.
- Avoid other Malvaceous crops in the vicinity of cotton crop.
- Timely earthing up and other agronomic practices should be done.
- Hand pick and burn periodically egg masses, visible larvae, affected and dropped squares, flowers and fruits and squash pink bollworm in the rosettes.
- Use locally fabricated light traps (modified Robinson type) with 125 Watt mercury lamps to determine the prevalence and insect population fluctuations.
- The magnitude of the activity of the moths of the cotton pink bollworm, the cutworm (*Spodoptera litura*) and the American bollworm can be assessed by setting up the species-specific sex pheromone trap each at the rate of 12 per ha.
- Apply insecticides only where it is absolutely necessary when pest population or damage reaches ET level.
- Intercropping with pulses viz., cowpea, greengram, blackgram, soybean and maize reduces the bollworm incidence and population of sucking pests of cotton, *viz*., aphid and leafhopper with the highest activity of natural enemies viz., spiders and predatory lady bird beetles.

**Economic threshold level for important pests**

| Pests | ETL |
|---|---|
| Thrips | 50 nymphs or adults/50 leaves |
| Aphids | 15% of infested plant |
| Leafhopper | 50 nymphs or adults/50 leaves |
| Mite | 10 mites/$cm^2$ leaf area |

| | |
|---|---|
| Boll-worms | |
| Spotted | 10% infested shoots / squares / bolls |
| Spiny | 10% infested shoots / squares / bolls |
| Pink | 10% infested fruiting parts |
| *Helicoverpa* | One egg or one larva /plant |
| Whiteflies | 5 - 10 /leaf |
| Stem weevil | 10% infestation |
| Tobacco cutworm | 8 egg masses/100 m row |

**Pest management strategies**

| Pests | Management strategies |
|---|---|
| **American bollworm**<br>*Helicoverpa armigera* | ▪ **Monitoring :**<br>Pest monitoring through light traps, pheromone traps and *in situ* assessments by roving and fixed plot surveys has to be intensified at farm, village, block, regional and State levels. For management, an action threshold of one egg per plant or 1 larva/ plant may be adopted.<br><br>**Cultural practices :**<br>▪ Synchronised sowing of cotton preferably with short duration varieties in each cotton ecosystem.<br>▪ Avoid continuous cropping of cotton both during winter and summer seasons in the same area as well as ratooning.<br>▪ Avoid monocropping. Growing of less preferred crops like greengram, blackgram, soyabean, castor, sorghum etc., along with the cotton as intercrop or border crop or alternate crop to reduce the pest infestation.<br>▪ Removal and destruction of crop residues to avoid carry over of the pest to the next season, and avoiding extended period of crop growth by continuous irrigation.<br>▪ Optimising the use of nitrogenous fertilisers which will not favour the multiplication of the pest.<br>▪ Judicious water management for the crop to prevent excessive vegetative growth and larval harbourage.<br>**Biological control :**<br>▪ Application of Nuclear Polyhedrosis Virus (NPV) at 3 x $10^{12}$ POB /ha in evening hours at 7th and 12th week after sowing.<br>▪ Conservation and augmentation of natural predators and parasitoids for effective control of the pest.<br>▪ Inundative release of egg parasitoid, *Trichogramma* spp., at 6.25 cc/ha at 15 days interval 3 times from 45 DAS<br>▪ Egg-larval parasitoid, *Chelonus blackburnii* and<br>▪ Predator *Chrysoperla carnea* at 1,00,000/ha at 6$^{th}$, 13$^{th}$ and 14$^{th}$ week after sowing.<br>▪ ULV spray of NPV at 3 x $10^{12}$ POB /ha with 10% cotton seed kernel extract, 10% crude sugar, 0.1% each of Tinopal and Teepol for effective control of *Helicoverpa*.<br>Note: Dicofol, endosulfan, methyl demeton, monocrotophos and phosalone are comparatively safer to *Chrysoperla* larva recording low egg mortality.<br>▪ **Chemical control :**<br>▪ Discourage the indiscriminate use of insecticides, particularly synthetic pyrethroids.<br>▪ Use of proper insecticides which are comparatively safer to natural enemies such as endosulfan, phosalone, etc., at the |

| | |
|---|---|
| | correct dosage and alternating different groups of insecticides for each round of spray.<br>▪ Avoid combination of insecticides as tank mix.<br>▪ Adopt proper delivery system using spraying equipments like hand compression sprayer, knapsack sprayer and mist blower to ensure proper coverage with required quantity of spray fluid and avoid ULV applications or Akela spray applications.<br>▪ Proper mixing and preparation of spray fluid for each filling of spray fluid tank.<br>▪ At early stages of square formation apply endosulfan 35 EC 2 l/ha.<br>▪ **During bolling and maturation stage, apply any one of the following insecticides (1000 l of spray fluid/ha):**<br>Phosalone 35 EC 2.5 l/ha<br>Quinalphos 25 EC 2.0 l/ha<br>Carbaryl 50 WP 2.5 kg/ha<br>Pyraclofos 50 EC 1.5 l/ha |

| | |
|---|---|
| **Pink bollworm**<br>*Pectinophora gossypiella* | ▪ Use pheromone trap to monitor the adult moth activity<br>▪ Spray triazophos 40 EC 2.5l/ha and endosulfan 35 EC 2.0 l/ha in alternation even after 100 DAS. |
| **Tobacco cutworm**<br>*Spodoptera litura* | ▪ Use of light trap to monitor and kill the attracted adult moths. Set up the sex pheromone trap Pherodin S.L. at 12/ha to monitor the activity of the pest and to synchronise the pesticide application, if need be, at the maximum activity stage.<br>▪ Growing castor along border and irrigation bunds.<br>▪ Removal and destruction of egg masses in castor and cotton crops.<br>▪ Removal and destruction of early stage larvae found in clusters which can be located easily even from a distance.<br>▪ Collection and destruction of shed materials.<br>▪ Hand picking and destruction of grown up caterpillars.<br>▪ **Spray any one of the following insecticides per ha using, a high volume sprayer covering the foliage and soil surface :**<br>Chlorpyriphos 20 EC 2.0 l<br>Dichlorvos 76 WSC 1.0 l<br>Phenthoate 50 EC 2.0 l<br>Chlorpyriphos 20 EC 1.25 l<br>Fenitrothion 50 EC 625 ml<br>▪ Spraying nuclear polyhedrosis virus at 1.5 x $10^{12}$ POB per ha.<br>▪ Spraying of insecticides should be done either in the early morning or in the evening and virus in the evening.<br>▪ Use of poison bait pellets prepared with rice bran 12.5 kg, jaggery 1.25 kg, carbaryl 50% WP 1.25 kg and water 7.5 litres. This bait can be spread in the fields in the evening hours so that the caterpillars coming out of the soil, feed and get killed. |
| **Stem weevil**<br>*Pempherulus affinis* | ▪ Soil application of carbofuran 3 G 30 kg/ha on 20 days after sowing and earthing up on $45^{th}$ day.<br>▪ Basal application of FYM 25 t/ha and 250 kg/ha of neem cake. |
| **Whitefly**<br>*Bemisia tabaci* | ▪ Avoid the alternate, cultivated host crops of the white fly in the vicinity of cotton crop.<br>▪ Growing cotton only once a year either in winter or summer season in any cotton tract.<br>▪ Adopting crop rotation with non-preferred hosts such as sorghum, ragi, maize etc., for the white fly to check the build up of the pest. |

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| | ▪ Removal and destruction of alternate weed hosts like *Abutilon indicum*, *Chrozophore rottlari*, *Solanum nigrum* and *Hibiscus ficulensus* from the fields and neighbouring areas and maintaining field sanitation.<br>▪ Timely sowing with recommended spacing, preferably wider spacing and judicious application of recommended dose of fertilizers, particularly nitrogenous and irrigation management is essential to arrest the excessive vegetative growth and pest build up. Late sowing may be avoided and the crop growth should not be extended beyond its normal duration.<br>▪ Field sanitation may be given proper attention.<br>▪ Cultivation of most preferred alternate host crops like brinjal, bhendi, tomato, tobacco and sunflower may be avoided. In case their cultivation is unavoidable, plant protection measures should be extended to these crops also.<br>▪ Monitoring the activities of the adult white flies by setting up yellow pan traps and sticky traps at 1 foot height above the plant canopy and also in situ counts.<br>▪ Collection and removal of whitefly infested leaves from the plants and those which were shed due to the attack of the pest and destroying them.<br><br>▪ **Chemical control :**<br>**Spray any one of the following plant products alone or in combination with the recommended dose of insecticide (at 2 ml/l of water)**<br>Neem seed kernel extract 5% (50 kg) and neem oil at 5 ml/l of water<br>Fish oil rosin soap 25 kg at 1 kg in 40 lit of water<br>Notchi leaves 5% extract<br>*Catharanthus rosea* extract 5%<br>▪ **Spray any one of the following in early stage** (500 l of spray fluid/ha)<br>Methyl demeton 25 EC 500 ml<br>Phosphamidon 40 SL 600 ml/ha<br>▪ **Spray any one of the following in mid and late stages** (1000 l spray liquid/ha)<br>Phosalone 35 EC at 2.5 l/ha<br>Quinalphos 25 EC at 2.0 l/ha<br>Ethion 50 EC 1.0 l/ha<br>Monocrotophos 36 SL 1.25 l/ha<br>Triazophos 40 EC 2.0 l/ha<br>Acephate 75 SP 1.30 kg/ha<br>▪ In the early stages with high volume sprayer, use a goose neck nozzle to cover the under surface of the foliage to get good control of the pest. If high volume sprayers are not available, 375 litres of spray fluid may be used per hectare for application in the low volume motorised knapsack mist blower.<br>▪ The use of synthetic pyrethroids should be discouraged in cotton to avoid the problem of whitefly. Cypermethrin, fenvalerte and deltamethrin cause resurgence of whiteflies. So avoid repeated spraying of pyrethroids.<br>▪ The plant protection measures should be adopted on a community basis in a specified cotton areas. |
| **Thrips**<br>*Thrips tabaci* | ▪ Seed treatment with imidacloprid 70 WS at 7 g/kg protect the crop from aphids, leafhoppers and thrips upto 8 weeks. |

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| **Aphids**<br>*Aphis gossypii* | **Spray any one of the following insecticides** (500 l spray fluid/ha) |
| **Leafhopper**<br>*Amrasca devastans* | Imidacloprid 200 SL at 100 ml/ha<br>Methyl demeton 25 EC 500 ml/ha<br>Dimethoate 30 EC 500 ml/ha<br>Phosphamidon 40 SL 600 ml/ha<br>Monocrotophos 36 SL 1000 ml/ha<br>NSKE 5% 25 kg/ha<br>▪ Where the leafhopper is a big menace apply Neem oil formulation 0.5 % or neem oil 3% thrice at fortnightly intervals |
| **Red spider mite**<br>*Tetranychus cinnabarinus* | ▪ **Apply any one of the following:**<br>Wettable sulphur 1.25 kg/ha<br>Dicofol 1.10 l/ha |

**Insecticide resistance**

In case of control failures monitor the insecticide resistance with following discriminating dose screen.

A. *Helicoverpa armigera*

(Topical assay with III instar larva weighing 30-40 mg)

1. Cypermethrin 0.1 µg/µl
2. Cypermethrin 1.0 µg/µl
3. Fenvalerate 0.2 µg/µl
4. Endosulfan 10 µg/µl
5. Quinalphos 0.75 µg/µl
6. Chlorpyriphos 1.0 µg/µl

B. Tobacco caterpillar - *Spodoptera litura*

(early III instar 8 day old larva weighing 30-40 mg and measuring 12±0.5 mm length)

1. Endosulfan topical 2.0 µg
2. Profenofos topical 3.0 µg
3. Chlorpyriphos topical 0.15 µg
4. Fenvalerate topical 0.2 µg/µl

C. Cotton leafhopper- *Amrasca devastans* (Distant)

(III instar larva of 0.14 mg weight, 1.30mm length)

1. Dimethoate IRAC method VIII (leaf disc) 400 ppm
2. Methyl demeton IRAC method VIII (leaf disc) 800 ppm
3. Acephate IRAC method VIII (leaf disc) 850 ppm

**Resurgence**

Repeated application of the following insecticides can cause resurgence of the insect pest of Cotton

- *Aphis gossypii : C*arbaryl, cypermethrin, deltamethrin, endosulfan, fenpropathrin, fenvaerate, flucythrinate, fluvalinate, monocrotophos, permethrin, phorate
- *Amrasca devastans : D*eltamethrin, dimethoate, disulfoton, methylparathion, phorate
- *Bemisia tabaci : C*ypermethrin, deltamethrin, dimethoate, endosulfan, fenvalerate, monocrotophos, phosalone
- *Ferrisia virgata : C*ypermethrin, deltamethrin, fenvalerate, permethrin
- *Tetranychus cinnabarinus : A*cephate, carbaryl, cypermethrin, deltamethrin, endosulfan, fenvalerate, fluvalinate, phosphamidon.

# RAINFED COTTON

## CROP PROTECTION

### A. Pest management

- The control measures recommended for irrigated cotton will hold good.
- **When water is not available, use any one of the following insecticides for the control of bollworms at 25 kg/ha :**
  Endosulfan 4 D
  Carbaryl 5 D
  Phosalone 4 D

### B. Disease management

| Name of the Disease | ***Management*** |
|---|---|
| Bacterial leaf blight<br>***Xanthomonas axonopodis*** **pv.** ***malvacearum*** | ▪ ***Avoid stacking of infected plants***<br>▪ Spray Streptomycin sulphate + Tetracycline mixture 100g + Copper oxychloride 1250g/ha<br>▪ Repeat spraying at 10 days interval twice or thrice if drizzling continues. |
| Alternaria leaf spot<br>***Alternaria macrospora*** | ➢ ***Spray any one of the following:***<br>▪ ***Copper Oxychloride 1250g***<br>▪ ***Mancozeb 1000g***<br>▪ ***Chlorothalonil 500g/ha*** |
| Grey Mildew<br>***Ramularia areola*** | ▪ *Spray Carbendazim 250 g/ha* |
| Boll rot<br>*Fusarium moniliforme, Colletotrichum capsici, Aspergillus flavus, A. niger, Rhizopus nigricans, Nematospora, Botryodiplodia* | ➢ ***Spray any one of the following:***<br>▪ *Spray Carbendazim 500g,*<br>▪ *Mancozeb 2000g,*<br>▪ ***Copper oxychloride 2500g/ha along with an insecticide recommended for bollworm from 45$^{th}$ day at fortnightly interval.*** |
| Root rot<br>*Macrophomina phaseolina Rhizoctonia bataticola* | ❖ **Cultural Method**<br>▪ Apply Neem cake @ 150 kg/ha to the soil and treat the seeds with talc based *T. viride* @ 4 g/kg to reduce the root rot incidence.<br>❖ **Biological control**<br>▪ Seed treatment with *T. viride* @ 4 g/kg followed by basal application of zinc sulphate @ 50 kg/ha<br>❖ **Chemical**<br>▪ Spot drench Carbendazim @ 1 g/lit at the base of affected plants as well as surrounding healthy plants. |

# JUTE ( *Corchorus olitorius* & *Corchorus capsularis* )

## CROP MANAGEMENT

Jute can be successfully grown in Coimbatore, Cuddalore, Villupuram, Vellore, Tiruvannamalai, Chengleput and parts of Thanjavur, Tiruvarur, Nagapattinam, Tiruchirapalli, Perambalur, Karur, Pudukkottai and Tirunelveli, Thoothukudi districts where assured supply of irrigation water is available for its cultivation and retting for fibre extraction.

**Soil type**: Alluvial sandy loam, clay loamy soils are best suited for jute production. Capsularis jute can grow even in standing water especially towards the latter part of its growth, but Olitorius jute will not thrive in standing water. The latter is more drought resistant and is therefore grown on lighter soils.

**Season**: February

**Land Preparation**: Fine tilth is required since the seeds are very small.

**Manures and fertilizer application**: Five tonnes of well decomposed farm yard manure is to be applied during last ploughing. Besides 20 kg per ha each of N, $P_2O_5$ and $K_2O$ are to be applied.

basally. Beds and channels are formed depending on water resources.

**Varieties:** Capsularis JRC 212, JRC 321, JRC 7447

Olitorius j JRO 524, JRO 878, JRO 7835

**Crop duration** : 120 to 140 Days

**Seed rate and sowing**: Seeds can be sown either by broadcasting or by line sowing.

| Jute type | Seed rate (kg/ha) | | Spacing (cm) | No. of Plants/ |
|---|---|---|---|---|
| | Line Sowing | Broad Casting | | Sq. Mtr. |
| Olitorius | 5 | 7 | 25 x 5 | 80 |
| Capsularis | 7 | 10 | 30 x 5 | 67 |

**Weed management**: Hand weeding twice on 20 - 25 DAS and 35 - 40 DAS. Fluchloralin can be sprayed at 3 days after sowing at the rate of 1.5 kg per hectare and is followed by irrigation. Further one hand weeding can be taken up at 30 - 35 DAS.

**Top dressing of fertiliser**: Apply 10 kg of N at 20 - 25 days after first weeding and then again on 35 - 40 days after second weeding as top dressing. During periods of drought and fertilizer shortage, spray 8 kg of urea as 2 per cent urea solution (20 g urea in one litre of water) on jute foliage on 40 - 45 as well as 70 - 75 DAS.

**Water Management**: Jute crop requires 500 mm of water. First irrigation is to be given after sowing and life irrigation on fourth day after sowing. Afterwards irrigation can be given once in 15 days.

**Harvest**: Jute crop can be harvested from 100 to 110 DAS but can be extended from 120 - 135 DAS depending on local cropping systems. Jute plants are left in the field for 3 - 4 days for leaf shedding. Then thick and thin plants are sorted out and bundled in convenient size.

**Yield**: The green plant weight yield is 45 to 50 tonnes per hectare whereas the fibre yield is 2.0 to 2.5 tonnes per hectare.

## AGAVE – CULTIVATION

Among the under-exploited resources, 'Agave' - a fibre yielding drought tolerant plant is one which can prosper the life of the dry land farmers without any risk. Agave is a short stemmed plant bearing a rosette of long erect pointed fleshy leaves. Agave is noted for its strong, coarse fibre, superior too and more flexible than Manila hemp. It is widely used for making ropes, cordage, twine, fishing nets, door mats and rugs and the short fibres are used for making mops, brushes. The waste material left after decorticating the leaves is used for making craft paper and paper boards. The fibres also contain about 73-78% of lignified form of cellulose. Apart from these wax from agave wastes and Hecogenin acetate a steroid useful for the pharmaceutical industry in India is obtained from agave juice. The genus Agave has about 275 species of which *A. sisalana, A.cantala* and *A.Americana* commonly occur in India. Agave blossoms only once during its life time and then dies. Agave plants are grown along railway line, road sides, river banks and as a hedge plant in dryland areas throughout the country. Till date it is grown in patches and as border crop in a neglected condition. The crop comes up on dry soils unsuitable for crop cultivation but grow vigorously on dry, well drained sandy loam soils.

**Nursery**: Agaves are usually propagated from bulbils or suckers. Grown up suckers can be dug out and planted during rainy months. In case of bulbils they are first sown in mother beds at close spacing @ 5000 bulbils per bed of 1 x 20 M. After 6 months the seedlings are pulled out and planted in the transplanting bed of size 20 x 1 m @ 500 Plants. In the second stage it is kept for three months. After 9 months from the date of planting bulbils suckers weighing 1/4 to 1/2 kg and 9 to 12" height are ready for planting.

**Main field planting**: In the main field they are planted at a space of 2 x 2 m in pits of size 30 $cm^3$. Planting is usually carried out during the rainy seasons for better establishment otherwise initial watering is quite essential for establishment.

**Harvesting**: The leaves are ready for harvesting from 3rd year onwards. The older leaves of length not less than a metre is harvested in the 3rd year. Each plant yields 40-50 leaves/year. The life cycle of the plant is upto 8 years. The content of fibre varies with variety from 2.5 to 4.5% and the highest

is reported under *A.sisalana* as 4.5%. Agave sisalana produces a better quality fibre than *Agave americana*. From 3rd year onwards leaf yield of 30-40 tonnes/acre could be harvested and a net profit of not less than Rs.2000/acre could be obtained out of this crop. Even as border crop Agave could fetch a revenue of not less than Rs.400/acre/year from 3rd year until 8th year. Instead of traditional hand scraping process now mechanical decorticators are available for the extraction of fibre. The extracted fibres are washed in water, cleaned and dried and packed in bales. As a precaution, fibre extraction is done on a bright sunny day and within 2 days of the harvesting of the leaves or else the quality of the fibre will be deteriorated. The fibre colour varies from milky white to golden yellow.

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